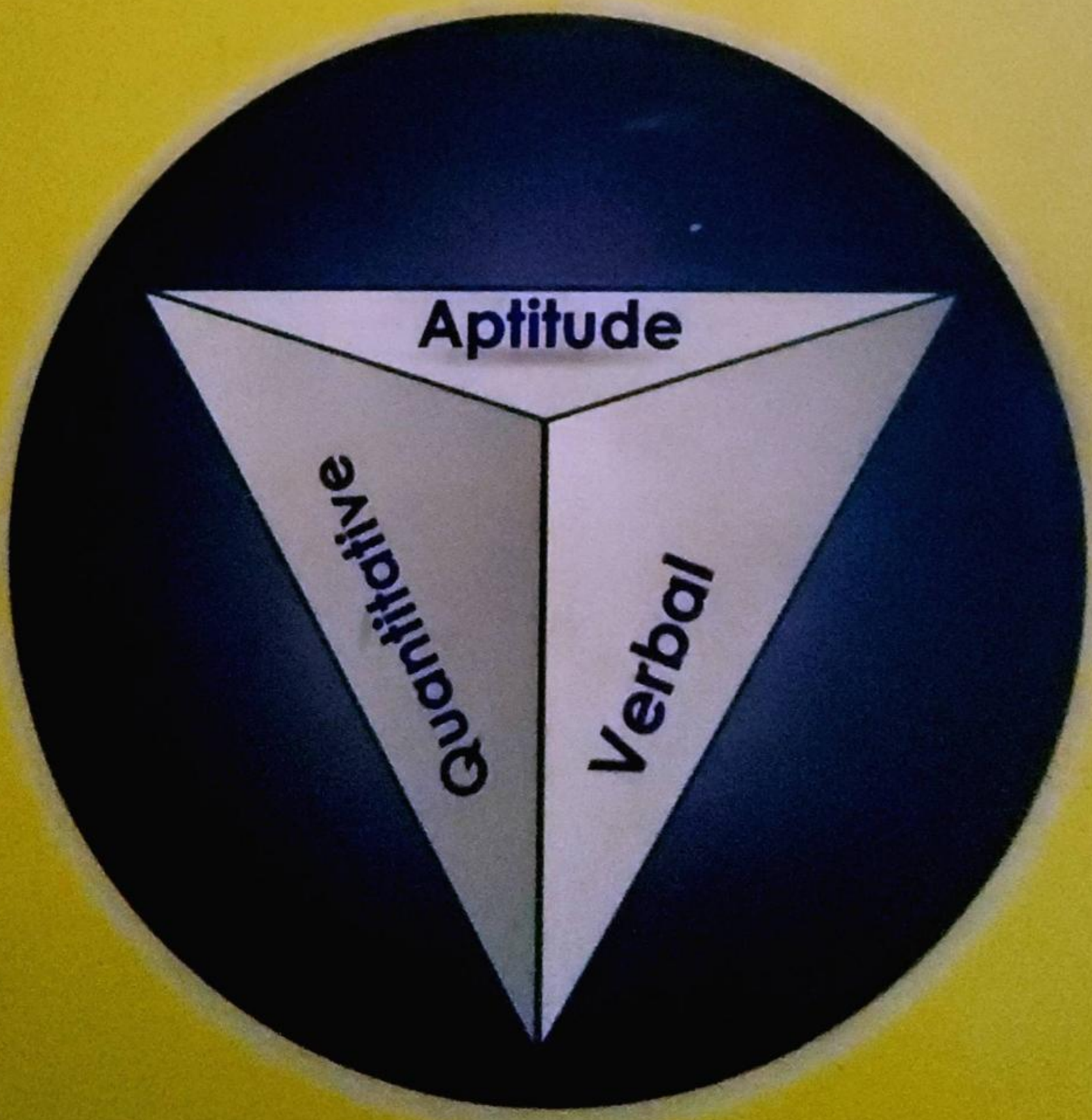
Extreme Series

Scholastic Aptitude Test

University Entrance Exam



Resume/CV Plus Job Interview

Extreme

(312)
2415

Extreme

Scholastic Aptitude Test

SAT

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1

SCHOLASTIC APTITUDE TEST (SAT)

A. What does the acronym (SAT) stand for?

S = refers to **Scholastic**

(i) Scholastic (adj)

- ✓ Your **scholastic** achievement or ability is your academic achievement or ability while you are at school.

(ii) Aptitude (N)

- ✓ Your aptitude for a particular kind of work or activity is your ability to learn it quickly to do it well.
- ✓ is a natural ability to acquire knowledge or skill.
- ✓ aptitude = talent /knack /appropriateness) suitability.

(iii) Aptitude Test

- ✓ Is a test that is specially designed to find out how easily and how well you can do something.
- ✓ Is one designed to show whether somebody has the natural ability for a particular job or educational course.
  E.g.: She showed a natural **aptitude** for the work.

B. Purpose of Aptitude Test

- Most colleges requires applicants to take entrance examinations. These include **aptitude tests** that measure basic skills used in many different subject areas or **achievement tests** that measure knowledge of a specific subject (eg. history, Biology…).
- So, colleges /schools use score from such examinations to **determine whether applicants (students)** meet admission requirements or should receive college credit for work completed in high school /preparatory.

C. How to Prepare for SAT

- The **SAT (Scholastic Aptitude Test)** is divided into sections on **verbal** and **mathematical** abilities. **SAT** is a type of test which requires talent and skills that are gained over a period of many years. Consequently, it is impossible to study for them in the usual sense. In other words, you can't simply sit down a few days before one of these exams and learn everything that you need to know for them. Instead, you must begin to develop the skills that these tests measure **(SAT)** i.e., your **vocabulary (antonym and synonym), analogy,** your **reading** and **computational skills,** your **ability to think reasonably** and **critically,** and your **understanding of the basic principles of English.**

  ☞ To prepare for university entrance examinations, familiarize yourself with the types of questions that they contain. These include **antonyms, synonyms, analogies, sentence completion, reading comprehension and usage questions.**

PART ONE

QUANTITATIVE APTITUDE/ REASONING

- *Arithmetic section*
  - ✓ Number system
  - ✓ Fraction
  - ✓ Percent, ratio, proportion
  - ✓ Average and Data Interpretation.
  - ✓ Word – problem
  - ✓ Probability
  - ✓ Counting Principle
  - ✓ Algebraic Expression
- *Geometry section*
  - ✓ Triangle
  - ✓ Area
  - ✓ Volume

PART TWO

VERBAL APTITUDE / REASONING

- ✓ Antonyms
- ✓ Analogy
- ✓ Synonyms
- ✓ Sentence completion
- ✓ Language usage

Part I: QUANTITATIVE APTITUDE / REASONING

Unit One: NUMBER SYSTEM 1

1.1 Composite numbers (c) 1

1.2 Divisibity of a number 1

1.3 Division algorithm 6

1.4 Number of divisors 10

1.5 Integers 12

1.6 Least common multiple (lcm) of integers 16

1.7 Greatest common factor (gcf) (or greatest common divisor) 17

1.8 Exponents 19

1.9 The arithmetic of inequalities 22

- Diagnostic/Review Test 34
- Answer and Explanation 37

Unit two: FRACTIONS 40

2.1 Equivalent fraction 40

- Diagnostic Test 51
- Answer and explanation 53

Unit three: PERCENT'S, RATIO AND PROPORTION 55

3.1 Percent's 55

3.2 **Percent Increase and Decrease** 60

3.3 **Ratio** 68

3.4 **Proportion** 71

3.5 Direct and inverse proportionality 71

- Review Exercise 75
- Answer Explanations 77

Unit four: AVERAGE 81

4.1 Average Speed 83

4.2 Data interpretation 86

- Review Exercise 92
- Answer and Explanation 92

Unit five: WORD PROBLEMN 95

5.1 Age problems 95
5.2 Distance problems 96
5.3 Work problems 100
5.4 Mixture Problems 102
5.5 Interest Problems 103
5.6 Miscellaneous collection of word problems 105
5.7 Discount 107
5.8 Profit 108

Unit Six: COUNTING AND PROBABILITY 109

6.1 Counting principle 109
6.2 Fundamental principle of counting 113
6.3 Permutation and combination 115
6.4 Combinations 117
6.5 Pro ability of an event 118
➢ Review Exercise 125
➢ Answer and Explanation 126

Unit Seven: ALGEBRAIC EXPRESSIONS 128

7.1 Operations of algebraic expressions 128
7.2 Solving equations 132
➢ Review Exercise 140
➢ Answerer and Explanation 142

Unit Eight: GEOMETRIC APTITUDE 144

8.1 Triangles 144
8.2 Solid geometry 155
8.3 Rhombus 158
➢ Review Exercise 158
➢ Answer and Explanation 159

Part II: Verbal Aptitude and Quantitative aptitude/Reasoning

Unit One: ANTONYMS 160
Unit two: SYNONYMS 225
Unit three: ANALOGIES 254
Unit four: SENTNCE COMPLETION 339
Unit five: LANGUAGE USAGE 387

Unit Nine: Resume/CV Plus Job Interview 410

9.1 Resume/Cv 410

9.2 Cover/Covering Letter 411

9.3 Job interview 412

- Preparatin and practice 412

9.4 Acting interview 415

- Act one (Entrance) 415
- Act two (performance) 415
- Act three (The exist) 434

UNIT – ONE

1. NUMBER - SYSTEM

Prime Numbers (P):
A natural number, which is greater than 1 and divisible by 1 and by itself only, is called a **prime number**

Prime numbers (P) = 2, 3, 5, 7, 11, 13, 17, ...

i. The smallest prime number is 2
ii. Except 2; all other prime numbers are odd.
iii. There are infinite prime numbers

1.1 COMPOSITE NUMBERS (C):

A natural numbers, which is greater than 1 and is not prime, is called **composite number**.

∴ Composite number (C) = 4, 6, 8, 9, 10, 12, 14, 15, ...

i. The smallest composite number is 4.
ii. A composite number can be even or odd
iii. There are infinite composite number.

Note: 1 and 0 are neither prime nor composite.

1.2 DIVISIBILITY OF A NUMBER

An integer is divisible:

- by 2 if it is even
- by 3 if the sum of its digits is divisible by 3

Example:2,145 is divisible by 3 because 2+1+4+5 =12 is divisible by 3.

Divisibility test for 4.

An integer is divisible by 4 if and only if, the last two digits of the integer is divisible by 4

Example:

a. 57,658 is divisible by 4, because 28 is divisible by 4.
b. 1,172 is divisible by 4, because 72 is divisible by 4.

Divisibility test for 5

An integer is divisible by 5; if and only if, its unit digit is divisible by 5, that is, if and only if the unit digit is 0 or 5.

Example:

a. 2,365 is divisible by 5, because the unit digit is 5.
b. 1270 is divisible by 5, because the unit digit is 0.

Divisibility test for 6

An integer is divisible by 6; if and only if the integer is divisible by both 2 and 3.

Example: 7,134 is divisible by 6, because 4 is divisible by 2 also
$7 + 1 + 3 + 4 = 15$ and 15 is divisible by 3
$\therefore$ 7, 134 is divisible by 6.

Divisibility Test for 7.

To determine whether a given number is divisible by 7, double the unit digit of the given number. Find the difference between this number and the number formed by omitting the unit digit from the given number. If necessary, repeat this procedure until you obtain a small difference. If the final difference is divisible by 7, then the given number is also divisible by 7. If the final difference is not divisible by 7, then the given number is not divisible by 7.

Illustrative Example

1. Determine whether each of the following number is divisible by 7
   a. 182 b. 1001 c. 273

Solution:

(a) To test divisibility of 182 for 7:

- Double of the unit digit: $2 \times 2 = 4$
- The number formed by omitting the unit digit is 18
- Subtract 4 from 18: $18 - 4 = 14$

Because 14 is divisible by 7, thus the original number 182 is divisible by 7.

(b) To test divisibility of 1001 for 7:

- Double of the unit digit yields $1 \times 2 = 2$
- Subtract 2 from 100 yields $100 - 2 = 98$
- Again, double the unit digit of 98. Yields $8 \times 2 = 16$

- Again subtract 9 from 16, yields $16 - 9 = 7$, because 7 is divisible by 7, therefore, 1001 is divisible by 7.

(c) To test divisibility of 273 for 7:

- Double of the unit digit, yields $2 \times 3 = 6$

The number formed by omitting the unit digit is 27

- Subtract 6 from 27, yields $27 - 6 = 19$

Because the final result 19 is not divisible by 7, thus 273 is **not** divisible by 7.

Divisibility Test for 8

A number is divisible by 8 if the last three digits of the number form a number that is divisible by 8

Illustrative Example

2. Determine whether each of the following numbers are divisible by 8

a. 97,136 b. 19,168 c. 16,278

Solution:

a. The last three digit of 97,136 form the number 136 which is divisible by 8.Thus, 97. 136 is divisible by 8.

b. The last three digit of 19, 168 form the number 168 which is divisible by 8.Thus, 19,168 is divisible by 8

c. The last three digit of 16,278 form the number 278, which is not divisible by 8. Thus 16,278 is not divisible by 8

Divisibility Test for 9

A number is divisible by 9 if the sum of all the digits of the number is divisible by 9

Illustrative Example

3. Determine whether each of the following number is divisible by 9.

a. 621,513 b. 732,624 c. 833,805

Solution:

a. The sum of the digit, $6 + 2 + 4 + 5 + 1 + 3 = 18$ and 18 is divisible by 9. Thus, 621, 513 is divisible by 9

b. The sum of the digit, $7 + 3 + 2 + 6 + 2 + 4 = 24$ and 24is not divisible by 9. Thus 732624 is not divisible by 9.

c. The sum of the digit $8 + 3 + 3 + 8 + 0 + 5 = 27$ and 27 is divisible by 9. Thus, 833, 805 is divisible by 9

A divisibility test for 13.

To determine whether a given number is divisible by 13, multiply the unit digit of the given number by 4. Find the sum of this multiple of 4 and the number formed by omitting the unit digit from the given number. If necessary, repeat this procedure until you obtain a final sum. If the final sum is divisible by 13, then given number is divisible by 13.

Illustrative Example

4. Determine whether each of the following number is divisible by 13.
   a. 1079 b. 1885 c. 14,507

To test divisibility by 13:

a. 1079: Four times the unit digit, yield $4\times9 = 36$. The number formed by omitting the unit digit is 107.
   - The sum of $36 + 107 = 143$
     Now repeat the procedure on 143.
   - Four times the unit digit yield $4\times3 = 12$
   - The sum of 14 and $12 = 26$

   The final sum 26 is divisible by 13.
   Thus 1079 is divisible by 13.

b. 1885 ⟵ Exercise left for you

c. 14,507 ⟵ Exercise left for you.

Divisibility Test for 11.

A natural number is divisible by 11, if and only if, the sum of the digit in the places that are even power of 10 **minus** the sum of the digits in the places that are odd powers of 10 is divisible by 11.

In short: To test divisibility for 11:
start at one end of the number and compute:
(The sum of every other digit) – (the sum of the remaining digits)

Illustrative Example

5. Determine whether each of the following number is divisible by 11.
   a. 57,729,364,583. b. 8, 471, 986 c. 4807

Solution:

To test divisibility for 11:

a. (Sum of every other digit) – (Sum of the remaining digit)
   $= (5 + 7 + 9 + 6 + 5 + 3) - (7 + 2 + 3 + 4 + 8)$
   $= 35 - 24 = 11$ ⟵ divisible by 11
   Thus 57, 729, 364, 583 is divisible by 11.

b. (8 + 7 + 9 + 6) (4 + 1 + 8) = 30 13 = 17 and 17 is not divisible by 11. Thus 8,471, 986 is not divisible by 11

c. (7 + 8) (4 + 0) = 11 and 11 is divisible by 11.
Thus 4807 is divisible by 11.

6. What least number must be added to 1056, so that the sum is completely divisible by 23?

A. 21 B. 2 C. 3 D. 18

Hint: 1056 = (23) (45) + 21
1058 = (23) (46), **Answer: B**

7. The greatest number of four digits which is divisible by 15, 25, 40 and 75 is

A. 59600 B. 9000 C. 9400 D. 9800

Answer: A

Conceptual Example

8. (b) If the remainder is 2, when a number n is divided by 7, what is the remainder when 3n is divided by 7

A. 2 B. 3 C. 6 D. 5

Solution:

- A number n divided by 7 with remainder 2 is expressed by $n = 7K + 2$, where, $K = 1, 2, 3, \ldots$
- When $n \longrightarrow 3n$, will be $3n = 21k + 6$

Note: For any integer d, x, and y, the following holds:

a. If d is divides by x and y, then:
   i. d is divisible by $x + y$
   ii. d is divisible by $x - y$
   iii. d is divisible by $ax + by$.

If d is divisible by xy then d is divisible by x and y

Answer: C

Illustrative Example

9. Classify each of the following as True or False:

a. If a number is divisible by 6, then it is divisible by 2 and by 3.
b. If a number is divisible by 2 and 3, then it is divisible by 6.
c. If a number is divisible by 8; then it is divisible by 2 and 4.
d. If a number is divisible by 2 and 4, then it is divisible by 8.
e. If a number is divisible by 15, then it is divisible by 3 and 5.

Solution: a. True
b. True
c. True

d. False, for example 12 divisible by 2 and 4 but 12 is not divisible by $\longrightarrow 8$

e. True

10. If $x = 14 \times 22 \times 39$, which of the following is NOT an integer?

A. $x \div 21$ C. $x \div 26$

B. $x \div 24$ D. $x \div 77$

Solution: The easiest way to answer is break x into prime factor.

Thus $x = 2 \times 7 \times 2 \times 11 \times 3 \times 13$

A. $\dfrac{x}{21} = \dfrac{2\times7\times2\times11\times3\times13}{3\times7} = 2\times2\times11\times13$

B. $\dfrac{x}{24} = \dfrac{2\times7\times2\times11\times3\times13}{2\times2\times2\times3} = \dfrac{7\times11\times13}{2}$

C. $\dfrac{x}{26} = \dfrac{2\times7\times2\times11\times3\times13}{2\times13} = 7\times2\times11\times3$

D. $\dfrac{x}{77} = \dfrac{2\times7\times2\times11\times3\times13}{7\times11} = 2\times2\times3\times13.$

Answer: B

1.3 DIVISION ALGORITHM

If x and y are positive integers, when **y** is divided by x, there exist unique integers q and r, called the quotient and remainder respectively, such that $\mathbf{y = xq + r}$ and $0 \le r < x$

Illustrative Example

11. Suppose $P = 173 \times 34 + 40$. **Find** the remainder when

a. P is divided by 173.

b. P is divided by 34.

c. P is divided by 17.

Solution: $P = 173 \times 34 + 40$

a. $\dfrac{P}{173} = \dfrac{173\times34+40}{173} = 34 + \dfrac{40}{173}$

$\therefore$ **Remainder r = 40.**

b. $\dfrac{P}{34} = \dfrac{173\times34+40}{34} = 173 + \dfrac{40}{34}$

$= 173 + \dfrac{34}{34} + \dfrac{6}{34} = 173 + 1 + \dfrac{6}{34} = 174 + \dfrac{6}{34}$

$\therefore$ **Remainder r = 6**

c. $\frac{P}{17} = \frac{173 \times 34 + 40}{17} = 246 + \frac{40}{17}$

$= 246 + 2 + \frac{6}{17} = 248 + \frac{6}{17}$

$\therefore$ **Remainder r = 6**

12. If the remainder is 1 when m is divided by 2 and the remainder is 3 when n is divided by 4, which of the following must be true?

A. m is even
B. n is even
C. m + n is even
D. mn is even

Solution:

- The remainder is 1;
  when m is divided by 2 means: $m = 2x + 1$ …*
- The remainder is 3;
  when n is divided by 4 means: $n = 4y + 3$ … (**)

Adding (*) and (**)

$\Rightarrow m + n = (2x + 1) + (4y + 3) = 2x + 4y + 4$

$\Rightarrow m + n = 2(x + 2y + 2) \longleftarrow$ even. **Answer: C**

13. If P divided by 9 leaves a remainder of 1, which of the following must be true.

I P is even II P is odd III $P = 3z + 1$ for some integer z.

A. I only
B. II only
C. III only
D. I and II only.

Solution:

P divided by 9 leaves a remainder of 1 means $P = 9m + 1$

Let $m = 1$, $p = 9 + 1 = 10$, even

Let $m = 2$, $p = 9(2) + 1 = 19$, odd.

But $p = 9m + 1 = 3(3m) + 1$, let $3m = z$

$\therefore p = 9m + 1 = 3z + 1$, **Answer: C**

14. How many positive integers less than 100 have a remainder of 3 when divided by 7.

Solution:

Using the division algorithm

$A_n = 7n + 3$

$A_1 = 7(1) + 3 = 10$

$A_2 = 7(2) + 3 = 17$
$\vdots \qquad \vdots \qquad \vdots$
$A_{10} = 7(10) + 3 = 73$
$A_{12} = 7(12) + 3 = 87$
$A_{13} = 7(13) + 3 = 94$
$A_{14} = 7(14) + 3 = 101$

Therefore, there are 13 positive integers less than 100 that have remainder of 3 when divided by 7.

15. Consider the following two quantities.

Quantity A	**Quantity B**
The remainder when 999 is divided by 7	The remainder when 777 is divided by 9

A. Quantity A is greater than quantity B.
B. Quantity A is less than quantity B.
C. The two quantities are equal.
D. The relationship cannot be determined from the information given.

Solution:

$$\text{Quantity A}: 999 \div 7 = \frac{999}{7} = \frac{9(111)}{7}$$

$$\text{Quantity B}: 777 \div 9 = \frac{777}{9} = \frac{7(111)}{9}$$

Therefore, $\frac{9}{7}(111) > \frac{7}{9}(111)$ **Answer: A**

16. When 10 is divided by the positive integer P, then the remainder is P – 4. Which of the following could be the value of P. **(UEE)**
A. 8 B. 7 C. 4 D. 3

Solution:

If 10 is divided by P, the remainder is p – 4
By division algorithm: $10 = Pz + p - 4$
Where z is an integer
$\Rightarrow 10 + 4 = Pz + P$ with $0 \le P - 4 < P$
$\Rightarrow 14 = P(z + 1)$ with $4 \le p < P + 4$

$$\Rightarrow P = \frac{14}{z+1}$$

Let $z = 1$; $P = \frac{14}{1+1} = \frac{4}{2} = 7$ Integer

Let $z = 2$; $P = \frac{14}{2+1}$ Not integer

Answer: B

17. If x and y are both prime and greater than 2, then which of the following CAN NOT be a divisor of xy

A. 2 B. 3 C. 11 D. 15

Solution:

The product of two odd numbers is odd. Hence 2 cannot be a divisor of xy.

Answer: A

18. Which of the following numbers is divisible by both 3 and 88

A. 88, 888,888
B. 8, 888, 888
C. 888,888
D. 88, 888

Solution: Here

A. Since 8 + 8 + 8 + 8 + 8 + 8 + 8 + 8 = 64 and 64 is not divisible by 3.

C. Since 8 + 8 + 8 + 8 + 8 + 8 = 48 and 48 is divisible by 3.
Also (8 + 8 + 8) – (8 + 8 + 8) = 0 and0 0 is divisible by 11.

Answer: C

19. Which of the following numbers is divisible by both 11 and 3.

A. 1, 111, 111
B. 11,111,111
C. 111,111,111
D. 111, 111

Solution:

C. Since sum of the digit 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 = 9 and 9 is divisible by 3 But (1+1 +1+1 + 1)–(1+1+1+1)= 1 and 1 is not divisible by 11.

D. Since 1+1+1+1+1+1 = 6 and 6 is divisible by 3.
Also (1+1+1) – (1+1+1) = 0 and 0 is divisible by 11.

Answer: D

20. If 2 is the greatest number that will divide evenly into both x and y, what is the greatest number that will divide evenly into both $5x$ and 5y?

A. 2 B. 4 C. 10 D. 8

Solution:

Since 2 divides evenly into x we get $x = 2z$.
Thus $5x = 5(2z) = 10z$. Also 2 divides evenly into y we get $y = 2k$.

Thus $5y = 5(2k) = 10k$. In other words, $5x$ and $5y$ are divisible by 10 **Answer: C**

21. If p and q are primes greater than 2. Which of the following must be true?

I. $p + q$ is even II. pq is odd III. $p^2 - q^2$ is even.

A. I, II and III
B. I and II only
C. I and III only
D. I only

Solution:

All primes greater than 2 are odd, so p and q are odd

- $P + q$ = even, example $3 + 7 = 10$, even
- $P \cdot q$ = odd, example, (3) (7) = 21, odd.
- $P^2 - q^2 = (q - q)(p + q) =$ (even) (even) = even **Answer: A**

1.4 NUMBER OF DIVISORS

How many divisors does 24 have?

The divisors of 24 are 1, 2, 3, 4, 6, 8, 12, 24

Thus 24 have 8 divisors.

If P is any prime and n is any natural number, then the divisors of $P^n = P^0 \cdot P^1 \cdot P^2 \cdot P^3 \ldots P^n$. Therefore there are (n + 1) divisors of P^n.

If p and q are different primes then $P^n q^m$ will have (n + 1) (m + 1) divisors.

Illustrative Example

22. Find the number of divisors of each of the following.

a. 36 b. 1000 c. 6^5

Solution

a. prime factor of $36 = 2 \times 2 \times 3 \times 3 = 2^2 \cdot 3^2$
Because 2^2 has 2+1 = 3 divisors and 3^2 has 2+1 = 3 divisors. Thus $36 = 2^2 \cdot 3^2$ has (2+1) (2+1) or 9 divisors

b. $1000 = 10^3 = (2 \times 5)^3 = 2^3 \cdot 5^3$
Thus $1000 = 2^3 \cdot 5^3$ has (3+1) (3+1) or 16 divisors

c. $6^5 = (2 \times 3)^5 = 2^5 \cdot 3^5$
Thus $65 = 2^5 \cdot 3^5$ has (5+1) (5+1) or 36 divisors

Example 22 and 23 refer to the following definition.

For any positive integer **n**, $\tau(n)$ represents the number of positive divisors of **n**.

23. Which of the following is (are) true?

I $\tau(5) = \tau(7)$

II $\tau(5)\cdot\tau(7) = \tau(35)$

III $\tau(5) + \tau(7) = \tau(12)$

A. I only C. I and II only

B. II only D. I and III only

Solution: Here is how to find.

I. Since 5 and 7 have two positive factors each, $\tau(5) = \tau(7)$. (I is true)

II. Since 35 have four divisors {1, 5, 7 and 35}

Also $\tau(5)\times\tau(7) = 2\times2 = 4$

$\therefore \tau(5)\cdot\tau(7) = \tau(35)$ is true

III. $\tau(5) = 2$ and $\tau(7) = 2$

Divisors of 12 are 1, 2, 3, 6, 12.

$\therefore \tau(12) = 6$

$\therefore \tau(5) + \tau(7) = \tau(12)$ is false **Answer: C**

24. What is the value of $\tau(\tau(\tau(12)))$?

A. 6 B. 2 C. 3 D. 4

Solution:

$\tau(\tau(\tau(12))) = \tau(\tau(6)) = \tau(4) = 3$

25. If x is the product of the positive integers from 1 to 8 inclusive, and if i, k, m, and p are positive integers such that $x = 2^i 3^k 5^m 7^p$, then i + k + m + p =

A. 4 B. 7 C. 8 D. 11

Solution: The product of the positive integer from 1 to 8 inclusive, is

$(1)(2)(3)(4)(5)(6)(7)(8) = (2^1)(3^1)(2^2)(5^1)(2^1)(3^1)(7^1)(2^3)$

$= (2^{1+2+1+3})(3^{1+1})(5^1)(7^1)$

$= (2^7)(3^2)(5^1)(7^1)$

So, i = 7, k = 2, m = 1, p = 1 and i + k + m + p

$= 7 + 2 + 1 + 1 = 11$ **Answer: D**

26. When positive integer '**x**' is divided by positive integer '**y**', the remainder is 6. If $\frac{x}{y} = 63.04$, what is the value of y.

A. 75 B. 150 C. 12 D. 48

Solution: r = 6, q = 63.

From division algorithm $x = yq + r$

$\Rightarrow \frac{x}{y} = \frac{yq}{y} + \frac{r}{y}$; But $\frac{x}{y} = 63.04 = 63 + 0.04$

$$\Rightarrow \frac{x}{y} = q + \frac{r}{y} = 63 + 0.04$$

$$\frac{x}{y} = 63 + \frac{6}{y}, \text{ thus } \frac{6}{y} = 0.04$$

$$\Rightarrow 6 = 0.04y, \text{ therfore, } y = \frac{6}{0.04} = 150$$

Answer: B

1.5 INTEGERS

The integers are {... −4, −3, −2, −1, 0, 1, 2, 3, 4, ...}
The positive integers are {1, 2, 3, 4, 5,...}
The negative integers are {..., −5, −4, −3, −2,−1}
Note: The integer 0 is neither positive nor negative.

Consecutive integers are two or more integers, written in sequence, each of which is 1 more than the preceding integer.
−3,−2, −1, 0, 1, n, n + 1, n + 2, n + 3...

Consecutive integers can be represented by, n, n + 1, n + 2, n + 3 ...
where n is an integer.

Consecutive even integers Can be represented by 2n, 2n + 2, 2n + 4 ...
where n is an integer

Consecutive odd integers can be represented by 2n + 1, 2n + 3, 2n + 5,... where n is an integer.

HOW TO COUNT CONSECUTIVE NUMBERS.
- The number of integers from A to B inclusive is B − A + 1.

Example: How many integers are there from 83 through 429 inclusive
Set up: 429 − 83 + 1 = 347.

Illustrative Example

27. If x, y and z are consecutive positive integers and $x < y < z$, which of the following must be true?

I. $z - x = 2$

II. xyz is an even integer

III. $\frac{x+y+z}{3}$ is an integer

A. I only C. II and III only

B. II only D. I, II and III.

Solution:

Since x, y and z are consecutive positive integers and $x < y < z$, then $y = x + 1$ and $z = x + 2$.

I. $z - x = (x + 2) - x = 2$, Must be true

II. (odd) (even) (odd) = even, Must be true

(even) (odd) (even) = even

III. $\frac{x+y+z}{3} = \frac{x+x+1+x+2}{3} = \frac{3x+3}{3} = x+1 = y$ True

Answer: D

28. The sum of three consecutive positive integers must be divisible by which of the following

A. 2 B. 3 C. 4 D. 5

Let the three consecutive positive integers be n, n + 1 and n + 2.

$\Rightarrow$ (n) + (n + 1) + (n + 2) = 3n + 3 = 3 (n + 1) Because

The sum 3 (n + 1) as a multiple of 3, it is divisible by 3.

Answer: B

29. If the sum of five consecutive integers is S, what is the largest of those integers interms of S?

A. $\frac{S-5}{4}$ B. $\frac{S-10}{5}$

C. $\frac{S+10}{5}$ D. $\frac{S+4}{4}$

Solution:

$S = n + (n + 1) + (n + 2) + (n + 3) + (n + 4) = 5n + 10$

$\Rightarrow 5n = S - 10$

$n = \frac{S-10}{5}$ ⟵ the smallest integer

The largest is $n + 4 = \frac{S-10}{5} + 4 = \frac{S-10+20}{5} = \frac{S+10}{5}$

Answer: C

30. If n is an integer, which of the following represents the products of two consecutive odd integer

A. $n^2 + 3n$

B. $4n^2 + 8n + 3$

C. $n^2 + 4n + 3$

D. $4n^2 + 8n$

Solution: **Here is to crack.** We know that (odd) (odd) = odd.

A. $n^2 + 3n = n(n + 3)$ not odd, let n = 1

B. $4n^2 + 8n + 3 = 4n^2 + 6n + 2n + 3$ ⟵ $8n = 6n + 2n$.

$= (4n^2 + 6n) + (2n + 3)$

$= 2n(2n + 3) + 1(2n + 3) = (2n + 1)(2n + 3)$

$(2n + 1)(2n + 3)$ is a product of consecutive odd integer

C. $N^2 + 4n + 3 = n(n+1) + 3(n+1) = (n+1)(n+3)$ is even

$4n2 + 8n = 4n(n+2)$ is even

Answer: B

31. If x is the product of the integers from 100 to 200, inclusive, and y is the product of the integer from 100 to 201 inclusive, what is $\frac{1}{x} + \frac{1}{y}$ in terms of y?

A. $\frac{(201)^2}{y}$

B. $\frac{(202)(201)}{y}$

C. $\frac{201}{y}$

D. $\frac{202}{y}$

Solution:

Since $x = (100)(101)(102)\ldots 200$ and

$y = (100)(101)(102)\ldots(199)(200)(201)$

$$\Rightarrow y = 201x \Rightarrow x = \frac{y}{201}$$

$$\therefore \frac{1}{x} + \frac{1}{y} = \frac{1}{\left(\frac{y}{201}\right)} + \frac{1}{y} = \frac{201}{y} + \frac{1}{y} = \frac{201+1}{y} = \frac{202}{y}$$

Answer: D

32. What is the units digit of $(13)^4 (17)^2 (29)^3$

A. 9 B. 7 C. 1 D. 5

Solution:

- The unit digit of 13^4 is 1, since $3 \times 3 \times 3 \times 3 = 81$

- The unit digit of 17^2 is 9, since $7 \times 7 = 49$
- The unit digit of 29^3 is 9, since $9 \times 9 \times 9 = 729$.

Therefore, the unit digit of
$(13)^4 (17)^2 (29)^3$ is 1, since $1 \times 9 \times 9 = 81$. **Answer: C**

33. How many prime numbers between 1 and 100 are factors of 7,150

A. One B. Two C. Three D. Four

Solution: To find the number of prime numbers between 1 and 100 that are factors of 7,150, find the prime factorization of 7,150.

$7, 150 = 10 \times 715 = (2 \times 5) \times (5 \times 143)$
$= (2 \times 5 \times 5 \times 11 \times 13)$

Thus, 7,150 has four prime factors: 2, 5, 11, and 13. **Answer: D**

34. If the sum of three consecutive integers is odd, then the first and the last integers must be

A. odd, even C. odd, odd
B. even, even D. even, odd

Solution: Let x, $x + 1$ and $x + 2$ consecutive integer.

Case 1: odd + even + odd = even
Case 2: even + odd + even = odd

Thus, from case 2, x and $x + 2$ are must be even. **Answer: B**

35. If the two digit integers M and N are positive and have the same digits, but in reverse order, which of the following **CANNOT** be the sum of M and N.

A. 99 B. 121 C. 165 D. 181

Solution:

It is given that M and N have the same digits in reverse order. Let $M = 10t + u$ and $N = 10u + t$
Where u and t are two digits.
Then, $M + N = (10t + u) + (10u + t) = 11t + 11u$
$\Rightarrow 11t + 11u = 11(t + u)$
This means that any sum of the two integers M and N must also be a multiple of 11.
Therefore, some multiplies of 11 are
{11, 22, 33, 44, 55, 66, 77, 88, 99, 110, 121, ...}

Answer: D

36. If a two- digit positive integer has its digits reversed, the resulting integer differs from the original by 27. By how much do the two **digits differ?**

A. 3 B. 4 C. 5 D. 6

Solution: Let the two digit integer be represented by

$M = 10t + u$ and when reversed $N = 10u + t$

$\Rightarrow N - M = (10u + t) - (10t + u) = 27 \Rightarrow 9u - 9t = 27$

$\Leftrightarrow 9(u - t) = 27$

$\therefore u - t = \frac{27}{9} = 3.$

Answer: A

37. If the sum of five consecutive odd integers is 735, what is the largest of these integers?

A. 155 B. 151 C. 145 D. 143

Solution:

The set of odd integers are described by {n/2n + 1, where n is integer}.

- Sum of five consecutive odd integers will be:

$(2n + 1) + (2n + 3) + (2n + 5) + (2n + 7) + (2n + 9) = 735$

$\Rightarrow 10n + 25 = 735$

$\Rightarrow 10n = 735 - 25 = 710$

$\Rightarrow n = \frac{710}{10} = 71$

Thus, largest of these: $2n + 9 = 2(71) + 9 = 151$ **Answer: B**

1.6 LEAST COMMON MULTIPLE (LCM) OF INTEGERS

The ***least common multiple*** **(LCM)** of two or more integers is the smallest positive integer that is a multiple of each of them.

Example: The LCM of 3 and 4 is 12

- Multiple of 3 = 3, 6, 9, 12, 15, … 3n
- Multiple of 4 = 4, 8, 12, 16, 20, … 4n.

Infinitely many positive integers are multiples of both 3 and 4 including 12, 24, 36, 48, … but 12 is the smallest one.

There are several methods for finding least common multiples.

a. **The intersection of set method,** we first find the set of all positive multiple of both numbers

Example: To find the LCM of 8 and 12 denote by M_8 and M_{12}

$M_8 = \{8, 16, 24, 32, \ldots\}$

$M_{12} = \{12, 24, 36, 48, \ldots\}$

The set of common multiples is $M_8 \cap M_{12} = \{24, 48, 72, \ldots\}$

Because the least number in $M_8 \cap M_{12}$ is 24, the LCM of 8 and 12 is 24 written LCM (8, 12) = 24.

b. The prime factorization method.

To find the LCM:

- List prime factorization of each number
- Multiply these primes, each raised to the highest power of the prime that occurs either of the prime factorization.

1.7 GREATEST COMMON FACTOR (GCF) (OR GREATEST COMMON DIVISOR)

The greatest common factor (divisor) of two or more integers is the largest integer that is a factor of each of them (or divides each of them).

Example: The greatest common divisor of 24 and 30 is 6.

$F_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}$

$F_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$

$F_{24} \cap F_{30} = \{1, 2, 3, 6\}$

Because the greatest number in the set of common positive divisor is 6

$\therefore$ GCF (24, 30) = 6

Method of finding GCF.

- **The prime factorization method.**

To final GCF of two or more positive integers, first find the prime factorizations of the given numbers and then identify each common prime factor of the given numbers.

The GCD is the product the common factors, each raised to the lowest power of each common prime factor

Illustrative Example

38. Find the GCD and the LCM for each of the following using the prime factorization.

a. 40 and 60

b. 30 and 45

Solution:

a. $40 = 2 \times 2 \times 2 \times 5 = 2^3 \times 5$

$60 = 2 \times 2 \times 5 \times 5 = 2^2 \times 5^2$

$\therefore$ LCM (40, 60) $= 2^3 \times 5^2 = 200$

$\therefore$ GCD (40, 60) $= 2^2 \times 5 = 20$.

b. $30 = 2 \times 3 \times 5 = 2^1 \times 3^1 \times 5^1$

$45 = 3 \times 3 \times 5 = 3^2 \times 5^1$

$\therefore$ LCM (30, 45) $= 2^1 \times 3^2 \times 5^1 = 90$

$\therefore$ GCD $(30, 45) = 3^1 \times 5^1 = 15$

Extreme fact

The product of the GCF and LCM of two numbers is equal to the product of the two numbers.

- That is GCD **(a, b)** × LCM **(a, b) = ab.**

$$\text{GCD}(a,b) = \frac{ab}{\text{LCM}(a,b)}$$

Illustrative Example

39. If the LCM (a, b) = 144 and a × b = 1728, then what is the GCD (a, b)?

Solution: use the extreme fact.

$$\text{GCD}(a,b) = \frac{ab}{\text{LCM}(a,b)} = \frac{1728}{144} = 12.$$

40. What is the smallest number that is divisible by

a. both 34 and 35

b. both 36 and 48.

Solution: We are being asked for the LCM of the given number.

a. $\text{LCM}(34,35) = \dfrac{34 \times 35}{\text{GCF}(34,35)} = \dfrac{1190}{1} = 1190$

$\therefore$ 1,190 is the smallest number that divides evenly into both 34and 35

b. $\text{LCM}(36,48) = \dfrac{36 \times 48}{\text{GCF}(36,48)} = \dfrac{36 \times 48}{12} = 144$

$\therefore$ 144 is the smallest number that is divisible by both 36 and 48.

41. Suppose $x = 2^5 \times 7^2 \times 11$

$y = 2^2 \times 7^4 \times 13$

$z = 2^3 \times 7 \times 17$

a. What is the GCD (x, y, z)?

b. What is the LCM (x, y, z)?

Solution:

a. GCD $(x, y, z) = 2^2 \times 7 = 28$

b. LCM $(x, y, z) = 2^5 \times 7^4 \times 11 \times 13 \times 17.$

1.8 EXPONENTS

Repeated multiplication of the same number is indicated by an exponent:

Example:

a. $5 \times 5 \times 5 = 5^3$

b. $3 \times 3 \times 3 \times 3 = 3^4$

c. $a \times a \times a \times \ldots \times a = a^n$, where **a** is used as a factor n – times.

Exponent Fact

For any numbers **b** and **c** positive integers m and n:

i. $b^m b^n = b^{m+n}$

ii. $\frac{b^m}{b^n} = b^{m-n}$

iii. $(b^m)^n = b^{mn}$

iv. $b^m c^m = (bc)^m$

Illustrative Example

42. If $3^x \times 3^y = 3^{100}$, what is the arithmetic mean of x and y.

Solution:

Since $3^x \times 3^y = 3^{x+y}$, we see that $x + y = 100$

$\Rightarrow$ Arithmetic means of x and y is

$$\frac{x+y}{2} = \frac{100}{2} = 50$$

43. If $50^{100} = k(100^{50})$, what is the value of k

A. 2^{50} B. $\left(\frac{1}{2}\right)^{50}$ C. 25^{50} D. 50^{50}

Solution:

$$50^{100} = k(100^{50}) \Rightarrow (50^{50})(50^{50}) = k(2^{50})(50^{50})$$

$$\Rightarrow k = \frac{50^{50}}{2^{50}} = \left(\frac{50}{2}\right)^{50} = 25^{50}$$

Answer: C

44. If $\frac{0.0015 \times 10^m}{0.03 \times 10^k} = 5 \times 10^7$, then $m - k =$

A. 9 B. 8 C. 6 D. 5

Solution:

$$\Rightarrow \frac{0.0015 \times 10^m}{0.03 \times 10^k} = \frac{15 \times 10^{-4} \times 10^m}{3 \times 10^{-2} \times 10^k} = \frac{15 \times 10^{m-4}}{3 \times 10^{k-2}} = 5 \times 10^7$$

$\Rightarrow 5 \times 10^{m-4-k+2} = 5 \times 10^{7}$

$\Rightarrow 10^{m-k-2} = 10^{7}$

$\Rightarrow m - k - 2 = 7$

$m - k = 7 + 2 = 9$

Answer: A

45. The value of $\dfrac{2^{-11} + 2^{-12} + 2^{-13} + 2^{-14}}{5}$ is how many times the value of $x\,2^{-14}$.

A. 3 B. 4 C. $\frac{3}{2}$ D. $\frac{5}{2}$

Solution: The value of $\dfrac{2^{-11} + 2^{-12} + 2^{-13} + 2^{-14}}{5}$ is x times the value of 2^{-14}, then

$$\Rightarrow (x)(2^{-14}) = 2^{-14}\left(\frac{2^{3} + 2^{2} + 2^{1} + 2^{0}}{5}\right) \Rightarrow x = \frac{8+4+2+1}{5} = \frac{15}{5} =$$

Answer: A

46. If $3^{x} = 81$, then $(2^{x+1})(5^{x-3})$

A. 50 B. 160 C. 40 D. 80

Solution:

$3^{x} = 81 \Rightarrow 3^{x} = 3^{4} \Leftrightarrow x = 4$

Then $(2^{x+1})(5^{x-3}) = (2^{4+1})(5^{4-3}) = (2^{5})(5) = (32)(5) = 160$

Answer: B

47. If $m = 2^{n-1}$ and $3^{2n-3} = 27$, what is the value of $\dfrac{m}{n}$?

A. $\frac{4}{3}$ B. $\frac{3}{4}$ C. 4 D. 3

Solution: $3^{2n-3} = 3^{3} \Leftrightarrow 2n - 3 = 3$

$\Rightarrow 2n = 6$, therefore, $n = 3$

Then $\dfrac{m}{n} = \dfrac{2^{3-1}}{3} = \dfrac{2^{2}}{3} = \dfrac{4}{3}$

Answer: A

Square Root of A number

If $\mathbf{b^2 = a}$, then **b** is a square root of **a**, denoted by $b = \sqrt{a}$.

For any positive numbers **a** and **b**:

$$\sqrt{ab} = \sqrt{a} \times \sqrt{b} \text{ and } \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

Illustrative Example

48. Simplify:

a) $\sqrt{\frac{32}{50}}$ b) $\sqrt{12} \times \sqrt{27}$

Solution:

a) $\sqrt{\frac{32}{50}} = \sqrt{\frac{2 \times 16}{2 \times 25}} = \sqrt{\frac{16}{25}} = \frac{\sqrt{16}}{\sqrt{25}} = \frac{4}{5}$

b) $\sqrt{12} \times \sqrt{27} = \sqrt{4 \times 3} \times \sqrt{3 \times 9} = \sqrt{4 \times 3 \times 3 \times 9} = \sqrt{4 \times 9 \times 9} =$
$2 \times 9 = 18$

49. $\sqrt{80} + \sqrt{125}$

A. $\sqrt{205}$ B. $20\sqrt{5}$ C. $41\sqrt{5}$ D. $9\sqrt{5}$

Solution:

$$\sqrt{80} + \sqrt{125} = \sqrt{16(5)} + \sqrt{25(5)} = \sqrt{16}\sqrt{5} + \sqrt{25}\sqrt{5}$$
$$= 4\sqrt{5} + 5\sqrt{5} = 9\sqrt{5}$$ **Answer: D**

50. If $7x + 17 = 49$, what is the value of $\sqrt{7x + 49}$? (EUEE SAT)

A. 7 B. $4\sqrt{7}$ C. $8\sqrt{5}$ D. 9

Solution: $7x + 17 = 49 \Rightarrow 7x = 49 - 17 = 32$

Now, $\sqrt{7x + 49} = \sqrt{32 + 49} = \sqrt{81} = 9$ **Answer: D**

51. a. If $x = 4$ then $-2^{2\sqrt{x}} + 2$ is equal to:

A. 18 B. -8 C. -14 D. -2

Solution:

Put $x = 4$ in expression $-2^{2\sqrt{x}}$

$\Rightarrow -2^{2(2)} + 2 = -2^4 + 2 = -16 + 2 = -14$ **Answer: C**

b. If $5x + 13 = 31$, what is the value of $\sqrt{5x + 31}$?

A. $\sqrt{13}$ B. $\sqrt{\frac{173}{5}}$ C. 7 D. 13

Solution: $5x + 13 = 31 \Rightarrow 5x = 31 - 13 = 18$

$\Rightarrow 5x + 31 = 18 + 31 = 49 \longleftarrow$ Add 31 to both side

Therefore $\sqrt{5x + 31} = \sqrt{49} = 7$ **Answer: C**

1.9 THE ARITHMETIC OF INEQUALITIES

For any number **a** and **b** exactly one of the following is true;

$a > b$ or $a = b$ or $a < b$.

i) **If $a < b$, then $a + c < b + c$ and $a - c < b - c$**

Example: Let $c = 100$

$5 < 8 \Rightarrow 5 + 100 < 8 + 100 \Rightarrow 105 < 108$

$5 < 8 \Rightarrow 5 - 100 < 8 - 100 \Rightarrow -95 < -92$

ii) **If $a < b$, and $c < d$ then $a + c < b + c$ and $a - c < b - c$**

Example: Let $C = 100$

$5 < 8$ and $7 < 10 \Rightarrow 5 + 7 < 8 + 10 \ldots (12 < 18)$

iii) **Multiplying or dividing an inequality by a positive number preserves it.**

If $a < b$, and c is positive, then $ac < bc$ and $\frac{a}{c} < \frac{b}{c}$.

Example, let $c = 100$

$$5 < 8 \Rightarrow 5 \times 100 < 8 \times 100 \ldots (500 < 800)$$

$$5 < 8 \Rightarrow \frac{5}{100} < \frac{8}{100}$$

iv) **Multiplying or dividing an inequality by a negative number reverses it.**

If $a < b$ and c is negative, then $ac > bc$ and $\frac{a}{c} > \frac{b}{c}$

Example: Let $c = -100$

$5 < 8 \Rightarrow 5 \times (-100) > 8 \times (-100) \Rightarrow -500 > -800$

$$5 < 8 \Rightarrow \frac{5}{-100} > \frac{8}{-100} \Rightarrow -0.05 > -0.08$$

v) **Taking negative reverses an inequality.**

If $a < b$, then $-a > -b$ and if $a > b$, $-a < -b$.

Example: $5 < 8 \Rightarrow -5 > -8$ and $8 > 5 \Rightarrow -8 < -5$

vi) **If a and b are both positives or both negatives and**

a < b then $\frac{1}{a} > \frac{1}{b}$

Example: $5 < 8 \Rightarrow \frac{1}{5} > \frac{1}{8}$

$$-8 < -5 \Rightarrow -\frac{1}{8} > -\frac{1}{5}$$

Extreme fact: Important inequalities for Numbers Between 0 and 1.

A. If $0 < x < 1$, and a is positive, then **(x) (a) < a**

Example: $0.8 \times 4 < 4 \Rightarrow 3.2 < 4$

B. If $0 < x < 1$, and m and n are integers with **m > n > 1** then

$x^m < x^n < x$

Example, let m = 6, n = 4, $x = \frac{1}{2}$

$$\Rightarrow \left(\frac{1}{2}\right)^6 < \left(\frac{1}{2}\right)^4 < \frac{1}{2}.$$

C. If $0 < x < 1$, then $\sqrt{x} > x$ and $x^2 < x$

Example a. let $x = \frac{1}{4}$, then $\sqrt{\frac{1}{4}} > \frac{1}{4} \Rightarrow \frac{1}{2} > \frac{1}{4}$

b. let $x = \frac{4}{9}$, then $\sqrt{\frac{4}{9}} > \frac{4}{9} \Rightarrow \frac{2}{3} > \frac{4}{9}$

D. If $0 < x < 1$, then $\frac{1}{x} > x$. In fact $\frac{1}{x} > 1$

Example, let $x = 0.4$

$$\Rightarrow \frac{1}{0.4} > 0.4 \Rightarrow \frac{10}{4} > \frac{4}{10}$$

Illustrative Example

52. If $0 < a < b < 1$, which of the following is (are) true.

I $a - b$ is negative II $\frac{1}{ab}$ is positive III $\frac{1}{b} - \frac{1}{a}$ is positive

A. I and II only
B. II only
C. I, II, and III
D. III only

Solution:

- $0 < a < b < 1 \Rightarrow a - b < 0$ (Negative)

  $a < b \Rightarrow \frac{1}{a} > \frac{1}{b} \Rightarrow \frac{1}{b} < \frac{1}{a} \Rightarrow \frac{1}{b} - \frac{1}{a} < 0$

- $a - b$ is negative (true)
- $\frac{1}{ab}$ is positive (true)
- $\frac{1}{b} - \frac{1}{a}$ is negative

Answer: A

53. If a and b are negative, and c is positive, which of the following is (are) true

I $a - b < a - c$ II if $a < b$, then $\frac{a}{c} < \frac{b}{c}$ III $\frac{1}{b} < \frac{1}{c}$

A. I only
B. II and III only
C. I, II and III
D. II only

***Solution*:** Since b is negative and c is positive

$b < c \Rightarrow -b > -c \Rightarrow a - b > a - c$

$b < c \Rightarrow -b > -c \Rightarrow a - b > a - c$ (I is false).

- Since c is positive, dividing by c preserves the inequality. (II is true)
- Since b is negative $\Rightarrow \frac{1}{b}$ is negative $\Rightarrow \frac{1}{b} < \frac{1}{c}$ . . . (III is true)

54. If $0 < x < 1$, which of the following lists the numbers in increasing order?

Answer: B

A. $x,\ x^2, \sqrt{x}$
B. $x^2,\ x, \sqrt{x}$
C. $\sqrt{x},\ x,\ x^2$
D. $x^2, \sqrt{x}, x$

Solution: For any number, x, between 0 and 1: $x^2 < x$ and $x < \sqrt{x}$

,let $x = \frac{1}{4}$ then , $x^2, x, \sqrt{x}$ increasing order

55. If $a = 0.99$, which of the following is (are) less than a

Answer: B

(I) $\sqrt{a}$ (II) a^2 (III) $\frac{1}{a}$

A) I only
B) II only
C) III only
D) II and III only

Solution:

Because $0 < a < 1$, then $\sqrt{a} > a$; let $a = \frac{1}{4}$.

Because $0 < a < 1$, then $a^2 < a$ (II is true)

Because $0 < a < 1$, then $\frac{1}{a} > a$ (III is false) **Answer: B**

56. If $\frac{a}{b} < 1$, and a and b are positive integers, which of the following must be greater than 1.

A. $\sqrt{\frac{a}{b}}$ B) $\frac{b}{a^2}$ C. $\frac{b}{a}$ D. $\frac{a}{2b}$

Solution: Since $a > 0$ and $b > 0$, $0 < \frac{a}{b} < 1$.

$$\Rightarrow \frac{a}{b} < 1 \Rightarrow b > a \Rightarrow \frac{b}{a} > 1$$ **Answer: C**

57. If $-1 < x < 0$, which of the following is the largest.

A. -1 B. x C. $2x$ D. x^3

Solution: Let $x = -\frac{1}{3}$

$$x = -\frac{1}{3}, 2x = -\frac{2}{3}, x^3 = -\frac{1}{27}$$ **Answer: D**

58. If $a = -0.4$, which of the following is true?

A. $a < a^2 < a^3$ C. $a < a^3 < a^2$

B. $a^2 < a < a^3$ D. $a^3 < a < a^2$

Answer: C

59. If $r = 0.345$, $S = (0.345)^2$, and $t = \sqrt{0.345}$, which of the following is the correct ordering of r, s, and t?

A. $s < t < r$ C. $r < s < t$

B. $s < r < t$ D. $r < t < s$

Solution: $s = r^2$ and $t = r^{\frac{1}{2}}$

Because $0 < r < 1$, the value of r^x decreases as x increases.

If $0 < r < 1$, then $\sqrt{r} > r$.

$r^{\frac{1}{2}} > r > r^2 \Rightarrow r^2 < r < r^{\frac{1}{2}} \Rightarrow s < r < t$ **Answer: B**

60. If **m** is number between 0 and 1, which of the following is NOT more than **m**?

A. m^2 B. $\frac{1}{m}$ C. $2m$ D. $\sqrt{m}$

Solution: Not more than m means "**less than m**"

- If $0 < m < 1$, then $m^2 < m$, $\frac{1}{m} > m$, and $\sqrt{m} > m$
- If $0 < m < 1$, and $a > 1$, then $ma > m$ **Answer: A**

Quantitative Comparison

Quantitative comparison question can be treated as an equation or inequality.

Either: Quantity A is greater **all the time no matter what;**

Quantity B is greater **all the time no matter what;**

Quantity A = Quantity B, **all the time**

Quantitative Comparison Problem

Illustrative Example

61. Compare the following two quantities.

Quantity A	Quantity B
$11 \times \sqrt{2}$	$9 \times \sqrt{3}$

A. The two quantities are equal.
B. Quantity A is greater than quantity B.
C. Quantity B is greater than A.
D. The two quantities cannot be compared.

Solution:

Quantity: $A = 11 \times \sqrt{2} = \sqrt{121} \times \sqrt{2} = \sqrt{242}$

Quantity: $B = 9 \times \sqrt{3} = \sqrt{81} \times \sqrt{3} = \sqrt{243}$

$\Rightarrow B > A$

Answer: C

62.

Quantity A	Quantity B
$5\sqrt{2}$	7

When you compare the two quantities.

A. The two quantities are equal.
B. Quantity A is greater.
C. Quantity B is greater.
D. The two qualities cannot be compared.

Solution:

Quantity: A $= 5\sqrt{2} = \sqrt{25}\sqrt{2} = \sqrt{50}$

Quantity: B $= 7 = \sqrt{49}$

Answer: B

63.

Quantity A	Quantity B
$\left(\frac{5}{13}\right)^2$	$\sqrt{\frac{5}{13}}$

A. The two quantities are equal.
B. Quantity A is greater.
C. Quantity B is greater.
D. The two quantities cannot be compared.

Solution: If $0 < x < 1$, then $\sqrt{x} > x > x^2$

Since $0 < \frac{5}{13} < 1$, then $\sqrt{\frac{5}{13}} > \frac{5}{13} > \left(\frac{5}{13}\right)^2$

Answer: C

64. Suppose C and d are any real number with $C = d + \frac{1}{5}$ and compare the two quantities.

Quantity A	Quantity B
$C - \frac{1}{5}$	$d - \frac{1}{5}$

A. The two quantities are equal.

B. Quantity A is greater.
C. Quantity B is greater.
D. The two quantities cannot be compared.

Solution: $C = d + \frac{1}{5}$

$$\text{Quantity A} = C - \frac{1}{5} = \left(d + \frac{1}{5}\right) - \frac{1}{5} = d$$

$$\text{Quantity B} = d - \frac{1}{5} = d - \frac{1}{5}$$

$$\therefore\ d > d - \frac{1}{5}$$

Answer: B

65. Suppose that $-1 \le x \le 0$.

Quantity A	Quantity B
$x^3 + 1$	$x^2 + 2x + 2$

A. Quantity A is greater.
B. Quantity B is greater.
C. The two quantities are equal.
D. The two quantities cannot be compared.

Solution:

Quantity A	Quantity B
Let $x = -1, (-1)^3 + 1 = 0$	$(-1)^2 + 2(-1) + 2 = 1$
Let $x = \frac{1}{2}, \left(-\frac{1}{2}\right)^3 + 1 = \frac{7}{8}$	$\left(-\frac{1}{2}\right)^2 + 2\left(-\frac{1}{2}\right) + 2 = \frac{5}{4}$
Let $x = 0, 0 + 1 = 1$	$0 + 0 + 2 = 2$

∴ Quantity B is greater,

Answer: B

66. Suppose n is a positive integer.

Quantity A	Quantity B
The reminder when n is divided by 11	The reminder when 100n is divided by 11.

Which of the two quantity are greater or equal?

Solution:

When n is divided by 11, there is quotient, q and a reminder, r such that $n = 11q + r$,

$\Rightarrow 100n = 1100q + 100r = 1100q + 99r + r = 11(100q + qr) + r$

Therefore the reminder is the same in both quantity.

$\therefore$ The two quantity are equal

67. Consider the following two quantities.

Quantity A	Quantity B
$3^{48} + 3^{48} + 3^{48} + 3^{48}$	3^{49}

Which quantity is greater?

Solution:

Quantity A $= 3^{48} + 3^{48} + 3^{48} + 3^{48} = 3^{48}(1 + 1+1 +1) = 4 \times 3^{48}$

Quantity B $= 3^{49} = 3^1 \times 3 = 3 \times 3^{48}$

$\therefore$ Quantity A is greater.

68. Suppose x and y are positive integers and $x > y$.

Quantity A	Quantity B
$\frac{x}{y}$	$\frac{x+1}{y+1}$

Which quantity is greater?

Solution: Let $x = 3$, $y = 2$

$\frac{3}{2} > \frac{3+1}{2+1} \Rightarrow \frac{3}{2} > \frac{4}{3}$

$\therefore$ Quantity A is greater.

Important - Fact

- (Divisor × Quotient) + Remainder = Dividend
- A number (DIVDEND) can be made completely divisible with the help of either of the following rules:

Rule I By *subtracting remainder from dividend.*

Rule II By *adding* (*divisor – remainder*) *to the dividend*

- **Remainder Rule**. This rule is applicable when the same number (dividend) is divided by two different divisors which are multiples of each other.

Suppose, the smaller divisor = x, then the larger divisor = kx, where K is integer greater than 1.

Now, when the number is divided by x the remainder r (say)
But, when the same number is divided by kx, remainder = R (say),
then by the remainder rule $\boxed{2x + r - R}$

Conceptual Example

69. Find the least number, that must be subtracted from 4,785, to get a number exactly divisible by 354
A. 183 B. 181 C. 201 D. 366
Solution:

- The least number to be subtracted is the remainder from dividend.
- ✓ On dividing 4,785 by 354, the remainder is 183.

Answer: A

70. What least number must be added to 38243 to get a number exactly divisible by 261
A. 137 B. 237 C. 124 D. 24
Solution:

- On dividing 38,243 by 261, the remainder is 137.
  By rule II: the least number to be added to the dividend = divisor – remainder = 261 – 137 = 124
- ∴ The least number to be added = 124

Answer: C

71. Find the greatest number of 3 digits, which is exactly divisible by 47.
A. 977 B. 987 C. 997 D. 957
Solution:

- The greatest number of 3 digit = 999
- ✓ On dividing 999 by 47, remainder is 12. Now by applying rule I. the required number = dividend – remainder = 999 – 12 = 987

Answer: B

72. Find the least number of 4- digits, which is exactly divisible by 29.
A. 986 B. 1014 C. 1015 d. 1029
Solution:

- The least number of 4 – digit = 1000.
- On dividing 1000 by 29, remainder = 14
- ∴ By rule II, the required number = dividend + (divisor – remainder).
  = 1000 + (29 – 14) = 1015

Answer: C

73. A number when divided by 558 leaves a remainder 42. What remainder would be obtained by dividing the same number by 18.
A. 28 B. 21 C. 9 D. 7

Solution: Here

$x = 18$ and $kx = 558$, where $k = \frac{558}{18} = 31$ since the value of

k is an integer greater than 1,
So, Remainder Rule is applicable.
$\therefore\ 2x + r = R \Rightarrow 2\times18 + r = 42 \Rightarrow r = 7$
Hence, the required remainder is 7 **Answer: D**

74. A number, when divided by 19, leaves a remainder 7. What remainder would be obtained by dividing the same number by 456.
A. 45 B. 31 C. 38 D. 9

Solution: Here

$x = 19$, and $kx = 456$, where $k = \frac{456}{19} = 24$

- Since the value of k is an integer >1,
- So, by applying Remainder rule, is applicable.
  $\therefore\ 2x + r = R \Rightarrow 2\times 19 + 7 = 45$

Hence, the required remainder is 45. **Answer: A**

75. Find the greatest number that will divide 1375 and 4935
a. exactly b. leaving in each case remainder 3.

Solution:

a. Note The greatest number that will exactly divide x, y, z is the H.C.F of x, y, z.
∴ H.C.F of 1375 and 4935 = 5
∴ The required number is 5
b. The greatest number that will divide x, y, and z leaving remainder a, b and c respectively then the greatest divisor = H.C.F of $(x - a)$, $(y - b)$ and $(z - c)$
⇒ H.C.F (1375 - 3) and (4935 – 3)
⇒ H.C.F (1372, 4932) 4

76. Find the greatest number that will divide 148,246 and 623 leaving remainder 4,6 and 11 respectively
A. 24 B. 12 C. 8 S. 36

Solution:

Required number = H.C.F (148 – 4, 246 – 6, 623 – 11)

= H.C.F (144, 240,612) = 12.

Answer: B

77. Find the LEAST NUMBER which is exactly divisible by 12, 15 and 40

A. 240 B. 120 C. 360 D. 60

Solution:

- Required number = L.C.M (12, 15,40)= 120

Answer: B

78. Find the least number which when divided by 27, 35, 45 and 49 leaves the remainder 6 in each cases.

A. 6621 B. 6615 C. 6609 D. None

Solution:

Required number = L.C.M (x, y, z, w) + r

$\Rightarrow$ L.C.M (27, 35, 45, 49) + 6 = 6615 + 6 = 6621

Answer: A

79. Find the LEAST NUMER which is when divided by 36, 48 and 64 leaves the remainders 23, 35 and 51 respectively.

A. 576 B. 589 C. 550 D. 563

Solution:

We observe that: 36 – 23 = 48 – 35 = 64 – 51 = 13

$\therefore$ Required number = L.C.M (36, 48, 64) – 13 = 576 – 13 = 563

Answer: D

80. Four bells first begins to tell together and then at interval of 6, 7, 8 and 9 seconds respectively find how many times the bells toll together in two hours and at what intervals they toll together?

Solution:

L.C.M (6, 7, 8, 9) = 504

$\therefore$ All the bells toll together after each interval of 504 second

$\therefore$ in two hours, number of times they toll together

$= \frac{2\times60\times60}{504} = 14$ times.

81. Find the GREATEST 4- digit number that will exactly divisible by 12, 15, 20 and 35.

A. 9,570 B. 9,660 C. 9,988 D. 9,970

Solution: L.C.M (12, 15, 20, 35) = 420

- ✓ On dividing 9999 by 420, the remainder is 339.
- Therefore, the required number 9999 – 339 = 9660

Answer: B

82. Find the SMALLEST 4 – digit number that will exactly divisible by 12, 15, 20 and 35

A. 1,260 B. 9,660 C. 1,840 D. 1,420

Solution:

- On dividing 1000 by 420, the remainder is 160.
- ✓ Required number
  = n digit smallest number + (L.C.M – R)
- ∴ Required number = 1000 + (420 – 160) = 1260

Answer: A

83. Find the GREATEST 4 – digit number which when divided by 12, 18, 21and 28 it leaves a remainder 4 in each case.

A. 9,828 B. 9,824 C. 9,832 D. 9,833

Solution: L.C.M (12, 18, 21, 28) = 252

- ✓ On dividing 9999 by 252 remainder 171.
- Required number = n digit greatest number – R) + k
- ∴ required number = (9999 – 171) + 4 = 9,832

Answer: C

84. Find the SMALLEST 4 – digit number which when divided which when divided by 12, 18, 21 and 28, it leaves a remainder 4 in each case.

A. 1004 B. 1008 C. 1016 D. 1012

Solution:

- ✓ On dividing 1000 by 252 remainder 244
- ∴ Required number = 1000 + (252 – 244) +4 = 1012

Answer: D

85. Find the maximum number of students among whom 429 mangoes and 715 oranges can be equally distributed.

A. 33 B. 55 C. 121 D. 143

Solution: H.C.F (429,715) = 143. **Answer: D**

86. The H.C.F and L.C.M of two numbers are 21 and 4,641 respectively. If one of the number lie between 200 and 300 what are the two possible number?

A. 273 and 357 C. 237 and 373

B. 221 and 441 D. 241 and 451

Solution:

L.C.M × H.C.F = product of two numbers

∴ $4641 \times 21 = x.y$

$\Rightarrow x.y = 21 \times 13 \times 17 \times 21$

Since H.C.F. is 21, the numbers must be the multiple of 21

$x = 21 \times 13 = 273.$

Hence, $y = 21 \times 17 = 357.$ **Answer: A**

Diagnostic/Review Test

1. If $0 < x < 1$, which of the following expression is greatest?

A. $\frac{1}{\sqrt{x}}$ B. $\sqrt{x}$ C. x^2 D. x^3

2. If $x > 0$ and $\sqrt{\frac{y}{x}} = x$ what is y in terms of x.

A. $\frac{1}{x}$ B. $\sqrt{x}$ C. $x\sqrt{x}$ D. x^3

3. P and q are integers. If P is divided by 2, the reminder is 1; and is q is divided by 6, the remainder 1. Which of the following must be true?

I) pq + 1 is even.

II) $\frac{pq}{2}$ is an integer.

III) pq is a multiplies of 12.

A. I only B. II only C. III only D. I and III only

4. If n^3 is an odd integer, which one of the following expression is an even integer?

A. $2n^2 + 2$ B. n^4 C. $n^2 + 1$ D. $n(n + 2)$

5. If $r < t$ and $r < 1$ and $rt = 1$ then which one of the following must be true?

A. $r > 0$ and $t < -1$ C. $r < -1$ and $t > -1$

B. $r > -1$ and $t < -1$ D. $r < 1$ and $t > 1$

6. If $x > y > 0$ and $p > q > 0$, then which of the following expression must be greater than 1.

A. $\frac{xq}{yp}$ B. $\frac{x}{P}$ C. $\frac{x+q}{y+p}$ D. $\frac{x+P}{y+q}$

7. Of the following which has the greatest value when $x = 0.0001$.

A. $\sqrt[3]{x}$ B. $\sqrt{x}$ C. x^2 D. x^3

8. If the sum of four consecutive odd integers is s, then in terms of s, what is the greatest of these integers?

A. $\frac{s-12}{4}$ B. $\frac{s+12}{4}$ C. $\frac{s+6}{4}$ D. $\frac{s-6}{4}$

9. If $x = 15 \times 42 \times 77$, which of the following is **Not** an integer?
   A. $x \div 33$ B. $x \div 28$ C. $x \div 21$ D. $x \div 35$
10. If x = 0.999, which of the following is smaller than x.
   A. x^3 B. x^2 C. $\sqrt{x}$ D. $\sqrt[3]{x}$
11. When positive integer x is divided by positive integer y the remainder is 6. If $\frac{x}{y} = 58.24$, what is the value of y:
   A. 58 B. 29 C. 25 D. 24
12. How many prime numbers between 1 and 100 are factors of 6930?
   A. Six B. Three C. Five D. Two
13. $\left(\frac{1}{2}\right)^{-3}\left(\frac{1}{4}\right)^{-2}\left(\frac{1}{16}\right)^{-1} =$ _____
   A. $\left(\frac{1}{2}\right)^{-48}$ B. $\left(\frac{1}{2}\right)^{-11}$ C. $\left(\frac{1}{2}\right)^{-6}$ D. $\left(\frac{1}{8}\right)^{-11}$
14. If n is a positive integer, then n(n + 1)(n + 2) is:
   A. Divisible by 3 only when n is odd.
   B. Odd whenever n is odd.
   C. Even only when n is odd.
   D. Divisible by 4 whenever n is even.
15. What is the smallest number that is divisible by both 18 and 24?
16. How many integers between 325 and 400, inclusive are divisible by 4.
   A. 18 B. 19 C. 20 D. 24
17. The sum of five consecutive integers is always divisible by:
   A. 2 B. 4 C. 5 D. 10
18. Find the smallest positive integer which gives a reminder of 3 when divided by 4, 6 or 8.
   A. 5 B. 27 C. 15 D. 26
19. Suppose you begin reading a book on page x and end on page y. if you read each page completely and the pages are numbered and read consecutively, then how many pages have you read?
   A. $y - x$ B. $y + x$ C. $y - x - 1$ D. $y - x + 1$

20. Suppose x is divisible by 8 but not by 3. Then which of the following **CANNOT** be an integer?

A. $\frac{x}{2}$ B. $\frac{x}{4}$ C. $\frac{x}{6}$ D. $\frac{x}{8}$

21. If $0 < m < 1$, then which of the following must be less than m.

A. $\sqrt{m}$ B. $\frac{3}{2}m$ C. m^2 D. $\frac{1}{m}$

22. The sum of two – digit number is 9. The number is 6 times the units digits. What is the number?

A. 63 B. 36 C. 72 D. 54

23. Let $x = 0.999$, $y = \sqrt{0.999}$ and $z = (0.999)^2$. Then which of the following is true?

A. $y < x < z$ C. $z < y < x$

B. $z < x < y$ D. $x < z < y$

24. Find the L.C.M of the following

a. 15, 25, 30 b. 56, 70, 112, 168

25. Find the number between 410 and 485 which are exactly divisibl by 12, 15, and 20

26. Find the smallest number which when added to 5000 is exactly divisible by 10, 12, 16 and 18

27. Find the greatest number which when subtracted from 3000, is exactly divisible by 7, 11 and 13.

28. Find the least number which when divided by 12, 15, 20 and 25 leaves no remainder (exactly divisible)

29. Find the least number which when divided by 12, 15, 20 and 25 leaves a remainder 7in each case.

30. Find the least number which when divided by 20, 25, 30, 36 and 48 leaves the remainders 15, 20, 25, 31 respectively.

31. Find the greatest number that will divide 134, 149 and 179 leaving remainders, 6, 5 and 3 respectively

Answer and Explanation

1. A; let $x = \frac{1}{4} \Rightarrow x^3 < x^2 < x < \sqrt{x} < \frac{1}{\sqrt{x}}$

2. D

3. A; $p = 2m + 1$ and $q = 6n + 1$
   $\Rightarrow pq = (2m + 1)(6n + 1) = (2m + 1)(2k + 1)$, where $k = 3n$
   $\Rightarrow$ pq is qn odd
   Therefore, pq + q is even

4. C; since n^3 is odd n must be odd $\Rightarrow n^2$ is odd, so $n^2 + 1$ is even

5. B; since rt = 1 and r > t, there are two possibilities.
   Case I (both positive): $0 < t < 1$ and $r > 1$
   Case II (both negative): $-1 < r < 0$ and $t < -1$

6. D; Adding the inequalities $x > y > 0$ and $p > q > 0$ yields $x + p > y + q > 0$
   $\Rightarrow$ Dividing the inequality by y + q
   $$\Rightarrow \frac{x+p}{y+q} > \frac{y+q}{y+q} \Rightarrow \frac{x+p}{y+q} > 1$$

7. A; if $0 < x < 1$, then $\sqrt[3]{x} > \sqrt{x} > x > x^2 > x^3$

8. B, let the four consecutive odd integers say n, n + 2, n + 4, n + 6
   $s = n + (n + 2) + (n + 4) + (n + 6) = 4n + 12$
   $$\Rightarrow s = 4n + 12 \Rightarrow n = \frac{s-12}{4} \longleftarrow \text{smallest odd integer}$$
   $$\Rightarrow n + 6 = \frac{s-12}{4} + 6 = \frac{s+12}{4} \longleftarrow \text{largest odd integer}$$

9. B; $$\frac{x}{28} = \frac{(3\times5)\times(2\times3\times7)\times(7\times11)}{2\times2\times7} = \frac{3\times3\times5\times7\times11}{2}$$

10. A; If $0 < x < 1$, then $x^3 < x^2 < \sqrt{x} < \sqrt[3]{x}$

11. C; The remainder is 6 when x is divided by y, so $x = yq + 6$ for positive integer q.
    $$\Rightarrow \frac{x}{y} = q + \frac{6}{y} = 58 + 0.24$$
    $$\Rightarrow \frac{6}{y} = 0.24 \Leftrightarrow y = \frac{6}{0.24} = 25$$

12. C; the prime factorization of 6,930 is
$6930 \quad 10 \cdot 693 = 2 \cdot 5 \cdot 3 \cdot 3 \quad 11 \cdot 7 \quad 2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$
Thus 6,930 has five prime factors: 2, 3, 5, 7 and 11.

13. B: $\left(\frac{1}{2}\right)^3\left(\frac{1}{4}\right)^2\left(\frac{1}{16}\right)^1 = \left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)^4 = \left(\frac{1}{2}\right)^{11}$

14. D

15. Determine the LCM the LCM (18, 24)

$$\Rightarrow \text{LCM}(18,24) = \frac{18\times 24}{\text{GCD}(18,24)} = \frac{18\times 24}{2\times 3} = 72$$

Thus 72 is the smallest number that is divisible by both 18 and 24.

16. B, divisible by 4 are {328, 332, 336, . . . 400}.
Use $A_n = A_1 + (n-1)(4)$
$\Rightarrow 400 = 328 + (n-1)(4)$

$$\Rightarrow 72 = 4n - 4 \Rightarrow 4n = 76 \Rightarrow n = \frac{76}{4} = 19$$

17. C, $(n) + (n+1) + (n+2) + (n+3) + (n+4) = 5n + 10$
$\Rightarrow 5(n+2) \longleftarrow$ divisible by 5

18. B

19. D, $A_n = y$, $A_1 = x$, $d = 1$, $A_n = A_1 + (n-1)d$
$\Rightarrow y = x + (n-1)(1)$
$n = y - x + 1$

20. C, $x = 8k$, A) $\frac{x}{2} = \frac{8k}{2} = 4k$ B) $\frac{x}{4} = \frac{8k}{4} = 2k$

C) $\frac{-}{6} = \frac{8k}{2\times 3} = \frac{4k}{3}$ (not divisible by 8)

21. C, if $0 < m < 1$, then $m^2 < m < \sqrt{m}$

22. B, let t = the tens digit and let u = the unit digit.
- Then any two digit number expressed as $10t + u$.
- Sum of the digit; $t + u = 9$.
- Since the number is 6 times its unit digit as: $10t + u = 6u$

$\Rightarrow 10(9-u) + u = 6u \Rightarrow 15u = 90$

$$\therefore u = \frac{90}{15} = 6, \text{ and } t = 3$$

∴ The number is 10(3) + 6 = 36

23. B
24. (a) L.C.M (15, 25, 30)=150 (b) L.C.M (56, 70, 112, 168) = 1,680
25. L.C.M (12, 15, 20) = 60, only (60×7) = 420 and (60×8) = 480 lie between 410 and 485.
26. Required number L.C.M (10, 12, 16, 18) – 5000 = 40.
27. Required number = 3000 – L.C.M (7, 11, 13) = 1999.
28. Required number = L.C.M (12, 15, 20, 25) = 300.
29. Required number = L.C.M + R = 300 + 7 = 307.
30. Here, (Divisor – Remainder) is the same in each case = 5
    ∴ Required number (least dividend)
    = L.C.M – 5 = 3600 – 5 = 3,595
31. Required number (greatest divisor) is H.C.F of (134 – 6), (149 – 5), (179 – 3) = H.C.F (128, 144, 176) = 16

UNIT – TWO

2. FRACTIONS

A fraction consists of two parts: a numerator and a denominator.

In a fraction $\frac{a}{b}$;

a is the numerator and

b is the denominator.

- If the numerator is less than the denominator, the fraction is called **proper** fraction.

Example: $\frac{1}{2}, \frac{5}{24}, \frac{3}{7}$... etc are proper fraction.

- If the numerator is more than denominator, the fraction is called an improper fraction.

Example: $\frac{4}{3}, \frac{2}{1}, \frac{5}{1}, \frac{7}{5}$,... are improper fraction.

2.1 Equivalent fraction

Two fractions are equivalent if multiplying or dividing both the numerator and denominator of the first fraction by the same number gives the same second fraction.

Example:

a. $\frac{1}{2}$ and $\frac{4}{8}$ are equivalent fraction because $\frac{1}{2} = \frac{1\times 4}{2\times 4} = \frac{4}{8}$

b. $\frac{3}{8}$ and $\frac{15}{40}$ are equivalent fraction because $\frac{3}{8} = \frac{3\times 5}{8\times 5} = \frac{15}{40}$.

Comparing fractions.

To compare two fractions, cross multiply. The larger number will be on the same side as the larger fraction.

Example: Which quantity is greater

Quantity A	Quantity B
$\frac{7}{8}$	$\frac{8}{9}$

Solution: Cross multiplying gives
7×9 Versus 8×8, which gives
63 Versus 64

Hence $\frac{8}{9}$ is greater than $\frac{7}{8}$ **∴ Answer:**

Quantity B.

- Always reduce a fraction to its lowerst terms.

Example: Which quantity is greater, smaller or equal

Quantity A	Quantity B
$\frac{2x^2+12x+18}{(x+3)^2}, x \neq -3$	2

Solution:

$$\text{Quantity A} = \frac{2(x^2+6x+9)}{(x+3)^2} = \frac{2(x+3)^2}{(x+3)^2} = 2$$

Quantity B = 2

Hence the two quantities are equal.

Illustrative Example

1. Which of the following is NOT equivalent to $\frac{15}{24}$

A. $\frac{45}{72}$ B. $\frac{75}{120}$ C. $\frac{195}{312}$ D. $\frac{3}{8}$

Solution: Short cut method.

To determine whether two fractions are equivalent convert them to decimal by dividing. For the fraction to be equivalent the two quotients must be the same.

A. $45 \div 15 = 3$ and $72 \div 24 = 3$, so the fractions are equivalent: $\frac{45}{72} = \frac{15 \times 3}{24 \times 3}$

B. $75 \div 15 = 5$ and $120 \div 24 = 5$, so the fractions are equivalent: $\frac{75}{120} = \frac{15 \times 5}{24 \times 5}$

C. $195 \div 15 = 13$ and $312 \div 24 = 13$, so to the fractions are equivalent $\frac{195}{312} = \frac{15 \times 13}{24 \times 13}$

D. $15 \times 3 = 5$ but $24 \div 8 = 3$, so the fractions are not equivalent.

Answer: D

2. What fractional part of a week is 124 hours

A. $\frac{31}{24}$ B. $\frac{24}{124}$ C. $\frac{1}{2}$ D. $\frac{31}{42}$

Solution: There are 24 hours in a day 7 days in a week, so there are $24 \times 7 = 168$ hours in a week: $\frac{124}{168} = \frac{31}{42}$, **Answer: D**

3. Which of the following is less than $\frac{5}{9}$

A. $\frac{5}{8}$ B. $\frac{21}{36}$ C. $\frac{25}{45}$ D. $\frac{11}{20}$

Solution:

A. $\frac{5}{8} > \frac{5}{9}$ because $9 \times 5 > 8 \times 5$.

B. $\frac{21}{36} = \frac{7}{12} > \frac{5}{9}$ because $7 \times 9 > 12 \times 5$

C. $\frac{25}{45} = \frac{5}{9}$

D. $\frac{11}{20} < \frac{5}{9}$ because $99 < 100$ **Answer: D**

4. Which of the following is the greatest?

A. $\frac{1}{2}$ B. $\frac{5}{11}$ C. $\frac{4}{9}$ D. $\frac{7}{13}$

Solution:

$$\frac{4}{9} < \frac{5}{11} < \frac{1}{2} < \frac{7}{13}$$

Answer: D

5. For how many odd integers, **a**, between 30 and 40 is true that $\frac{5}{a}, \frac{8}{a}$, and $\frac{13}{a}$ are all in lowest terms?

A. 1 B. 2 C. 3 D. 4

Solution: The only possibilities are 31, 33, 35, 37, and 39, but

$\frac{5}{35} = \frac{1}{7}$ and $\frac{13}{39} = \frac{1}{3}$, so only 31, 33 and 37 (that is, 3 integers) remain. **Answer: C**

6. Which group of numbers is arranged in descending order?

A. $\frac{5}{7}, \frac{7}{12}, \frac{6}{11}, \frac{3}{13}$ B. $\frac{7}{12}, \frac{5}{7}, \frac{6}{11}, \frac{3}{13}$

C. $\frac{7}{12}, \frac{5}{7}, \frac{6}{11}, \frac{3}{13}$ D. $\frac{3}{13}, \frac{7}{12}, \frac{6}{11}, \frac{5}{7}$

Solution: descending order means decreasing order.

$$\frac{5}{7} > \frac{7}{12} > \frac{6}{11} > \frac{3}{13}$$

Answer: A

7. If $\frac{1}{3}$ of a number is 3 more than $\frac{1}{4}$ of the number, then what is the number?

A. 18 B. 24 C. 30 D. 36

Solution: Let x = the number

$$\frac{1}{3}x = 3 + \frac{1}{4}x \Leftrightarrow \frac{1}{3}x - \frac{1}{4}x = 3$$

$$\Rightarrow \frac{4x-3x}{12}=3 \text{ implies } \frac{x}{12}=3$$

Therefore, $x = 12\times3 = 36$, **Answer: D**

8. If $\frac{3}{5}$ of a number is 4 more than $\frac{1}{2}$ of the number, then what is the number?

A. 20 B. 40 C. 28 D. 35

Solution: Let x = a number.

$$\Rightarrow \frac{3}{5}x = 4 + \frac{1}{2}x \Rightarrow \frac{3}{5}x - \frac{1}{2}x = 4$$

$$\Rightarrow \frac{6x-5x}{10} = 4 \Leftrightarrow x = 10\times4 = 40$$ **Answer: B**

9. If $\frac{3}{11}$ of a number is 22, what is $\frac{6}{11}$ of that number?

A. 33 B. 12 C. 44 D. 6

Solution: Let x = a number is

$$\Rightarrow \frac{3}{11}x = 22, \text{ then } \frac{6}{11}x = ?$$

$$\text{Since } \frac{6}{11}x = 2\left(\frac{3}{11}x\right) = 2(22) = 44$$ **Answer: C**

10. Let $x = \frac{7}{11}$, $y = \sqrt{\frac{7}{11}}$, and $z = \left(\frac{7}{11}\right)^2$ Then which of the following must be true?

A. $x < z < y$ B. $z < y < x$

C. $z < x < y$ D. $y < x < z$

Solution:

Proper Fractions often behave in unusual ways: Squaring a fraction makes it smaller, and taking the square root of a fraction makes it larger

$$\therefore \left(\frac{7}{11}\right)^2 < \left(\frac{7}{11}\right) < \sqrt{\frac{7}{11}} \Leftrightarrow z < x < y$$ **Answer: C**

11. $\dfrac{6^4-6^3}{5}$ is equal to

A. $\dfrac{1}{5}$ B. 6^3 C. $\dfrac{6}{5}$ D. $\dfrac{6}{5}$

Solution: $\dfrac{6^4-6^3}{5}=\dfrac{6^3(6-1)}{5}=\dfrac{6^3(5)}{5}=6^3$ **Answer: B**

12. $\dfrac{1}{10^9}-\dfrac{1}{10^{10}}$ is equal to

A. $\dfrac{-1}{10}$ B. $\dfrac{-1}{10^9}$ C. $\dfrac{-1}{10^{19}}$ D. $\dfrac{9}{10^{10}}$

Solution:

$$\frac{1}{10^9}-\frac{1}{10^{10}}=\frac{10}{10^{10}}=\frac{10-1}{10^{10}}=\frac{9}{10^{10}}$$ **Answer: D**

13. $\dfrac{1}{1-\dfrac{1}{1-\dfrac{1}{2}}}$ is equal to

A. 1 B. –1 C. 2 D. –2

Solution:

$$\frac{1}{1-\dfrac{1}{1-\dfrac{1}{2}}}=\frac{1}{1-\dfrac{1}{\dfrac{2}{2}-\dfrac{1}{2}}}=\frac{1}{1-\dfrac{1}{\dfrac{1}{2}}}$$

$$=\frac{1}{1-2}=\frac{1}{-1}=-1$$ **Answer: B**

14. $\dfrac{1+\dfrac{1}{6}}{2-\dfrac{3}{4}}$ is equal to

A. $\frac{14}{15}$ B. $\frac{16}{33}$ C. $\frac{8}{11}$ D. $\frac{15}{22}$

Solution:

$$\frac{1+\frac{1}{6}}{2-\frac{3}{4}}=\frac{\frac{6+1}{6}}{\frac{8-3}{4}}=\frac{\frac{7}{6}}{\frac{5}{4}}=\frac{7}{6}\times\frac{4}{5}=\frac{28}{30}=\frac{14}{15}$$

Answer: A

15. If $\frac{x+2}{x-2}=y$, what is the value of x interms of y?

A. $2-y$ B. $\frac{2}{y}$ C. $\sqrt{y+8}$ D. $\frac{-2y-2}{1-y}$

Solution:

$$\frac{x+2}{x-2}=y \Leftrightarrow (x-2)\left(\frac{x+2}{x-2}\right)=y(x-2)$$

$$\Rightarrow x+2=xy-2y$$

$$\Rightarrow x-xy=-2y-2$$

$$\Rightarrow x(1-y)=-2y-2$$

$$\therefore x=\frac{-2y-2}{1-y},$$

Answer: D

16. $\frac{1}{1+\frac{1}{3}}-\frac{1}{1+\frac{1}{2}}=$

A. $-\frac{1}{3}$ B. $-\frac{1}{6}$ C. $\frac{-1}{12}$ D. $\frac{1}{12}$

Solution: the expression becomes

$$\frac{1}{\frac{3+1}{3}}-\frac{1}{\frac{2+1}{2}}=\frac{3}{4}-\frac{2}{3}=\frac{9-8}{12}=\frac{1}{12}$$

Answer: D

17. $\frac{1}{4}$ is the average (arithmetic mean) of $\frac{1}{5}$ and what number?

A. $\frac{1}{3}$ B. $\frac{3}{10}$ C. $\frac{9}{20}$ D. $\frac{9}{40}$

Solution: Let x = the number

$$\frac{\frac{1}{5}+x}{2}=\frac{1}{4}\Rightarrow\frac{1}{5}+x=\frac{1}{2}$$

$$\Rightarrow x=\frac{1}{2}-\frac{1}{5}=\frac{5-2}{10}=\frac{3}{10}$$

Answer: B

18. A piece of paper with an area of $60cm^2$ is divided into two pieces so that the area of one is $\frac{2}{3}$ of the area of the other. What is the area of one of the pieces (EUEE 2005 .E.C)

A. $20cm^2$ B. $24cm^2$ C. $30cm^2$ D. $45cm^2$

Solution: $\frac{1}{3}$ of the pieces $=\frac{1}{3}(60)=20cm^2$ **Answer: A**

19. If $\frac{1}{5}$ of a certain number is 6, then $\frac{1}{6}$ of the number is (EUEE 2006)

A. 5 B. 6 C. 30 D. 36

Solution: Let x = a number then $\frac{1}{5}x=6\Rightarrow x=30$

$$\therefore\ \frac{1}{6}\text{ of the number }=\frac{1}{6}(30)=5$$ **Answer: A**

20. A Mathematics class has 16 boys and 24 girls. What fraction of the class are boys

A. $\frac{2}{5}$ B. $\frac{3}{5}$ C. $\frac{2}{3}$ D. $\frac{3}{4}$

Solution: The class has 40 students of whom 16 are boys.

The boys makeup $\frac{16}{40}=\frac{2}{5}$ of the class **Answer: A**

21. If the senior class has 3600 students, of whom $\frac{5}{12}$ are girls, and the junior class has 350 student, of whom $\frac{4}{7}$ are girls, how many more girls are there in the junior class than in the senior class.

Solution: $\left(\frac{4}{7}\times 350\right)-\left(\frac{5}{12}\times 360\right)=200-150=50$

22. In a jar, $\frac{1}{2}$ are red, $\frac{1}{4}$ are white and $\frac{1}{5}$ are blue. What fraction of the marbles are neither red, white, nor blue?

A. $\frac{19}{20}$ B. $\frac{3}{11}$ C. $\frac{1}{20}$ D. $\frac{7}{11}$

Solution:

The red, white and blue marbles constitute:

$\frac{1}{2}+\frac{1}{4}+\frac{1}{5}=\frac{10}{20}+\frac{5}{20}+\frac{4}{20}=\frac{19}{20}$ of the total, so

$1-\frac{19}{20}=\frac{20-19}{20}=\frac{1}{20}$ of the marbles are neither red, white nor blue.

Answer: C

23. Mr. Mosisa spends $\frac{1}{3}$ of his work day in meeting, $\frac{1}{6}$ on the phone, and $\frac{1}{8}$ answering questions. What fraction of his work day can be devoted to other things?

A. $\frac{5}{8}$ B. $\frac{3}{8}$ C. $\frac{4}{7}$ D. $\frac{3}{7}$

Solution: $1-\left(\frac{1}{3}+\frac{1}{6}+\frac{1}{8}\right)=1-\left(\frac{8}{24}+\frac{4}{24}+\frac{3}{24}\right)$

$=\frac{24}{24}-\left(\frac{8+4+3}{24}\right)=\frac{24-15}{24}=\frac{9}{24}=\frac{3}{8}$

Answer: B

24. A man reads $\frac{3}{10}$ of a book on the first day and $\frac{3}{4}$ of the remainder, on the second day. If the number of pages still unread are 42, how many pages did the book contain?

Solution: Use $\left(1-\frac{a}{b}\right)\left(1-\frac{x}{y}\right)\times$ total = balance

$\left(1-\frac{3}{10}\right)\left(1-\frac{3}{4}\right)\times$ total page = 42

$\Rightarrow\left(\frac{7}{10}\right)\left(\frac{1}{4}\right)\times$ Total page = 42

$\therefore$ Total page = $= 42\times\frac{40}{7}=240$

25. A man spends $\frac{1}{3}$ of his income on food, $\frac{1}{4}$ of the rest on house rent and $\frac{1}{5}$ of the rest on clothes. He still has birr 1,760 left with him. Find his income.

Solution:

Since it is dependent activity.

Here, we use

$\left(1-\frac{a}{b}\right)\left(1-\frac{x}{y}\right)\left(1-\frac{m}{n}\right)\times$ total = balance

$\Rightarrow\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{5}\right)\times$ total = 1760

$\Rightarrow\left(\frac{2}{3}\right)\left(\frac{3}{4}\right)\left(\frac{4}{5}\right)(\text{total})=1760\Leftrightarrow\frac{5}{2}\times1760=4400$

$\therefore$ Total income = 4400 birr

26. A man spends $\frac{1}{3}$ of his income on food, of the rest $\frac{1}{4}$ on house rent and $\frac{1}{5}$ on clothes. He still has birr 1760 left with him. Find his income.

Solution:

After spending $\frac{1}{3}$ on food, $\frac{1}{4}$ is spent on house rent and $\frac{1}{5}$ is spent on clothes. So, spending on these two items are independent of each other, but dependent on the expenditure incurred on the first item. It is a problem both on dependent and independent activities.

$$\therefore \left(1-\frac{a}{b}\right)\left[1-\left(\frac{x}{y}+\frac{m}{n}\right)\right] \times \text{total} = \text{balance}$$

$$\Rightarrow \left(1-\frac{1}{3}\right)\left[1-\left(\frac{1}{4}+\frac{1}{5}\right)\right] \times \text{total} = 1,760$$

$$\Rightarrow \left(\frac{2}{3}\right)\left(\frac{11}{20}\right)(\text{total}) = 1,760$$

$$\therefore \quad \text{total income} = 1,760 \times \frac{3}{2} \times \frac{20}{11} = 4800 \text{birr}$$

27. A student while returning from his school, travels $\frac{2}{3}$ of the distance by bus and $\frac{3}{4}$ of the rest, partly by mini – bus taxi, and partly by foot. If he travels 2km on foot.
Find the distance covered by him.
Solution:

$$\left(1-\frac{2}{3}\right)\left(1-\frac{3}{4}\right) \; x \text{ total distance} = 2\text{km}$$

$$\Rightarrow \left(\frac{1}{3}\right)\left(\frac{1}{4}\right) \text{ (total distance)} = 2\text{km}$$

$$\Rightarrow \text{total distance} = 2\text{km} \times 12 = 24\text{km}$$

28. $\frac{2}{5}$ of a pole is in the mud. When $\frac{1}{4}$ of it is pulled out, 210 cm of the pole is still in the mud. Find the full length of the pole.
Solution:

$$\text{Total length of pole} = \frac{\text{length in mud}}{\text{part in mud}} = \frac{210\text{cm}}{\frac{2}{5}-\frac{1}{4}} = 1400\text{cm}.$$

29. A vessel, full of water, weight is 18.5kg. When the vessel is $\frac{1}{4}$ full, it's weights 7.25 kg. Find the weight of empty vessel.

Solution:

Let weight of empty vessel = V_o

Let weight of water (full) = V

$\therefore$ Vo + V = 18.5 kg … (*)

$$V_o + \frac{1}{4}V = 7.25\text{kg} \ldots (**)$$

$$\Rightarrow (*)-(**)=V-\frac{1}{4}V=\frac{3}{4}V=11.25\text{kg}$$

$$\therefore\ V=\frac{4}{3}\times 11.25 = 15\text{kg and } V_o = 18.5-15 = \mathbf{3.5kg}$$

Hence, weight of empty vessel is **3.5 kg**

Diagnostic Test

1. Which of the following lists the fractions $\frac{2}{3}, \frac{5}{8}, \frac{7}{11}$ and $\frac{13}{20}$ in order from least to greatest.

A. $\frac{2}{3},\frac{5}{8},\frac{7}{11},\frac{13}{20}$ B. $\frac{5}{8},\frac{7}{11},\frac{13}{20},\frac{2}{3}$

C. $\frac{5}{8},\frac{13}{20},\frac{7}{11},\frac{2}{3}$ D. $\frac{13}{20},\frac{7}{11},\frac{5}{8},\frac{2}{3}$

2. Which of the following fractions is the LEAST

A. $\frac{12}{119}$ B. $\frac{1}{10}$ C. $\frac{2}{21}$ D. $\frac{4}{39}$

3. $\frac{1}{10^{35}}-\frac{1}{10^{36}}=$

A. $\frac{1}{10}$ B. $\frac{9}{10^{36}}$ C. $\frac{1}{10^{36}}$ D. $\frac{-1}{10}$

4. Which of the following sum is the LARGEST. (EUEE 2007 E.C)

A. $\frac{5}{6}+\frac{7}{17}$ B. $\frac{5}{6}+\frac{7}{17}$ C. $\frac{6}{7}+\frac{5}{11}$ D. $\frac{8}{9}+\frac{1}{3}$

5. Which of the following has the greatest value?

A. $\sqrt{\frac{7}{10}}$ B. $\left(\frac{7}{10}\right)^2$ C. $\frac{7}{10}$ D. $\frac{3}{5}$

6. $1-\left(\frac{1}{2}-\frac{2}{3}\right)=$ ___

A. $\frac{6}{5}$ B. $\frac{7}{6}$ C. $\frac{6}{7}$ D. $\frac{5}{6}$

7. A man spends $\frac{5}{6}$ of his income on food and $\frac{1}{12}$ on house rent and the rest he saves. If he save birr 50. Find his income

8. A man spends $\frac{1}{6}$ of his salary on food and $\frac{1}{3}$ of the remaining on clothing and $\frac{1}{4}$ of the remaining on entertainment. He is still left with birr 400. **Find** his salary.

9. A lamp post has half of its length in mud, $\frac{1}{3}$ of its length in water and $3\frac{1}{3}$ above the water. **Find** the total length on the post

10. A man distribute 0.375 of his money to his wife and 0.4 to his son. He has still birr 4,200 left with him. How much initial money the man have? How much did his wife get?

11. In an Examination, a student was asked to find $\frac{4}{15}$ of a certain number. By mistake, he found $\frac{4}{5}$ of it. His answer was 200 more than the correct answer. The given number is
A. 350 B. 375 C. 240 D. 175

12. One of the numbers between $\frac{2}{7}$ and $\frac{3}{4}$ is:
A. $\frac{5}{14}$ B. $\frac{3}{49}$ C. $\frac{1}{4}$ D. $\frac{1}{2}$

13. Which one of the following fraction is less than $\frac{1}{3}$
A. $\frac{22}{63}$ B. $\frac{33}{98}$ C. $\frac{4}{11}$ D. $\frac{15}{46}$

ANSWER AND EXPLANTATION

1. B

2. C, since $\frac{1}{10}=\frac{12}{120}\Rightarrow\frac{1}{10}<\frac{12}{119}$

$$\frac{1}{10}=\frac{2}{20}\Rightarrow\frac{2}{21}<\frac{1}{10}$$

$$\frac{1}{10}=\frac{4}{40}\Rightarrow\frac{1}{10}<\frac{4}{39}$$

3. B 4. B 5. A 6. B

7. $1-\left(\frac{5}{6}+\frac{1}{12}\right)(\text{total})=50\Rightarrow \text{total}=12\times50=600$

8. $\left(1-\frac{1}{6}\right)\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)(\text{total})=400$

$\Rightarrow$ that is total $=\left(\frac{12}{5}\right)(400)=960$ birr

9. $\left[1-\left(\frac{1}{2}+\frac{1}{3}\right)\right]$ x length of post $=3\frac{1}{3}$

$$\therefore \text{ length of post } = \frac{10}{3} \times 6 = 20\text{m.}$$

10. $[1-(0.375+0.4)]$ x total money $= 4200$ birr

$$\therefore \text{ Total money } = \frac{4200}{1-0.775} = \frac{4200}{0.225} = 18666.6$$

11. D: $\left(\frac{4}{5} - \frac{4}{15}\right)$ x number $= 200 \Rightarrow$ number $= 375$

12. C: $\frac{3}{14} < \frac{1}{4} < \frac{2}{7}$

13. D

UNIT – THREE

3. PERCENT'S, RATIO AND PROPORTION

3.1 PERCENTS

Percent means *per hundred* or number out of 100.

- A percent can be represented as a fraction with a denominator of 100, or as a decimal.

Example: $56\% = \frac{56}{100} = 0.56$

Common fractional equivalents of percents and decimal

Percent	Fraction	Decimal
50%	$\frac{50}{100} = \frac{1}{2}$	0.5
25%	$\frac{25}{100} = \frac{1}{4}$	0.25
75%	$\frac{75}{100} = \frac{3}{4}$	0.75
20%	$\frac{20}{100} = \frac{1}{5}$	0.2

Solving percent Problems

Percent problems often require you to translate a sentence into a mathematical equation.

Example: (a) What is 40% of 60.

(b) 24 is 40% of what number?

(c) 24 is what percent of 60?

Translate the sentence into a mathematical equation as follows:

Let x = the unknown number.

(a) What is 40% of 60.

$$x = 40\% \text{ of } 60 = \frac{40}{100} \times 60 = 24$$

(b) 24 is 40% what number

$$24 = 40\% \text{ of } x \Rightarrow \frac{40}{100}x = 24$$

$$\therefore x = \frac{100}{40}(24) = 60$$

(c) 24 is what percent of 60?

$$24 = (x\%) \text{ of } 60 \Rightarrow 24 = \frac{x}{100}(60)$$

$$x = (24)\left(\frac{100}{60}\right) = 40$$

Illustrative Example

1. What percent of 30 is 6?

A. 10% B. 20% C. 30% D. 35%

Solution:

$$x\% \text{ of } 30 = 6 \Rightarrow \left(\frac{x}{100}\right)(30) = 6$$

$$x = \left(\frac{10}{3}\right)(6) = 20$$

Answer: B

2. 7 is 28% of what number

A. 20 B. 21 C. 25 D. 35

Solution:

$$7 = 28\% \text{ of } x \Rightarrow \frac{28}{100}\times = 7$$

$$\therefore x = \frac{100 \times 7}{28} = 25$$

Answer: C

3. If 15% of a number is 4.5, then 45% of the same number is

A. 1.5 B. 3.5 C. 13.5 D. 15

Solution:

$$15\% \text{ of } x = 4.5 \Rightarrow \frac{15}{100}x = 4.5$$

$$\Rightarrow x = \frac{4.5}{15}(100) = 30, \text{ then } 45\% \text{ of } 30 = \frac{45}{100}(30) = 13.5,$$

Answer: C

4. If a = 4b, what percent of 2a is 2b?
A. 10% B. 20% C. 25% D. 26%

Solution:

$$x\% \text{ of } 2a = 2b \Rightarrow \left(\frac{x}{100}\right)(2a) = 2b$$

$$\Rightarrow \left(\frac{x}{100}\right)(2(4b) = 2b \Rightarrow \frac{x}{25} = 1$$

$\therefore x = 25\%$ **Answer: C**

5. If m > 0, and x is m percent of y, then, interms of m, y is what percent of x.

A) 100m B) $\frac{1}{m}$ C) $\frac{10}{m}$ D. $\frac{10,000}{m}$

Solution: x = m% of y

$$\Rightarrow x = \frac{m}{100}(y) \Rightarrow x = \frac{my}{100}$$

$y = \frac{100x}{m}$, to find the percent of $\frac{100}{m}$ is multiply

by 100, thus, $y = \frac{10,000}{m}$ **Answer: D**

6. What is 10% of 20% of 30%?
A. 0.006% B. 0.6% C. 6% D. 60%

Solution:

10% of 20% of 30% = (0.1) (0.2)(0.3) = 0.006

$\Rightarrow$ 0.006 = 0.006 × 100% = 0.6% **Answer: B**

7. If a is a positive number, 400% of a is what percent of 400a.
A. 0.01% B. 0.1% C. 1% D. 10%

Solution:

400% of a = x% of 400a

$$\left(\frac{400}{100}\right)a = \frac{x}{100}(400a)$$

$$x = \left(\frac{400}{100}\right)a\left(\frac{100}{400a}\right) = 1$$

Answer: C

8. If a jar contains 24 white marbles and 48 black marbles. What percent of black marbles in the jar?

A. $66\frac{2}{3}\%$ B. 60% C. 25% D. 20%

Solution: We create a fraction.

$$\frac{\text{black marble}}{\text{of all marble}} = \frac{48}{24+48} = \frac{48}{72} = \frac{2}{3} \times 100\% = 66\frac{2}{3}\%$$

Answer: A

9. In a company with 180 employees, 108 of the employees are female. What percent of the employees are male?

A. 5% B. 25% C. 35% D. 40%

Solution:

There are 180 – 108 = 72 male employees.

What percent of the employees are male?

$$x\% \text{ of } 180 = 72 \Rightarrow \frac{x}{100}(80) = 72$$

$$x = \frac{72 \times 100}{180} = 40,$$

Answer: D

10. If there are 15 boys and 25 girls in a class, what percent of the boys in the class?

A. 10% B. 18% C. 25% D. 37.5%

Solution: There are 15 + 25 = 40 student in the class.

What percent of the class is boys $x\%$ of $40 = 15$

$$\left(\frac{x}{100}\right)(40) = 15 \Rightarrow x = \frac{15}{40}(100) = 37.5\%$$ **Answer: D**

11. In a class of 200 students, forty percent are girls. Twenty – five percent of the boys and 10 percent of the girls signed up for a tour to Bahirdar. What percent of the class signed up for the tour?

A. 19% B. 23% C. 25% D. 27%

Solution:

Girls are 40 percent of the class.

G = 40% of 200

$$\Rightarrow G = \left(\frac{40}{100}\right)(200) = 80$$

∴ **The number of boys: 200 – 80 = 120.**

- 25 percent of boys: $\frac{25}{100}(120) = 30$
- 10 percent of girls: $\left(\frac{10}{100}\right)(80) = 8$, signed up for the tour.

Thus $30 + 8 = 38$ students signed up. Next,
What percent of the students in the class who signed up for the tour?

$$\Rightarrow x\% \text{ of } 200 = 38 \Rightarrow \left(\frac{x}{100}\right)(200) = 38$$

$$x = \frac{38}{2} = 19$$

Answer: A

12. In a certain school, 40 percent of the students are boys. If there are 80 boys in the school. What is the total number of students in the school?

A. 32 B. 50 C. 120 D. 200

Solution:

Let x = total number of student

$\Rightarrow 40\%$ of $x = 80$

$$\Rightarrow \left(\frac{40}{100}\right)(x) = 80$$

$$\therefore x = (80)\left(\frac{100}{40}\right) = 200$$

Answer: D

13. If w is 10% less than x, and y is 30% less than z, then wy is what percent less than xz?

A. 10% B. 20% C. 37% D. 40%

Solution:

- w is 10% less than x: $w = x - 0.1x = 0.9x$
- y is 30% less than z: $y = z - 0.3z = 0.7z$

$\Rightarrow wy = (0.9x)(0.7z) = 0.63xz$

$\Rightarrow wy = xz - 37xz = 0.63xz$

That means wy is 37% less than xz **Answer: C**

3.2 Percent Increase and Decrease

Often you will need to find the percent of increase (or decrease). To find it, calculate the increase (or decrease) and divide it by the original amount:

$$\text{Percent of change} = \frac{\text{Amount of change}}{\text{original amount}} \times 100\%$$

The percent increase of quantity is $\frac{\text{actual increase}}{\text{orignial amount}} \times 100\%$

The percent decrease of quantity is $\frac{\text{actual decrease}}{\text{orginal amount}} \times 100\%$

Illustrative Example

14. A price decreased from birr 60 to birr 45. What was the percent decrease in the price?

A. 15% B. $33\frac{1}{3}\%$ C. 25% D. 40%

Solution:

$$\text{Percent decrease} = \frac{\text{actual decrease}}{\text{original amount}}$$

$$= \frac{(60-45)}{60} = \frac{15}{60} \times 100\% = 25\%$$ **Answer: C**

15. A price increased from birr 50 to birr 80. What was the percent increase in the price?

A. 60% B. 30% C. 50% D. 40%

Solution:

$$\text{Percent increase} = \frac{\text{actual increase}}{\text{original amount}} \times 100\%$$

$$= \frac{80-50}{50} \times 100\% = \frac{30}{50} \times 100\% = 60\%$$

Answer: A

16. What is 40% more than 40?

A. 44 B. 48 C. 56 D. 16

Solution: 40 + 40% of 40

$$= 40 + (40)\left(\frac{40}{100}\right) = 40 + 16 = 56$$ **Answer: B**

17. Fourteen is 50% less than what number?
A. 7 B. 28 C. 35 D. 14

Solution: $14 = x - (50\% \text{ of } x)$

$$14 = x - \frac{1}{2}x \Rightarrow 2x - x = 28$$

$$\therefore x = 28$$ **Answer: B**

18. A town has a population growth rate of 10% per year. The population in 1990 was 2000. What was the population in 1992?
A. 1600 B. 2200 C. 2400 D. 2420

Solution:

Since the population increased at a rate of 10% per year, the population of any year is the population of the previous year + 10% of that same year.

Hence the population in 1991 is the population of 1990 + 10% of the population of 1990:

2000 + 10% of 2000 = 2000 + 200 = 2200

Similarly, the population in 1992 is the population of 1991 + 10% of the population of 1991:

2200 + 10% of 2200 = 2200 + 220 = 2420 **Answer: D**

19. The price of an item increased by 25%. If the price of the item after the increase is birr 2.00, what was the original price?
A. 1.50 birr
B. 1.60 birr
C. 1.75 birr
D. 2.50 birr

Solution:

Let x = the original price

$x + 25\% \text{ of } x = 2 \Rightarrow x + 0.25x = 2$

$x(1 + 0.25) = 2 \Rightarrow 1.25x = 2$

$$\therefore x = \frac{2}{1.25} = 1.60 \text{ birr}$$ **Answer: B**

20. If Kedija must have a mark of 70% to pass a test of 40 items, the number of items she may miss and still pass the test is (EUEE 2003)
A. 6 B. 8 C. 12 D. 28

Solution: 30% of $40 = \left(\frac{30}{100}\right)(40) = 12$ **Answer: C**

21. In one class of a given school 30% of the students are boys. In a second class that is twice the size of the first, 45% of the students are boys. What percent of both classes are boys? (EUEE 2003)

A. $33\frac{1}{3}\%$ B. 36% C. $36\frac{2}{3}\%$ D. 40%

Solution: It is given that 1st class = 30% of x are boys = $0.3x$
2nd class = 45% of $2x$ are boys = $0.9x$
The percent of boys in both class.

$$\Rightarrow \frac{0.3x + 0.9x}{3x} \times 100\% = \frac{1.2x}{3x} \times 100\% = 40\%$$

Answer: D

22. What is x percent of y divided by y percent of x? (EUEE 2003)

A. 10 B. 1 C. y D. x

Solution:

$$\frac{x\% \text{ of } y}{y\% \text{ of } x} = \frac{\frac{xy}{100}}{\frac{yx}{100}} = 1$$

Answer: B

23. By how much is $\frac{3}{7}$ larger than 20 percent of 2? (EUEE 2004)

A. $\frac{1}{35}$ B. $\frac{4}{7}$ C. $\frac{3}{7}$ D. $\frac{2}{25}$

Solution:

$$x + 20\% \text{ of } 2 = \frac{3}{7} \Rightarrow x + \frac{40}{100} = \frac{3}{7}$$

$$x = \frac{3}{7} - \frac{2}{5} = \frac{15-14}{35} = \frac{1}{35}$$

24. If $x > 0$, $\frac{x}{25} + \frac{x}{50}$ is what percent of x?

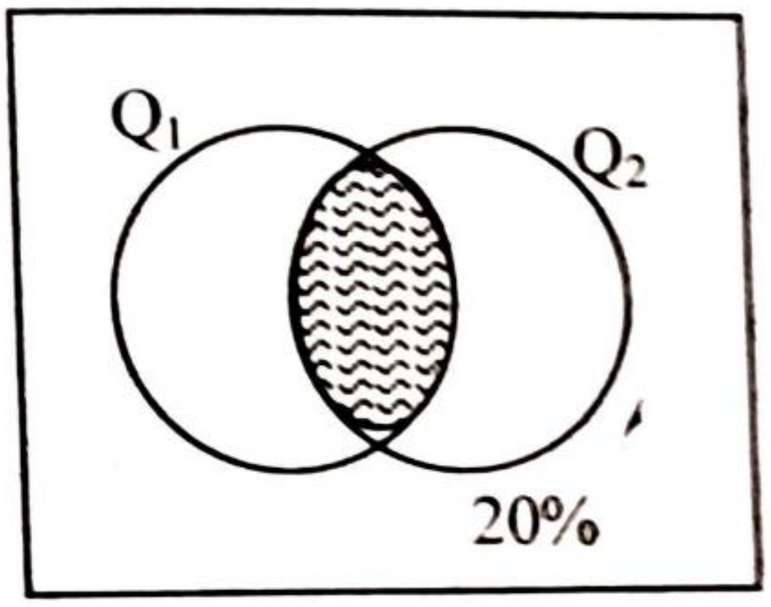
A. 6% B. 25% C. 37% D. 60%

Solution: $\frac{x}{25}+\frac{x}{50}=\frac{4x}{100}+\frac{2x}{100}=\frac{6x}{100}=\left(\frac{6}{100}\right)x=6\%\ \text{of}\ x$

25. If 75% of a class answered the first question on a certain test correctly; 55% answered second question on the test correctly, and 20% answered neither of the question correctly what percent answered both correctly?

A. 10% B. 20% C. 30% D. 50%

Solution:



$Q_1 = 75\%,$

$Q_2 = 55\%$

$Q_1 \cap Q_2 = x,$

$Q_1 \cup Q_2 = 80\%$

Use, $n(A) + n(B) = n(A \cap B) + n(A \cup B)$

$\Rightarrow Q_1 + Q_2 = (Q_1 \cap Q_2) + (Q_1 \cup Q_2)$

$75\% + 55\% = x + 80\%$

$\Rightarrow x = 75\% + 55\% - 80\% = 50\%$ **Answer: D**

26. On a test containing 60 questions Azeb answered 75% of the first 40 questions correctly. What percent of the other 20 questions does she need to answer correctly for her grade on the entire examination to be 80%? (EUEE)

A. 80% B. 85% C. 90% D. 100%

Solution:

$75\%\ \text{of}\ 40 + x\%\ \text{of}\ 20 = 80\%\ \text{of}\ 60$

$$\left(\frac{75}{100}\right)(40)+\frac{x}{100}(20)=\left(\frac{80}{100}\right)(60)$$

$$\Rightarrow 30 + 0.2x = 48 \Rightarrow 0.2x = 48 - 30 = 18$$

$$\therefore x=\frac{18}{0.2}=90$$

Answer: C

27. In 1995 the population of town A and town B were the same. From 1995 to 2005 however, the population of town A increased

by 60% while the population of town B decreased by 60%. In 2005, the population of town B was what percent of the population of town A?

A. 25% B. 36% C. 40% D. 60%

Solution: In 2005, the population of

- Town A $= x + 60\%$ of $x = x + 0.6x = 1.6x$
- Town B $= x - 60\%$ of $x = x - 0.6x = 0.4x$

Then in 2005, town B's population was

$$\frac{0.4x}{1.6x} \times 100\% = \frac{4}{16} \times 100\% = 25\%$$

Answer: A

28. If x is 25% of y, then y is what percent of x?

A. 400% B. 300% C. 250% D. 125%

Solution:

$$x = \left(\frac{25}{100}\right) y = \frac{1}{4} y \Rightarrow y = 4x$$

$\Rightarrow$ y = 400% of x

Answer: A

29. Which of the following items does not belong?

A. 20% of 60 C. 10% of 120

B. 40% of 30 D. 5% of 30

Solution:

$$\text{A. } 20\% \text{ of } 60 = \frac{20}{100}(60) = 12$$

$$\text{B. } 40\% \text{ of } 30 = \left(\frac{40}{100}\right)(30) = 12$$

$$\text{C. } 10\% \text{ of } 120 = \left(\frac{10}{100}\right)(120) = 12$$

$$\text{D. } 5\% \text{ of } 30 = \left(\frac{5}{100}\right)(30) = 1.5$$

Answer: D

30. If 120% of a is equal to 80% of b which of the following is equal to a + b?

A. 1.5a B. 2a C. 2.5a D. 3a

Solution:

- 120% of a = 80% of b $\Rightarrow$ 1.2a = 0.8b

$$\Rightarrow b = \frac{1.2a}{0.8} = \frac{12a}{8} = \frac{3}{2}a = 1.5a \text{ thus; } a+b = a+1.5a = 2.5a$$

Answer: C

31. What is a divided by a% of a

A. $\frac{a}{100}$ B. $\frac{100}{a}$ C. $\frac{a^2}{100}$ D. $\frac{100}{a^2}$

Solution:

$$a \div (a\% \text{ of } a) = a \div \left(\frac{a}{100} \times a\right) = a \div \left(\frac{a^2}{100}\right)$$

$$\Rightarrow a \times \frac{100}{a^2} = \frac{100a}{a^2} = \frac{100}{a}$$

Answer: B

32. If x is 80% of y, then y is what percent of x.

A. 80% B. 90% C. 120% D. 125%

Solution:

$$x = 80\% \text{ of } y \Rightarrow x = \left(\frac{80}{100}\right)y = 0.8y$$

$$\therefore\ y = \frac{x}{0.8} = \left(\frac{1}{0.8}\right)x = (1.25)x$$

$$\therefore\ y = 125\% \text{ of } x$$

Answer: D

33. 20% of 48 is equal to 16% of what number?

A. 54 B. 56 C. 60 D. 64

Solution:

$$20\% \text{ of } 48 = 16\% \text{ of } x \Rightarrow \left(\frac{20}{100}\right)(48) = \left(\frac{16}{100}\right)(x)$$

$$\therefore\ x = \left(\frac{20}{100}\right)(48)\left(\frac{100}{16}\right) = 60$$

Answer: C

34. A jar contains 2000 marbles. If 61.5% of them are red, 27.2% of them are white, and 10% of them are blue, how many are neither red, white, nor blue?

A. 13 B. 26 C. 12 D. 24

Solution:
Since 61.5 + 27.2 + 10 = 98.7, then 98.7% of the marbles are red, white or blue and the other 100% − 98.7% = 1.3% are some other colors.
Therefore: 1.3% of 2000 = 0.013 × 2000 = 26

Answer: B

35. If x is 150 percent of y, then y is what percent of x + y?
A. 60% B. 40% C. 100% D. 75%

Solution: x = 150% of y = 1.5y, but

$$y = t\% \text{ of } x + y \Rightarrow t\% = \frac{y}{x+y}$$

$$t\% = \frac{y}{1.5y + y} = \frac{y}{2.5y} = \frac{1}{2.5} = 0.4$$

$$\therefore\ t = 0.4 \times 100 = 40\%$$

Answer: B

36. 120 is increased by 10% and then decreased by 20%. What is the new number?
A. 132 B. 105.6 C. 36 D. 110

Solution: Here at

a) $129 + 0\% \text{ of } 120 = 120 + \frac{100}{100}(20) = 132$

Again 132 is decreased by 20%

$$132 - 20\% \text{ of } 132 = 132 - \left(\frac{20}{100}\right)(132) = 132 - 26.4 = 105.6$$

Answer: B

37. 250 increased by 40% is the same as what percent of 250?
A. 1.4% B. 14% C. 140% D. 40%

Solution: Here at,

250 increased by 40% = x% of 250

$$\Rightarrow 250 + 40\% \text{ of } 250 = (x\%)\,(250)$$

$$\Rightarrow 250 + \frac{40}{100}(250) = \left(\frac{x}{100}\right)(250)$$

$$350 = 2.5x \Rightarrow x = \frac{350}{2.5} = 140\%$$

Answer: C

38. A radio costs birr 690 including VAT at 15%. What is the cost without VAT?

Solution: Let x = cost without VAT.

Total cost = x + VAT = x + 15% of x

$\Rightarrow$ 690 = x + 15% of x = $x + 0.15x = 1.15x$

$$\therefore \quad x = \frac{690}{1.15} = 600\text{birr}$$

39. The price of an item was reduced by 10%, then later reduced by 20%. The two reductions were equivalent to a single reduction of:

A. 15% B. 28% C. 30% D. 70%

Solution: Let x be the price of an item.

- First reduction: x – 10% of $x = x - 0.1x = 0.9x$
- 2nd reduction: $0.9x$ – 20% of $0.9 = 0.9x - 0.18x = 0.72x$

$\Rightarrow$ Equivalent reduction: $\dfrac{x - 0.72x}{x} \times 100\% = 28\%$

Answer: B

40. From 2007 to 2008, the number of applicants to a certain job increased 12% to 6020. How many applicants were there in 2007 E.C.

A. 5,375 B. 6,840 C. 722.4 D. 7224

Solution:

The number of applicants in 2007 was $6020 \div 1.12 = 5375$

Answer: A

41. The population of a town doubled every 10 year from 1970 to 2000. What was the percent increase in population during this time?

A. 300% B. 800% C. 600% D. 700%

Solution: Let x = the number of population

- 1970 – 1980 = $2x$
- 1980 – 1990 = $4x$
- 1990 – 2000 = $8x$

Therefore, the % increase: $\dfrac{8x - x}{x} \times 100\% = 700\%$

Answer: D

42. Last year Hann's annual salary was 20,000 birr. This year's raise brings her to an annual salary of 25,000 birr. If she gets the same percent raise every year, what will be her salary next year?

A. 30,000 birr
B. 31,250 birr
C. 35,000 birr
D. 32,500 birr

Solution:

$$\text{Percent increase} = \frac{25,000 - 20,000}{20,000} \times 100\% = 25\%$$

Her raise next year will be 25% × 25,000 birr = 6,250 birr.
Her salary next year will be 25,000 + 6,250 = 31,250 birr

3.3 RATIO

A **ratio** is a comparison of two numbers or quantities by division.
Ratio can be expressed in the following ways:

a to b **a:b** $\frac{\mathbf{a}}{\mathbf{b}}$

Ratio must expressed in the same units.

Example: What is the ratio of 2m to 400cm?

Solution: The units must be the same, so change meter to centimeter
2m = 200cm.

$$\text{The ratio is } \frac{200\text{cm}}{400\text{cm}} = \frac{2}{4} = \frac{1}{2}$$

Note: *If* a set of objects is divided into two groups in the ratio of **a:b**, then:

i) First group contains: $\frac{a}{a+b}$ of the object.

ii) Second group contains $\frac{b}{a+b}$ of the object.

Illustrative Example

43. Last year, the ratio of the number of mathematics tests Daniel passed to the number of mathematics test he failed was 3: 2. What percent of his mathematics tests did Daniel pass?

A. 150% B. $66\frac{2}{3}\%$ C. 60% D. 40%

Solution:

$$\text{Daniel pass } \frac{3}{3+2} = \frac{3}{5} = 60\% \text{ of his mathematics tests.}$$

Answer: C

44. If the ratio of orange to apple to Mango fruit juice sold at shop was 9: 16: 15 on a particular day, what percent of the fruits sold were apple?

A. 40% B. 16% C. 60% D. 25%

Solution:

$$\frac{16}{9+16+15} = \frac{16}{40} \times 100\% = 40\%$$

45. If **a, b** and **c** are numbers such that **a: b: c** = 2: 3: 5 and b = 18, then what is the value of a + b + c?

A. 10 B. 30 C. 60 D. 900

Solution: $\frac{a}{b} = \frac{2}{3}$ and $\frac{b}{c} = \frac{3}{5}$

$$\Rightarrow \frac{a}{18} = \frac{2}{3} \Rightarrow a = \frac{2}{3}(18) = 12$$

$$\Rightarrow \frac{18}{c} = \frac{3}{5} \Rightarrow c = 18\left(\frac{5}{3}\right) = 30$$

Therefore a + b + c = 12 + 18 + 30 = 60. **Answer: C**

46. The ratio of Abebe's weight to Bekele's weight is 3: 4 and the ratio of Abebe's weight to chala's weight is 2: 5. What is the ratio of Chala's weight to Bekele's weights?

A. 8: 15 B. 15: 8 C. 1: 2 D. 2: 1

Solution: $\frac{\text{Abebe's wt}}{\text{Bekele's wt}} = \frac{A}{B} = \frac{3}{4}$ and $\frac{\text{Abebe's wt}}{\text{Chala's wt}} = \frac{A}{C} = \frac{2}{5}$

$$\frac{\text{Chala's wt}}{\text{Bekele's wt}} = \frac{C}{B} = \frac{A}{B}.\frac{C}{A} = \frac{3}{4}.\frac{5}{2} = \frac{15}{8}$$ **Answer: B**

47. If a : b = 3: 5 and a: c = 5 : 7 what is the value of b: c

A. 3: 7 B. 21:35 C. 21:25 D. 25

Solution: $\frac{a}{b} = \frac{3}{5}$ and $\frac{a}{c} = \frac{5}{7}$

$$\Rightarrow a = \frac{3}{5}b \text{, so that } \frac{a}{c} = \frac{\frac{3}{5}b}{c} = \frac{5}{7}$$

$$\Rightarrow \frac{b}{c} = \frac{5}{7} \times \frac{5}{3} = \frac{25}{21}$$ **Answer: D**

48. If $\frac{2}{3}$ of the workers in a office are women, what is the ratio of men to women?

A. 1: 2 B. 2: 5 C. 3: 5 D. 2: 3

Solution:

If every 3 workers, 2 are women, and 1 is men. Then, the ratio of women's to men is 1: 2.

Answer: A

49. From 2007 to 2008 E.C the number of boys in the school basket ball club decreased by 25%, and the number of girls in the club increased by 25%. The ratio of girls to boys in the club in 2008 was how many times the ratio of girls to boys in the club in 2007?

A. $\frac{2}{3}$ B. $\frac{5}{3}$ C. $\frac{4}{5}$ D. $\frac{3}{5}$

Solution:

Year	Number of girls	Number of boys	Ratio of girls to bys
2007	100	100	$\frac{100}{100}=1$
2008	125	75	$\frac{125}{75}=\frac{5}{3}$

Answer: B

50. If $\frac{x}{y}=\frac{4}{5}$ and $\frac{z}{y}=\frac{2}{5}$, then $\frac{x}{z}$

A. $\frac{2}{5}$ B. $\frac{1}{2}$ C. $\frac{8}{25}$ D. 2

Solution:

$\frac{x}{y}=\frac{4}{5}\Rightarrow x=\frac{4}{5}y$ and $\frac{z}{y}=\frac{2}{5}\Rightarrow z=\frac{2}{5}y$

Thus; $\frac{x}{z}=\frac{\frac{4}{5}y}{\frac{2}{5}y}=\frac{4}{5}.\frac{5}{2}=\frac{4}{2}=2$

Answer: D

3.4 PROPORTION

A proportion is an equation that states the two ratios are equivalent.
Example: The ratio of x to y is equal to the ratio of 5 to 4 is translated as $\frac{x}{y}=\frac{5}{4}$.

Example: What is the fourth proportion to 3, 4, 5?

Set up: $\frac{3}{4}=\frac{5}{x} \Rightarrow 3x=20 \Rightarrow x=\frac{20}{3}$

$\therefore$ The 4^{th} proportion is $x=\frac{20}{3}$.

3.5 DIRECT AND INVERSE PROPORTIONALITY

Direct Proportionality
As one quantity increase (decreases) another quantity also increases (decreases).
This type of problem can be solved by setting up a direct proportion.
Two variables are direct proportional if one is a constant multiple of the other.

$\frac{y}{x}=k \Rightarrow y=kx$ where k is a constant

Illustrative Example

51. A machine makes 18 articles in 5 hours. How many articles will it make in 15 hours?
A. 45 B. 54 C. 6 D. 9

Solution: Use direct proportionality $\frac{18}{5}=\frac{x}{15} \Rightarrow x=15\left(\frac{18}{5}\right)=54$

Answer: B

52. A certain machine fills a bag with 7 kg of a potato chips in 3.5 seconds. At this rate, how many seconds will it take the machine to fill a bag with 15 kg of potato chips? (UEE 2005)
A. 6.5 B. 7.0 C. 7.5 D. 8

Solution:

$\frac{7\text{ kg}}{3.5\text{ second}}=\frac{15\text{ kg}}{x} \Rightarrow x=\left(\frac{3.5}{7}\right)(15)=7.5$

Answer: C

53. If **a** apples cost **c** cents, how many apples can be bought for **d** birr?

A. $\frac{100d}{ac}$ B. $\frac{ad}{100}$ C. $\frac{c}{100ad}$ D. $\frac{100ad}{c}$

Solution: $\frac{\text{apple}}{\text{cents}} = \frac{a}{c} = \frac{x}{100\,d} \Rightarrow x = \frac{100\,ad}{c}$ **Answer: D**

54. A car travels 240 km on 15 liters of petrol. How much petrol will be needed for a journey of 408 km?

A. 17ℓt B. 34ℓt C. 28.9ℓt D. 32ℓt

Solution:

Set up a proportion $\frac{240\text{ km}}{15\ell t} = \frac{408\text{km}}{y}$

$\Rightarrow y = \frac{15}{240}(408) = 28.9\ell t$ **Answer: C**

55. At a certain school, 45% of the students purchased a year book. If 540 students purchased year books, how many students did not busy a year book?

A. 243 B. 540 C. 575 D. 660

Solution:

Set up a proportion

Since 45% bought a year book, 55% did not.

$\Rightarrow \frac{45\%}{55\%} = \frac{540}{x} \Rightarrow x = 540\left(\frac{55}{45}\right) = 660$ **Answer: D**

Inverse Proportionality

If an increase in one quantity produce a decrease in a second quantity or if a decrease in one quantity produce an increase in a second quantity, the two quantities are in **inverse proportional.**

The statement "y is inversely proportional to x" is written as $y = \frac{k}{x}$, where k is a constant $xy = k$.

Illustrative Example

56. If 12 men can do a job in 15 days how long will it take in 9 men to do the same job.

A. 30 days B. 20 days C. $11\frac{1}{4}$ days D. 18 days

Solution:

Set up: Inverse proportion

- Since it is the same job, the number of men increases working day decrease.
- (12 men, 15 days) = $12 \times 15 = 180$ constant
- (9 men, y days) = $9y = 180$, constant.

$\Rightarrow 9y = 180$ implies $y = \frac{180}{9} = 20$ days **Answer: B**

57. If $\frac{ab}{c} = 7 = \frac{rs}{t}$, $r = 3a$, $c = 2t$, and $s = kb$, what is k

A. 6 B. $\frac{1}{6}$ C. 3 D. $\frac{1}{3}$

Solution: $\frac{ab}{c} = \frac{r6}{t}$, replace r by 3a, c by 2t, and s by kb:

$$\Rightarrow \frac{ab}{2t} = \frac{(3a)(kb)}{t} \Rightarrow abt = 6abtk$$

$$\Rightarrow k = \frac{abt}{6\,abt} = \frac{1}{6}$$

Answer: B

58. The decimal 0.1 is how many times greater than the decimal $(0.001)^3$?

A. 10^8 B. 10^5 C. 10^2 D. 10

Solution:

$0.1 = 10^{-1}$ and $(0.001)^3 = (10^{-3})^3 = 10^{-9}$

Now, $\frac{0.1}{(0.001)^3} = \frac{10^{-1}}{10^{-9}} = 10^{-1-(-9)} = 10^{-1+9} = 10^8$

Answer: A

59. The difference between the number obtained by increasing a certain number by 8% and that obtained by diminishing it by 4% is 534. Find the original number

A. 13350 B. 4450 C. 6675 D. 2136

Solution:

$$\text{Original number} = \frac{\text{Difference of result value}}{\text{Difference of percents}}$$

$$\frac{534}{8-(-4)} \times 100 = 4450$$

Answer: B

60. The price of sugar is increased 25% before budget and in budget 15% is increased again.

Find the total % increase in the price of the sugar.

A. 40% B. 20% C. 43% D. 35%

Solution: Here, the price is increased in two steps.

$$\therefore \text{The net \% change} = x + y + \frac{xy}{100}\text{; let } x = 25,\ y = 15$$

$$\therefore \text{The net \% change in price} = 25 + 15 + \frac{(25)(15)}{100} = 43$$

Answer: C

61. A number X exceeds Y by 20%. By what percent is y shorter *x*?

A. 20 B. 16.7 C. 80 D. 18

Solution:

$$\text{\% of shortness} = \frac{\text{\% excess}}{100 + \text{\% excess}} \times 100$$

$$= \frac{20}{100+20} \times 100 = 16.666\ldots = 16.7$$

Answer: B

62. A number is by mistake divided by $\frac{2}{3}$ instead of being multiplied by $\frac{3}{2}$. What is the percentage error in the calculation?

A. $55.\dot{5}\%$ B. 150% C. $66.\dot{6}\%$ D. $83.\dot{3}\%$

Solution:

$$\%\text{ of error} = \frac{\frac{3}{2} - \frac{2}{3}}{\frac{3}{2}} \times 100 = \frac{5}{9} \times 100 = 55.5\% \qquad \textbf{Answer: A}$$

63. One liter of water is evaporated from 9 liters of a solution containing 8% salt. Find the percentage of salt in the remaining solution.

Solution: From the solution of 9 liters only water is evaporated so the amount of salt remains the same even after evaporation. Therefore, in the solution, % of

$$\text{Salt} = \frac{\text{amount of salt}}{\text{rest amount of solution}} \times 100$$

$$= \frac{8\% \text{ of } 9}{(9-1)} \times 100 = 9\%$$

64. If 'a' is x% more than b and b is y% less than 'a', then find a relation between x and y.

Solution:

'b' is y% less than 'a' but

'a' is x% more than 'b'

$$\Rightarrow \text{y\% = \% less of 'b' over 'a'}$$

$$= \frac{\text{\% more of 'a'}}{100 + \text{\% more of a}} \times 100$$

$$\Rightarrow y = \frac{x}{100+x} \times 100 \Leftrightarrow \frac{1}{y} = \frac{100+x}{x} \times \frac{1}{100}$$

$$\Rightarrow \frac{1}{y} = \frac{100+x}{100x} = \frac{1}{x} + \frac{1}{100} \Rightarrow \frac{1}{y} - \frac{1}{x} = \frac{1}{100}$$

Review Exercise

1. Abebe bought a birr 560 on sale at 20% off. How much did he pay, including 15% sales tax?
   A. 112 birr B. 448 birr C. 89.6birr D. 537.6 birr
2. 80 is 16% of what number?
   A. 400 B. 500 C. 128 D. 12.8
3. 63 is what percent of 42?
   A. 150% B. $66\frac{2}{3}\%$ C. 15% D. 18%

4. What number is 38% of 70
   A. 266 B. 26.6 C. 2660 D. 184.2
5. On a test consisting of 80 questions Heran answered 75% of the first 60 questions correctly. What percent of the other 20 questions did she need to answer correctly for her grade on the entire exam to be 80%.
   A. 85% B. 87.5% C. 90% D. 95%
6. If 30 students took an exam and 12 of them failed, what percent of them passed?
   A. 80% B. 40% C. 60% D. $66\frac{2}{3}\%$
7. There are twice as many girls as boys in mathematics class. If 30% of the girls and 45% of the boys have already handed in their book reports, what percent of the students have not yet handed in their reports?
   A. 35% B. 65% C. 75% D. 25%
8. 10 is what percent of A?
   A. $\frac{1000}{A}\%$ B. $\frac{100}{A}\%$ C. $\frac{10\%}{A}$ D. $\frac{1}{10\,A}\%$
9. 120 is increased by 10% and then decreased by 20%. What is the percent of decrease?
   A. 10% B. 20% C. 12% D. 30%
10. 600 decreased by what percent is the same as 60% of 600.
    A. 600% B. 36% C. 40% D. 30%
11. 150% of 5C is b. What percent of 2b is c?
    A. $5\frac{1}{2}\%$ B. $6\frac{2}{3}\%$ C. 15% D. 75%
12. If $3a = 2b$ and $3b = 5c$. what is the ratio of a to c.
    A. 3: 5 B. 2:15 C. 9:10 D. 10: 9
13. Three associates agreed to split the profit of an investment in the ratio of 2: 5: 8. If the profit was 3000 birr, what is the difference between the largest and the smallest share?
    A. 200 B. 1200 C. 1000 D. 800
14. Abera can read 72 pages per hour. At this rate, how many pages can he read in 72 minutes?
15. If $a = 2b$, $3b = 4c$ and $5c = 6a - 7$. What is the value c?
16. If $a = 2b$, $3b = 4c$, and $5c = 6d$. What is the ratio of a to d?

17. If a, b and c are positive numbers such that $3a = 4b = 5c$ and if $a + b = kc$, what is the value of k?

A. $\frac{35}{12}$ B. $\frac{5}{7}$ C. $\frac{10}{7}$ D. $\frac{12}{35}$

18. If $\frac{3}{4}$ of a number is 7 more than $\frac{1}{6}$ of the number, what is $\frac{5}{3}$ of the number?

A. 12 B. 15 C. 18 D. 20

19. The ratio of a to b is 7:3. The ratio of b to c is 2:5. What is the ratio of a to c?

A. 14: 6 B. 6: 15 C. 14:15 D. 15:14

Answer Explanations

1. D. Since 20% of 560 is 112. Abebe saved birr 112, and thus paid 560 - 112 = 448 birr for the radio. He then had to pay 15% sales tax (VAT) on the birr 448: $\left(\frac{20}{100}\right)(448) = 89.6$ birr, so the total cost was birr: 448 + 89.6 = 537.6

2. B, Since 80 = 16% of x $\Rightarrow x = \frac{80}{0.16} = 500$

3. A, 63 = x% of 42 $\Rightarrow x = \frac{63}{42} \times 100\% = 150\%$

4. B, 38% of 70 $= \left(\frac{38}{100}\right)70 = 26.6$

5. D, 75% of 60 + x% of 20 = 80% of 80

$\Rightarrow 45 + 0.20x = 64 \Rightarrow 0.20x = 64 - 45 = 19$

$\therefore x = \frac{19}{0.2} = 95\%$

6. C, 30 – 12 = 18 student passed

$\Rightarrow \frac{18}{30} \times 100\% = 60\%$

7. B, since 30% of 2x + 45% of x = 0.6x + 0.45x = 1.05x Then, 1.05x of the 3x students have handed in the reports, and 3x –1.05x = 1.95x have not.

$$\Rightarrow \frac{1.95x}{3x} \times 100\% = 65\%$$

8. A, solve the equation, $10 = \left(\frac{x}{100}\right)A \Rightarrow xA = 1000$

$$\Rightarrow x = \frac{1000\%}{A}$$

9. C, since $120 + 10\%$ of $120 = 120 + \frac{10}{100}(120) = 132$

Again 132 is decreased by 20%.

$$132 - 20\% \text{ of } 132 = 132 - \frac{20}{100}(132) = 132 - 26.4 = 105.6$$

$$\text{Percent of decrease} = \frac{120 - 105.6}{120} \times 100\% = \frac{14.4}{120} \times 100\% = 12\%$$

10. C, 600 decreased by $x\%$ = 60% of 600

$600 - x\%$ of $600 = 60\%$ of 600

$$600 - \left(\frac{x}{100}\right)(600) = \left(\frac{60}{100}\right)(600)$$

$$600 - 6x = 360 \Rightarrow 6x = 240$$

$$\therefore x = \frac{240}{6} = 40\%$$

11. B, $\left(\frac{150}{100}\right)(5c) = b \Rightarrow 7.5c = b$

$$\therefore x\% \text{ of } 2b = c \Rightarrow \left(\frac{x}{100}\right)(2b) = c$$

$$\Rightarrow \frac{x}{100} = \frac{c}{2b} = \frac{c}{2.5(7.5c)} = \frac{1}{15}$$

$$\therefore x = \frac{1}{15} \times 100\% = 6\frac{2}{3}\%$$

12. D, since $3a = 2b \Rightarrow b = \frac{3}{2}a$, then $3\left(\frac{3}{2}a\right) = 5c$

$$\Rightarrow \frac{9}{2}a = 5c \Rightarrow 9a = 10c$$

Therefore, $\frac{9a}{c} = 10 \Rightarrow a:c = \frac{10c}{9} = 10:9$.

13. B since $2x + 5x + 8x = 3000$, then $15x = 3000 \Rightarrow x = 200$
$\therefore 8x - 2x = 200(8) - 2(200) = 1600 - 400 = 1200$

14. $\frac{72 \text{ pages}}{\text{hour}} = \frac{72 \text{ page}}{60\text{min}} = \frac{x \text{ page}}{72 \text{ minute}}$

$$\therefore x = \frac{72 \times 72}{60} = 86.4 \text{ pages}$$

15. Since, a = 2b, then 3a = 6b; and since 3b = 4c, then 6b = 8c.
Therefore, 3a = 8c and 6a = 16c, so $5c = 6a - 7 \Rightarrow 5c = 16c - 7$

$-11c = -7$ implies $c = \frac{7}{11}$

16. Since $a = 2b \Rightarrow b = \frac{a}{2}$ and $3b = 4c \Rightarrow 3\left(\frac{a}{2}\right) = 4c \Rightarrow 3a = 8c$

Because $5c = 6d \Rightarrow c = \frac{6}{5}d$

Therefore, $3a = 8\left(\frac{6}{5}d\right) \Rightarrow a = \frac{16}{5}d$.

Therefore, a to d = 16: 5.

17. A, since $3a = 5c \Rightarrow a = \frac{5}{3}c$ and $4b = 5c \Rightarrow b = \frac{5}{4}c$

Thus, $a + b = \frac{5}{3}c + \frac{5}{4}c = \frac{35}{12}c$

18. D, since $\frac{3}{4}$ of $x = 7 + \frac{1}{6}$ of x

$$\Rightarrow \frac{3}{4}x = \frac{1}{6}x + 7 \Rightarrow \frac{3}{4}x - \frac{1}{6}x = 7$$

$$\Rightarrow \frac{18x - 4x}{24} = 7 \Rightarrow \frac{14x}{24} = 7$$

$$x = 7 \times \frac{24}{14} = 12$$

Therefore $\frac{5}{3}(12) = 20$

19. C, since $\frac{a}{c} = \frac{a}{b} \cdot \frac{b}{c} = \frac{7}{3} \cdot \frac{2}{5} = \frac{14}{15}$

UNIT – FOUR

4. AVERAGE

How to find the Average

- Average $= \dfrac{\text{sum of terms}}{\text{number of terms}}$
- Sum of terms = (Average) (number of terms)

How to find combined average.

> If the average of **n** item is A and the average of **m** item is B then the average of **n + m** item is equal to $\dfrac{\mathbf{nA + mB}}{\mathbf{n + m}}$

Illustrative Example

1. If the average of 38, 41, 44, 46, and x is 40, what is x

   A. 28 B. 30 C. 31 D. 34

   Solution: Using the general idea of average:

$$\frac{38 + 41 + 44 + 46 + x}{5} = 40$$

$$\Rightarrow 169 + x = 5x40 \Rightarrow x = 200 - 169 = 31$$ **Answer: C**

2. For a certain student, the average of ten test scores is 70. If the high and low scores are dropped, the average is 74. What is the average of the high and low scores?

   A. 72 B. 73 C. 54 D. 58

   Solution:

   - The sum of the ten scores is $70 \times 10 = 700$
   - The sum of the eight scores after the two scores have been dropped is $8 \times 74 = 592$. So the two scores that were dropped total $700 - 592 = 108$. Then the average of the two dropped is $\dfrac{108}{2} = 54$. **Answer: C**

3. In a certain shipment, the average weight of six packages is 750 kg. If another package is added to the shipment, the average weight of seven packages is 764 kg. What is the weight of the additional package?

A. 848 kg B. 900 kg C. 850 kg D. 757 kg

Solution:

- Total weight of six package is $6 \times 750 = 4500$ kg.
- Total weight of seven packages is $7 \times 764 = 5348$ kg

So the weight of the final package is 5348 kg – 4500 kg = 848 kg

Answer: A

4. In a certain group of student, five of the students are 16 years old and three of the students are 18 years old. What is the average age in years of the student in the group.

A. 17.5 B. 14.25 C. 4.25 D. 16.75

Solution:

We use the combined average.

$$\frac{5\times16+3\times18}{5+3}=\frac{134}{8}=16.75$$

Answer: D

5. If the average of seven numbers is 3. After one number is removed the new average is – 2. What number was removed?

A. 9 B. 33 C. 21 D. 30

Solution:

- Sum of seven number = $7 \times 3 = 21$
- Sum of six number = $6 \times (-2) = -12$

  Thus, the number removed = $21 - (-12) = 33$.

Answer: B

6. On Thursday, 20 of the 25 students in a chemistry class took a test, and their average was 80. On Friday, the other 5 students took the test, and their average was 90.
What was the average for the entire class?

A. 80 B. 82 C. 84 D. 85

Solution:

- Sum of the first 20 student = $20 \times 80 = 1600$ pt.
- Sum of the other 5 student = $5 \times 90 = 450$ pt.

  Therefore, the average of the entire

$$\text{Class} = \frac{20\times80+5\times90}{20+5}=\frac{2050}{25}=82$$

Answer: B

7. The girls' average score is 60. The boys' average score is 57. If there are twice as many boys as girls, what is the overall average?

A. 58.5 B. 58 C. 59 D. 57.5

Solution:

Set up: Weighted Avg $= \dfrac{1 \times 60 + 2 \times 57}{3} = \dfrac{174}{3} = 58$

Answer: B

4.1 Average Speed

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

Illustrative Example

8. For the first 3 hours of her trip, Heran drove at 50km per hour. Then, because of construction delays, she drove at only 40 km per hour for the next 2 hours.
What was her average speed, in kilometer per hour, for the entire trip?

Solution:

$$\text{Average speed} = \frac{(3)(50)+2(40)}{5} = \frac{230}{5} = 46 \text{ km/h}$$

9. Roza drove 120 km one way at an average speed of 40 km/hr and returned by the same 120 km route at an average speed of 60 km/hr. what was Roza's average speed for the entire 240 km round trip?

A. 48 km/hr
B. 100 km/ hr
C. 50 km/hr
D. 44 km/hr

Solution:

Set up: To drive 120 km at 40 km/hr takes 3 hours.
To return at 60km/hr takes 2 hours. The total time, then is 5 hours.

$$\text{Average speed} = \frac{240 \text{ km}}{5 \text{ hours}} = 48 \text{ km/hr}$$

Answer: A

How to find the AVERAGE of CONSECUTIVE NUMBERS.

- The average of consecutive numbers is **the average of the smallest number and the largest number.**

Illustrative Example

10. What is the average of all integer from 11 to 75.

Solution: $\dfrac{11+75}{2} = \dfrac{86}{2} = 43$

11. What is the average of all integer:

a. from 0, 1, 2, … 100

b. from 1, 2, 3 … 100

Solution:

a. $\dfrac{0+100}{2} = \dfrac{100}{2} = 50$

b. $\dfrac{1+100}{2} = \dfrac{101}{2} = 50.5$

12. What is the sum of the integers from 10 through 60 inclusive?

A. 1200 B. 1,230

C. 1,750 D. 1,785

Solution:

Set up: Average $= \dfrac{10+60}{2} = 35$

Numbers of terms = 60 – 10 + 1 = 51

Sum = (Average) × number of terms

= 35 × 51 = 1, 785

13. The sum of the first nine natural numbers (1, 2, 3, … 9) is (EUEE)

A. 90 B. 55 C. 45 D. 100

Solution: Sum= (Average) (number of terms)

$= \left(\dfrac{1+9}{2}\right)(9) = 45$ **Answer: C**

14. The sum of five consecutive odd integers is 425. What is the greatest of these integers?

A. 85 B. 87 C. 89 D. 83

Solution: Sum = (average) (number of terms)

425 = (average) (5)

∴ Average = 425 ÷ 5 = 85 (middle integer)

∴ 81, 83, 85, 87, 89 **Answer: C**

15. If Adane reads y pages on Wednesday, twice as many pages on Thursday as on Wednesday, and 3 pages more on Friday than on Thursday, what was Adane's average number of pages read over those three days?

A. $3y + 3$ B. $y + 1$ C. $\frac{5}{3}y + 1$ D. $5y + 3$

Solution: Adane read: y pages on Wednesday,

- 2y pages on Thursday
- 2y + 3 pages on Friday

So, average $= \frac{y + 2y + 2y + 3}{3} = \frac{5y}{3} + 1$ **Answer: C**

16. The average of nine number is 16. The average of four of these numbers is 11. What is the average of the other five numbers? **(EUEE)**

A. 14.46 B. 20 C. 1 D. 5

Solution: $9 \times 16 = 4 \times 11 + 5x \Rightarrow 5x = 144 - 44 = 100$

$$\therefore x = \frac{100}{5} = 20$$ **Answer: B**

17. The average of 8 numbers is 10, and one of the number is 14. If 14 is replaced with 30, then what is the new average.

A. 21 B. 12 C. 14 D. 24

Solution: Let the seven unknown number be $x_1, x_2, x_3, \ldots x_7$.

$\Rightarrow$ Average of 8 numbers

$$= \frac{x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + 14}{8} = 10$$

Replacing 14 with 30 (= 14 + 16)

$$\text{New average} = \frac{x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + (14 + 16)}{8}$$

$$= \frac{x_1 + x_2 x + x_3 + x_4 + x_5 + x_6 + x_7 + 14}{8} + \frac{16}{8}$$

$= 10 + 2 = 12,$ **Answer: B**

4.2 DATA INTERPRETATION

Occasionally a question or set of questions will be based on data provided in a table or graph. Some examples of tables and graphs are given below.

Illustrative Example

18. Refer to the following table.

Population by Age Group (in thousands)	
Age	Population
15 years and under	52, 464
16 – 40 years	74, 593
41 – 60 years	38, 720
60 years and over	21, 084

How many people are 40 years old or younger?

A. 74,593,000 C. 127,057,000

B. 59,804,000 D. 134,397,000

Solution: The figures in the table are given in thousands.

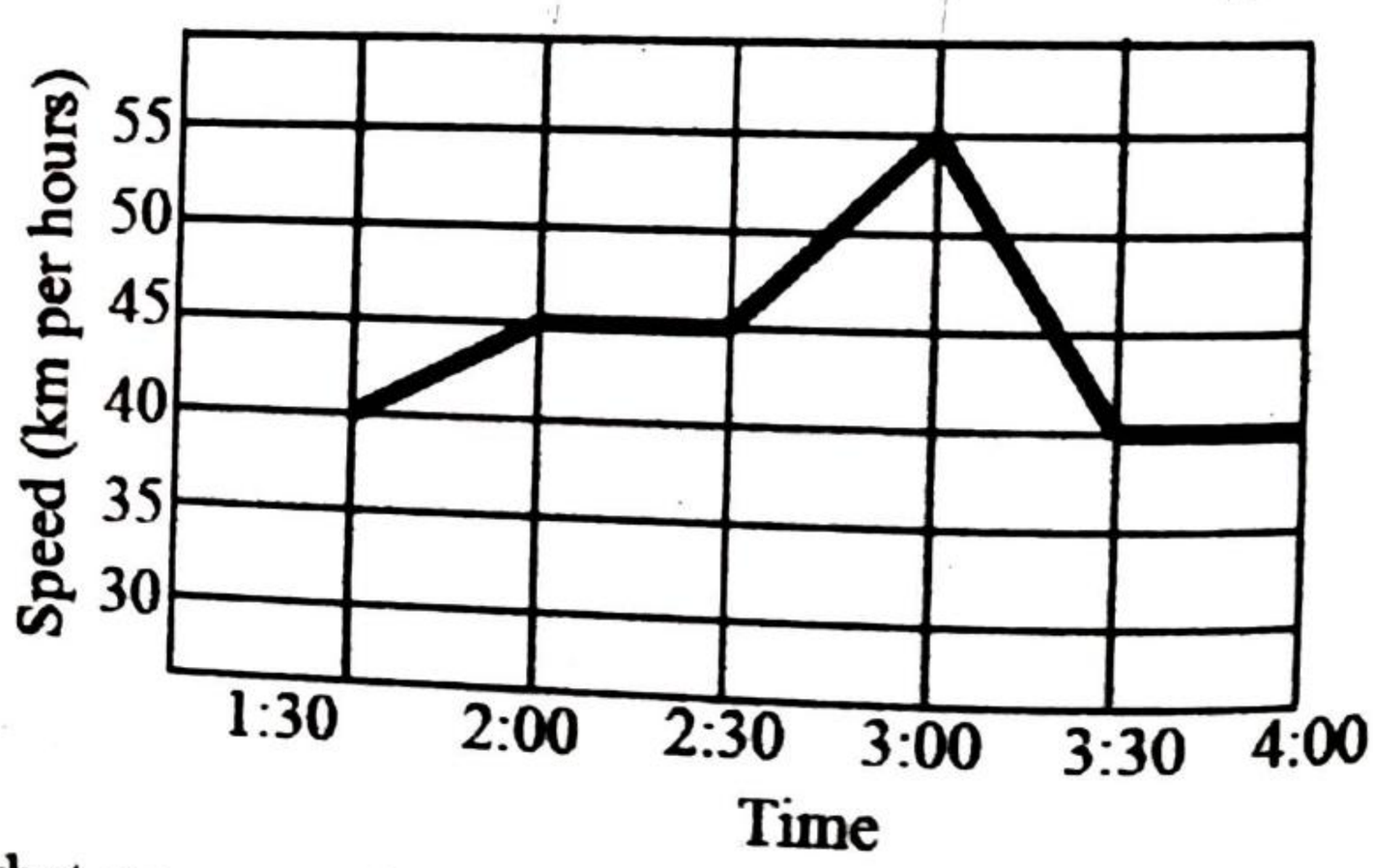
Therefore, 52,464 + 74,593 = 127,057 thousand

⇒The result is 127,057 thousand, which is 127,057,000.

Answer: C

From question 19 – 20 refer to the following graph.

Speeds at which Robel Drove on Saturday Morning



19. For what percent of the time was Robel diriving at 45 km/hr or faster?

A. 40% B. 50% C. 25% D. 20%

Solution:

Of the $2\frac{1}{2}$ hours (from 1:30 until 4:00) that Robel was driving; he was going 45 km/hr or faster for $1\frac{1}{4}$ hour s (from 2: 00 until 3:15).

Therefore, he was driving 45 km/hr or faste is $\frac{1\frac{1}{4}\text{hr}}{2\frac{1}{2}\text{hr}} \times 100\%$

$$= \frac{5}{4} \times \frac{2}{5} \times 100\% = \frac{1}{2} \times 100\% = 50\%$$ **Answer: B**

20. How far, did Robel drive between 3:30 and 4:00?

A. 35km B. 17.5km C. 0 D. 37

Solution:

During the half hour between 3:3 0 and 4 hour Robel was driving at constant rate of 35 km/hr so, he drove:

$\frac{1}{2}\text{hr} \times 35\text{km/hr} = 17.5\text{km}$ **Answer: B**

21. What was Robel's average speed, between 1: 30 and 3:00?

A. 45 km/hr
B. 47.5km/hr
C. 48km/hr
D. 45.83km/hr

Solution:

From the graph it is clear that, from 1: 30 to 2:00 Robel's speed increased steadily fom 40 to 45 kilometer per hour,

So his average was $\frac{40+25}{2} = 42.5\text{km/hr}$

From 2:00 to 2:30 his average speed was 45 km/hr.

From 2:30 to 3:00 his average speed was

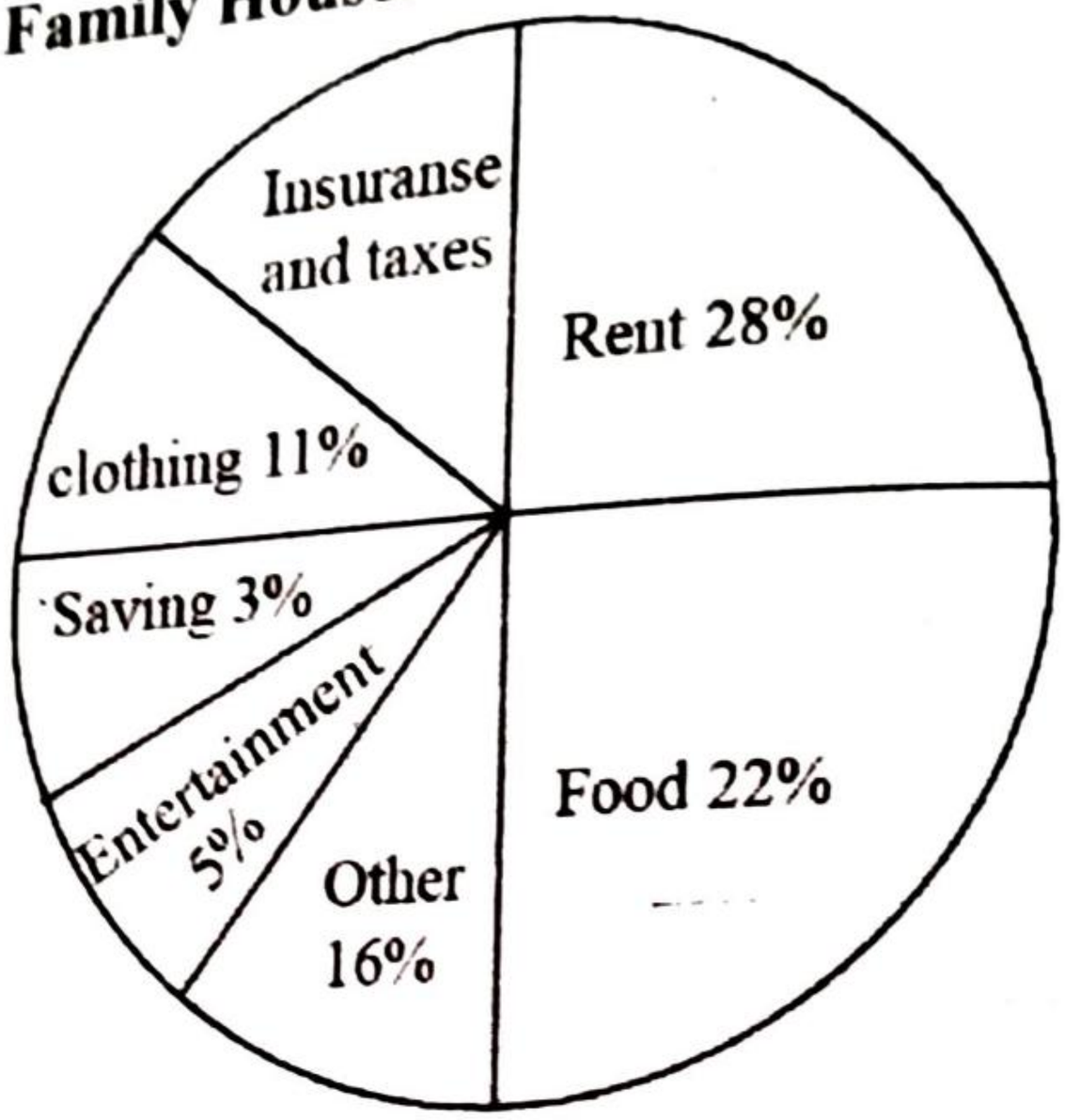
$\frac{45+55}{2} = 50\text{km/hr}$.

Therefore, the entire hour, he averaged

$\frac{42.5+45+50}{3} = 45.83$ **Answer: D**

From **Questions 22 – 23** refer to the following graph.

2017 Engineer Family Household Budget



22. If the engineers' in come in 2017 was 12,000 birr, how much more did they spend on insurance and taxes than they did on clothing?

A. 1,800 birr C. 2,000 birr
B. 1,320 birr D. 480 birr

Solution:
The total percent for the six catagories for which percents are given is 28% + 22% + 16% + 5% + 3% + 11% = 85%.
So the percent of their income that the engineers spend on insurance and taxes is 100%−85% = 15%. Since they spend 11% on clothing, the difference between the two catagries 15% −11% = 4%. Therefore, 4% of 12,000 is $0.04 \times 12{,}000 = 480$.

Answer: D

23. What is the degree measure of the central angle of the sector representing insurance and taxes?

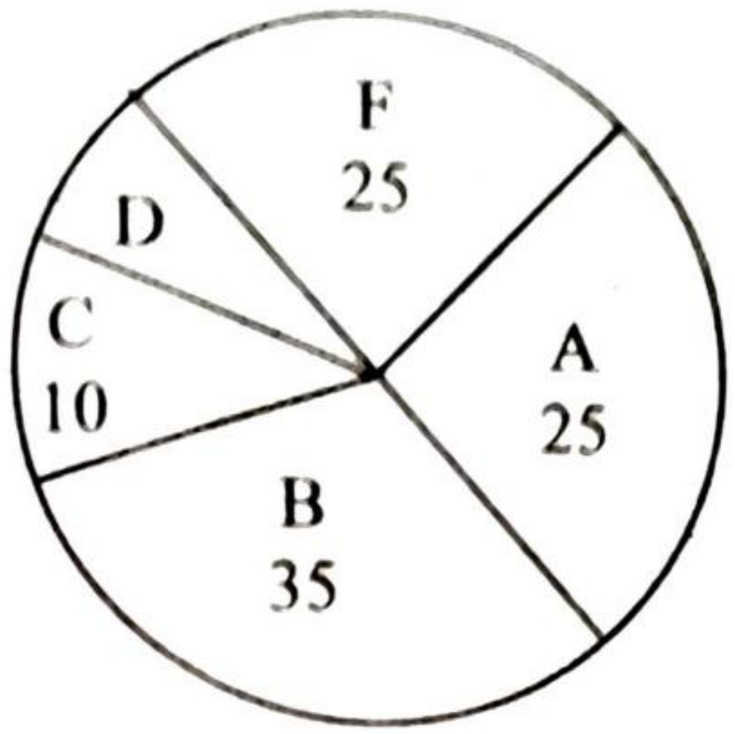
A. 72° B. 60° C. 54° D. 45°

Solution: 15% of $360° = 0.15 \times 360° = 54°$

Answer: C

Questions 24 refer to the following graph.

Distribution of Grades on the Final Exam in Mathematics.



24. If 800 students took the exam how many earned grades of D.

A. 50 B. 25 C. 40 D. 80

Solution:

Since 20% + 30% + 35% + 10% = 95%

5% of the students earned grades of D

so that 5% of 800

$= 0.05 \times 800 = 40$

Answer: C

Questions 25 – 26 refer to the following Tables

DISTRIBUTION OF IMPORTS AND EXPORTS FOR COUNTRY X IN 2005 E.C.

Imports 200 millions item	
Autos	50 %
Textiles	30 %
Food	5 %
Techno	15 %

Exports 100 millions item	
Auto	10 %
Textiles	20%
Food	40 %
Techno	30 %

25. How many textiles did country X export in 2005 E.C

A. 20 million B. 25 million

C. 30 million D. 24 millions

Explanation:

The table shows that 100 millions item where exported in 2005 and 20% were textiles. Hence $100 \times 20\% = 100 \times 0.2 = 20$ millions.

Answer: A

26. In how many categories did the total number of items. (import and export) exceed 75 millions.
A. 1 B. 3 C. 4 D. 2

Explanation:

	Import	Export	Total
Autos	100	10	110
Textiles	60	20	80
Food	10	40	50
technology	30	30	60

Hence autos and textiles exceeded 75 million total items,

Answer: D

Questions 27 – 29 refer to the following table.
Number of automobile and automobile accidents town (*x*)

Year	Number of car in town *x*	Number of car accidents in town *x*.
1990	2100	1500
1991	5200	3400
1992	12,000	8,400
1993	11,000	7,500
1994	13,000	8,500

27. For which specific year is the ratio of number of car accidents to the number of car highest?
A. 1994 B. 1990 C. 1993 D. 1992

Solution:

In 1994: $\frac{8500}{13,000} = 0.654$

In 1990: $\frac{1500}{2100} = 0.714$

In 1993: $\frac{7500}{11,000} = 0.681$

In 1992: $\frac{8400}{12,000} = 0.700$

Answer: B

28. The number of accidents in 1991 was what percentage of the number of cars?

A. 71% B. 65% C. 65.4% D. 68%

Solution: $\frac{3400}{5200} \times 100 = 65.4\%$ **Answer: C**

29. If no car in 1990 was involved in more than five accidents, what is the minimum number been in accidents in 1990?

A. 420 B. 1040 C. 680 D. 300

Solution:

The number of car involved in accidents will be minimized when each car has exactly 5 accidents.

Now, divide 1500 by 5 yields

$\frac{1500}{5} = 300$ **Answer: D**

From questions 30 – 31 refer the following table.

v	2	7	12	17	22	27	32
f	4	6	8	18	6	5	3

30. What will be the value of the resulting mean?

A. 24.3 B. 16.3 C. 21 D. 15

Solution:

The sum all observations: $\sum_{i=1}^{n} f_i \, V_i$

$\Rightarrow 2\times 4 + 7x6 + 12\times 8 + 17\times 18 + 22\times 6 + 27\times 5 + 32\times 3 = 815$

The total number of observation:

$\sum_{i=1}^{n} f_i = 4 + 6 + 8 + 18 + 6 + 5 + 3 = 50$

$\therefore$ Mean of the data $= \frac{815}{50} = 16.3$ **Answer: B**

31. What will be the value of the resulting mean, if each observation is increased by 20%

A. 19.56 B. 19.3 C. 3.26 D. 29.16

Solution:

Resulting mean = 16.3 + 20% of 16.3 = 19.56 **Answer: A**

32. The mean of 7 observation is 15. If the mean of the first 4 observation is 14 and the mean of the last 4 observation is 15.5 find the value of the 4^{th} observation.

A. 14.75 B. 15.25 C. 13.25 D. 13

Solution:

$\Rightarrow$ Total of 7 observations $= 7 \times 15 = 105$

Total of the first 4 observation $= 4 \times 14 = 56$

Total of the last 4 observation $= 4 \times 15.5 = 62$

$\therefore$ The 4^{th} observation: $56 + 62 - 105 = 13$

Answer: D

33. The following data have been arranged in ascending order of their values: 14, 17, 33, 35, m – 5, m + 7, 57, 63, 69, and 80. If the median of the data is 48. Find the value of **m**

A. 66 B. 47 C. 48 D. 23

Solution:

Since, the number of observations = 10, that is $n = 10$

Median = Average of $\left(\frac{n}{2}\right)^{th}$ and $\left(\frac{n}{2}+1\right)^{th}$ observations

$$=\frac{1}{2}\left[\left(\frac{10}{2}\right)^{th}\text{item} + \left(\frac{10}{2}+1\right)\text{item}\right]$$

$$= \frac{1}{2}\left[(m-5)+(m+7)\right] = \frac{1}{2}(2m+2) = 48$$

$\Rightarrow m + 1 = 48 \Rightarrow m = 47$

Answer: B

Review Exercise

1. The average of six numbers is 18. If five of the numbers are 17, 39, 18 11and 25, what is the sixth number.

   A. –20 B. –2 C. 2 D. 8

2. The average of five numbers is 8. If the average of two of these numbers is –6, what is the sum of the other three numbers?

   A. 28 B. 46 C. 52 D. 34

3. Two classes, one with 50 students and the other with 30, take the same exam. The combined average of both classes is 84.5. If the larger class average is 80, what is the average of the smaller class?

   A. 87.2 B. 89 C. 92 D. 94.5

4. What is the average of the first 80 positive integers

   A. 38 B. 39.5 C. 40 D. 40.5

5. The average of four numbers is 30. If one of the number is removed, the average of the remaining number is 22. What number was removed?
A. 35 B. 54 C. 52 D. 26

6. The average of 6 numbers is A, and one of the number is 13. If 13 is replaced with 31, then what is the new average interms of A.
A. $A + \frac{31}{6}$ B. $A + 13$ C. $2A + 1$ D. $A + 3$

7. If the average of five number is 12, and the sum of three of the number is 27, then what is the average of the other two numbers?
A. 15 B. 19.5 C. 16.5 D. 16

8. If $3a - 2b = 13$ and $a + 3b = 30$
A. 8 B. 16 C. 21.5 D. 13.5

9. The average of eight numbers is 24. The average of three of these numbers is 15. What is the average of the other five number.
A. 9 B. 29 C. 39 D. 29.4

10. Abdulnasir's average (arithmetic mean) on four test is 80. What grade does he need on his fifth test to raise his average to 84
A. 82 B. 84 C. 100 D. 96

11. If $a + b = 3(c + d)$, which of the following is the average of a, b, c, and d?
A. $c + d$ B. $\frac{3}{8}(c + d)$ C. $\frac{c + d}{2}$ D. $\frac{c + d}{4}$

12. If $x + y = 6$, $y + z = 7$, and $z + x = 9$, what is the average of x, y and z?
A. $\frac{11}{2}$ B. $\frac{11}{3}$ C. $\frac{22}{3}$ D. 11

Answer and Explanation

1. B, $(6 \times 18) - (17 + 39 + 18 + 11 + 25) = 108 - 110 = -2$
2. C, $(5 \times 8) - (2 \times (-6)) = 40 - (-12) = 52$.
3. C, $50 \times 80 + 30y = 84.5 \times 80 \Rightarrow y = \frac{80}{30}(84.5 - 50) = 92$
4. D, $Av = \frac{1 + 80}{2} = \frac{81}{2} = 40.5$
5. B, $(4 \times 30) - (3 \times 22) = 120 - 66 = 54$

6. D, $\frac{a+b+c+d+e+13}{6} = A,$

$\Rightarrow \frac{a+b+c+d+e+(13+18)}{6} = \frac{a+b+c+d+e+13}{6} + \frac{18}{6} = A+3$

7. C, $\frac{(5x12)-(27)}{2} = \frac{60-27}{2} = \frac{33}{2} = 16.5$

8. A

9. D, $\frac{(8\times 24)-(3\times 15)}{5} = \frac{192-45}{5} = \frac{147}{5} = 29.4$

10. C, $(5 \text{ x } 84) - (4\ x\ 80)\ 80 = 420 - 320 = 100.$

11. A, $\frac{a+b+c+d}{4} = \frac{3c+3d+c+d}{4} = \frac{4c+4d}{4} = c+d$

12. B, $x+y+y+z+z+y = 2x+2y+2z = 22 \Rightarrow x+y+z = 11$

$\text{Average} = \frac{x+y+z}{3} = \frac{11}{3}$

UNIT – FIVE

5. WORD PROBLEMS

Translating words into Mathematical Symbols.

Before we begin solving word problems we need to be very comfortable with translating words into Mathematical symbols.

Following is a partial list of words and their mathematical equivalents.

English words	Mathematical Meaning
Is, was, will be, had, has, will have, is equal to, is the same as	Equals (=)
Plus, more than, sum, increased by, added to, exceeds,	Addition (+)
Fewer, less than, difference, decreased by, younger than, gave, lost.	Subtraction (–)
Times, of, product, twice, double, half	Multiplication (×)
Divided by, ratio quotient, per, for	Division (÷)
More than, greater than	$>$
At least	$\geq$
Fewer than, less than	$<$
At most	$\leq$

5.1 Age problems

In problems involving ages, remember that "years ago" means you need to subtract. "years from now" means you need to add.

Illustrative Example

1. Abebe is 20 years older than Towfik In 10 years, Towfik's age will be half that of Abebe's. What is Towfik's age?
   A. 10 years B. 8 years C. 25 years D. 5 years

   Solution: Let x = Towfik's age, and then $x + 20$ is Abebe's age.
   Ten years from now, Towfik's age will be $x + 10$ and
   Abebe's age will be $x + 30$
   Summarizing this information in a table:

	Towfik	Abebe
Age now	x	x + 20
Age in 10 years	x + 10	x + 30

Since "in 10 years, Towfik's age will be half that of Abebe's

we get $\frac{1}{2}(x+30) = x+10 \Rightarrow x+30 = 2(x+10)$

$\Rightarrow 2x+20 = x+30 \Rightarrow 2x-x = 30-20$

$\therefore \quad x = 10$ years old

Answer: A

2. In 1990, Sara was three times as old as Habtamu, but in 1994 she was only twice as old as he was. How old was Habtamu in 2000?

A. 16 B. 12 C. 14 D. 8

Solution: Let x be Habtamu's age in 1990.

	1990	1994
Habtamu	x	$x+4$
Sara	$3x$	$3x+4$

Since Sara's age in 1994 was twice Habtamu's age in 1994.

$\Rightarrow 3x+4 = 2(x+4) \Leftrightarrow 3x+4 = 2x+8$

$\Rightarrow 3x-2x = 8-4 \Rightarrow x = 4$

Since he was 4 years in 1990, then 10 years later Habtamu's age in 2000, he was 14 years

Answer: C

3. Henok is 10 years older than Faruk. However, 5 years ago Henok was twice as old as Faruk. How old is Faruk?

A. 25 B. 12 C. 10 D. 15

Solution: Let x = Faruk's age

Then, Henok's age is $x + 10$.

Now, 5 years ago, Faruk's age was $x - 5$ and Henok's age was $(x + 10) - 5 = x + 5$ since at that time, Henok was twice as old as Faruk.

$\Rightarrow x + 5 = 2\,(x-5) \Rightarrow 2x - x = 15$

$\therefore \quad x = 15$ years,

Answer: D

4. In 7 years Heran will be twice as old as she was 6 years ago. How old is Heran now?

A. 19 years B. 23 years C. 6 years D. 32 years

Solution: Let x = Heran's age now, six years ago she was $x - 6$ and 7 years from now she will be $x + 7$.

Then; $x + 7 = 2(x - 6)$

$\Rightarrow 2x - 12 = x + 7 \Rightarrow 2x - x = 12 + 7 = 19$

$\therefore x = 19$ years,

Answer: A

5. Seven years ago, Solomon was 3 times as old as Martha was at that time. If Solomon is now 5 years old than Martha, how old is Solomon?

A. $13\frac{1}{2}$ B. $14\frac{1}{2}$ C. 15 D. 13

Solution: Let x = Solomon's age and let y = Martha's age If Solomon is now 5 years older than Martha's age.

$\Rightarrow x = y + 5 \ldots (*)$

Seven years ago: $x - 7 = 3(y - 7) \ldots (**)$

From (*) and (**) we have

$x - 7 = 3(x - 5 - 7) \Rightarrow x - 7 = 3x - 36$

$\Rightarrow 2x = -7 + 36 = 29$

$\therefore x = \frac{29}{2} = 14\frac{1}{2}$ years **Answer: C**

6. The age of Bethawit (B) is half the sum of the ages of Azeb (A) and Semira (S). If Bethawit (B) is 2 years younger than Azeb, and Semira is 32 years old then the age of Bethawit (B) must be:

A. 28 years B. 30 years C. 32 years D. 34 years

Solution: Let A = the age of Azeb and

S = the age of Semira and

B = the age of Bethawit.

$B = \frac{A+S}{2}$ Since Bethawit (B) is two years younger than Azeb (A).

$\Rightarrow$ B = A – 2 implies that A = B + 2

- We know that age of Simira (S) = 32

$$\Rightarrow B = \frac{A+S}{2} = \frac{(B+2)+32}{2}$$

$$\Rightarrow 2B = B + 34 \Rightarrow B = 34 \text{ years}$$

Answer: D

7. The ages of three people are such that the age of one person is twice the age of the second person and three times the age of the third person. If the sum of the ages of the three people is 33, then the age of the youngest person is

A. 18 B. 6 C. 9 D. 3

Solution: Let x = the age of the oldest, y = the age of the second person, and z = the age of the youngest person.

$$\Rightarrow x = 2y \text{ and } x = 3z \Leftrightarrow y = \frac{x}{2}, z = \frac{x}{3}$$

The sum of the ages of the three people is $x + y + z = 33$

$$\Rightarrow x + \frac{x}{2} + \frac{x}{3} = 33 \Leftrightarrow 6x + 3x + 2x = 198$$

$$\Rightarrow 11x = 198 \Rightarrow x = \frac{198}{11} = 18$$

$$\therefore y = \frac{x}{2} = \frac{18}{2} = 9 \text{ and } z = \frac{x}{3} = \frac{18}{3} = 6$$

$$\therefore z = 6$$

Answer: B

5.2 Distance - problems

All distance problems involve one of three variations of the same formula:

$$\text{Distance} = \text{rate} \times \text{time} \Rightarrow d = rt$$

$$\Rightarrow \text{rate} = \frac{\text{distance}}{\text{time}} \Rightarrow r = \frac{d}{t}$$

$$\Rightarrow \text{time} = \frac{\text{distance}}{\text{rate}} \Rightarrow t = \frac{d}{r}$$

Illustrative Example

8. Elyas drove to a meeting at 60 kilometer per hour. Returning over the same route, he encountered heavy traffic, and was able to drive at only 40 km per hour. If the return trip took 1 hour longer, how many kilometer did he drive each way?

A. 2 B. 240 C. 120 D. 5

Solution: Let x = number of hours Elyas took to go, and make a table.

	Rate	Time	Distance
Going	60 km/h	x	$60x$
Returning	40 km/h	$x + 1$	$40(x + 1)$

Since he drove the same distance going and returning

$60x = 40(x + 1) \Rightarrow 60x = 40x + 40$

$$\Rightarrow 20x = 40 \Rightarrow x = \frac{40}{20} = 2\text{hour}$$

$$d = rt = \left(\frac{60\text{ km}}{\text{hr}}\right)(2\text{ hr}) = 120\text{ km.}$$ **Answer: C**

9. Ajet uses 80 gallons of fuel to fly 320 miles. At this rate, how many gallons of fuel are needed for a 700 mile flight?

A. 175 B. 170 C. 160 D. 155

Solution: $x = \frac{80\text{ gallons}}{320\text{ mile}} x\ 700\text{mile} = 175$ **Answer: A**

10. Acyclist travels 20 miles at a speed of 15 miles per hour. If he returns along the same path and the entire trip takes 2 hours, at what speed did he return?

A. 30 mph B. 15 mph C. 20 mph D. 22 mph

Solution: $D = rt \Rightarrow t = \frac{D}{r} = \frac{20}{15} = \frac{4}{3}\text{hr.}$

Since the entire trip take 2 hours, the return trip takes

$2 - \frac{4}{3}$ hours $= \frac{6-4}{3} = \frac{2}{3}$ hours. Now, the return trip is also 20 miles, so solving $D = R \times T \Rightarrow R = \frac{D}{T}$.

$$\Rightarrow R = \frac{20}{\frac{2}{3}} = 20 \times \frac{3}{2} = 30 \text{ mph},$$

Answer: A

Motion in Opposite Directions.

11. Two students start jogging at the same point and time but in opposite directions. If the rate of one jogger is 2 mi/ hr faster than the other and after 3 hours they are 30 miles apart, what is the rate of the faster jogger?

A. 3 B. 4 C. 5 D. 6

Solution: Let r = the rate of slower jogger.

then r + 2 = the rate of faster jogger.

since they are jogging for 3 hour the distance traveled by the slower jogger is d = rt = 3r and distance traveled by faster jogger is 3 (r + 2) = 3r + 6 since they are 30 miles apart, the total distance traveled is the sum of each.

$\Rightarrow$ 3r + 3 (r + 2) = 30 $\Rightarrow$ 6r = 24, hence $r = \frac{24}{6} = 4$

$\therefore$ the rate of faster jogger is **r + 2 = 4 + 2 = 6.**

Answer: D

5.3 WORK PROBLEMS.

In work problem, the rates at which certain persons or machines work alone are usually given, and it is necessary to compute the rate at which they work together (or vice versa).

The basic formula for solving work problems is $\frac{1}{r} + \frac{1}{t} = \frac{1}{h}$, where **r** and **t** are, the number of hour it takes to complete a job when working alone, and **h** is the number of hours it takes to do the job when working together.

Illustrative Example

12. If Machine A can produce 600 bolts in 2 hours and Machine B can produce 600 bolt in 3 hours, in how many hours can Machine A and B, working together at these constant rates, produce 1,000 bolts?

A. $1\frac{1}{2}$ hour B. $1\frac{1}{5}$ hour

C. $2\frac{1}{2}$ hour D. $2\frac{1}{5}$ hours

Solution:

Machine A and B

Working together = Machine A rate + Machine B rate

$$\text{rate} = \frac{1}{A} + \frac{1}{B}$$

$$\Rightarrow \frac{1}{h} = \frac{1}{2} + \frac{1}{3} = \frac{5}{6} \Rightarrow h = \frac{6}{5} = 1\frac{1}{5}\text{ hr}$$

$\therefore$ Working together machine A and B can produce 600 bolts in $1\frac{1}{5}$ hr

Answer: B

13. If two workers can assemble a car in 8 hours and a third worker can assemble the same car in 12 hours, then how long would it take the three workers together to assemble the car?

A. $2\frac{2}{5}$ hrs B. $2\frac{4}{5}$ hrs C. $3\frac{1}{2}$ hrs D. $4\frac{4}{5}$ hrs

Solution:

The time taken by the people working together to complete the job is $\frac{1}{h} = \frac{1}{r} + \frac{1}{s} \Rightarrow \frac{1}{h} = \frac{1}{8} + \frac{1}{12} = \frac{3+2}{24}$

$$\Rightarrow \frac{1}{h} = \frac{5}{24} \Rightarrow h = \frac{24}{s} = 4\frac{4}{5}\text{hr.}$$

Answer: D

14. Two printing presses working together can complete a job in 2.5 hours. Working alone, press A can do the job in 10 hours. How many hours will press B take to do the job by itself?

A. $3\frac{1}{3}$ hr B. 4 hr C. 5 hr D. $6\frac{1}{4}$ hr

Solution: Let x = number of hours press B would take working alone.

$$\frac{1}{2.5} = \frac{1}{10} + \frac{1}{x} \Rightarrow \frac{1}{x} = \frac{1}{2.5} - \frac{1}{10}$$

$$\Rightarrow \frac{1}{x} = \frac{10 - 2.5}{25} = \frac{7.5}{25}$$

$$\therefore x = \frac{25}{7.5}\text{hr} = 3\frac{1}{3}\text{hr}$$

Answer: A

15. A tank is being drained at constant. If it takes 3 hours to $\frac{6}{7}$ of its capacity, how much longer will it take to drain the tank completely

A. $\frac{1}{2}$ hr B. $\frac{3}{4}$ hr C. 1 hr D. $\frac{3}{2}$ hr

Solution: Rate $= \frac{w}{t} = \frac{\frac{6}{7}}{3} = \frac{6}{21} = \frac{2}{7}$

Now, since $\frac{6}{7}$ of the work has been completed, $\frac{1}{7}$ of the work remains.

$\Rightarrow W = R\,x\,t$ gives $\frac{1}{7} = \frac{2}{7}\,x\,t \Rightarrow t = \frac{1}{2}$hr.

Answer: A

5.4 Mixture Problems

In mixture problems, substances with different characteristics are combined, and it is necessary to determine the characteristics of the resulting mixture.

Illustrative Example

16. How many liters of a solution that is 15% salt must be added to 5 liters of a solution that is 8% salt so that the resulting solution is 10% salt?

A. 3 liters B. 2 liters C. 4 liters D. 5 liters

Solution: Let x = the number of liters of the 15% solution.

The amount of salt in the 15% solution

[0.15 x] plus the amount of salt in the 8% solution

[(0.08) (5)] must be equal to the amount of salt in the 10% mixture [0.10 (x + 5)]. Therefore

$0.15x + 0.08(5) = 0.10(x + 5)$

$\Rightarrow 15x + 40 = 10x + 50$

$\Rightarrow 15x - 10x = 50 - 40 = 10$

$\therefore\ 5x = 10 \Leftrightarrow x = \frac{10}{5} = 2$ liters.

That is two liters of the 15% salt solution must be added to the 8% solution to obtain the 10% solution. **Answer: B**

5.5 Interest Problems

A. Simple Interest Problems

Interest = Amount × Time × Rate

I = PRT, where rate (R) is in percent and the time (T) is given in year.

Illustrative Example

17. If 8000 birr is invested at 6% simple annual interest, how much interest is earned after 3 months?

Solution: P = 8000 birr, rate = 6% $T = \frac{3}{12} = \frac{1}{4}$

$$I = PRT = (8000)(0.06)\left(\frac{3}{12}\right) = 120 \text{birr.}$$

18. Find the annual interest rate if a principal of 10,000 birr increa to 10,900 birr at the first year.

Solution: we use $I = PRT$

$$900 = (10{,}000)(R)(1) \Rightarrow R = \frac{900}{10{,}000} \times 100\%$$

$\therefore$ rate = 9%.

19. A total of 1,800 birr is deposited in two saving accounts for one year, part at 6% and the remainder at 8%. If 128 birr was earne interest, how much was deposited at 6%?

A. 480 B. 640 C. 800 D. 600

Solution: Let x = the amount deposited at 6%

Then $1800 - x$ is the amount deposited at 8%.

Since the total interest is 128 birr, we get:

$$0.06x + 144 - 0.08x = 128$$

$$\Rightarrow -0.02x = 128 - 144 = -16$$

$$\therefore x = \frac{-16}{-0.02} = 800 \text{ birr.}$$

Answer:(

How To Solve Compounded Interest Problem

$$\text{Final balance} = (\text{Principal})\left(1 + \frac{\text{interest rate}}{C}\right)^{(\text{time})(C)}$$

Inshort: $A = P\left(1 + \frac{r}{c}\right)^{tc}$

20. If 20,000 birr is invested at 10% annual interest, compounded semiannually? What is the balance after 1 years?

Solution: The interest rate for each 6 – month period is half of th 10% annual rate which is 5% = 0.05.

The balance after the first 6 months would be:

20,000 + 20, 000 (0.05) = 21, 000 birr.

The balance after one year would be

21, 000 + 21, 000 (0.05) = 22,050 birr. Or in short:

$$A = P\left(1 + \frac{r}{c}\right)^{tc}$$

$$\Rightarrow A = 20{,}000\left(1+\frac{0.1}{2}\right)^{1(2)} = 20{,}000(1.05)^2 = 22{,}050 \text{ birr}$$

5.6 Miscellaneous collection of word problems.

1. Two students appeared at an examination. One of them secured 9 marks more than the other and his mark was 56% of the sum of their mark. The marks obtained by them are:

A. 39, 30 B. 43, 34 C. 42, 33 D. 41, 32

Solution: Let x an $x + 9$ was their mark.

$x + (x + 9) = 2x + 9$

$x + 9$ is 56% of $2x + 9$

$x + 9 = 0.56\,(2x + 9) = 1.12\,x + 5.04$

$1.12\,x - x = 9 - 5.04 = 3.96$

$$\therefore x = \frac{3.96}{0.12} = 33 \quad \text{and } 33 + 9 = 42$$

Answer: C

2. The product of 3 and 6 more than a certain number is 5 times that number. What is the number ?

A. 18 B. 9 C. 12 D. 45

Solution: Let x = the number

$\Rightarrow 3\,(6 + x) = 5x \Leftrightarrow 18 + 3x = 5x$

$\Rightarrow 2x = 18$

$\therefore x = 9$

Answer: B

3. In one month, Hana used $\frac{1}{6}$ of her monthly salary for a car payment and $\frac{1}{4}$ more than the car payment for rent. What fraction of her monthly salary did Hana use that month for the car payment and rent combined

A. $\frac{7}{12}$ B. $\frac{3}{8}$ C. $\frac{5}{12}$ D. $\frac{5}{24}$

Solution: Let x = Hana's monthly salary.

$$\Rightarrow \frac{\text{Car payment + rent}}{\text{Total}}$$

$$\Rightarrow \frac{\frac{1}{6}\text{ of } x + \left(\frac{1}{4}\text{of } x + \frac{1}{6}\text{ of } x\right)}{x}$$

$$\Rightarrow \frac{\frac{1}{6}x + \frac{1}{4}x + \frac{1}{6}x}{x} = \frac{\frac{7}{12}x}{x} = \frac{7}{12}.$$

Answer: A

24. At 7: 00 P.M, the hostess of the party remarked that only $\frac{1}{5}$ of her guests had arrived so far, but that, as soon as 12 more showed up, $\frac{1}{3}$ of the guests would be there.How many people were invite?

A. 300 B. 150 C. 90 D. 120

Solution: Let x = number of people invited.

$$\Rightarrow \frac{1}{3}x - \frac{1}{5}x = 12 \Leftrightarrow 5x - 3x = 180$$

Answer: C

$$x = \frac{180}{2} = 90$$

25. In a family of three, the father weighted 5 times as much as the child, and the mother weighed $\frac{3}{4}$ as much as the father. If the three of them weighed a total of 156 kg. how much did the mother and child weighed.

A. 96 kg B. 76 kg C. 80 kg D. 92 kg

Solution: Let x = weight of the child,

Then x = weight of the father and $\frac{3}{4}(5x)$ = weight of the mother

$$\Rightarrow x + 5x + \frac{15x}{4} = 156 \Leftrightarrow \frac{39x}{4} = 156$$

$$\therefore x = \frac{4 \times 156}{39} = 16$$

$$\therefore 16+\frac{15}{4}(16)=16+60=76\text{kg}$$

Answer: B

26. In the afternoon, sara read 100 pages at the rate of 60 pages per hour, in the evening, when she was tired. She read another 100 pages at the rate of 40 pages per hour. In pages per hour, what was her average rate of reading for the day?

A. 52 B. 50 C. 45 D. 48

Solution:

- In the afternoon she read for $\frac{100}{60}=\frac{5}{3\text{ hr}}$
- In the evening, she read for $\frac{100}{40}=\frac{5}{2}$ hr.

$$\Rightarrow \text{Total time}=\frac{5}{3}\text{hr}+\frac{5}{2}\text{ hr}=\frac{25}{6}\text{hr.}$$

$$\therefore \text{Average rate}=\frac{100+100}{\frac{25}{6}}=\frac{200\times 6}{25}=48\text{ pages per hour}$$

Answer: D

5.7 DISCOUNT

If a price is discounted by x%, then the price becomes $(100-x)\%$ of the original price.

Illustrative Example

27. A certain customer paid 240 birr for a dress. If that price represented a 25% discount on the original price of the dress, what was the original price of the dress?

A. 180 birr B. 300 birr C. 320 birr D. 200 birr

Solution: Let x = the original price

$$x-0.25\,x=240\Rightarrow 0.75\,x=240$$

$$\therefore x=\frac{240}{0.75}=320\text{ birr}$$

Answer: C

28. The price of an item is discounted by 10% and then this reduced price is discounted by an additional 20%. These two discounts are equal to an overall discount of what percent?

A. 28% B. 72% C. 70% D. 30%

Solution: Let x = the original price of the item.

- Then after the first 10% discount will be $x - 10\%\, x = 0.9x$.
- The price after the 2^{nd} 20% discount will be $0.9x - (0.9x)(0.2) = 0.72x$

$\therefore$ The overall discount is $100\% - 72\% = 28\%$

Answer: A

5.8 PROFIT

Gross profit = (Revenue) – (Expense) or
= (Selling price) – (Cost price).

Illustrative Example

29. A certain appliance costs a marchant 600 birr. At what price should the marchant sell the appliance in order to make a gross profit of 40% of the cost of the appliance?

A. 360 birr
B. 240 birr
C. 840 birr
D. 960 birr

Solution: If x is the selling price of the appliance, then

$$x - 600 = (0.4)(600) \Rightarrow x = 600 + 240 = 840$$

Answer: C

UNIT – SIX

6. COUNTING AND PROBABILITY

6.1 COUNTING PRINCIPLE

How to solve a group problem involving Neither/Both.

Some SAT problems involve two groups with overlapping elements, and possibly elements that belong to neither group. It is easy to identify this type of question because the words "both" and/ or "neither" appear in the question. These problems are quite easy if you just memorize the following formula:

$$\text{Total} = \text{Group}_1 + \text{Group}_2 + \text{Neither} - \text{Both}.$$

Illustrative Example

1. Of the 120 students at a certain school, 54 students are taking mathematics and 38 are taking chemistry and 43 are taking neither mathematics nor chemistry. How many are taking both mathematics and chemistry?

A. 28 B. 15 C. 14 D. 30

Solution: use, $\text{group}_1 + \text{group}_2 + \text{neither} - \text{both} = \text{Total}$

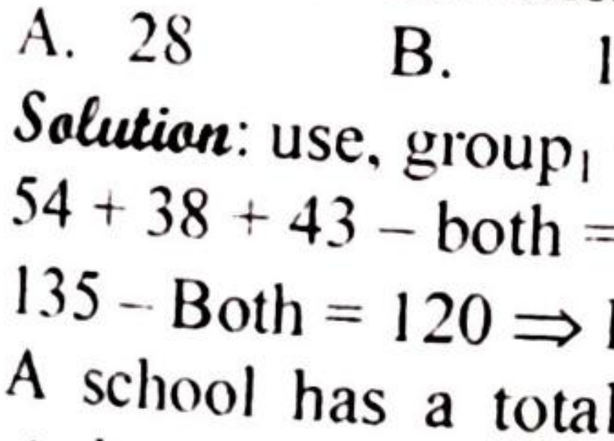
$54 + 38 + 43 - \text{both} = 120$

$135 - \text{Both} = 120 \Rightarrow \text{Both} = 135 - 120 = 15$ **Answer: B**

2. A school has a total enrollment of 90 students. There are 30 students taking physics, 25 taking English, and 13 taking both. What percentage of the students are taking either physics or English?

A. 30% B. 58% C. 36% D. 47%

Solution:



The number of students enrolled in either physics or English or both is $30 + 25 - 13 = 42$

$$\Rightarrow \frac{\text{Physics or English enrollment}}{\text{Total enrollment}} = \frac{42}{90} = 0.47$$

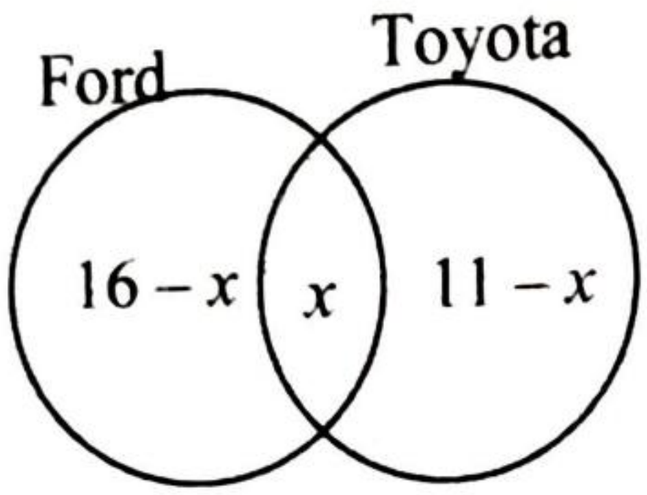
$\Rightarrow 47\%$

Answer: D

3. In a small town, 16 people own Fords and 11 people own Toyotas. If exactly 15 people own only one of the two types of cars, how many people own both types of cars.

A. 2 B. 6 C. 7 D. 12

Solution: Let x = the number of people who own both types of cars.



$16 - x$ = people who own only Ford

$11 - x$ = people who own only Toyotas

$\Rightarrow 16 - x + 11 - x = 15 \Rightarrow 27 - 2x = 15$

$-2x = 15 - 27 = -12$

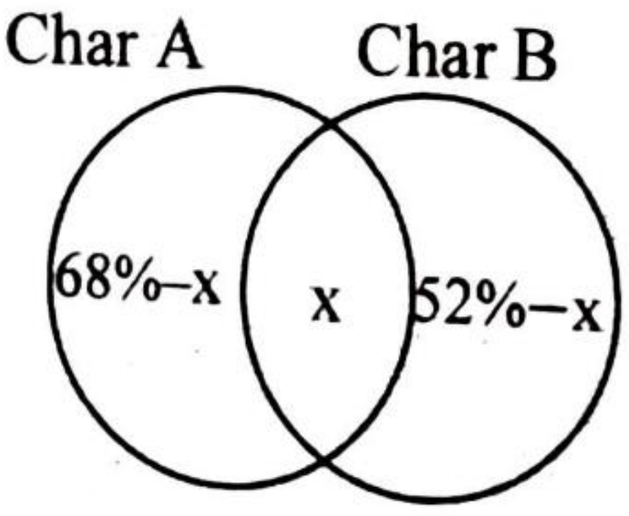
$$\therefore x = \frac{-12}{-2} = 6$$

Answer: B

4. In a certain population group 68% of the people have characteristic A and 52% have characteristic B. If every person in the group has at least one of the two characteristic, what percent of the people have both A and B?

A. 20% B. 32% C. 48% D. 40%

Solution: Let x = have characteristic both A and B.



$68\% - x + 52\% - x = 100\%$

$120\% - x = 100\%$

$\therefore x = 120\% - 100\% = 20\%$

Answer: A

5. The following tables shows the number of students in three clubs at certain preparatory school. Although no student is in all three clubs, 10 students are in both chess and Drama, 5 students are in both chess and Math, and 6 students are in both Drama and Math.

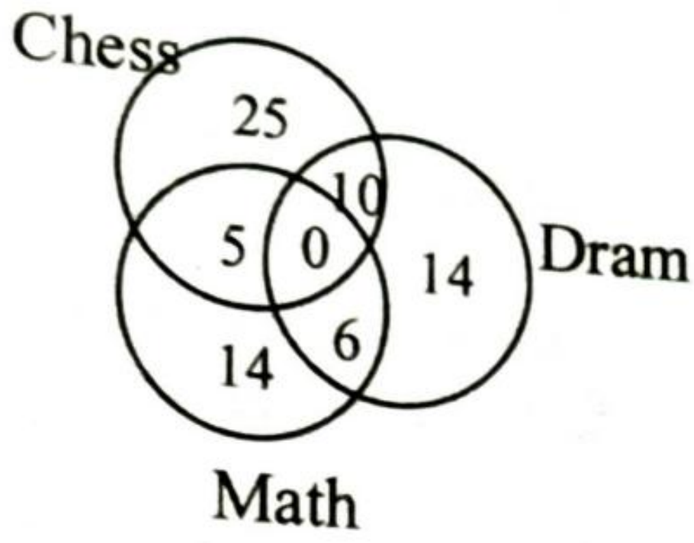
Club	Chess	Drama	Math
Number of student	40	30	25

How many different students are in the three clubs?

A. 68 B. 69 C. 74 D. 84

Solution:

- 10 in chess and drama
- 5 in chess and Maths
- 6 in Drama and Math
- For chess club:
  $40 - 10 - 5 = 25$
  member.
- Math club has:
  $25 - 5 - 6 = 14$ members.
- Drama club has: $30 - 10 - 6 = 14$ members



Therefore, the three clubs have $25 + 14 + 14 + 10 + 5 + 6 = 74$ students.

Answer: C

6. At a certain professional conference with 150 attendees, 90 of the attendees are doctors and the rest are dentists. If 40 of the attendees are women, and $\frac{1}{5}$ of the dentists in attendance are women, how many of the attendance are male doctors?

A. 62 B. 60 C. 48 D. 110

Solution:

- (Number of doctor) + Number of dentists) = 150
  $\Rightarrow$ 90 + (Number of dentists) = 150
  $\therefore$ Number of dentists = $150 - 90 = 60$
- If 40 are women then 110 are male.
- $\frac{1}{5}$ of dentists = $\frac{1}{5}(60) = 12$ dentists are women.

$\Rightarrow$48 dentists are male.

	Doctors	**Dentists**	**Total**
Male	62	48	110
Female	28	12	40
Total	90	60	150

Male doctors have 62 and female doctors are 28. **Answer: A**

7. On a certain Ethiopian – Eritrean Committee, $\frac{2}{3}$ of the members are men, and $\frac{3}{8}$ of the men are Ethiopians. If $\frac{3}{5}$ of the committee members are Eritrean, what fraction of the members are Ethiopian women?

A. $\frac{3}{20}$ B. $\frac{11}{60}$ C. $\frac{1}{4}$ D. $\frac{5}{12}$

Solution: The LCM of all the denominator $\left(\frac{2}{3},\frac{3}{8},\frac{3}{5}\right)$ is 120.

- So assume that the committee has 120 members.

  → Then $\frac{2}{3}\times 120 = 80$ men and 40 women

- Since, either $\frac{3}{5}$ of the committee members

  $= \frac{3}{5}(120) = 72$ are Eritrean,

  there are $120 - 72 = 48$ Ethiopian.

  →30 men and 18 women are Ethiopian

  $\Rightarrow \frac{18}{120} = \frac{3}{20}$ are Ethiopian women **Answer: A**

> The number of integers between two integers inclusive is one more than their difference.

Illustrative Example

8. How many integers are there between 59 and 201 inclusive?

A. 140 B. 142 C. 143 D. 141

Solution: $(201 - 59) + 1 = 143$ **Answer: C**

9. Callers 49 through 91 to a radio show won a prize. How many callers won a prize?

A. 41 B. 43 C. 42 D. 44

Solution: $(91 - 49) + 1 = 43$ **Answer: B**

10. From 2: 10 to 2: 14 Mahder read pages 309 through 313 in her English book. What was her rate of reading, in pages per minute?

A. $\frac{5}{4}$ B. $\frac{4}{5}$ C. $\frac{3}{4}$ D. $\frac{3}{5}$

Solution:

She read $313 - 309 + 1 = 5$ pages

She read for 2: $14 - 2: 1 = 4$ minutes

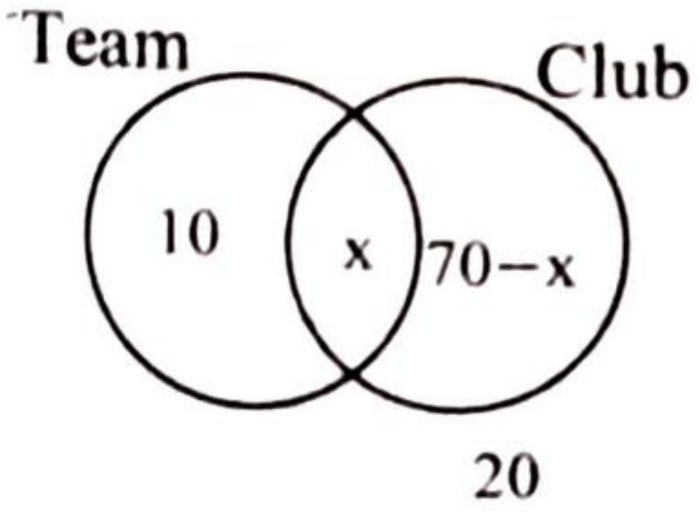
So she read at the rate of $\frac{5}{4}$ pages per minute.

Answer: A

11. In a group of 100 students, more students are on a team than are members of a club. If 70 are in clubs and 20 are neither on a team nor in a club, what is the minimum number of students who could be both on a team and in a club?

A. 60 B. 61 C. 80 D. 40

Solution: $100 - 20 = 80$ are either.



Since 70 are in club, so that 10 are in team.

$\Rightarrow 10 + x > 70 \Rightarrow x > 60$ since x must be integer, the least it can be 61.

Answer: B

6.2 FUNDAMENTAL PRINCIPLE OF COUNTING

If an event occurs **m** times, and each of the **m** event s is followed by a second event which occurs **k** times, then the first event follows the second event **m** × **k** times.

Illustrative Example

12. How many four – digit numbers have only even digits?

A. 480 B. 500 C. 400 D. 320

Solution:

- The first digit can be 2, 4, 6 or 8 $\Rightarrow$ 4 ways.
- The 2^{nd}, 3^{rd} and fourth digits can be chosen in any of 5 ways (0, 2, 4, 6, 8).

Therefore, the total number of four digits numbers with only even digits is $4 \times 5 \times 5 \times 5 = 500$

Answer: B

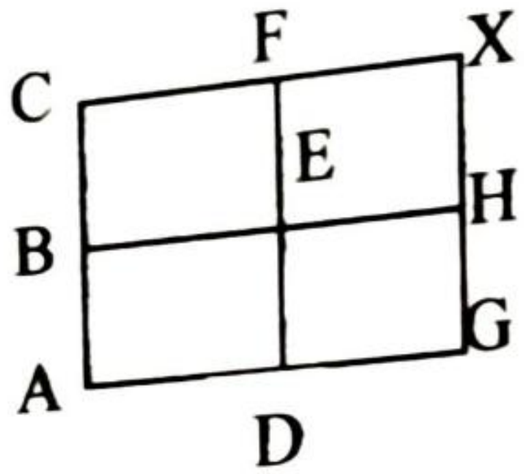
13. In the figure at the right how many paths are there form A to x if

A. 9 B. 8 C. 6 D. 5

Solution: It is better labeling all the vertices and systematically:

→ ABCFX, ABEFX, ABEHX

→ ADEFX, ADEHX, ADGHX



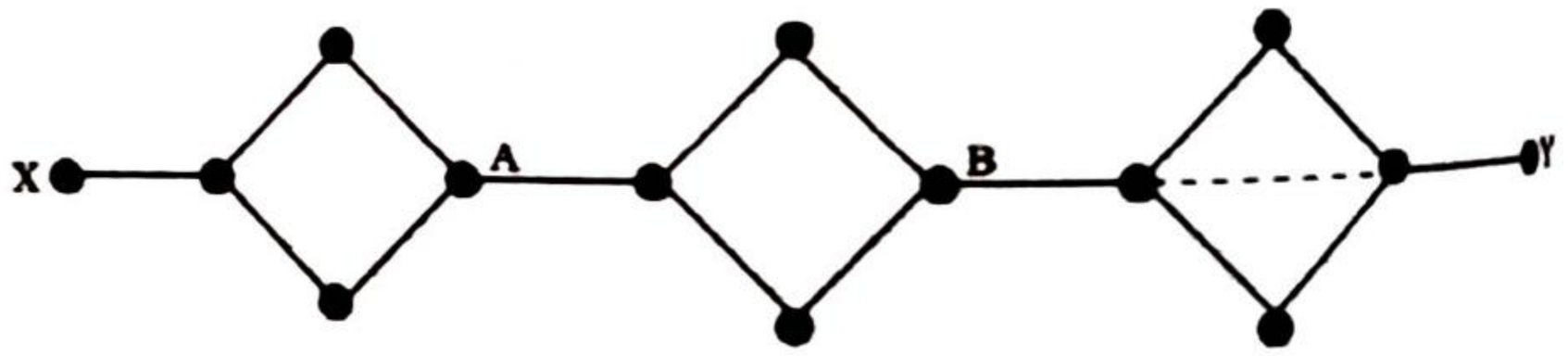
In all, there are 6 paths from A to X. **Answer:** C

14. How many three – digit numbers can be formed with the digits 1,3 and 5?

A. 6 B. 4 C. 12 D. 8

Solution: 135, 153, 315, 351, 513, 531 **Answer:** A

15. The diagram below shows the various paths along which a cat can travel from point X, where it is released to point Y, where it is rewarded with a food pellet.



How many different paths from X to Y can the cat take if it goes directly from X to Y without retracing any point along a path?

A. 6 B. 7 C. 12 D. 14

Solution:

The total number of ways to:

→ get from X to A is 2 path = 2 ways

→ get from A to B is 2 path = 2 ways

→ get from B to y is 3 path = 3 ways

Thus, the total number of different path is $2 \times 2 \times 3 = 12$. **Answer:** C

16. Let A be the set of primes less than 7, and B be the set of positive odd numbers less than 7. How many different sums of the form a + b are possible if a in A and b is in B?

A. 6 B. 7 C. 8 D. 9

Solution: A = {2, 3, 5}, B = {1, 3, 5}

List the sums systematically; First add 2 to each number in B, then 3, and then 5: 3, 5, 7, 4, 8, 6, 8, 10 = {3, 4, 5, 6, 7, 8, 10}.

There are 7 different sums.

Answer: B

6.3 PERMUTATION AND COMBINATION

An arrangement of things in a definite order with no repetition is a **permutation.**

Example: BON, BNO, OBN, ONB, NBO, and NOB are all different arrangements of the three letters, B, O, and N.

Definition of n Factorial

n factorial (n!) is the product of the natural numbers **n** through **1**. That is $n! = n\times(n-1)\times(n-2)\ldots 3\times 2\times 1$

Here are some examples

$S! = 5\times 4\times 3\times 2\times 1$

$8! = 8\times 7\times 6\times 5\times 4\times 3\times 2\times 1$

$0! = 1$

Permutation Formula for Distinct Objects

The number of permutations P(n, k) of n distinct objects selected k at a time is $P(n,. k) = \dfrac{n!}{(n-k)!}$

Permutation of N objects, k of which are identical

The number of arrangements of **N** objects, in a straight line, where $\mathbf{k_1}$ objects of one kind are the same, $\mathbf{k_2}$ objects of antoehr kind are the same and the rest are different, is given by $\dfrac{N!}{k_1!k_2!}$ where $N = k_1 + k_2$.

Illustrative Example

17. A 20 km Marathon has 10 people entered. In how many different ways can the first, second, and third place prizes be awarded?

A. 720 B. 360 C. 90 D. 1080

Solution:

The order in which the runners finish is important, so the number of ways to place first, second, and third is $P(10,3) = \frac{10!}{(10-3)!}$

$= 10 \times 9 \times 8 = 720$ ways. **Answer:** A

18. A college golf team consists of five players who are ranked from 1 through 5. If golf coach has seven players from which to choose, how many different golf teams can the coach select?

A. 6720 B. 5040 C. 42 D. 2520

Solution: Because the players are ranked, the number of different golf teams possible is the number of permutation of seven players selected five at a time.

$$P(7,5) = \frac{7!}{(7-5)!} = \frac{7!}{2!} = \frac{7\times6\times5\times4\times3\times2!}{2!} = 5040$$ **Answer:** B

Illustrative Example

19. Find the number of rearrangements of the letters in each of the following words:

A. BANANA B. STATISTICS

Solution:

a) There are six letters in the word BANANA with:

A repeated three times

N repeated two times

Hence, the number of arrangements is

$$\frac{6!}{3!2!} = \frac{6\times5\times4\times3\times2\times1}{3\times2\times1\times2\times1} = 60 \text{ way}$$

b) There are ten letters in the word STATISTICS, with three Ss and three Ts, and two I's duplicated in the word. Hence, the number of arrangement is

$$\frac{10!}{3!3!2!} = \frac{10.9.8.7.6.5.4.3.2.1}{3.2.1.3.2.1.2.1} = 50400$$

20. From the letters a, b, c, d and e how many four letter groups can be formed if each letter can be formed if each letter can be used exactly once?

A. 625 B. 120 C. 125 D. 240

Solution: $5 \times 4 \times 4 \times 2 = 120$ **Answer:** B

21. Three women and four men are to be seated in a row of seven chairs. How many different seating arrangements are possible if there are no restrictions on the seating arrangements?

A. 480 B. 5040 C. 720 D. 840

Solution:

Since there are no restriction on seating arrangement then the number of seating arrangement is $P(7,7) = \frac{7!}{(7-7)!} = 7! = 5040$

Answer: B

22. Three women and four men are to be seated in a row of seven chairs. How many different seating arrangements are possible if the women sit together and the men sit together?

A. 144 B. 30 C. 288 D. 72

Solution:

There are 3! Ways to arrange the women and 4! Ways to arrange the men. We must also consider that either the women or the men could be seated at the beginning of the row. There are two ways to do this. There are $2(3!\times4!)$ ways to seat the women together and the men together $2(3!\times4!) = 2(3\times2\times1\times4\times3\times2\times1) = 288$.

Answer: C

6.4 COMBINATIONS

A **combination** is a collection of objects for which the order is not important.

The number of combinations of n objects chosen k at a time is

$$C(n,k) = \frac{n!}{k!(n-k)!}.$$

Illustrative Example

23. A basket ball team consists of 11 players. In how many different ways can a coach choose the five starting players, assuming the position of a player is not considered?

A. 55440 B. 462 C. 11088 D. 55

Solution:

$$C(11,5) = \frac{11!}{5!(11-5)!} = \frac{11\times10\times9\times8\times7\times6!}{5\times4\times3\times2\times1\times6!} = 462$$

Answer: B

24. At the beginning of semester (I) of a mathematics class for preparatory school teachers, each of the class's 27 students shook

hands with each of the other students exactly once. How many handshakes took place?

A. 54 B. 108 C. 216 D. 351

Solution:

Since the handshake between persons A and B is the same as that between persons B and A, so that

$$C(27,2) = \frac{27!}{2!(27-2)} = \frac{27 \times 26}{2} = 351$$

Answer: D

6.5 PRO ABILITY OF AN EVENT

For an experiment with sample space S of equally likely out comes, the probability P (E)

of an event E is given by: $P(E) = \frac{n(E)}{n(S)}$ where n(E)

is the number of elements in the event and n(s) is the number of elements in the sample space.

Probability of Mutually Exclusive Events

Two events A and B are **mutually exclusive** if $A \cap B = \phi$

Properties of Probability

1. **Impossible event:** if P (E) = 0
2. **Certain even** if P(E) = 1
3. For any event E, $0 \leq P(E) \leq 1$
   - **Likely event** if $\frac{1}{2} < P(E) < 1$
   - **Unlikely event** if $0 < P(E) < \frac{1}{2}$
4. **Addition Rule for probability.**
   If A and B are events, then $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
5. **Condition Probability**
   Let A and B be two events in a sample space S. Then the conditional probability of B given that A has occurred is

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

6. **Product Rule for probability**
   - $P(A \text{ and } B) = P(A \cap B) = P(A).\ P(B|A)$

- If A and B are independent event then
  $P(A \text{ and } B) = P(A \cap B) = P(A).\ P(B)$

7. **Odd in favor and odd against**

a) The **odds in favor** of an event A are given by:

$$\frac{P(A)}{P(A')} = \frac{P(A)}{1 - P(A)}$$

b) The **odds against** an event **A** are given by

$$\frac{P(A')}{P(A)} = \frac{1 - P(A)}{P(A)}$$

Illustrative Example

A bag contains 2 red balls, 4 blue balls, and 3 white balls. (for 25-27) questions

25. What is the probability of the event R that a ball drawn at random is red?

A. $\frac{2}{9}$ B. $\frac{1}{5}$ C. $\frac{4}{9}$ D. $\frac{1}{3}$

Solution:

The bag contains a total of $2 + 4 + 3 = 9$

$\therefore P(R) = \frac{2}{9}$ **Answer: A**

26. What is the probability of the event "**not red**"?

A. $\frac{4}{9}$ B. $\frac{4}{5}$ C. $\frac{7}{9}$ D. $\frac{5}{9}$

Solution: $P(\text{not R}) = 1 - P(R) = 1 - \frac{2}{9} = \frac{7}{9}$ **Answer: C**

27. What is the probability of the event that a ball drawn at random is either red **(R)** or blue **(B)**?

A. $\frac{2}{3}$ B. $\frac{5}{9}$ C. $\frac{4}{9}$ D. $\frac{1}{5}$

Solution: Since $R \cap B = \phi$, so $P(R \cup B) = P(R) + P(B)$

$$= \frac{2}{9} + \frac{4}{9} = \frac{6}{9} = \frac{2}{3}$$

Answer: A

28. Find the probability of rolling a sum of 7 or 11 when rolling a fair pair of dice.

A. $\frac{1}{9}$ B. $\frac{2}{9}$ C. $\frac{1}{6}$ D. $\frac{5}{36}$

Solution:

Number on First Die	Number on second Die					
	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Possible sum

We see that a sum of 7 appears six times, that is, in 6 ordered pairs. Hence the event "a sum of 7" is subset of the set of ordered pairs.

$\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$

$$P(\text{sum of } 7) = \frac{6}{36}$$

Sum of 11 appears 2 times that is $\{(5, 6), (6, 5)\}$

$$P(\text{sum of } 11) = \frac{2}{36}$$

$$P(\text{sum of 7 or sum of 11}) = \frac{6}{36} + \frac{2}{36} = \frac{8}{36} = \frac{2}{9}$$

Answer: B

29. A fair pair of dice is rolled. Let E be the event of rolling a sum that is an even number and P the event of rolling a sum that is a prime number. Find the probability of rolling a sum that is even or prime.

A. $\frac{1}{9}$ B. $\frac{2}{9}$ C. $\frac{1}{12}$ D. $\frac{8}{9}$

Solution:

Outcomes row represents the sum of the number that appears on the two dice.

Outcome	2	3	4	5	6	7	8	9	10	11	12

sum											
Probability	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Even sum: E = {2, 4, 6, 8, 10, 12}

Prime sum: P = {2, 3, 5, 7, 11} and E ∩ P = {2}

P (E ∪ P) = P(2)+P(4)+P(6)+P(8)+P(10)+P(12)+P(3)+P(5)+P(7)+P(11)

$$=\frac{1}{36}+\frac{3}{36}+\frac{5}{36}+\frac{5}{36}+\frac{3}{36}+\frac{1}{36}+\frac{2}{36}+\frac{4}{36}+\frac{6}{36}+\frac{2}{36}=\frac{32}{36}=\frac{8}{9}$$

Another approach is use P(E∪P) = P(E) + P(P) – P(E∩P)

$$=\frac{18}{36}+\frac{15}{36}-\frac{1}{36}=\frac{32}{36}=\frac{8}{9}$$

Answer: D

30. An integer between 100 and 999, inclusive, is chosen at random. What is the probability that all the digits of the number are odd?

A. $\frac{9}{10}$ B. $\frac{5}{36}$ C. $\frac{4}{9}$ D. $\frac{1}{3}$

Solution:

Since both endpoints are include, there are 999–100+1= 900 integers between 100 and 999. To form three odd digits:

→ the first digit can be 1, 3 5, 6 or 9 (5 ways)

→ the second digit can be 1, 3, 5, 7 or 9 (5 ways)

→ the third digit can be 1, 3, 5, 7 or 9 (5 ways)

There are 5 × 5 × 5 = 125 three digit numbers are formed.

Therefore, the probability is $P(E)=\frac{125}{900}=\frac{5}{36}$ **Answer: B**

31. A die is tossed two times. What is the probability that a 6 will show on the upward face at least once.

A. $\frac{1}{6}$ B. $\frac{11}{36}$ C. $\frac{25}{36}$ D. $\frac{1}{3}$

Solution:

Event 6: E = {(1, 6), (2, 6), (3, 6), (4, 6),(5, 6),(6, 6),(6, 1), (6, 3), (6, 4), (6, 5)}

$$\therefore P(E)=\frac{11}{36}$$

Another approach: $n(s) = 6 \times 6 = 36$

$n(E') = \{\text{no } 6'5\} = 5 \times 5 = 25$

$$n(E) = 1 - P(E') = 1 - \frac{25}{36} = \frac{36-25}{36} = \frac{11}{36}$$ **Answer: B**

32. A die is tossed three times. What is the probability that a 6 will show on the upward face at least once?

A. $\frac{25}{36}$ B. $\frac{125}{216}$ C. $\frac{91}{216}$ D. $\frac{1}{216}$

Solution: Let E = {at least one 6}. Then E' = {no 6's}

Because on each toss of the die there are six possible outcomes $n(S) = 6.6.6 = 216$

On each toss of the die there are five numbers that are not 6's.

Therefore, $n(E') = 5 \times 5 \times 5 = 125$

$$\therefore P(E) = 1 - P(E') = 1 - \frac{125}{136} = \frac{216-125}{216} = \frac{91}{216}$$ **Answer: C**

33. A pair of dice are tossed twice. What is the probability that the first roll is a sum of 7 and the second roll is a sum of 11?

A. $\frac{1}{108}$ B. $\frac{2}{91}$ C. $\frac{12}{36}$ D. $\frac{25}{36}$

Solution: The rolls of a pair of dice are independent.

Let A = {sum of 7 on the first roll} and B = {sum of 11 on the second roll}.

Then $P(A \cap B) = P(A).P(B) = \frac{6}{36} \times \frac{2}{36} = \frac{1}{108}$

Answer: A

34. Two dice are tossed, one after the other. What is the probability that the result is a sum of 6, given that the first die is not a 3.

A. $\frac{5}{36}$ B. $\frac{5}{6}$ C. $\frac{2}{15}$ D. $\frac{1}{9}$

Solution:

Let A = {the sum is 6}={(5, 1), (1, 5),(4, 2), (2, 4)}

B = {the first die is not a 3} $= \frac{30}{36} = \frac{5}{6}$

$$\text{Then } P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{4}{36}}{\frac{5}{6}} = \frac{\frac{1}{9}}{\frac{5}{6}} = \frac{2}{15}$$

35. A box contains four red, three white, and five green balls. Suppose three balls are randomly selected from the box in a succession without replacement. **Answer: C**

a) What is the probability that first a red, second a white and then a green ball are sleeted?

b) What is the probability that two white balls followed by one green balls are selected?

Solution:

a) Let A = {a red ball is selected first}.
B = {a white ball is selected 2nd}
C = {a green ball is selected 3rd}
Then,
P (A followed by B followed by C)

$$= P(A).\ P(B|A).\ P(C|A \text{ and } B) = \frac{4}{12}.\frac{3}{11}.\frac{5}{10} = \frac{1}{22}$$

b) Let x = {a white ball is selected first}
y = {a white ball is selected 2nd}
z = {a green ball is selected third}
Then, P(x followed by y followed by z)

$$= P(x).P(y|x)\ .P(z|x \text{ and } y) = \frac{3}{12}.\frac{2}{11}.\frac{5}{10} = \frac{1}{44}$$

36. If a pair of fair dice are rolled once, what are the odds in favor of rolling a sum of 7?

A. $\frac{1}{6}$ B. $\frac{1}{5}$ C. $\frac{5}{36}$ D. $\frac{5}{6}$

Solution: Let E = the event of rolling a sum of 7.
E = {(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)}

$$\text{Odd in favor of E} = \frac{P(E)}{P(\text{not E})} = \frac{P(E)}{1 - P(E)}$$

$$= \frac{\frac{6}{36}}{1-\frac{6}{36}} = \frac{1}{6} \times \frac{6}{5} = \frac{1}{5}$$

Answer: B

37. If three red, four white, and five blue balls are placed in a box and one ball is randomly selected from the box, what are the odds against the ball being blue?

A. $\frac{2}{5}$ B. $\frac{5}{7}$ C. $\frac{7}{12}$ D. $\frac{7}{5}$

Solution: Let B = the event selected blue ball.

$$\text{Odd against B} = \frac{\text{P(not B)}}{\text{P(B)}} = \frac{1-\text{P(B)}}{\text{P(B)}}$$

$$= \frac{1-\frac{5}{12}}{\frac{5}{12}} = \frac{7}{12} \times \frac{12}{5} = \frac{7}{5}$$

Answer: D

38. The probability of tossing five coins with a result of three heads and two tails is $\frac{5}{16}$. What are the odds in favor of three heads and two tails?

A. $\frac{5}{11}$ B. $\frac{5}{21}$ C. $\frac{16}{21}$ D. $\frac{9}{16}$

Solution:

$$\text{The odds in favor of E are } \frac{\text{P(E)}}{1-\text{P(E)}} = \frac{\frac{5}{16}}{1-\frac{5}{16}} = \frac{5}{16} \times \frac{16}{11} = \frac{5}{11}$$

Answer: A

39. Six students at a school picnic are to stand in a row for a group photograph. In how many ways can this be done if:

A. Heran and Betty insist on standing next to each other.

B. Heran and Betty refuse to stand next to each other.

Solution:

Since the order in which the student stand is important we use permutations. Since Heran and Betty insist on standing together, let us think of them as one object. Thus we have five objects to arrange in a row and there are P(5, 5) ways of doing this. For each of these arrangements there are 2 ways of having **Heran and Betty** together.

→ Heran – Betty or Betty – Heran

Thus by fundamental counting principle;

$2 \times P(5, 5) = 2 \times 5! = 240$.

b) There are P (6, 6) ways or arranging the six student. Of these, $2 \times P(5, 5)$ have Heran and Betty stand together by part (a). All the rest have Heran and Betty standing apart. So the number of arrangement with Heran and Betty apart is

$P(6, 6) - 2 \times P(5, 5) = 720 - 240 = 480$

Review Exercise

1. Find the number of ways of choosing three initials from the alphabet if none of the letters can be repeated.
2. Suppose four different colored squares are arranged in a row. How many different ways are there to order the colors?
3. How many 4 letter code symbols can be formed with letters A, B, C and D with repetition?
4. How many permutations are there of the letters of the word **QUALIFY** if the letters are taken 3 at a time.
5. How many 4 digit numbers can be named using the digits 5, 6, 7 and 9.
   a. Without repetition? b. with repetition
6. In how many ways can 5 students be arranged in a straight line?
7. In how many ways can 5 students be arranged in a round table.
8. How many ways can a president, vice president, secretary and treasurer be chosen from a committee of 12 people?
9. In how many ways can 3 men and 3 women be seated in a row of 6 seats:
   a. With no seating restrictions?
   b. If men and women must alternate?
   c. If a particular man and women must sit together.
   d. If a particular man and woman must not sit together?
10. There are 15 students in a business club. How many sets of four officers can be selected?

11. On a test, a student is to select 4 out of 10 questions. In how many ways can the student do this?
12. How many lines are determined by 8 points, no three of which are collinear? How many triangles are determined by the same points?
13. A true/false test consists of 10 questions. How many ways to answer the first three questions?
14. Of the first 10 questions on a test a student must answer 7. On the next 5 questions the student must answer 3. In how many ways can this be done?
15. A university offers 5 natural science courses, 7 social science courses, and 4 literature courses. In how many ways can a student choose, 3 natural science courses, 4 social science courses and 1 literature courses?
16. In a certain state, automobile license plates display three letters followed by three digits. How many such plates are possible:
    a. if repetition of the letters is allowed.
    b. if repetition of the letters is Not allowed.
17. A team of ten teachers met for a discussion. Each teacher shook hands exactly once with each of the others present. How many handshakes took place?
18. At a party, 28 handshakes took place. Each person shook hands exactly once with each of the others present. How many people were at the party?
19. Find the number of ways to rearrange the letters in the following words.
    a. ABCA b. LETTER

Answer and Explanation

1. There are 26 ways of choosing the first letter, 25 ways of choosing the second, and 24 ways of choosing the third letter. Hence, there are $26\times25\times24 = 15600$.
2. $4! = 4\times3\times2\times1 = 24$ ways
3. There are four choices for the first letter and since we allow repletion 4 choices for 2^{nd}, 3^{rd} and 4^{th}.
   Thus, $4\times4\times4\times4 = 256$ ways of code
4. $$P(7,3) = \frac{7\times6\times5\times4\times3\times2\times1}{4\times3\times2\times1} = 210$$
5. a) $4\times3\times2\times1 = 24$ b) $4\times4\times4\times4 = 256$

6. $5! = 5\times4\times3\times2\times1 = 120$

7. $(5-1)! = 4! = 4\times3\times2\times1 = 24$

8. $P(12, 4) = \dfrac{12!}{8!} = 12\times11\times10\times9 = 1180$

9. a) $6! = 6\times5\times3\times2\times1 = 720$
   b) $3!\times3! + 3!\times3! = 3!\times3!\times2 = 72$
   c) $2!\times5! = 240$
   d) $6! - (2!\times5!) = 720 - 240 = 480$

10. $C(15, 4) = \dfrac{15!}{(15-4)4!} = \dfrac{15\times14\times13\times12}{4\times3\times2\times1} = 1365$

11. $C(10, 4) = \dfrac{10!}{6!4!} = \dfrac{10\times9\times8\times7}{4\times3\times2\times1} = 210$

12. There are $C(8, 2) = \dfrac{8!}{6!2!} = 28$ lines

    There are $C(8, 3) = \dfrac{8!}{5!3!} = 56$ triangles

13. Using fundamental principles of counting $2\times2\times2 = 8$ ways.

14. Possible ways: $C(10, 7)\times C(5, 3) = 1200$.

15. $C(5, 3)\times C(7, 4)\times C(4, 1) = 10\times35\times4 = 1400$ ways

16. a) There are six choices. At first 26 possible choices, at second 26, at third 26 choices, at fourth 10 possible digit, at fifth 10 choice, and at sixth stage 10 choices. By fundamental counting principle, the number of possible license plate is $26\times26\times26\times10\times10\times10 = 17{,}576{,}000$.
    b) If repetition of letters is not allowed:
    $26\times25\times24\times10\times10\times10 = 15{,}600{,}000$.

17. $C(10, 2) = \dfrac{10!}{8!2!} = \dfrac{10\times9}{2} = 45$ handshakes

18. $C(x, 2) = 28 \Rightarrow \dfrac{(x)(x-1)}{2} = 28$

    $\Rightarrow x^2 - x - 56 = 0 \Leftrightarrow (x-8)(x+7) = 0$

    $\therefore\ x = 8$ people were at the party.

19. a) $\dfrac{4!}{2!} = 4\times3 = 12$

    b) $\dfrac{6!}{2!\times2!} = \dfrac{6\times5\times4\times3}{2} = 180$

UNIT – SEVEN

7. ALGEBRAIC EXPRESSIONS.

A mathematical expression that contains a variable is called an algebraic expression. Some examples of **algebraic expressions** are x^2, $4x+3y$, $2\left(y^3-\frac{1}{z^4}\right)$. Two algebraic expressions are called **like terms** if both the variable parts and the exponents are identical.

7.1 OPERATIONS OF ALGEBRAIC EXPRESSIONS.

When simplifying algebraic expressions; Perform operations within parentheses first and then exponents and then multiplication and then division and then addition and lastly subtraction. This can be recall as **PEMDAS**. (**P**lease **E**xcuse **M**y **D**ear **A**unt **S**elam)

Illustrative Example

1. $7-\left(9-2^3\left[6\div3+1\right]\right)=$

   A. -8 B. 8 C. 22 D. 10

   Solution:

   $7-\left(9-2^3\left[6\div3+1\right]\right)=$

   $\Rightarrow 7-(9-8[2+1])$ ⟵ By performing the exponential and the division within the inner most parentheses

   $\Rightarrow 7-(9-8[3])$ ⟵ By performing the addition within the inner most parentheses.

   $\Rightarrow 7-(9-24)$ ⟵ By performing the multiplication

   $\Rightarrow 7-(-15)$ ⟵ By performing the subtraction within parentheses.

   $\Rightarrow 7+15=22$

   Answer: C

2. If $x=-4$ and $y=3$, then $x^2-\left(y-\left[x+\frac{1}{2}\right]\right)-12$

   A. $\frac{3}{2}$ B. $\frac{-3}{2}$ C. $\frac{21}{2}$ D. $\frac{-5}{2}$

Solution:

$$x^2 - \left(y - \left[x + \frac{1}{2}\right]\right) - 12 =$$

$$\Rightarrow (-4)^2 - \left(3 - \left[-4 + \frac{1}{2}\right]\right) - 12 = 16 = \left(3 - \left[\frac{-7}{2}\right]\right) - 12$$

$$= 16 - \left(3 + \frac{7}{2}\right) - 12$$

$$= 4 - \frac{13}{2} = \frac{-5}{2},$$

Answer: D

The three most important binomial products on SAT course are these:

- $(x - y)(x + y) = x^2 + x y - y x - y^2 = x^2 - y^2$
- $(x - y)^2 = (x - y)(x - y) = x^2 - x y - x y + y^2$
  $= x^2 - 2 x y + y^2$
- $(x + y)^2 = (x + y)(x + y) = x^2 + x y + x y + y^2$
  $= x^2 + 2 x y + y^2$

Illustrative Example

3. If $a - b = 18.4$ and $a + b = 15$, what is the value of $a^2 - b^2$?

   A. 276 B. 33.4 C. 184 D. 270.4

   Solution:

   $a^2 - b^2 = (a - b)(a + b) = (18.4)(15) = 276$ **Answer: A**

4. If $x = y + \frac{1}{y}$ and $z = y - \frac{1}{y}$, where $z \neq 0$ then $(x - z)(x + z)$ is equal to

   A. $\frac{1}{y^2}$ B. y^2 C. 2 D. 4

   Solution:

$$x - z = \left(y + \frac{1}{y}\right) - \left(y - \frac{1}{y}\right) = \frac{1}{y} + \frac{1}{y} = \frac{2}{y}$$

$$x + z = \left(y + \frac{1}{y}\right) + \left(y - \frac{1}{y}\right) = y + y = 2y$$

Then the product of * and **

$\Rightarrow (x-z)(x+z) = \left(\frac{2}{y}\right)(2y) = 4$

Answer: D

5. If $a^2 + b^2 = 6$ and $(a-b)^2 = 14$, what is the value of ab

A. 4 B. -4 C. -8 D. 8

Solution: $(a-b)^2 = a^2 - 2ab + b^2 = a^2 + b^2 - 2ab$

$\Rightarrow 14 = 6 - 2ab \Rightarrow -ab = 8$

$\therefore ab = \frac{8}{-2} = -4$

Answer: B

6. If $\left(\frac{1}{x} + x\right)^2 = 64$; what is the value of $\frac{1}{x^2} + x^2$?

A. 60 B. 62 C. 58 D. 16

Solution:

$$\left(\frac{1}{x} + x\right)^2 = \frac{1}{x^2} + 2 + x^2 = 64$$

$$\Rightarrow \frac{1}{x^2} + x^2 = 62$$

Answer: B

7. Which of the following is equal to $\left(\frac{1}{m} + m\right)^2 - \left(\frac{1}{m} - m\right)^2$

A. $\frac{2}{m^2} - 3m^2$ B. $\frac{1}{m^2} - m^2$ C. 4 D. 0

Solution:

$$\left(\frac{1}{m} + m\right)^2 - \left(\frac{1}{m} - m\right)^2 = \left(\frac{1}{m^2} + 2 + m^2\right) - \left(\frac{1}{m^2} - 2 + m^2\right)$$

$$= 2 - (-2) = 4$$

Answer: C

8. If $x^2 - y^2 = 56$ and $x - y = 16$, what is the average of x and y?

A. 3.5 B. 7 C. 1.75 D. 0.875

Solution: $x^2 - y^2 = (x-y)(x+y) = 56$

$$\Rightarrow (16)(x+y) = 56 \Rightarrow x + y = \frac{56}{16} = 3.5$$

Average of x and $y = \frac{x+y}{2} = \frac{3.5}{2} = 1.75$

Answer: C

9. If $x^2 + y^2 = 2ab$ and $2xy = a^2 + b^2$, with $a > 0$, $b > 0$, $x > 0$, and $y > 0$, then $x + y =$

A. $a + b$ B. $a - b$ C. ab D. $\sqrt{a^2 + b^2}$

Solution:

$$\left.\begin{array}{l} x^2 + y^2 = 2ab \\ 2xy = a^2 + b^2 \end{array}\right\} \text{ adding together}$$

$$x^2 + y^2 + 2xy = a^2 + b^2 + 2ab$$

$$\Rightarrow (x + y)^2 = (a + b)^2$$

$$\therefore x + y = a + b$$

Answer: A

10. If $x^2 + y^2 = xy$, then $(x + y)^4$ is equal to

A. xy B. $9x^2 y^2$ C. $x^2 y^2$ D. $(x^2 + y^2)^2$

Solution:

Adding $2xy$ to both sides of the equation $x^2 + y^2 = xy$ yields $x^2 + y^2 + 2xy = 3xy \Rightarrow (x + y)^2 = 3xy$

$\therefore (x + y)^4 = ((x + y)^2)^2 = (3xy)^2 = 9x^2 y^2$ **Answer: B**

11. What is the value of $(1,000,001)^2 - (999,999)^2$?

A. 4,000,000 B. 2,000,000

C. 1,000,000 D. 200,000

Solution:

We know that $a^2 - b^2 = (a - b)(a + b)$

$\Rightarrow (1,000,001)^2 - (999,999)^2$

$= (1,000,001 - 999,999)(1,000,001 + 999,999)$

$= (2)(2,000,000) = 4,000,000$ **Answer: A**

12. If $r^2 - 2rs + s^2 = 3$, then $(r - s)^6 =$

A. 3 B. 9 C. 27 D. 81

Solution:

$$r^2 - 2rs + s^2 = (r - s)^2 = 3$$

$$\Rightarrow (r - s)^6 = [(r - s)^2]^3 = 3^3 = 27$$ **Answer: C**

13. If $x - y = K$, then $3x^2 - 6xy + 3y^2 =$

A. K B. $3K^2$ C. $9K^2$ D. K^2

Solution:

$$3x^2 - 6xy + 3y^2 = 3(x^2 - 2xy + y^2)$$

$$= 3(x - y)^2 = 3(K)^2 = 3K^2$$ **Answer: B**

14. If x and y are prime numbers such that $x > y > 2$, then $x^2 - y^2$ must be divisible by one of the following numbers

A. 3 B. 4 C. 5 D. 9

Solution: let $x = 5$, $y = 3$

$$x^2 - y^2 = (x - y)(x + y),$$
$$= (5 - 3)(5 + 3) = (2)(8) = 16$$

$\Rightarrow 16 = (4)(4)$

$\Rightarrow$ It must multiple of 4

Answer: B

14. If $a^2 - b^2 = 21$ and $a^2 + b^2 = 29$, which of the following could be the value of ab

I -10 II $5\sqrt{2}$ III 10

A. I only B. II only
C. III only D. I and III only

Solution:

Adding the two equation

$$+\begin{cases} a^2 - b^2 = 21 \\ a^2 + b^2 = 29 \end{cases}$$

$2a^2 = 50 \Rightarrow a^2 = 25$ and $b^2 = 4$

$\therefore a = \pm 5$ and $b = \pm 2$

Thus, $ab = (\pm 5)(\pm 2) = \pm 10$

Answer: D

7.2 SOLVING EQUATIONS

The basic principle to which you must adhere in solving any equation is that you can manipulate the equation in any way, as long as you do the same thing to both sides. For example, you may always add the same number to each side, subtract the same number from each side, multiply or divide each side by the same number (except 0), square each side, take the square root of each side.

Illustrative Example

15. If $x - 8 = 7$, what is the value of $x - 12$?

A. -15 B. -7 C. -1 D. 3

Solution:

$x - 8 = 7 \Rightarrow x = 8 + 7 = 15$

$\therefore x - 12 = 15 - 12 = 3$

16. If $4x + 6 = 30$, what is the value of $3x + 1$?

A. 9 B. 19 C. 15 D. 18

Answer: D

Solution:

$$4x + 6 = 30 \Rightarrow 4x = 2x = 24 \Rightarrow x = \frac{24}{4} = 6$$

$$\therefore 3x + 1 = 3(6) + 1 = 19$$ **Answer: B**

17. If $x + 2y = a$ and $x - 2y = b$, which of the following expression for xy?

A. $\frac{a+b}{2}$ B. $\frac{a-b}{2}$

C. $\frac{a^2-b^2}{4}$ D. $\frac{a^2-b^2}{8}$

Solution:

Adding and subtracting the two equation:

$$+\begin{cases} x + 2y = a \\ x - 2y = b \end{cases}$$

$$2x = a + b$$

$$\Rightarrow x = \frac{a+b}{2}$$

$$-\begin{cases} x + 2y = a \\ x - 2y = b \end{cases}$$

$$0 + 4y = a - b$$

$$\Rightarrow y = \frac{a-b}{4}$$

Thus, $xy = \left(\frac{a+b}{2}\right)\left(\frac{a-b}{4}\right) = \frac{a^2-b^2}{8}$ **Answer: D**

18. If $4y - 3x = 5$, what is the smallest integer value of x for which $y > 100$

A. 131 B. 75 C. 132 D. 133

Solution:

First, solve for y in terms of x:

$$\Rightarrow 4y = 3x + 5 \Rightarrow y = \frac{3x+5}{4}$$

Then, since $y > 100 \Rightarrow \frac{3x+5}{4} > 100$

$$3x + 5 > 400 \Leftrightarrow x > \frac{400-5}{3}$$

$$x > \frac{395}{3} \Rightarrow x > 131.666$$

The smallest integer value of x is 132 **Answer: C**

19. If $\frac{2}{x-y}=6$ then $x=$

A. $y+3$ B. $y-6$ C. $y+\frac{1}{3}$ D. $y-\frac{1}{6}$

Solution:

$$\frac{2}{x-y}=6\Rightarrow 6(x-y)=2$$

$$x-y=\frac{2}{6}=\frac{1}{3}\Rightarrow x=y+\frac{1}{3}$$

Answer: C

20. If the average of **3a** and **4b** is less than 50, and **a** is twice **b**, what is the largest integer value of z?

A. 9 B. 10 C. 11 D. 19

Solution:

Since a = 2b, then 2a = 4b

$\therefore$ The average of 3a and 4b $=\frac{3a+4b}{2}$

$$\Rightarrow \frac{3a+4b}{2}=\frac{3a+2a}{2}=\frac{5a}{2}=2.5a$$

Therefore, $2.5a<50\Rightarrow a<\frac{50}{2.5}\Rightarrow a<20$, So the largest integer value of a is 19

Answer: D

21. If a = 3b, 4b = 9c, and 12c = 6a – 19 what is the value of c

A. $\frac{3}{2}$ B. $\frac{19}{12}$ C. $\frac{1}{3}$ D. $\frac{2}{3}$

Solution:

a = 3b, 4b = 9c, 12c = 6a – 19

$\Rightarrow$ $12c=18b-19 \longleftarrow a=3b\Rightarrow 6a=18b$

$\Rightarrow$ $12c=18\times\frac{9}{4}c-19 \longleftarrow 4b=9c\Rightarrow b=\frac{9}{4}c$

$$12c=\frac{81c}{2}-19\Leftrightarrow 24c=81c-38$$

$$\Rightarrow 57c=38\Leftrightarrow c=\frac{38}{57}=\frac{2}{3}$$

Answer: D

22. If $a = 3b$, $b^2 = 2c$, $9c = d$, then $\frac{a^2}{d} =$

A. $\frac{1}{2}$ B. 2 C. $\frac{10}{3}$ D. 5

Solution:

$$\frac{a^2}{d} = \frac{(3b)^2}{9c} = \frac{9b^2}{9c} = \frac{b^2}{c} = \frac{2c}{c} = 2$$

Answer: B

23. If $x + 3y = 5$ and $3x + y = 7$, then $x + y =$

A. 1 B. 2 C. 3 D. 4

Solution:

Adding the two equation

$$+\begin{cases} x + 3y = 5 \\ 3x + y = 7 \end{cases}$$

$$4x + 4y = 12 \Rightarrow 4(x + y) = 12 \Rightarrow x + y = \frac{12}{4} = 3$$

Answer: C

24. If $x + y = \frac{4a}{5}$, $y + z = \frac{7a}{5}$ and $z + x = \frac{9a}{5}$, $x + y + z =$ _____

A. $\frac{17a}{15}$ B. a C. 2a D. 3a

Solution:

Adding the three equations vertically yields:

$$(x + y) + (y + z) + (z + x) = \frac{4}{5}a + \frac{7}{5}a + \frac{9a}{5} = \frac{20}{5}a = 4a$$

$$\Rightarrow \quad 2x + 2y + 2z = 4a \Leftrightarrow 2(x + y + z) = 4a$$

$$\therefore \quad x + y + z = 2a$$

Answer: C

25. If $2x + 6x$ is 8 less than $y + 7y$, then $x - y =$

A. –8 B. $\frac{1}{5}$ C. 1 D. –1

Solution:

$$2x + 6x = y + 7y - 8$$

$$\Rightarrow 8x - 8y = -8 \Leftrightarrow 8(x - y) = -8$$

$$\therefore \quad x - y = \frac{-8}{8} = -1$$

Answer: D

26. If $z \neq 0$ and $z = 3x = \sqrt{3}y$, what is the value of $z + x$ in terms of y?

A. $\frac{4\sqrt{3}}{3}y$ B. $\frac{2\sqrt{3}y}{3}$ C. $\sqrt{3y}$ D. $\frac{4}{\sqrt{3}}y$

Solution:

We have $z = 3x = \sqrt{3}y$

$$z + x = \sqrt{3}y + \frac{\sqrt{3}}{3}y = \frac{3\sqrt{3}y + \sqrt{3}y}{3} = \frac{4\sqrt{3}y}{3}$$

Answer: A

27. If $\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$ and $ab = c$, what is the average of a and b?

A. $\frac{a+b}{2c}$ B. $\frac{1}{2}$ C. 0 D. $\frac{c}{2}$

Solution: $\frac{1}{a} + \frac{1}{b} = \frac{1}{c} \Leftrightarrow abc\left(\frac{1}{a} + \frac{1}{b} = \frac{1}{c}\right)$

$= bc + ac = ab \Leftrightarrow c(b+a) = ab$

$\Rightarrow a + b = \frac{ab}{c} = \frac{c}{c} = 1$

$\Rightarrow$ Average of a and b $= \frac{a+b}{2} = \frac{1}{2}$

Answer: B

28. If percent of $(x + y)$ is 6 and x is a positive integer, what is the greatest possible value of y?

A. 146 B. 150 C. 148 D. 149

Solution:

4% if $x + y$ is $6 \Rightarrow 0.04(x + y) = 6$

$\Rightarrow x + y = \frac{6}{0.04} = 150$

$\Rightarrow y = 150 - x$

This expression will be greatest when x is as small as possible. This is when $x = 1$

$\Rightarrow y = 150 - x = 150 - 1 = 149$

Answer: D

29. If $x + y = 12$ and $x - y = 4\%$ of $(x + y)$, then $x^2 - y^2 =$

A. 48 B. 5.76 C. 4.8 D. 57.6

Solution:

$x - y = 4\%$ of $(x + y) \Rightarrow x - y = (0.04)(12) = 0.48$

Hence $x^2 - y^2 = (x - y)(x + y) = (0.48)(12) = 5.76$

Answer: B

If $x > 0$ and y is 3 more than the square of x, which one of the following expresses x in terms of y

A. $x = \sqrt{y^2 - 3}$ B. $x = \sqrt{y+3}$

C. $x = \sqrt{y^2 - 3}$ D. $x = -\sqrt{y-3}$

Solution: y is 3 more than the square of x.

$$y = x^2 + 3 \Leftrightarrow x^2 = y - 3$$

$$x = -\sqrt{y-3}$$

Answer: D

If the operation * is defined for all non – zero x and y by the equation $x * y = (xy)^2$, then $(x*y) * z =$

A. $x^4 y^4 z^2$ B. $x^2 y^4 z^2$

C. $x^2 y^2 z^2$ D. $x^4 y^2 z^2$

Solution: Since $x * y = (xy)^2$, then

$$(x * y) * y = (xy)^2 * z = [(xy)^2 z]^2$$
$$= x^4 y^4 z^2$$

Answer: A

If x, y and k are positive numbers such that

$\left(\frac{x}{x+y}\right)(10) + \left(\frac{y}{x+y}\right)(20) = k$ and if $x < y$. which of the following could be the value of K?

A. 10 B. 18 C. 12 D. 30

Solution:

$$k = \frac{10x}{x+y} + \frac{20y}{x+y}$$

$$= \frac{10x + 10y + 10y}{x+y} = \frac{10(x+y)}{x+y} + \frac{10y}{x+y}$$

$$= 10 + \frac{10y}{x+y}$$

Since $0 < x < y$

$\Rightarrow 0 + y < x + y < y + y$

$\Rightarrow y < x + y < 2y$

$$\Rightarrow \frac{1}{2y} < \frac{1}{x+y} < \frac{1}{y} \Leftrightarrow \frac{10y}{2y} < \frac{10y}{x+y} < \frac{10y}{y}$$

$$\Rightarrow 5 < \frac{10y}{x+y} < 10$$

$$\Rightarrow 15 < 10 + \frac{10y}{x+y} < 20$$

33. If $x - y + z = 7$ and $x + y - z = 11$, which of the following statement **MUST** be true?

I. x is positive II. $x > y$ III. $xy < 0$

A. I only B. II only
C. III only D. I and II only

Solution: Adding the two equations:

$$+\begin{cases} x - y + z = 7 \\ x + y - z = 11 \end{cases}$$

$$2x = 18 \Rightarrow x = 9 \text{ (I is true)}$$

- Replace x by 9 in each equation

  $\Rightarrow 9 - y + z = 7 \Leftrightarrow z - y = -2$, and

  $9 + y - z = 11 \Rightarrow y - z = 2$.

  Since $y - z = 2$, then $x > y$. (II is true)

Answer:

34. If $a - b = 3$, $b - c = 4$, and $c - a = d$, what is the value of d?

A. -3 B. 1 C. -7 D. 3

Solution: Add the three equations:

$$+\begin{cases} a - b = 3 \\ b - c = 4 \\ c - a = d \end{cases}$$

$$0 = 7 + d \Rightarrow d = -7$$

Answer:

35. If $5x + 13 = 31$, what is the value of $\sqrt{5x+31}$?

A. $\sqrt{13}$ B. $\sqrt{\frac{173}{5}}$ C. 7 D. 13

Solution:

$$5x + 13 = 31 \Rightarrow 5x + 13 + 18 = 31 + 18 = 49$$

$$\Rightarrow 5x + 31 = 49 \Leftrightarrow \sqrt{5x+31} = \sqrt{49} = 7$$

Answer:

36. If $a < b$ and c is the sum of a and b, which of the following is the positive difference between a and b?

A. $2a - c$ B. $2b - c$ C. $c - 2b$ D. $c - a + b$

Solution:

$$c = a + b \Rightarrow a = c - b$$

$$\Rightarrow b - a = b - (c - b) = 2b - c$$

37. If $4^x = 5^y = 20^z$, then z equals to

Answer: B

A. xy B. $\frac{x+y}{xy}$ C. $\frac{xy}{x+y}$ D. $\frac{1}{xy}$

Solution:

$4^x = 20^z \Rightarrow 20^{\frac{z}{x}} = 4$ and $5^y = 20^z \Rightarrow 20^{\frac{z}{y}} = 5$

Multiplying both, $4 \times 5 = \left(20^{\frac{z}{x}}\right)\left(20^{\frac{z}{y}}\right)$

$$\Rightarrow 20^{\frac{z}{x}+\frac{z}{y}} = 20^1 \Rightarrow \frac{z}{x} + \frac{z}{y} = 1$$

$$\Rightarrow z\left(\frac{1}{x} + \frac{1}{y}\right) = 1 \Leftrightarrow \frac{1}{z} = \frac{1}{x} + \frac{1}{y} = \frac{x+y}{xy}$$

$$\therefore z = \frac{xy}{x+y}$$

Answer: C

38. If $\left(\frac{1}{2}\right)^k = \sqrt{3}$ and $\left(\frac{1}{3}\right)^m = \sqrt{2}$, then find the value of $\left(\frac{mk}{2}\right)$

A. $\frac{1}{8}$ B. $\frac{1}{6}$ C. $\frac{1}{\sqrt{6}}$ D. $\frac{1}{2\sqrt{2}}$

Solution:

a. $\left(\frac{1}{2}\right)^k = \sqrt{3} \Rightarrow 2^{-k} = 3^{\frac{1}{2}}$ and $\left(\frac{1}{3}\right)^m = \sqrt{2} \Rightarrow 3^{-m} = 2^{\frac{1}{2}}$

$\Rightarrow 2^{-2k} = 3$ and $3^{-2m} = 2$

Since $3 = 2^{-2k}$ thus $(2^{-2k})^{-2m} = 2 \Rightarrow 2^{4mk} = 2$

$4mk = 1 \Rightarrow mk = \frac{1}{4}$

$\therefore \frac{mk}{2} = \frac{1}{8}$

Answer: A

39. If $A = x - 2^{-1}$, $B = x + 2^{-1}$ and $A^2 - B^2 > 0$, then the value of x is
A. $X = A + B$ B. $x = 0$ C. $x < 0$ D. $x > 0$

Solution:

$$A^2 - B^2 > 0 \Rightarrow \left(x - \frac{1}{2}\right)^2 - \left(x + \frac{1}{2}\right)^2 > 0$$

$$\Rightarrow -2x > 0 \Rightarrow x < 0$$

Answer: C

40. If x and y are real numbers such that $x > y$ and $mx = my$, then 'm' is equal to

A. 0 B. $\frac{y}{x}$ C. $\frac{x}{y}$ D. 1

Solution:

for $x > y$

$mx = my$

$mx - my = 0$

$m(x - y) = 0$

$$m = \frac{0}{x - y} = 0$$

Answer: A

41. When x is a natural number, $3^x + 3^{x+1} + 3^{x+2}$ is divisible at least by
A. 39 B. 3 C. 13 D. 10

Solution:

$3^x + 3^{x+1} + 3^{x+2} = 3^x + 3^x . 3 + 3^x . 3^2 = 3^x (1 + 3 + 9) = 3^x . 13$

Since x is natural number ≥ 1.

Hence, $3^x . 13 = 39 . 3^{x-1}$ it is divisible by 39 at least.

Answer: A

Review Exercise

1. $3 - (15 - 2^3 [12 \div 6 + 2]) =$
A. 20 B. 60 C. 25 D. 14

2. $(3 + x)(3 + y) - (3 + x)(y + 5)$
A. $2x + 6$ B. $-2x - 6$
C. $x^2 + 3xy + 6$ D. $x^2 + 9$

3. If $x + y = 7$ and $x - y = 4$, then $x^2 - y^2 =$
A. 56 B. 23 C. 65 D. 28

4. What is the value of $\frac{x^2 - y^2}{y - x}$ when $x = 15.7$ and $y = 17.9$
A. 33.6 B. −33.6 C. 32.6 D. −32.6

5. If $a^2 + b^2 = 10$ and $(a - b)^2 = 2$, what is the value of ab?
A. 8 B. cvc C. 4 D. 5

6. If $x = \frac{1}{3 - 2\sqrt{2}}$ and $y = \frac{1}{3 + 2\sqrt{2}}$ then what is the value of $xy^2 + x^2y$?
A. 9 B. 1 C. 6 D. $3\sqrt{2}$

7. If $x^2 - y^2 = 27$ and $x - y = 6$, what is the average of x and y
A. 9 B. $\frac{9}{2}$ C. $\frac{9}{4}$ D. 3

8. If $x = \frac{4 - x}{2 + x}$, what is the value of $x^2 + 3x - 4$?
A. –4 B. –1 C. 0 D. 1

9. If $x = -1$, then $\frac{x^4 - x^3 + x^2}{x - 1} =$
A. $\frac{1}{2}$ B. $\frac{3}{2}$ C. $\frac{-1}{2}$ D. $\frac{-3}{2}$

10. If $x = 1 - 3t$ and $y = 2t - 1$, then for what values of t does $x = y$
A. $\frac{5}{2}$ B. $\frac{2}{5}$ C. $\frac{3}{2}$ D. $\frac{2}{3}$

11. If $x + y = a$ and $x - y = b$, then $2xy =$ _____
A. $\frac{a^2 - b^2}{2}$ B. $\frac{b^2 - a^2}{2}$
C. $\frac{a - b}{2}$ D. $\frac{ab}{2}$

12. If $a + b + c = 110$, $a = 4b$, and $3a = 2c$, then $b =$
A. 8 B. 9 C. 14 D. 10

13. If x is an integer, and the average of 3, 4 and x is less than 15, what is greatest possible value of x?
A. 38 B. 37 C. 39 D. 36

14. If $x - 6 = 21$, what is the value of $x - 12$
A. 15 B. 27 C. –15 D. 9

15. For what value of x is $\frac{4}{x} + \frac{3}{5} = \frac{10}{x}$

16. If $x + 2y = 14$ and $3x + y = 12$, then $x =$ _____
A. –2 B. 1 C. –1 D. 2

17. $a * b = a(a - b)$ for all integers a and b, then $4 * (3 * 5)$ equals:
A. 20 B. 40 C. 2 D. −2

Answerer and Explanation

1. A: $3 - (15 - 8(4)) = 3 - (-17) = 20$
2. B: $(3 + x)[(3 + y) - (y + 5)] = (3 + x)(3 + y - y - 5) = -2x - 6$
3. D: $x^2 - y^2 = (x + y)(x - y) = (7)(4) = 28$
4. B: $\dfrac{x^2 - y^2}{y - x} = \left(\dfrac{x - y}{y - x}\right)(x + y) = -(x + y) = -(15.7 + 17.9) = -33.6$
5. C: $(a - b)^2 = a^2 - 2ab + b^2 = 2$

$\Leftrightarrow a2 + b2 - 2ab = 10 - 2ab = 2$

$\Rightarrow 2ab = 10 - 2 \Rightarrow ab = \dfrac{8}{2} = 4$

6. C: $xy^2 + x^2y = (xy)(y + x)$

$= \left(\dfrac{1}{3 - 2\sqrt{2}}\right)\left(\dfrac{1}{3 + 2\sqrt{2}}\right)\left(\dfrac{1}{3 + 2\sqrt{2}} + \dfrac{1}{3 - 2\sqrt{2}}\right)$

$= \left(\dfrac{1}{9 - 8}\right)\left(\dfrac{3 - 2\sqrt{2} + 3 + 2\sqrt{2}}{9 - 8}\right)$

$= 6$

7. C: $x^2 - y^2 = (x - y)(x + y) = (6)(x + y) = 27$

Average, $\dfrac{x + y}{2} = \dfrac{27}{2x6} = \dfrac{27}{12} = \dfrac{9}{4}$

8. C: If $x = \dfrac{4 - x}{2 + x} \Rightarrow x^2 + 2x = 4 - x$

$\Rightarrow x^2 + 3x - 4 = 0$

9. D: $\dfrac{x^4 - x^3 + x^2}{x - 1} = \dfrac{x^2(x^2 - x + 1)}{x - 1} = \dfrac{(1)(1 + 1 + 1)}{-1 - 1} = \dfrac{-3}{2}$

10. B: $x = y \Leftrightarrow 1 - 3t = 2t - 1 \Rightarrow 2 = 5t,\ t = \dfrac{2}{5}$

11. A: Adding the two, $2x = a + b \Rightarrow x = \dfrac{a + b}{2}$ and subtracting,

$2y = \dfrac{a - b}{a} \Rightarrow y = \dfrac{a - b}{2}$

$$\therefore 2xy = 2\left(\frac{a+b}{2}\right)\left(\frac{a-b}{2}\right) = \frac{a^2-b^2}{2}$$

12. D: $4b + b + 6b = 110 \Rightarrow b = 10$

13. B: $\frac{3+4+x}{3} < 15 \Rightarrow 7 + x < 45 \Rightarrow x < 38$

14. $x - 6 = 21$ subtract both side by 6, $x - 12 = 21 - 6 = x - 12 = 15$

15. Multiply each side by $5x$:

$$5x\left(\frac{4}{x}\right) + 5x\left(\frac{3}{5}\right) = \left(\frac{10}{x}\right)5x \Leftrightarrow 20 + 3x = 50$$

$$\Rightarrow 3x = 50 - 20 = 30 \Leftrightarrow x = \frac{30}{3} = 10$$

16. D: $2(3x + y) = 2(12) \Rightarrow 6x + 2y = 24$

$$-\begin{cases} 6x + 2y = 24 \\ x + 2y = 14 \end{cases}$$

$5x = 10 \Rightarrow x = 2$, so that

17. $4*(3*5) = 4*(3(3-5)) = 4*(-6) = 4(4-(-6)) = 40$

UNIT - EIGHT

8. GEOMETRIC APTITUDE

8.1 TRIANGLES

A closed and plane figure, bounded by three lines segments is called a **triangle**. A triangle is divided in to 3 based on their sides

- **Scalene;** if all its sides are different lengths and all its angles are of different measures.

Isosceles Triangle

A triangle is **an isosceles;** if its any two sides are equal and the angels opposite to these two sides are also equal.

Equilateral triangle

A triangle is an equilateral triangle; if all its sides are equal and all its angles are also equal in such away that each angle is 60°.

A triangle is divided into 3 based on their angles;

- **acute angles triangle:** if each of its angles is **less than 90°**
- **right angles triangle:** if only one of its angle is 90°.
- **obtuse angles triangle**: if only one of its angles is greater than 90°.

Triangle Fact (A):

Let a, b, and c be the sides of ΔABC, with $a \leq b \leq c$.

- $a^2 + b^2 = c^2$ if and only if $\angle c = 90^\circ$.
- $a^2 + b^2 < c^2$ if and only if angle C is obtuse.
- $a^2 + b^2 > c^2$ if and only if angle ($\angle c$) is acute.

Illustrative Example

Which of the following are **NOT** the sides of a right triangle?

A. 5, 12, 13
B. $1, 1, \sqrt{2}$
C. $1, \sqrt{3}, 2$
D. $\sqrt{5}, \sqrt{12}, \sqrt{13}$

Solution:

A: $5^2 + 12^2 = 25 + 144 = 169 = 13^2$ ⟵ sides of a right angles triangle

B: $1^2 + 1^2 = 1 + 1 = 2 = \left(\sqrt{2}\right)^2$ ⟵ sides of a right angles triangle

C: $1^2 + \sqrt{3}^2 = 1 + 3 = 4 = 2^2$ ⟵ sides of a right angles triangle

D: $\left(\sqrt{5}\right)^2 + \left(\sqrt{12}\right)^2 = 5 + 12 = 17 \neq \left(\sqrt{3}\right)^2$ ⟵ Not a right angles triangle

Answer: D

2. What type of a triangle has side length of 6, 8 and 11?

A. Acute Scalene
B. Right Scalene
C. Obtuse Scalene
D. None of the above

Solution:

$6^2 + 8^2 = 36 + 64 = 100 < 11^2$

$\Rightarrow 6^2 + 8^2 < 11^2$ ⟵ obtuse scalene

Answer: C

Triangle Fact (B)

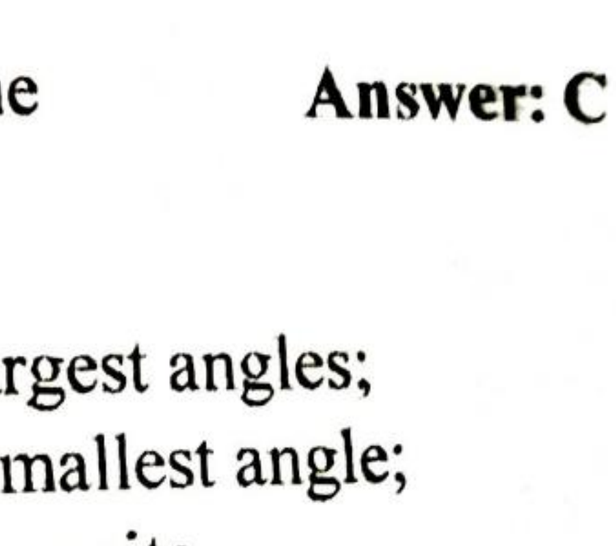
In any triangle:

- the longest side is opposite the largest angles;
- the shortest side is opposite the smallest angle;
- sides with the same length are opposite angels with the same measure.

Illustrative Example

3. Which of the following statements concerning the length of side BC is true?

A. $BC < 7$
B. $BC = 7$
C. $7 < BC < 8$
D. $BC > 8$



Solution:

By triangle fact B,

$m(\angle A) + 70° + 50° = 180°$

$\Rightarrow m(\angle A) = 180^\circ - 120^\circ = 60^\circ$

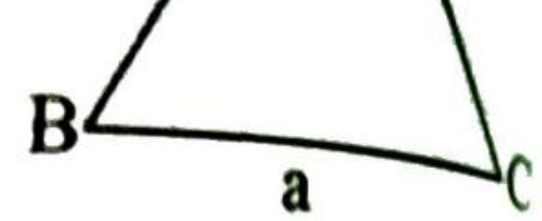
Then B is the largest angle, C is the smallest, and A is in between.

Therefore $AB < BC < AC \Rightarrow 7 < BC < 8$

Answer: C

Triangle Fact (C) (Triangle Inequality)

- The sum of the lengths of any two sides of a triangle is greater than the length of the third side. That is $a + b > c$, $b + c > a$. and $c + a > b$.



The difference between the length of any two sides of a triangle is less than the length of the third side.

Illustrative Example

4. Is it possible to have a triangle with sides:

   A. 3 cm, 5 cm and 9 cm?
   B. 6 cm, 4cm and 10cm?
   C. 4cm, 7cm, and 11cm?
   D. 8cm, 7cm and 12cm?

Solution:

A. No; $3 + 5 = 8 < 9$

B. No; $6 + 4 = 10$

C. No; $4 + 7 = 11$

D. Yes; $8 + 7 = 15 > 12$, $7 + 12 = 19 > 8$

$8 + 12 = 20 > 7$

Answer: D

5. If the lengths of two sides of a triangle are 5 and 7, which of the following could be the length of the third side?

(I) 2 (II) 4 (III) 13

A. I only
B. II only
C. I and II only
D. I, II and III

Solution:

Let x be the third side. The third side must be greater than $7 - 5 = 2$ and must be less than $7 + 5 = 12$. That is $2 < x < 12$.

Answer: B

If the perimeter of a triangle is 18, then the length of one of sides **CANNOT** be:

A. 8 B. 9 C. 3 D. 1

Solution:

Let a, b, and c are side length of a triangle then semi- perimeter of a triangle is greater than any side length of a triangle.

ie $\frac{a+b+c}{2} > a$, $\frac{a+b+c}{2} >$ band $\frac{a+b+c}{2} > c$

Therefore, $\frac{18}{2} > 8$, $\frac{18}{2} >$ and $\frac{18}{2} > 1$

Therefore. 9 cannot be the side length of a triangle. **Answer: B**

If the lengths of two sides of a triangle are 7 and 8, then the perimeter of a triangle **CANNOT** be:

A. 20 B. 28 C. 17 D. 16

Solution:

Let x = the third side length

$\Rightarrow 8 - 7 < x < 8 + 7 \Rightarrow 1 < x < 15$

Any number between 1 and 15 could be the length of third side.

$\Rightarrow$ Perimeter $= 7 + 8 + 1 < x < 15 + 7 + 8$

$\Rightarrow$ Perimeter $= 16 < P < 30$ **Answer: D**

Two sides of a right triangle are 12 and 13. Which of the following could be the length of the third side?

I. 5 II. 11 III. $\sqrt{313}$

A. I only C. I and II only

B. II only D. I and III only

Solution: Let x be the third side so that x: $1 < x < 25$

Case i: Let 13 be the hypotenuse:

$\Rightarrow x^2 + 12^2 = 13^2 \Rightarrow x^2 = 169 - 144 = 25$

$\therefore x = 5$

Case ii: Let x be the hypotenuse:

$x^2 = 12^2 + 13^2 = 144 + 169 = 313$

$\therefore x = \sqrt{313}$ **Answer: D**

Triangle fact C (Heron's Formula)

The area of a triangle is given by $A = \frac{1}{2}bh$, where b = base and h = height or $A = \sqrt{S(S-a)(S-b)(S-c)}$, where

$$S = \frac{a+b+c}{2}$$

Illustrative Example

9. What is the area of an equilateral triangle whose sides are 8?

A. $16\sqrt{3}$ B. 32 C. $32\sqrt{3}$ D. 16

Solution:

If A represents the area of an equilateral triangle with side length x, then $A = \frac{x^2\sqrt{3}}{4}$.

$$\therefore A = \frac{8^2\sqrt{3}}{4} = \frac{64\sqrt{3}}{4} = 16\sqrt{3}$$

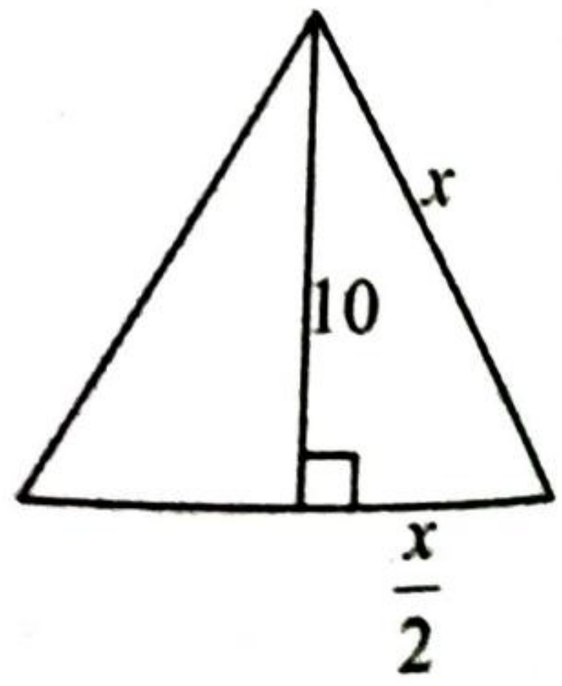
Answer: A

10. What is the area of an equilateral triangle whose altitude is 10?

A. 30 B. $\frac{100\sqrt{3}}{3}$ C. $30\sqrt{3}$ D. $\frac{10\sqrt{3}}{3}$

Solution: Let x be the side length

$$\Rightarrow \left(\frac{x}{2}\right)^2 + 10^2 = x^2$$



$$\Rightarrow x^2 - \frac{x^2}{4} = 100$$

$$\Rightarrow 3x^2 = 400$$

$$\Rightarrow x^2 = \frac{400}{3} \Leftrightarrow x = \frac{20}{\sqrt{\ }}$$

$$\therefore A = \frac{x^2\sqrt{3}}{4} = \frac{400\sqrt{3}}{12} = \frac{100\sqrt{3}}{3}$$

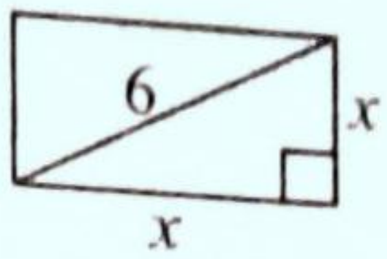
Answer: B

11. What is the area of square whose diagonal is 6?
A. 12 B. 24 C. 18 D. 36

Solution: Let x = side length

$$x^2 + x^2 = 6^2 \Leftrightarrow 2x^2 = 36$$

$$x^2 = \frac{36}{2} = 18$$

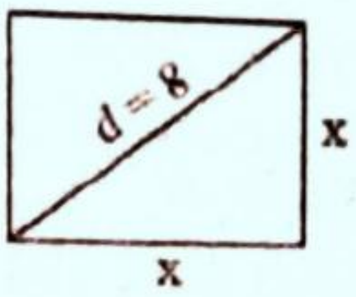


Answer: C

12. What is the length of each side of a square if its diagonals are 8?
A. $4\sqrt{3}$ B. $8\sqrt{2}$ C. $4\sqrt{2}$ D. $12\sqrt{2}$

Solution:

$$x^2 + x^2 = d^2$$
$$2x^2 = 8^2 = 64$$
$$\Rightarrow x^2 = 32 \Rightarrow x = \sqrt{32} = 4\sqrt{2}$$



Answer: C

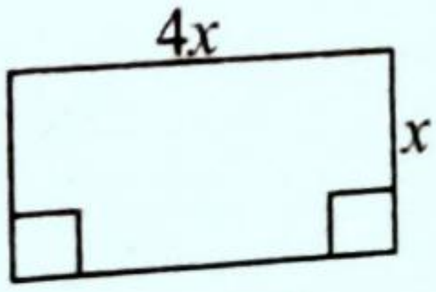
Important Fact

Here are the area formula you need to know:

- Area of a parallelogram: A = bh
- Area of a rectangle: A = ℓw
- Area of a square: $A = s^2$ or $A = \frac{1}{2}d^2$ where d – is diagonal
- Area of a circle: $A = \pi r^2 = \frac{\pi d^2}{4}$ where d – is diameter

Illustrative Example

13. If the length of a rectangle is 4 times its width, and if its area is 144, what is its perimeter?
A. 96 B. 60
C. 30 D. 24



Solution:

Since the area is 144, then $(4x)(x) = 144 \Leftrightarrow x^2 = 36 \Rightarrow x = 6$

The width is 6, the length is 24, and the perimeter is

$2(l + w) = 2(6 + 24) = 60$.

Answer: B

14. The length of a rectangle is 5 more than the side of a square, and the width of the rectangle is 5 less than the side of the square. If the area of square is 45. What is the area of the rectangle?
A. 20 B. 25 C. 45 D. 50

Solution: Let x = the side of the square

Then, width (W) of rectangle: $W = x - 5$

length (ℓ) of rectangle: $\ell = x + 5$

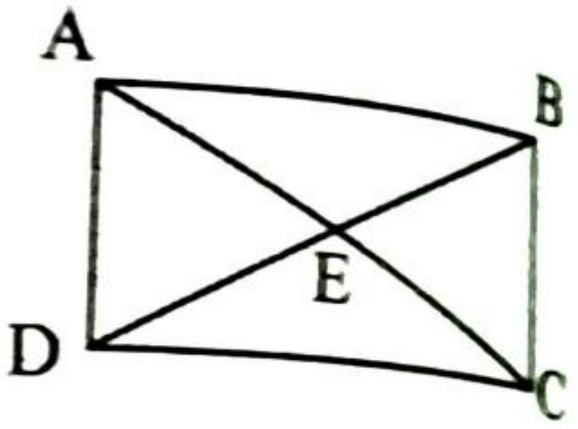
Area of rectangle = $(\ell)(W) = (x + 5)(x - 5) = x^2 - 25$

Since 45 is the area of square, $x^2 = 45$, and
so $x^2 - 25 = 45 - 25 = 20$ is area of rectangle

Answer: A

Quantitative Comparison

15. Suppose that ABCD is a rectangle.



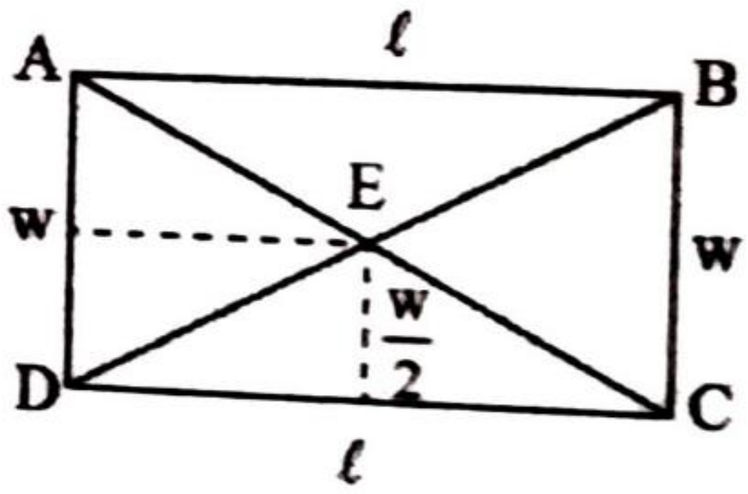
Quantity A	Quantity B
The area of ΔAED	The area of ΔEDC

A. Quantity A is greater than quantity B.
B. Quantity A is less than Quantity B.
C. The two quantities are equal.
D. The relationship cannot determine from the information given.

Solution:

- Area of $\Delta AED = \frac{1}{2} W \times \left(\frac{\ell}{2}\right) = \frac{\ell W}{4}$
- Area of $\Delta EDC = \frac{1}{2}\ell = \left(\frac{W}{2}\right) = \frac{\ell W}{4}$

} the two quantities are equal



Answer: C

16. What is the circumference of a circle, whose area is 100π?
A. 20π B. 10π C. 20 D. 10

Solution:

$A = \pi r^2 = 100\pi \Rightarrow r^2 = 100 \Rightarrow r = 10$
$C = 2\pi r = 2\pi(10) = 20\pi$

Answer: A

17. What is the area of a circle whose circumference is π?

A. 2π B. π C. $\frac{\pi}{2}$ D. $\frac{\pi}{4}$

Solution:

$$C = 2\pi r = \pi \Rightarrow r = \frac{1}{2}$$

$$A = \pi r^2 = \pi\left(\frac{1}{2}\right)^2 = \frac{\pi}{4}$$

Answer: D

18. A circular region has circumference C centimeter and area K cm^2. If $c = 3k$, what is the radius of the circle in centimeter?

A. $\frac{3}{2}$ B. $\frac{1}{3}$ C. $\frac{2}{3}$ D. 3

Solution:

$$\frac{\text{Area}}{\text{cirumference}} = \frac{k}{C} = \frac{\pi r^2}{2\pi r} = \frac{r}{2}$$

$$\Rightarrow \frac{k}{C} = \frac{r}{2} \Leftrightarrow \frac{r}{2} = \frac{k}{3k} = \frac{1}{3} \Rightarrow r = \frac{2}{3}$$

Answer: C

19. A wire when bent in the form of a square, encloses an area of $121cm^2$. If the same wire is bent in the form of a circle, find the area of the circle. $\left(\text{use } \pi = \frac{22}{7}\right)$

Solution:

- Area of the square $= 121cm^2$
  $\Rightarrow (\text{side})^2 = 121 \Rightarrow \text{side} = 11cm$
- Perimeter of the square $= 4 \times 11 = 44$ cm
  $\Rightarrow$ Length of the wire $= 44cm$

Now, the same wire is bent in the form of the circle, therefore, the circumference of the circle = length of the wire

$$\Rightarrow 2\pi r = 44 \Rightarrow 2\left(\frac{22}{7}\right)r = 44 \Rightarrow r = 7 \text{ cm}$$

$$\therefore \text{ Area of the circle} = \pi r^2 = \left(\frac{22}{7}\right)(49) = 154cm^2$$

20. Find the number of trees which can be planted in a triangular piece of ground having its sides 16m, 14m and 6m respectively and each tree occupies $2\sqrt{3}\ m^2$ space.

A. 24 B. 18 C. 12 D. 9

Solution: Here, a = 16, b = 14, C = 6 and

$$S = \frac{a+b+c}{2} = \frac{16+14+66}{2} = 18$$

Using Heron's the formula:

Area: $A = \sqrt{S(S-a)(S-b)(S-c)} = \sqrt{18(18-16)(18-14)(18-6)} = 24\sqrt{3}m^2$

So, number of tree planted $= \frac{\text{Area of ground}}{\text{Area of each tree}} = \frac{24\sqrt{3}m^2}{2\sqrt{3}m^2} = 12$

Answer: C

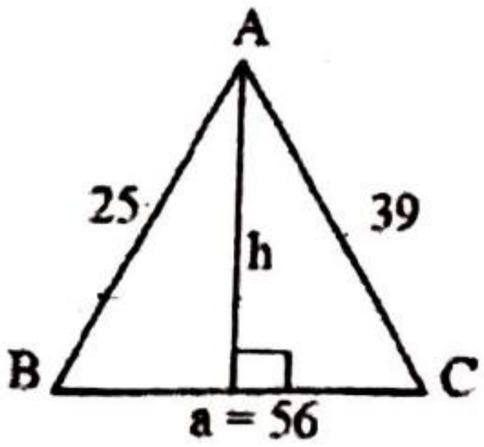
21. The sides of a triangle are 56m, 39m, and 25m respectively. Find the perpendicular from the opposite angle on the greatest sides.

A. 10m B. 15m C. 7.5m D. 20m

Solution: Here, a = 56, b = 39, c = 25

and $S = \frac{a+b+c}{2} = \frac{56+39+25}{2} = \frac{120}{2} = 60m.$

Let h = the perpendicular from < A on side a = 56m.



Using Heron's formula:

Area $= \sqrt{S(S-a)(S-b)(S-c)}$

$\Rightarrow A = \sqrt{60(60-56)(60-39)(60-25)} = \frac{1}{2}(\text{base})(\text{h}$

$\Rightarrow \frac{1}{2}(56)h = \sqrt{60(4)(21)(35)} = 420$

$\therefore h = \frac{420\times 2}{56} = 15m$

Answer: B

22. If the perimeter of a right angled triangle measure 112m and the hypotenuse measures 50m, find the other two side.

Solution: It is given that a + b + c = 112

$\Rightarrow a + b + 50 = 112 \Rightarrow a + b = 62 \ldots (*)$

Also, $a^2+b^2 = c^2 \Leftrightarrow a^2 + b^2 = 50^2$.

We have $(a + b)^2 = a^2 + b^2 + 2ab$

$$\Rightarrow 62^2 = 50^2 + 2ab \Rightarrow 2ab = 62^2 - 50^2 = 1344$$

Again $(a-b)^2 = a^2 + b^2 - 2ab = 50^2 - (1344) = 1156$

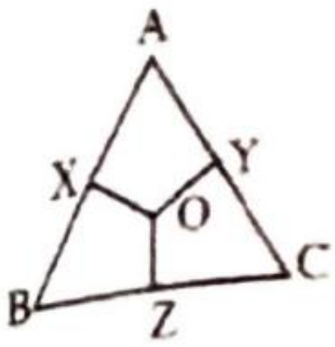
$$\Rightarrow a - b = \sqrt{1156} = 34 \ldots (**)$$

Solving (*) and (**) we have

$$\begin{cases} a + b = 62 \\ a - b = 34 \end{cases} \Rightarrow 2a = 96 \Rightarrow a = 48, b = 14$$

Therefore, the two sides of a given right triangle are 14 and 48.

23. In the adjoining, equilateral triangle ABC, three perpendicular **OX, OY** and **OZ** are drawn from point **O** to the three sides.



If the perpendiculars measures 4m, 5m, and 6m respectively.
Find the area of the triangle.

A. $75\sqrt{3}m^2$ B. $300m^2$ C. 75m2 D. $100\sqrt{3}m^2$

Solution: Joining point O to the three vertices, A, B, and C:

Area of ΔABC = area of $(\Delta OBC + \Delta OCA + \Delta OAB)$

$$\frac{\sqrt{3}}{4}a^2 = \left(\frac{1}{2}\times a\times 6\right) + \left(\frac{1}{2}\times a\times 5\right) + \left(\frac{1}{2}\times a\times 4\right) = \frac{15a}{2}$$

$$\Rightarrow a = \frac{15\times 4}{2\sqrt{3}} = 10\sqrt{3}m \longleftarrow \text{side length}$$

Hence, area of $\Delta ABC = \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4}\left(10\sqrt{3}\right)^2 = 75\sqrt{3}m^2$

Answer: A

24. What is the side of that equilateral triangle whose area costs as much to pave at 20birr per square meter as it would cost to fence the three sides at birr 45 a meter?

Solution: Let a = the side of equilateral triangle

$$\text{Area}(A) = \frac{\sqrt{3}}{4}a^2 m^2$$

$$\Rightarrow \text{Cost of paving} = 20\times\frac{\sqrt{3}}{4}a^2 \text{ birr}$$

according to the equation $20\times\frac{\sqrt{3}}{4}a^2 = 45\times 3a$

$$\Rightarrow a = \frac{45\times3\times4}{20\times\sqrt{3}} = 9\sqrt{3}m$$

Hence, the side of triangle is $9\sqrt{3}m$ **Answer: (**

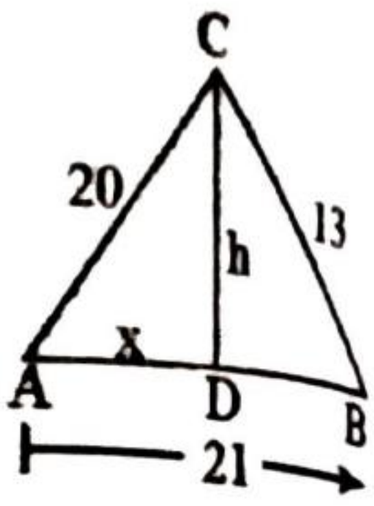
25. The side of a triangle are 21m, 20m and 13m respectively. Find the areas of the triangles into which it is divided by the perpendicular up on the longest side from opposite vertex:

Solution:

$$\frac{1}{2}h\times21=\sqrt{S(S-a)(S-b)(S-c)}$$

$$\frac{1}{2}h\times21=\sqrt{27\times6\times7\times14}$$

$$\Rightarrow h = 12m$$



$$AD = X = \sqrt{20^2-h^2} = \sqrt{20^2-12^2} = 16m$$

- Area of $\Delta ACD = \frac{1}{2}\times h\times AD = \frac{1}{2}\times12\times16 = 96m^2$
- Area of $\Delta BCD = \frac{1}{2}\times h\times BD = \frac{1}{2}\times12\times(21-16) = 30m^2$

26. If the length of each side of an equilateral triangle were increased by 4m, the area would be increased by $10\sqrt{13}m^2$. Find the length of each side.

Solution: Let a = the side length

Then after increment will be: a + 4

$$\Rightarrow \frac{\sqrt{3}}{4}(a+4)^2 - \frac{\sqrt{3}}{4}a^2 = 10\sqrt{3} \Rightarrow \frac{\sqrt{3}}{4}(8a+16)$$

$$\Rightarrow 8a = 40 - 16 = 24 \Rightarrow a = 3$$

27. The area of parallelogram is $288m^2$. If the altitude is twice the corresponding base, determine the base and the altitude.

Solution:

Area: $A = bh \Rightarrow 228 = (2b)(b) = 2b^2$

$$\Rightarrow b^2 = 144 \Rightarrow b = 12m$$

Height = 2b = 2(12) = 24m.

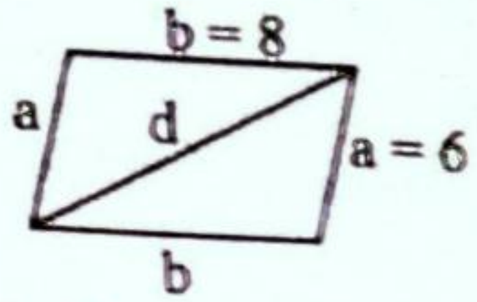
28. Find the area of a parallelogram whose adjacent sides are 6cm and 8cm and the diagonal connecting the ends is 12cm

Solution:

Use: $A = 2\sqrt{S(S-a)(S-b)(S-d)}$ where $S = \frac{a+b+d}{2}$

$\Rightarrow A = 2\sqrt{13(13-6)(13-8)(13-12)}$

$\therefore A = 2\sqrt{455}m^2$



29. If the perimeter of a triangle is 27m, then the length of one of sides **CANNOT** be:

A. 1 B. 10 C. 15 D. 13

Solution: Let a, b and c the side length of a triangle then semi-perimeter of a triangle is greater than any side length of a triangle therefore $\frac{27}{2} = 13.5$

$\Rightarrow$ 13.5 > 1, 13.5 > 10, and 13.5 > 13

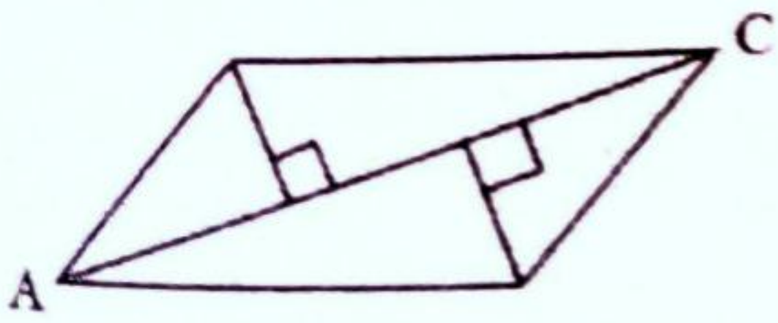
Answer: C

30. Find the area of a quadrilateral of whose diagonal is 24m long and the lengths of perpendicular from the other two vertices are 19m and 11m respectively.

A. $150m^2$
B. $360m^2$
C. $240m^3$
D. $180m^2$

Solution: Let the length of perpendicular be h_1 and h_2 respectively

$$\therefore \text{Area} = \frac{1}{2}(AC)h_1 + \frac{1}{2}(AC)h_2$$



$$= \frac{1}{2}(AC)(h_1 + h_2)$$

$$= \frac{1}{2}(24m)(19m + 11m) = 360m^2$$

Answer: B

8.3 SOLID GEOMETRY

Important fact

Here are the area formula you need to know:

Surface – Area and Volume

B = area of base, P = perimeter

C = Circumference, h = height

ℓ = slant height, r = radius

	Surface area	Volume
Prism	$S = 2B + ph$	$V = Bh$
Cylinder	$S = 2B + Ch$ $= 2\pi r^2 + 2\pi rh$	$V = \pi r^2 h$
Regular Pyramid	$S = B + \frac{1}{2}P\ell$	$V = \frac{1}{3}Bh$
Right cone	$S = B + \frac{1}{2}C\ell$ $= \pi r^2 + \pi r\ell$	$V = \frac{1}{3}\pi r^2 h$
Sphere	$S = 4\pi r^2$	$V = \frac{4}{3}\pi r^3$

31. The diagonals of a rhombus are 15m and 20m, find the side, area and the height of the rhombus

Solution:

To find the side we use the formula: $_2d_1^2 + d_2^2 = 4a^2$

$$\Rightarrow a^2 = \frac{d_1^2 + d_2^2}{4} = \frac{(15m)^2 + (20)^2}{4} = \frac{625}{4}$$

$$\therefore a = \sqrt{\frac{625}{4}} = \frac{25m}{2}$$

- Area $= \frac{1}{2}d_1d_2 = \frac{1}{2}(15)(20) = 150m^2$
- To find the height: $A = bh \Rightarrow 150 = \frac{25m}{2}xh$

$$\Rightarrow h = \frac{2\times150}{25} = 12m.$$

32. The diagonals of a rhombus are 10m and 24m respectively. Find the perimeter of the rhombus.

Solution: Each side: $a = \sqrt{\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2}$

$$\therefore a = \sqrt{5^2 + 12^2} = \sqrt{169} = 13m$$

Hence the perimeter: $P = 4a = 4(13) = 52m.$

Illustrative Example

33. What is the volume of cube whose surface area is 150?
A. 25 B. 100 C. 125 D. 500

Solution:

Since each of the six faces of the cube are equal whose area $A = \frac{150}{6} = 25$, then $S^2 = 25 \Rightarrow S = 5$

$\text{Volume} = S^3 = 5^3 = 125$

Answer: C

34. What is the surface area of a cube whose volume is 64?
A. 16 B. 64 C. 128 D. 96

Solution: Let a = side length

$V = a^3 \Rightarrow a^3 = 64 \Rightarrow a = 4$

$\text{Surface area} = 6a^2 = 6(4^2) = 96$

Answer: D

35. The height, h of a cylinder is equal to the edge of a cube. If the cylinder and the cube have the same volume. What is the radius of the cylinder?

A. $\frac{h}{\sqrt{\pi}}$ B. $h\sqrt{\pi}$ C. $\frac{\sqrt{\pi}}{h}$ D. $\frac{h^2}{\pi}$

Solution:

Since the volumes are equal, $\pi r^2 h = h^3$

$\Rightarrow \pi r^2 = h^2 \Rightarrow r^2 = \frac{h^2}{\pi} \Rightarrow r = \frac{h}{\sqrt{\pi}}$

Answer: A

36. If the height of a cylinder is 4 times its circumference, what is the volume of the cylinder in terms of its circumference C?

A. $\frac{C^3}{\pi}$ B. $\frac{2C^3}{\pi}$ C. $\frac{2C^2}{\pi^2}$ D. $\frac{\pi C^2}{4}$

Solution:

$V = \pi r^2 h$ and $h = 4C$

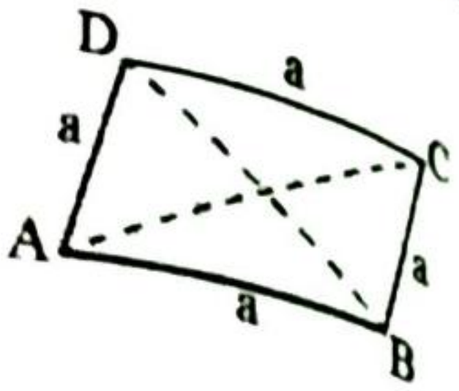
$C = 2\pi r \Rightarrow r = \frac{C}{2\pi}$

$\therefore V = \pi\left(\frac{C}{2\pi}\right)^2 (4C) = \frac{C^3}{\pi}$

Answer: A

8.4 RHOMBUS

- A rhombus is a quadrilateral whose all sides are equal.
- Diagonals AC = d_1 and BD = d_2 are bisect each other at 90^o, but AC ≠ BD



Important fact on Rhombus.

- Area: $A=\frac{1}{2}\times$ product of its diagonals $=\frac{1}{2}d_1d_2$

$$A=\frac{1}{2}\times d_1\times\sqrt{a^2-\left(\frac{d_1}{2}\right)^2}\text{, since } d_2{}^2=a^2-\left(\frac{d_1}{2}\right)^2.$$

$$A=\frac{1}{2}\times d_2\times\sqrt{a^2-\left(\frac{d_2}{2}\right)^2}\text{, since } d_1{}^2=a^2-\left(\frac{d_2}{2}\right)^2.$$

$$d_1^2+d_2^2=4a^2 \text{ and } (d_1+d_2)^2=4(a^2+A)$$

$$(d_1-d_2)^2=4(a^2-A)$$

Review Exercise

1. Which of the following sets of numbers **cannot** represent the sides of a triangle?
   A. 9, 40, 41 B. 7, 7, 3 C. 4, 5, 1 D. 6, 6, 6
2. If the lengths of two sides of a triangle are 12 and 20 and the third side is represented by x, then:
   A. $x = 32$ B. $x > 32$ C. $x < 8$ D. $8 < x < 32$
3. The diagonals of rhombus are 18 and 24. Find
   a) the area of the rhombus
   b) the length of a side of the rhombus
4. Find the length of a side of a square if a diagonal has a length of 8.
5. The circumference of the base of a cone is 8π cm. If the volume of the cone is $16\pi cm^3$. What is the height?
6. The sides of a triangle are 7cm, 8cm and 9cm. find its area.
7. The sides of a triangle are 13cm, 14cm and 15cm. find the perpendicular distance from the vertex to the longest side.
8. The diagonals of a rhombus are 8m and 6cm respectively. Find the side and the height

9. Find the area of a quadrilateral one of whose diagonals is 14m and the lengths of perpendiculars drawn from other two vertices are 8m and 10m respectively.

Answer and Explanation

1. C. $4 + 1$ is not greater than 5.

2. D. $20 - 12 < x < 20 + 2 \Rightarrow 8 < x < 32$

3. Area of rhombus $= \frac{1}{2}(d_1 d_2) = \frac{18 \times 24}{2} = 216$

4. Let x side length $\Rightarrow x^2 + x^2 = 8^2 \Rightarrow 2x^2 = 64$
$\Rightarrow x^2 = 32 \Rightarrow x = \sqrt{32} = 4\sqrt{2}$

5. Since $C = 8\pi cm \Rightarrow 2\pi 4 = 8\pi cm \Rightarrow r = 4cm$
$v = \frac{1}{3}\pi r^2 h \Leftrightarrow 16\pi = \frac{1}{3}\pi(4^2)h$
$\Rightarrow 48\pi = 16\pi h \Rightarrow h = 3$

6. Area $= \sqrt{(12)(12-7)(12-8)(12-9)} = 12\sqrt{5}cm^2$

7. Area $= \frac{1}{2}(base)(h) = \sqrt{21(21-13)(21-14)(21-15)}$
$\Rightarrow \frac{1}{2}(15)h = \sqrt{21 \times 8 \times 6} = 12\sqrt{17}$
$\Rightarrow h = \frac{24}{15}\sqrt{7} = \frac{8}{5}\sqrt{7}$

8. Each side $= a = \sqrt{\left(\frac{d_1}{2}\right)^2 + \left(\frac{d_2}{2}\right)^2} = \sqrt{9+16} = 5m$
Area $= \frac{1}{2}d_1 d_2 = \frac{1}{2}ah \Rightarrow (5h) = 48 \Rightarrow h = \frac{48}{5}$

9. $A = \frac{1}{2}d(h_1 + h_2) = \frac{1}{2}(14)(8+10) = 126m^2$

UNIT ONE

ANTONYMS

(Verbal Tactics)

- Besides meaning, words have another element called **relationship**. In the family of words, words can relate closely to one another or be distinctly opposite.
- A word that means the **opposite** of another word is called **antonym**. (i.e. An **antonym** is a word that has the **opposite** meaning). Nearly all words have a synonyms. Not all words have antonym. When a word doesn't have a suitable **antonym**, the space provided for the response has been removed. In most cases, you will find a single best answer available among the options. In a few cases there may be two synonyms that would be suitable for one word.

Some of words in the list of **synonyms** and **antonyms** may be difficult or unfamiliar. For these words, consult your dictionary or **thesaurus.**

Antonym Questions

- The **antonym** questions are always the **first group of questions** in a verbal section.
- **Antonym** questions provide a single word and ask you to select from a list of words the one that is **most opposite** in meaning.
  - To answer an **antonym question,** use the following strategies.

1. Look for a word that is **opposite** in meaning. Do not be thrown off by any synonyms - words that are **similar** in meaning-that are included among the choices.
2. Before looking at the choices, be sure that you know the meaning of the first word. Define it using any of the following methods:
   a. Think of another word or group of words that means the same thing.

b. Think of a sentence or sentences that use the word then try to arrive at an exact definition.
c. Try analyzing the parts of the word.

Once you know the meaning of the first word, look at the choices. If none of these is obviously the correct antonym, try either or both of the following strategies.

a. Eliminate any obviously incorrect answers.
b. Remember that many words have more than one meaning. If none of the choices seems to be opposite in meaning, think of other meanings of the first word.

♦ **A typical antonym question looks like this:**

DEPRESS: A. force C. clarify E. loosen
B. allow D. elate

✧ Based on the tactics given above, the answer to the sample question is **D.**

▲ Bekan knew the **joy** of a hard workout would come later from the sheer **pleasure** of victory.

▲ Bekan knew the **pain** of a hard workout would be rewarded by the sheer **pleasure** that comes from victory.

✧ The first sentence displays **"pleasure"** as a synonym for **"joy,"** permitting the author to get an idea across without being dull and repetitions. In the second sentence, the use of **"pain"** as an antonym helps the author to state an entirely different thought by changing the sentence only slightly.

✧ Thus, **synonyms** and **antonyms** become important elements in the word mastery process.

UEE /EHEECE and Other Questions with Detailed Explanations Examples and Answer. (2001 – 2009)

I. Choose the word or phrase that is most nearly <u>opposite</u> in meaning to the word in capital letters.

1. **REBEL**

A. appreciate
B. comply
C. simplify
D. strive

<u>Hints</u>: **Rebel** is someone who fights against the government of their country.

Comply: to obey a rule, or order, etc. of government (they are **opposite**).

Answer: B

2. **IMMEDIATE**

A. complex
B. multiple
C. limited
D. future

<u>Hints</u>: **immediate (N)** happening or done at once.

future (N) not now or not present.

(The time comes after present).

(They are <u>opposite</u> words)

Answer: D

3. **REASONABLE**

A. rapid
B. authentic
C. indirect
D. absurd

<u>Hints</u>: **reasonable (adj):** fair and sensible

absurd: something which doesn't give sense, or which is ridiculous. (Not reasonable).

Answer: D

4. **IMPLAUSIBLE**

A. credible
B. crucial
C. mercenary
D. penurious

<u>Hints</u>: **implausible (adj):** not readily believable

credible (adj): believable (which can be believed).

Answer: A

DISPARAGE

A. fascinate B. praise C. retire D. abuse

Hints: **disparage (v):** to speak critically of. (belittle)
It is the opposite of **praise**.

praise (v): to express approval of or admiration .
to express thanks to or respect. **Answer: B**

7. LAXITY

A. vivacity C. strictness
B. coyness D. probability

Hints: **laxity (N):** not strict /sever/ careful
strictness(N): being rigid. (follow the rules rigidly)

Answer: C

EXHAUSTIVE

A. incomplete C. indecisive
B. strong D. conserving

Hints: **exhaustive (adj):** very through; covering all aspects fully
its opposite is incomplete.

Answer: A

8. HETEROGENEOUS

A. mixed C. homogeneous
B. monastic D. tangential

Hints: **heterogeneous (adj):** composed of different kinds, varied in composition.

homogeneous (adj): formed of parts that are all the same type.

Answer: C

9. ERRONEOUS

A. careful C. accurate
B. thoughtful D. vigorous

Hints: **erroneous (adj):** not correct. or wrong
accurate (adj): correct and exact

Answer: C

10. **DYNAMIC**

A. automatic
B. listless
C. sympathetic
D. unfortunate

Hints: **dynamic (adj):** energetic (full of energy); active.
listless (adj): lacking energy or enthusiasm.

Answer: B

11. **RESTRAIN**

A. unleash
B. criticize
C. ignore
D. combine

Hints: **restrain (v):** keep under control; hold back.
unleash (v): release, or set loose.

Answer: A

12. **PERSIST**

A. approach
B. quit
C. flourish
D. change slowly

Hints: **persist (v):** means to continue /to keep on
quit (v): is to mean cease /stop.

Answer: B

13. **INFINITESIMAL**

A. gargantuan
B. atomic
C. terminal
D. perpetual

Hints: **infinitesimal (adj):** is to mean extremely small; very minute.
gargantuan (adj): is to mean very huge or gigantic

Answer: A

14. **EBULLIENT**

A. soluble
B. superficial
C. glittering
D. apathetic

Hints: **ebullient (adj):** enthusiastic
apathetic (adj): uncaring

Answer: D

15. **AMIABILITY**
A. lack of wealth
B. irritability
C. inconvenience
D. strength of mind

Hints: **amiability (N):** a good humor
irritability (N): anger, annoyance

Answer: B

16. **PERMANENT**
A. spontaneous
B. subterranean
C. transient
D. desultory

Hints: **permanent (adj):** meant lasting for a long time or forever without change.
transient (adj): lasting only for a short time. **Answer: C**

17. **GENERAL**
A. fallacious
B. chronological
C. disinterested
D. topical

Hints: **general (adj):** not limited to one part.
topical (adj): specific subject of discussion at a time.

Answer: D

18. **GROUP**
A. abandon B. isolate C. restrain D. reveal

Clue: **group (N):** association, cluster, aggregate
isolate (N): alone /solitude /separate **Answer: B**

19. **EXTINGUISH**
A. distinguish
B. ignite
C. switch off
D. turn on

Hints: **extinguish (v):** to cause or to stop burning /bring to an end.
distinguish (v): to cause /to start burning. **Answer: A**

20. **ACCUMULATE**
A. weaken
B. permit
C. permit
D. scatter

Clues: **accumulate (v):** to collect or gather /assemble.
scatter (v): to throw in various random directions.
to separate and move off different directions.

Answer: D

21. **PROXIMITY**

A. diversity
B. complexity
C. substitution
D. distance

Hints: **proximity (N):** nearness
distance (N): farness

Answer: D

22. **ROADBLOCK**

A. disorder
B. separation
C. aid to progress
D. quick reaction

Hints: **roadblock (N):** obstacle to progress.
Its **opposite** word is **aid to progress.**

Answer: C

23. **BLEAK**

A. cheery
B. uniform
C. talkative
D. insignificant

Clues: **bleak (adj):** dismal; gloomy.
cheery (adj): happy and optimistic.

Answer: A

24. **FOMENT**

A. exacerbate
cmulate
B. suppress
C. belittle
D.

Hints: **foment (v):** stir up conflict; incite; arouse.
suppress (v): forcibly put and end to.

Answer: B

25. **FLAGRANT**

A. unappealing
B. indefinite
C. concise
D. inconspicuous

Clues: **flagrant (adj):** shockingly glaring; noticeable
inconspicuous (adj): not clearly visible (invisible).

Answer: D

26. **CONFRONT**

A. squander
B. take seriously
C. avoid
D. misplace

Hints: **confront (v):** to face

Answer: C

27. **FLAMBOYANT**

A. restrained
B. contemporary
C. unattractive
D. authentic

Hints: **flamboyant (adj):** showy; lacking restraint.
restrained (adj): kept undercontrol.

Answer: A

28. **DESTITUTE**

A. conclusive
B. subtle
C. affluent
D. meddlesome

Hints: **destitute (adj):** lacking something: **devoid**; especially lacking money; **poor.**

- **affluent (adj)** wealthy

Answer: C

29. **BARREN**

A. skillful
B. landscape
C. arid
D. fertile

Hints: **barren (adj):** unproductive, sterile, desolate, arid, bare.
fertile (adj): productive, fruitful

Answer: D

30. **DISRUPT**

A. organize
B. disorder
C. displace
D. disturb

Hints: **disrupt (v):** upset, disturb
organize (v): arrange

Answer: A

31. **ADJACENENT**

A. alongside
B. nearby
C. adjoining
D. faraway

Clues: **adjacent (adj):** near, next to, neighboring
☞ **antonym:** far away, distant, remote

Answer: D

32. **ALIGHT**

A. disappear
B. descend
C. ascend
D. touchdown

Clues: **alight** (vb: to get down from; to come down from; to land, descend, touch down, disappear.

☞ **antonym:** mount, **ascend**, board, take off

Answer: C

33. **GERMINATE**

A. sprout
B. shoot up
C. grow
D. stagnate

Clues: **germinate** (vb: to begin to grow, come into being, shoot up, sprout, burgeon.

☞ **antonym:** wither, die, shrivel up, **stagnate.**

Answer: D

34. **HUMDRUM**

A. monotonous
B. boring
C. prosaic
D. exciting

Clues: **humdrum (adj):** lacking variety or excitement; dull, boring.

Answer: D

35. **VICIOUS**

A. savage
B. spiteful
C. good
D. malicious

Clues: **vicious (adj):** wicked, bad, spiteful, having an ugly disposition, savage, malicious.

☞ **opposite:** good, kind, kindly, mild, harmless.

Answer: C

36. **SULLEN**

A. grumpy B. surly C. peevish D. sociable

Clues: **sullen (adj):** silent or brooding because of ill humor, anger, or resentment, slow moving, sluggish (grumpy, surly, peevish, morose).

☞ **antonyms:** cheerful, blithe, **sociable**, vivacious.

Answer: D

HURTLE

A. speed B. fly C. race D. crawl

Hints: hurtle (v): speed, fly, race, catapult, fling
☞ **Opposite: crawl**, creep

Answer: D

INTERMINABLE

A. never-ending
B. ceaseless
C. short
D. non-stop

Hints: **interminable (adj):** never-ending, non-stop, ceaseless
☞ **antonyms:** brief, **short**, fleeting

Answer: C

TRICKELE

A. dribble B. drizzle C. drip D. pour

Hints: **trickle (v):** dribble, drizzle, drip
☞ **antonyms:** gush, **pour**, flood; (n) deluge **Answer: D**

TRIVIAL

A. important
B. insignificant
C. petty
D. trifling

Hints: **trivial (adj):** insignificant, petty, trifling
☞ **antonyms: important**, weighty, momentous. **Answer: A**

TRUCE

A. warfare
B. cease-fire
C. armistice
D. a pause in fighting

Clues: **Truce (N):** temporary peace /a pause in fighting (cease-fire, armistice).
☞ **antonyms:** war, **warfare**, fighting **Answer: A**

AVALABLE

A. obtainable
B. at hand
C. not to be used
D. ready for use

Clues: **available (adj):** at hand, obtainable, ready for use.
☞ **opposite:** unobtainable, **not to be used**, not at hand.

Answer: C

43. **CATER**

A. provide B. frustrate C. pamper D. gratify

Clues: **cater(v):** to satisfy the need of, try to make things easy and pleasant, to supply food and service (provide, **pamper**, indulge, gratify).

☞ **antonyms: frustrate**, deny, refuse

Answer: B

44. **CUSTOMARY**

A. regular
B. strange
C. normal
D. traditional

Clues: **customary (adj):** usual, expected, routine, normal, traditional, regular.

☞ **opposite: strange**, odd, usual, untraditional

Answer: B

45. **DISSUADE**

A. talk into C. discourage E. A and D
B. discuss D. persuade

Clues: **dissuade (v):** to persuade not to do something (discourage, talk out of).

☞ **antonyms: persuade, talk into**

Answer: E

46. **FIREBRAND**

A. peacemaker
B. agitator
C. rabble-rouser
D. hothead

Clues: **firebrand (N):** a piece of burning wood, a trouble maker, an extremely energetic or emotional person (hothead, agitator, rabble-rouser).

☞ **antonyms: peacemaker**, pacifier, conciliator.

Answer: A

47. **HAZARD**

A. danger B. venture C. safe D. hardship

Clues: **hazard (n):** risk, peril
hazard (v): to expose to danger or harm; to gamble

☞ **opposite: safe**, comfort

Answer: C

48. **INDIFFERENCE**

A. apathy
B. unconcern
C. uninterest
D. enthusiasm

Clues: **indifference (N):** a lack of interest or concern (apathy, unconcern)

antonyms: interest, concern, **enthusiasm**. **Answer: D**

49. **INDIGNANT**

A. offended B. pleased C. resentful D. outraged

Clues: **indignant (adj):** filled with resentment or anger over something unjust, unworthy, or mean.

opposite: **pleased**, delighted, overjoyed, elated.

Answer: B

50. **INDISPENSABLE**

A. essential B. crucial C. vital D. unnecessary

Clues: **indispensable (adj):** absolutely necessary, not to be neglected (choice A, B, C are synonyms.)

Answer: D

51. **MUTUAL**

A. one-sided B. shared C. joint D. reciprocal

Clues: **mutual (adj):** shared, felt, or shown equally by two or more. (two-sided, joint, shared, reciprocal).

antonyms: one-sided, unilateral **Answer: A**

52. **PLAGUE**

A. epidemic B. pestilence C. blessing D. pester

Clues: **plague (n):** an easily spread disease causing a large number of deaths, a wide spread evil.

plague (v): to annoy or bother, pester, vex.

Answer: C

opposite: boon, **blessing**

53. **POISED**

A. nervous
B. self-confidence
C. calm
D. control

Hints: **poised (adj):** balanced, suspended, calm, controlled, ready for action, self-confident.

☞ **antonyms:** nervous, tense

Answer: A

54. **RETARD**

A. slow B. delay C. holdback D. hasten

Clues: **retard (v):** to make slow, delay, hold back, slow down, restrain, impede.

☞ **antonyms: hasten**, speed up

Answer: D

55. **TRANSPARENT**

A. translucent B. obvious C. smoky D. clear

Hints: Choice A B D are **synonyms** for **transparent**.

The **antonyms are:** frosted, story, **smoky**, indistinct, unclear.

Answer: C

56. **UNSCATHED**

A. unhurt B. unimpaired C. injured D. intact

Clues: **unscathed (adj):** wholly unharmed, not injured (unhurt, sound, unimpaired).

☞ **antonyms: injured**, damaged, harmed, hurt

Answer: C

57. **IMPROMPTU (2007 E.C).**

A. natural
B. unrehearsed
C. planned
D. spontaneous

Clues: **impromptu (adj):** done without being planned or rehearsed.

☞ **antonyms: planned**, intended (done intentionally)

Answer: C

58. **ALIEN (2007 E.C)**

A. inimical B. affluent C. parochial D. indigenous

Clues: **alien (adj):** foreigner /not native, unfamiliar
Opposite: indigenous, native, familiar. **Answer: D**

59. **ALLY**
A. enemy B. fugitive C. prisoner D. mediator
Clues: **ally (adj):** is one who **helps**.
enemy (adj) is one who seeks to **harm**. **Answer: A**

60. **PROTECT**
A. challenge B. damage C. reward D. respect
Clues: To **protect** is to shied from injury or distraction.
To **damage** is to destroy. **Answer: B**

61. **BRITTLE**
A. genuine B. aggressive C. curious D. pliant
Clues: **Brittle (adj):** means easily shattered.
Pliant (adj): means flexible. **Answer: D**

62. **APPREHEND**
A. release B. defend C. enforce D. acceleration
Clues: To **apprehend (v)** is to arrest.
To **release (v)** is to set free. **Answer: A**

63. **BLUNDER**
A. pack tightly
B. await anxlously
C. work diligently
D. act intelligently
Clues: To **blunder (v)** is to make an error.
The opposite of to act intelligently. **Answer: D**

64. **SOOTH**
A. annoy B. isolate C. defend D. refuse
Clues: To **sooth (v)** is to relieve or comfort.
To **annoy (v)** is to irritate or disturb. **Answer: A**

65. **CONTAMINATE**
A. evaluate B. persuade C. purify D. energize

Clues: **contaminate (v)** means pollute.
purify (v) means cleanse.

Answer: C

66. **LENIENCY**
A. defense B. severity C. strength D. justification
Clues: **leniency (N)** is tolerance or permissiveness.
severity (N) is strictness or harshness.

Answer: B

67. **DEBASE**
A. enhance C. conclude
B. work alone D. expose to danger
Clues: To **debase (v)** is to lower in quality or value.
To **enhance (v)** means to raise in quality or value.

Answer: A

68. **CALAMITY**
A. outcome C. occurrence
B. evasion D. good fortune
Clues: A **calamity (N)** is a disaster; its opposite is **good fortune.**

Answer: D

69. **CLAMOR**
A. innocent C. tranquility
B. ambition D. meekness
Clues: **clamor (N)** means noise.
tranquility (N) means free of disturbance.

Answer: C

70. **COMPLIANCE**
A. disobedience C. honesty
B. commitment D. detachment
Clues: **compliance (N)** means yielding to the demands.
disobedience (N) is to mean the refusal to obey.

Answer: A

71. **AGITATION**
A. criticism B. calmness C. devotion D. authority

Clues: **agitation (N)** means disturbance.
calmness (N) means lack of disturbance. **Answer: B**

72. **CHOAS**
A. courage B. opinion C. praise D. order
Clues: **chaos (N)** is utter disorder. Its opposite is **order**.

Answer: D

73. **COMPLEX**
A. simple B. boring C. trivial D. unique
Clues: **complex (adj)** means complicated.
simple (adj) means plain, uncomplicated. **Answer: A**

74. **DRAWBACK**
A. evidence B. essential C. benefit D. contraction
Clues: **drawback (N)** is an objectivable future.
benefit (N) is a useful future. **Answer: C**

75. **GRIM**
A. skillful B. cheerful C. modern D. discreet
Clues: **grim (adj)** means stern, harsh, the opposite of **cheerful**.
Answer: B

76. **HUMANE**
A. vulnerable B. spontaneous C. attentive D. cruel
Hints: **humane (adj)** means kind, its antonym is **cruel**.
Answer: D

77. **IMPROVE**
A. impair B. deny C. borrow D. comply
Clues: **improve (v)** is to make better.
impair (v) is to make worse. **Answer: A**

78. **CLARITY**
A. certainty B. ambiguity C. tenderness D. loudness
Clues: **clarity (N)** means non-ambiguity, certainty.
ambiguity (N) means obscurity. **Answer: B**

79. **HUMDRUM**

A. interesting B. disappointing C. futile D. mournful

Clues: **humdrum (adj)** means monotonous or boring, or not interesting.

Answer: A

80. **INDISPUTABLE**

A. injust
B. indelicate
C. questionable
D. indescribable

Clues: something **indisputable** can't be denied. It is not questionable.

Answer: C

81. **ADORATION**

A. strength
B. abhorrence
C. absolution
D. vilification

Clues: **adoration (N)** is to mean love, and **abhorrence (N)** is to mean hatred.

Answer: B

82. **CONVENTIONAL**

A. extraordinary
B. obvious
C. kindly and warm
D. useful and pleasing

Clues: Something **conventional** is typical. It has become common place or ordinary. Its opposite is extraordinary.

abhorrence (N) is to mean hatred.

Answer: A

83. **SOPHISTICATED**

A. clever B. integrated C. ingenuous D. partial

Clues: **sophisticated (worldly)** is the opposite of **ingenuous (naive).**

Answer: C

84. **TRAVAIL**

A. relaxation
B. ineffectiveness
C. suffrage
D. sedation

Hints: **travail (adj)** is painful, arduous labor, the **antonym of relaxation (N).**

Answer: A

85. **AMITY**

A. exhaustion
B. verification
C. orientation
D. hostility

Clues: **amity (N)** friendliness, the **opposite** of **hostility**.

Answer: D

86. **CONCISE**

A. sudden
B. lengthy
C. inaccurate
D. extraneous

Hints: **concise (N)** is to mean expressing much in few words is the **autonym** of lengthy.

Answer: B

87. **SAVORY**

A. unappetizing
B. wise
C. warm
D. uncivilized

Clues: **savory (adj)** means **tasty** is the opposite of **unappetizing**.

Answer: A

88. **PROPITIOUS**

A. ambitious
B. antagonistic
C. unfavorable
D. innocuous

Clues: **propitious (adj)** means favorable is the opposite of **unfavorable**.

Answer: C

89. **ABROGATE**

A. institute B. punish C. launch D. discharge

Clues: **abrogate (v)** is to abolish, the opposites **institute (v)** which means to begin.

Answer: A

90. **VINDICATIVE**

A. figurative B. passive C. forgiving D. courageous

Clues: **vandicative (adj)** means revengeful is the opposite of **forgiving.**

Answer: C

91. **RELENTLESS**

A. unwise B. fearless C. yielding D. bizzarre

Clues: **relentless (adj)** means **unyielding**. **Antonym = yielding.**

Answer: C

92. **ASSIMILATE**

A. penetrate B. imitate C. pretend D. reject

Clues: **assimilate (v)** means to take in as one's own. Its antonym is **reject**.

Answer: D

93. **EVANESCENT**

A. permanent B. sparkling C. frantic D. segregated

Clues: **evanescent (adj)** means vanishing, not lasting long.

Answer: A

94. **PUNY**

A. strong B. humorous C. studious D. incorrect

Clues: **puny (adj)** means tiny and weak is **the opposite** of strong (powerful).

Answer: A

95. **SPURIOUS**

A. authentic
B. predictable
C. comprehensive
D. incandescent

Clues: **spurious (adj)** means false, not genuine.

Answer: A

96. **ERUDITE**

A. immalerial
B. illegible
C. illiterate
D. ill-mannered

Clues: **erudite (adj)** means learned and **illiterate** are antonyms.

Answer: C

97. **CANDID**

A. critical B. insincere C. hasty D. sufficient

Clues: **candid (adj)** is to mean frank, sincere.

Answer: B

98. **INANIMATE**

A. emotional B. alive C. cruel D. ridiculous

Clues: **inanimate (adj)** to mean **not alive; lifeless**

Answer: B

99. **EXOTIC**

A. involved B. ambiguous C. emotional D. ordinary

Clues: **exotic (strikingly unusual)** and **ordinary** are **opposites.**

Answer: D

100. **INCITE**

A. exclude B. vandicate C. subdue D. destroy

Clues: **incite (v)** means to stirrup, provoke. :
subdue (v) means to tone down.

Answer: C

101. **ABSTRACT**

A. apathetic B. cooperative C. diligent D. concrete

Clues: **abstract (adj)** means nonspecific and detached.
concrete (adj) means specific and tangible.

Answer: D

102. **LETHARGY**

A. alertness
B. foolishness
C. caution
D. persistence

Hints: **lethargy (N)** means sluggishness, drowsiness and **alertness (N)** are antonyms.

Answer: A

103. **VILIFY**

A. analyze B. eulogize C. violate D. magni

Clues: To **vilify (v)** means to make abusive and slanderous statements; to **eulogize** is to praise.

Answer: A

104. **CACOPHONY**

A. evasion B. futility C. harmony D. selfishness

Clues: **cacophony (adj)** means discord and **harmony** are antonyms.

Answer: C

105. **PRODIGAL**

A. loyal B. frugal C. ignorant D. deceitful

Clues: **prodigal (adj)** means extravagant.
frugal (adj) means economical.

Answer: B

106. **NOXIOUS**

A. healthful
B. lively
C. uncertain
D. unprepared

Clues: **noxious (adj)** means harmful to health and **healthful** are opposites.

Answer: A

107. **ERRATIC**

A. harmless
B. constant
C. indifferent
D. unprepared

Clues: **erratic** (adj) means inconsistent, irregular; its opposite is **constant** unchanging.

Answer: B

108. **IMPRUDEN**

A. powerful careful
B. wealthy
C. excitable
D.

Clues: **imprudent (adj)** means not prudent or careful it's the **opposite** of **careful**.

Answer: D

109. **AFFLUENCE**

A. impoverishment
B. beauty
C. evil
D. neutrality

Hints: **affluence (N)** means wealth; **impoverishment (N)** means poor.

Answer: A

110. **BEWAIL**

A. shine brightly
B. feel joyous
C. make effective
D. abandon hope

Clues: to **bewail (v)** is to express sorrow, not joy. **Answer: B**

111. **RIGOR**

A. padding
B. mold
C. excavation
D. lazity

Clues: **rigor (N)** means strictness, severity.
lazity (N) means lacking in strictness. **Answer: D**

112. **ADULTERATE**

A. make pure
B. send away
C. grow up
D. turn loose

Clues: **adulterate (v)** means making impure by adding inferior things.

Answer: A

113. **MAMMOTH**

A. minute
B. modern
C. constructive
D. dramatic

Clues: **mammoth (adj)** means large, huge.
minute (adj) means extremely small-tiny. **Answer: A**

Antonyms

114. **ENHANCE**
A. fascinate B. sympathize C. weaken D. resist
Clues: **enhance (v)** is to improve in quality, strengthen.
weaken (v) is to make something weak. **Answer: C**

115. **ENIGMATIC**
A. obvious C. blasphemous
B. extraordinary D. extraneous
Clues: That which is **enigmatic** (puzzling) cannot be obvious.
Answer: A

116. **LUCRATIVE**
A. indolent B. refined C. unprofitable D. somber
Hints: **lucrative (adj)** means profitable. **Answer: C**

117. **BANKRUPT**
A. embezzled B. solvent C. bereft D. quantitative
Clues: A person who is **bankrupt (adj)** is not solvent.
Answer: B

118. **ADVISABLE**
A. formidable C. imprudent
B. appropriate D. inconsiderate
Clues: **imprudent (adj)** means not expedient or advisable.
Answer: C

119. **INTANGIBLE**
A. incomplete B. careless C. uninjured D. concrete
Clues: **intangible (adj)** means imperceptible to the sense of touch.
concrete (adj) means definite, perceptible. **Answer: D**

120. **ABUSIVE**
A. intricate B. cordial C. flexible D. pliant
Clues: **abusive (adj):** characterized by verbal abuse.
cordial (adj): warm and pleasant. **Answer: B**

121. **STAMINA**
A. weakness B. darkness C. hast D. reliability
Clues: **stamina (N)** means vigor, power to endure suffering.
Answer: A

122. **AMINALBE**
A. religious B. feminine C. proud D. intractable
Clues: an **aminable** person is yielding and submissive.
an **intractable** person refuses to be managed or led.
Answer: D

123. **TENTATIVE**
A. permanent B. absent C. forgotten D. concerned
Clues: something that is **tentative** is done as a trial or experiment; its opposite is permanent.
Answer: A

124. **FORTUTIONS**
A. disastrous C. beneficial
B. premeditated D. sagacious
Clues: **fortuitous (adj)** is to mean happening by chance.
premeditated (adj) is to mean planned in advance.
Answer: B

125. **DESULTORY**
A. bland C. methodological
B. agreeable D. honorable
Clues: **desultory (adj)** means lacking in method or organization.
Answer: C

126. **GALVANIZE**
A. deaden B. coalesce C. disapprove D. restrict
Clues: **galvanize (v)** is to stirrup or stimulate by shock, as opposed to **deaden** or lose life.
Answer: A

127. **STEADFAST**
A. objective B. scrupulous C. equitable D. capricious
Clues: **steadfast (adj)** means firm, unwavering.
capricious (adj) means changing.
Answer: D

ADHERENT
A. anarchist
B. conspirator
C. dissenter
D. predecessor

Clues: an **adherent (adj)** is a <u>follower</u> or <u>believer</u>, the **opposite** of **dissenter**.

Answer: C

29. **INNOCUOUS**
A. harmful
B. noticeable
C. weak
D. prejudiced

Hints: **innocuous (adj)** means **harmless.**

Answer: A

30. **ACCLAMATION**
A. negotiation
B. condemnation
C. imitation
D. nepotism

Clues: **acclamation (N)** is demonstration of <u>approval</u> and **condemnation** is expression of <u>disapproval</u>.

Answer: B

131. **VIVACIOUS**
A. obstinate B. decorous C. articulate D. inert

Hints: **vivacious (adj)** <u>animated</u>, <u>spirited</u>
inert (adj) sluggish

Answer: D

132. **SEDATE**
A. active B. erect C. buried D. timid

Clues: **sedate (adj): undisturbed; calm**

Answer: A

133. **ANTIQUATED**
A. limited
B. prophetic
C. inevitable
D. contemporary

Clues: **antiquated (adj)** means <u>old</u>.
contemporary (adj) means <u>modern</u>.

Answer: D

134. **DEBILITATE**
A. persuade B. invigorate C. inebriate D. fascinate

Clues: **debilitate (v)** is to mean to weaken, enfeeble is the **opposite** of **invigorate**.

Answer: B

135. **ORNATE**

A. original B. obsolete C. systematic D. plain and plain

Clues: ornate (adj) elaborately decorated

antonyms

Answer: D

136. **SUBSERVIENT**

A. prolific C. supercilious

B. magnificent D. overcoming

Clues: subservient (adj): servilely submissive

supercilious (adj): haughty

Answer: C

137. **ANIMATED**

A. energetic B. vigorous C. dull D. lively

Clues: animated (adj) which means full of life, lively, energetic, vigorous.

☞ **antonyms: dull**, lifeless, dead, flat

Answer: C

138. **CALMINATE**

A. to end B. climax C. conclude D. commence

Clues: culminate (v) means to reach a high point of development; to end, climax, conclude, terminate.

☞ **opposite:** begin, initiate, kick off, **commence**.

Answer: D

139. **DRONE**

A. bum B. do-nothing C. hum D. hard worker

Clues: drone (N) a loafer, idler, do nothing.

☞ **antonym:** hard worker work a holic.

Answer: D

140. **GOAD**

A. prod B. spur on C. restrain D. incite

Clues: goad (v) to drive or urge on; (n) something used to drive or urge on (prod, spurn on, incite).

☞ **antonym:** curb, check, **restrain**.

Answer: C

141. **INDULGE**

A. deny B. oblige C. humor D. coddle

Clues: **indulge (v)** to give into a wish or desire, give oneself up to. (oblige, humor, coddle, pamper).

Antonym: deny, refuse.

Answer: A

142. **LITERATE**

A. educated B. ignorant C. trained D. intellectual

Clues: **literate (adj)** able to read and write, having knowledge or training.

☞ **opposite: unlettered, unschooled, ignorant**

Answer: B

143. **LUSTER**

A. gloss B. sheen C. dullness D. shine

Clues: **luster (N)** the quality of giving off light, brightness, glitter, brilliance (gloss, sheen, shine).

☞ **antonyms:** furnish, **dullness**

Answer: C

144. **MISCELLANEOUS**

A. mixed B. assorted C. uniform D. varied

Clues: **miscellaneous (adj)** mixed, of different kinds (varied, assorted, mostly).

☞ **antonyms:** identical, **uniform**, homogeneous.

Answer: C

145. **PREEVISH**

A. stubborn B. agreeable C. crabby D. cranky

Clues: **preevish (adj)** cross, complaining, irritable, contrary, crabby, cranky, testy, stubborn.

☞ **antonyms: agreeable**, amiable, even-tempered pleasant.

Answer: A

46. **SINGE**

A. incinerate B. scorch C. char D. sear

Clues: **singe (v)** to burn slightly

singe (N) a burn at the ends or edges.

☞ **opposite: incinerate**

Answer: A

147. **UNIQUE**

A. unequaled B. unusual C. ordinary D. distinctive

Clues: **unique (adj)** one of a kind, unequaled, usual.

☞ **Antonyms: ordinary**, common place, run-of-the-mill

Answer: A

148. **UPRIGHT**

A. vertical

B. horizontal

C. straight

D. perpendicular

Clues: **upright (adj)** vertical, straight, perpendicular, or virtuous, honest, good.

☞ **antonyms: horizontal**, prone, or dishonest, corrupt.

Answer: B

149. **VERIFY**

A. prove

B. validate

C. substantiate

D. discredit

Clues: **verify (v)** to establish the truth or accuracy of, confirm (choice A B C are its **synonyms**).

☞ **antonyms:** disprove, refute, **discredit**. **Answer: D**

150. **ALLIANCE**

A. pact B. leage C. coalition D. split

Clues: **alliance (N):** a joining together for some common purpose (choice A B C are its **synonyms**).

☞ **antonyms:** rift, **split** **Answer: D**

151. **BEWILDER**

A. set straight B. baffle C. perplex D. confuse

Clues: **bewilder (v)** to puzzle completely, confuse.

☞ **antonyms: set straight, enlighten.** **Answer: A**

152. **DISHEARTEN**

A. dismay

B. demoralize

C. to discourage

D. encourage

Clues: **dishearten (v)** to discourage, dismay, demoralize, dispirit

☞ **antonyms: encourage, hearten** **Answer: D**

FRUITLESS

A. useless B. productive C. vain D. futile

Hints: **fruitless (adj):** means not producing the desired results, unsuccessful. (useless, vain, unproductive, futile).

antonyms: productive, effective **Answer: B**

HOSTILE

A. friendly B. unfavorable C. aggressive D. warlike

Clues: **hostile (adj):** unfriendly, unfavorable, warlike, aggressive

antonyms: **friendly**, **cordial, peaceful** **Answer: A**

INFLAMMABLE

A. combustible C. fire-resistant

B. excitable D. warm

Clues: **inflammable (adj):** easily set on fire; easily angered or aroused (combustible).

antonyms: fireproof, **fire-resistant**, calm. **Answer: C**

INFLICT

A. deal out C. suffer

B. visit upon D. feel happy

Clues: **inflict (v)** means to give or cause something unpleasant, impose (**deal out, visit upon)**

opposite: suffer, undergo, sustain **Answer: C**

MALIGNANT

A. wicked B. beneficial C. lethal D. harmful

Clues: **malignant (adj)** means deadly, extremely harmful, spiteful, malicious.

opposite: wholesome, **beneficial**, begin **Answer: B**

ORTHODOX

A. traditional C. customary

B. standard D. unconventional

Clues: **orthodox (adj)** in agreement with established or generally accepted beliefs or ways of doing things.

opposite: unusual, **unconventional**, heretical **Answer: D**

160. **SCURRY**

A. rush B. crawl C. dash D. scramble

Clues: **scurry (v)** to run quickly, scamper, hurry
☞ **antonyms:** trudge, plod, creep, **crawl**

Answer:

160. **SODDEN**

A. parched C. water logged
B. drenched D. saturated

Clues: **sodden (adj)** means soaked with liquid or moisture; expressionless, dull; spiritless, listless.
☞ **antonyms: parched**, arid

Answer: A

161. **SPIRITED:**

A. lively B. animated C. lifeless D. gallant

Clues: **spirited (adj)** is to mean full of life and vigor courageous. (lively, animated, gallant)
☞ **opposite: lifeless**, dull, lackluster

Answer: (

162. **VOID:**

A. invalid B. vacant C. bare D. teeming with

Hints: **void (adj)** completely empty, having no legal force effect.
☞ **antonyms:** in effect, **teeming with** (v) confirm.

Answer: I

163. **WAYWARD**

A. perverse C. willful
B. disobedient D. well-behaved

Clues: **wayward (adj):** disobedient, willful; unpredictable capricious, perverse.
☞ **antonyms:** docile, **well-behaved**, predictable.

Answer: I

166. CONSOLIDATE
A. strength B. firm up C. merge D. scatter
Clues: **consolidate (v)** to combine, unite, to make solid to firm, merge, firm up.
antonyms: scatter, disperse, dissipate, separate.
Answer: D

165. COUNTERFEIT
A. genuine B. false C. phony D. bogus
Clues: **counterfeit (n)**: an imitation designed to deceive.
counterfeit (adj): not genuine, fake.
counterfeit (v): to make an illegal copy.
antonyms: genuine, real, authentic. **Answer: A**

164. DOCILE
A. teachable B. manageable C. pliant D. unruly
Clues: **docile (adj)**: easily taught, led, or managed; obedient.
antonyms: unruly, wayward, intractable, disobedient.
Answer: D

167. ENTREAT
A. clamor for C. beseech
B. plead D. implore
Clues: **entreat (v)**: to beg, implore, ask earnestly
opposite: clamor for **Answer: A**

168. FALLIBLE
A. flawless C. errant
B. imperfect D. mistaken
Clues: **fallible (adj)**: capable of being wrong, mistaken, or inaccurate (imperfect, errant).
antonyms: foolproof, unfailing, **flawless** **Answer: A**

169. **FUGITIVE**

A. inconstant C. capricious
B. faithless D. steady

Clues: **fickle (adj)** means liable to change very rapidly, erratic; marked by a lack of constancy or steadiness, inconsistent.
antonyms: constant, **steady**, invariable **Answer: D**

170. **FUGITIVE**

A. enduring B. fleeting C. runaway D. deserter

Clues: **fugitive (N):** One who flees or runs away.
fugitive (adj): fleeting, lasting a very short time; wandering; difficult to grasp.
antonyms: lasting, **enduring**, permanent. **Answer: A**

171. **GRIMY**

A. filthy B. sooty C. spotless D. soiled

Hints: **grimy (adj):** very dirty, covered with dirt or soot.
antonyms: spotless, spick-and span, immaculate.

172. **IOTA** **Answer: C**

A. speck B. dab C. jot D. deluge

Clues: **iota (N):** a very small part or quantity (bit, smidgen)
antonyms: flood, **deluge**, avalanche, glut. **Answer: D**

173. **RADIANT**

A. glowing, tarnished B. brilliant C. dazzling D.

Clues: **radiant (adj):** shining, bright, giving forth light or energy. (glowing, brilliant, dazzling, resplendent.
antonyms: dull, tarnished, lackluster **Answer: D**

174. **RURAL**

A. citified B. countrified C. rustic D. village

Hints: **rural (adj):** relating to farm areas and life in the country (countrified, rustic).
opposite: urban, metropolitan, **citified** **Answer: A**

SUBSTANTIAL

A. considerable B. big C. minor D. large

Hints: **substantial (adj):** large, important; major, significant, prosperous; not imaginary, material.

opposite: **minor**, insignificant, negligible **Answer: C**

TACTFUL

A. skillful B. clumsy C. diplomatic D. discreet

Hints: **tactful (adj):** skilled in handling difficult situations or people, polite. (synonym with A C D)

antonyms: **clumsy**, gauche, boorish, indiscreet

Answer: B

ULTIMATE

A. last B. final C. terminal D. initial

Clues: **ultimate (adj):** last, final; most important or extreme; eventual; basic, fundamental, farthest.

antonyms: first, initial, most mediate nearest

Answer: D

BROADEN

A. restrict C. terminal

B. abandon D. initial

Clues: **broaden (v)** to make widen is the **opposite** of **restrict** or to make narrow.

Answer: A

BRILLIANT

A. thoughtless C. cautious

B. dull D. acceptable

Clues: **brilliant (adj)** is brightly shining is the opposite of **dull**.

Answer: B

REJECTION

A. awareness C. open mind

B. indifference D. approval

Hints: **rejection (N)** means refusal is the opposite of **approval**.

Answer: D

181. **FORTUITOUS**
A premeditated
B impassive
C illogical
D fickle

Clues **fortuitous (adj)** means accidental; it is the opposite of **premeditated** which means intended.
Answer: A

182. **ASSUAGE**
A. antagonize
B. eradicate
C. aggravate
D. deceive

Clues: **assuage (v)** is to make something less severe.
aggravate (v) is to make something worse (worsen).
Answer: C

183. **SKIMPY**
A. abundant B. prudent C. rigid D. shallow

Clues: **skimpy (adj):** lacking in quantity; meager is the opposite of **abundant**.
Answer: A

184. **OVERRULE**
A. decide in favor
B. eliminate
C. fail to state
D. undertake

Clues: **overrule (v)** rule or decide against is the opposite of **decide in favor.**
Answer: A

185. **FORTITUDE**
A. intensity B. humility C. ugliness D. cowardice

Hints: **fortitude (adj):** patient courage is the opposite of **cowardice.**
Answer: D

186. **TACITURN**
A. tactful B. verbose C. garrulous D. desultory

Clues: **taciturn (adj)** inclined to silence is the opposite of **verbose (wordy).**
Answer: B

188. **BADGER**

A. equip B. let alone C. deceive D. keep clean

Hints: **badger (N)** paster. It is the opposite of **let alone**.

Answer: B

189. **DIFFIDENCE**

A. monotony
B. indifference
C. boldness
D. fame

Clues: **diffidence (N)** means lack of confidence; timidity, the opposite of **boldness**.

Answer: C

190. **FLIPPANCY**

A. charity
B. choherence
C. seriousness
D. rivalry

Clues: **flippancy (N)** disrespect, lack of seriousness. **Answer: C**

191. **GUSTY**

A. calm B. pale C. dry D. slender

Clues: **gusty (adj)** means marked by bursts of wind; windy. Its opposite is **calm**.

Answer: A

192. **OPPONENT**

A. athlete
B. supervisor
C. hard worker
D. ally

Hints: **opponent (adj)** means adversary or foe.
ally (adj) means supporter.

Answer: D

193. **SLUMP**

A. criticize
B. move rapidly
C. stand erect
D. befriend

Hints: to **slump (v)** is to slouch or droop in posture.
Its opposite is to stand erect.

Answer: C

194. **LETHARGIC**

A. erratic B. alert C. immune D. dignified

Clues: **lethargic (adj)** means drowsy, slow to react.
Its opposite is **alert** or wide-awake.

Answer: B

194. **LEVITY**

A. slowness
B. complexity
C. insistence
D. earnestness

Clues: **levity (N)** means frivolity lack of seriousness, constancy. Its opposite is **earnestness.**

Answer: D

195. **EXTRANEOUS**

A. destitute
B. enthusiastic
C. pertinent
D. diverse

Clues: **extraneous (adj)** means irrelevant. Its opposite is **pertinent** or to the point.

Answer: C

196. **DEFUNCT**

A. extant
B. prone
C. indebted
D. erect

Clues: **defunct (adj)** means no longer existence. Its opposite is **extant** (existing).

Answer: A

197. **RIBALD**

A. proper
B. ironic
C. ridiculous
D. impulsive

Clues: **ribald (adj)** means crude or indecent. Its opposite is decent or **proper.**

Answer: A

198. **MENDACITY**

A. punishment
B. destruction
C. truthfulness
D. insignificant

Clues: **mendacity (N)** means dishonesty or untruthfulness. Its opposite is **truthfulness.**

Answer: C

199. **DRAWBACK**

A. careful plan
B. desirable future
C. gradual advance
D. hidden purpose

Clues: **a drawback (N)** is a hindrance or disadvantage. **a desirable future** is a help.

Answer: B

200. **ENDEAR**

A. disgust
B. introduce
C. solve
D. vanish

Clues: **endear (v):** to endear is to a rouse affection. Its opposite is to **disgust** or provoke dislike.

Answer: A

201. **MIND**

A. repeat B. keep clean C. stay near D. disobey

Clues: **mind (v):** to mind means to **obey**. Its opposite is disobey.

Answer: D

202. **GULLIBLE**

A. skeptical B. credible C. crucial D. extravagant

Clues: **gullible (adj):** a gullible person is too trusting, easy to deceive.

A **skeptical** person in doubting, hard to convince.

Answer: A

203. **EASE**

A. aggravate B. preserve C. approve D. magnify

Clues: **easy (v):** to **ease** is to lighten or lessen.

aggravate (v): to **aggravate** is to intensify or make more severe.

Answer: A

204. **COLOSSAL**

A. relevant
B. infinitesimal
C. haggard
D. victorious

Clues: **colossal (adj):** means **huge**.

infinitesimal (adj): means **extremely small**. **Answer: B**

205. **MUTABLE**

A. talkative B. frantic C. incapable D. unalterable

Hints: **mutable (adj)** means changeable.

unalterable (adj) means fixed or unchangeable.

Answer: D

206. **RETROGRESS**

A. continue B. fashion C. advance D. introduce

Clues: **retrogress (v):** to **retrogress** is to go backward,

Its opposite is to **advance.**

Answer: C

207. **MORBID**

A. impatient B. rapid C. greedy D. wholesome

Clues: **morbid (adj)** means diseased or unhealthy.
Its opposite is **wholesome.**

Answer: D

208. **SUCCINT**

A. wordy B. inadequate C. original D. profound

Clues: **succinct (adj)** means brief or concise.
wordy (adj) means long-winded or redundant.

Answer: A

209. **ENIGMATIC**

A. obvious
B. quantitative
C. qualitative
D. pertinent

Clues: **enigmatic (adj)** means obscure or ambiguous. Its opposite of **obvious**.

Answer: A

210. **EXHORT**

A. intend B. dissuade C. despair D. misquote

Clues: **exhort (v):** to **exhort** is to encourage someone to do something.
To **dissuade (v)** is to discourage someone from doing something.

Answer: B

211. **ASSIDUOUS**

A. limited
B. saccharine
C. lackadaisical
D. obscure

Clues: **assiduous (adj)** means very careful and attentive.
lackadaisical (adj) means listless and indifferent.

Answer: C

212. **PENURY**

A. candor B. civility C. affluence D. liberty

Clues: **penury (N)** is poverty, the lack of money.
affluence (N) is wealth, having plenty of money.

Answer: A

197

213. **TURPITUDE**

A abundance C evanescence

B intelligence D integrity

Clues **turpitude (N)** means immorality or depravity

integrity (N) means honesty or being ethical

Answer: D

214. **RECOLLET**

A. forget B. dislike C. insist D. answer

Clues: **recollect (v)** means **remember** is the opposite of **forget**.

Answer: A

215. **INHUMANE**

A. stingy B. compassionate C. anxious D. smug

Clues: **inhuman (N)** lacking kindness, is the opposite of **compassionate**.

Answer: B

216. **DOWNHEARTED**

A. vain B. attractive C. upstanding D. cheerful

Clues: **downhearted (adj)** dejected, depressed is the opposite of **cheerful**.

Answer: D

217. **GRUF**

A. generous B. gentle C. complex D. colorful

Hints: **gruf (adj)** means harsh, rough is the opposite of **gentle**.

Answer: B

218. **HINDER**

A. value B. support C. prefer D. applaud

Clues: **hinder (v)** is to mean block, hamper, is the opposite of **support** or help.

Answer: B

219. **ASSURANCE**

A. doubt B. support C. prefer D. applaud

Clues: **assurance (N)** certainty, freedom, from doubt, is the opposite of **doubt**.

Answer: A

220. **INCOMPTABLE**

A. painstaking B. beneficial C. congenial D. parallel

Clues: **incompatible (adj)** inharmonious; disagreement, is the opposite of **congenial** (compatible; like-minded).
Answer: C

221. **BANTER**
A. lawful act
B. deceptive appearance
C. ready inquiry
D. serious discussion
Clues: **banter (N) playful joking,** is the opposite of **serious discussion.**
Answer: D

222. **FLAG**
A. rise
B. soothe
C. conceal
D. seem indifferent
Clues: **flag (v)** droop, decline in energy, etc). is the opposite of **rise.**
Answer: A

223. **INCISIVE**
A. arrogant
B. outspoken
C. dull-witted
D. small-minded
Clues: **incisive (adj)** perceptive; keen, is the opposite of **dull-witted.**
Answer: C

224. **CENSURE**
A. recall B. approve C. gather D. heed
Clues: **censure (v)** criticize; blame, is the opposite of **approve.**
Answer: B

225. **SCANTINESS**
A. emphasis B. acuity C. propriety D. profusion
Clues: **scantiness (N)** scarcity; lack, is the opposite of **profusion** (wealth).
Answer: D

226. **TRANQUILITY**
A. cowardice
B. agitation
C. disinclination
D. opulence
Hints: **tranquility (N)** peacefulness; calm, is the opposite of **agitation.**

227. **SQUELCH**

A. polarize
B. speak truthfully
C. encourage
D. delay temporarily

Clues: **squelch (v)** suppress, crush, is the opposite of **encourage.**

Answer: C

228. **SUPERCILIOUS**

A. excellent
B. omniscient
C. ridiculous
D. obsequious

Clues: **supercilious (adj)** arrogant; scornful is the opposite of **obsequious** (subservient; faw)

Answer: D

229. **BROWSE**

A. skim B. scan C. graze D. scrutinize

Clues: **browse (v)** nibble, graze; to read casually; to win down-shop, dip into.

☞ **antonyms:** pore over, scrutinize

Answer: D

230. **DUPE**

A. hoodwink B. disabuse C. fool D. mislead

Hints: **dupe (N)** a person easily tricked or deceived.
dupe (V) to deceive (delude).

☞ **opposite:** undeceive, disabuse

Answer: B

231. **DYNAMIC**

A. vigorous
B. high-powered
C. energetic
D. lazy

Clues: **dynamic (adj)** active, energetic, forceful.

☞ **antonyms:** lazy, lackadaisical, lethargic, sluggish.

Answer: D

232. **ERADICATE**

A. wigout
B. uproot
C. get rid of
D. promote

Clues: **eradicate (v)** to root out, get rid of, destroy, completely, wig out, up root.

☞ **antonyms:** implant, instill, foster, promote.

Answer: D

233. **FRUSTRATE**

A. thwart B. foil C. baffle D. assist

Hints: **frustrate (v)** to prevent from accomplishing a purpose or fulfilling a desire; to cause feeling of discouragement.

☞ **antonyms:** help, assist, abet

Answer: D

234. **GRIM**

A. merciful B. dreadful C. frightful D. ferocious

Clues: **grim (adj)** stern, merciless, fierce, savage, cruel, ferocious.

☞ **antonyms:** mild, merciful, delightful.

Answer: A

235. **RENAISSANCE**

A. reawaking B. regressing C. revival D. rebirth

Answer: B

236. **MARGINAL**

B. board line B. minimal C. central D. peripheral

Clues: **marginal (adj)** in, at, or near the edge or marginal; only barely good, large, or important enough for the purpose. (A B D) are its synonyms.

☞ **antonyms: central**, pivotal, focal

Answer: C

237. **PENDING**

A. undecided B. settled C. unsettled D. reserved

Clues: **pending (adj);** waiting to be settled.
pending (prep): until

☞ **antonyms:** settled, **decided**, resolved.

Answer: B

238. **PRESCRIBE**

A. generalize B. specify C. appoint D. recommend

Clues: **prescribe (v)** to order as a rule or course to be followed; to order for medical purposes. (specify, appoint, recommend).

Answer: A

39. **PROMINENT**

A. unnoticeable
B. conspicuous
C. curious
D. picturesque

Clues: **prominent (adj)** standing out so as to be easily seen; important, well-known.

↝ **antonyms: inconspicuous, unnoticeable, obscure**

Answer: A

40. **QUAINT**

A. familiar
B. strange
C. curious
D. picturesque

Clues: **quaint (adj):** odd or old-fashioned in a pleasing way; clever, ingenious; skill fully made.

↝ **antonyms: familiar**, commonplace, modern, contemporary

Answer: A

41. **RELUCTANT**

A. unwilling
B. holding back
C. hesitant
D. inclined

Clues: **reluctant (adj):** doing something without enthusiasm. (resistant, un willing, loath, disinclined).

↝ **antonyms:** willing, eager, **inclined**.

Answer: D

42. **SCRIMP**

A. splurge B. economize C. modify D. simplify

Clues: **scrimp (v):** to handle very economically or stingily; to supply in a way that is small, short, or scanty.

Answer: A

↝ **opposite:** splurge

43. **SNARE**

A. pitfall B. entrap C. liberate D. catch

Hints: **snare (v):** to trap, catch
snare (N): a trap or entanglement

Answer: C

↝ **antonym:** liberate

244. **UTMOST**

A. least B. greatest C. highest D. farthest

Hints: **utmost (adj):** maximum, supreme, best, the extreme (greatest, highest, farthest).

☞ **antonyms: least, minimum**

245. **VENGEANCE**

A. revenge C. reprisal

B. retaliation D. forgiveness

Hints: **vengeance (N):** punishment in return for an injury or wrong; unusual force or violence.

☞ **antonyms: forgiveness, pardon, excuse**

246. **AMISS**

A. properly B. faulty C. imperfect D. awry

Clues: **amiss (adj):** faulty, imperfect, not as it should be.

amiss (adv): in a mistaken or improper way, wrongly

247. **DETEST**

A. dispise B. abhor C. to hate D. esteem

Clues: **deterst (v)** to hate, dislike very much, loathe

☞ **antonyms:** relish, love, admire, esteem

248. **DOMESTIC**

A. foreign B. alien C. native D. A and B

Clues: **domestic (adj)** native to a country, not foreign, relating to the life or affairs of a household.

domestic (N): a household servant.

☞ **antonyms:** foreign, alien

249. **FLAGRANT**

A. blatant B. petty C. gross D. outrageous

Clues: **flagrant (adj):** extremely bad, glaring, scandalous notorious.

☞ **antonyms: petty**, piddling, trifling, inconsequential

FLAW

A. imperfection
B. blemish
C. faultlessness
D. defect

Clues: **flaw (N)** a slight fault, defect, crack

antonyms: perfection, **faultlessness**

Answer: C

FLUSTER

A. agitate
B. rattle
C. reassure
D. disconcert

Clues: **fluster (v)** to make or become confused, agitated, or nervous; (N): a state of confusion or agitation.

antonyms: **reassure**, soothe, quiet

Answer: C

FOREMOST

A. leading
B. principal
C. hindmost
D. paramount

Clues: **foremost (adj):** chief, most important, primary.

foremost (adv): in the first place.

antonyms: **hindmost**, last, secondary

Answer: C

NOTABLE

A. noteworthy
B. unremarkable
C. impressive
D. attractive

Clues: **notable (adj):** striking, remarkable

notable (N): a person who is well known, distinguished, or outstanding in some way.

antonyms: undistinguished, **unremarkable**, unknown

Answer: B

NURTURE

A. neglect
B. foster
C. raise
D. rear

Clues: **nurture (V)** to bring up, care for, train, nourish.

nurture (N) rearing, training, upbringing.

opposite: **neglect**, ignore, discourage, hinder

Answer: A

255. **PRIOR**

A. earlier B. former C. previous D. later

Answer: D

Clues: **prior (adj)** anterior, previous

☞ **antonyms:** subsequent, **later**, ensuing, following

256. **PROFICIENT**

A. competent B. unskilled C. adept D. able

Answer: B

Clues: **proficient (adj)** choice A C D are **synonyms.**

☞ **antonyms:** incompetent, inept, **unskilled**, ignorant.

257. **VIGILANT**

A. sleepy
B. attentive
C. on one's toes
D. watchful

Answer: A

Clues: **vigilant (adj)** wide-awake, alert, watchful

☞ **antonyms: sleepy**, inattentive, unobservant

258. **WRATH**

A. rage B. pleasure C. choler D. fury

Answer: B

Clues: **wrath (N)** intense anger, indignation

☞ **antonyms:** favor, approval, **pleasure**, blessing

259. **CATASTORPHE**

A. calamity B. tragedy C. success D. cataclysm

Answer: C

Clues: **catastrophe (N)** a large-scale disaster, misfortune, failure. (See A B D).

☞ **antonyms:** triumph, victory, **success**

260. **CAPSIZE**

A. overturn B. upend C. remain D. tip over

Answer: A

Clues: **capsize (v)** to turn bottom side up, upset.

☞ **antonyms:** remain upright

Answer: C

EJECT

A. oust B. admit C. expel

D. insert E. B and D

Clues: **eject (v)** to drive or throw out, evict.

antonyms: **admit**, **let in**, **insert**

Answer: E

DISPUTATIOUS

A. peaceable C. argumentative

B. contentious D. quarrelsome

Clues: **disputatious (adj):** inclined to argue or debate; provoking debate.

antonyms: non argumentative, **peaceable**, pacific

Answer: A

INCENTIVE

A. stimulus B. spur C. motive D. hindrance

Clues: **incentive (N)** a reason for doing something that stimulates action (stimulus, spur, motive, inducement).

antonyms: curb, check, restraint, **hindrance** **Answer: D**

FLOORISH

A. fade B. increase C. prosper D. burgeon

Clues: **flourish (v)** to grow, thrive, be prosperous; to wave in the air. **(See B C D)**

antonyms: whither, die, **fade**, shrivel up **Answer: A**

LEGIBLE

A. readable C. clear

B. indecipherable D. easily read

Clues: **legible (adj)** easily read **(see A C D)**

opposite: unreadable, **indecipherable** **Answer: B**

NUB

A. core B. nucleus C. fringe D. crux

Clues: **nub (N)** is the central point or heart of a matter, a knob.

antonyms: **fringe**, periphery, edge **Answer: C**

267. **ORDAIN**

A. forbid B. anoint C. decree D. consecrate

Clues: **ordain (v)** to establish by law; to order or command; to appoint as a priest or minister; to destine.

antonyms: forbid, veto, cancel

Answer: A

268. **QUENCH**

A. douse B. ignite C. stifle D. slake

Clues: **quench (v)** to put out, extinguish, end.

antonyms: ignite, kindle

Answer: B

269. **SIMULTANEOUS**

A. occurring at the same time
B. concurrent
C. occurring at different times
D. parallel action

Clues: **simultaneous (adj)** happening or existing **at the same time**.

Answer: C

270. **BYSTANDER**

A. active participant C. spectator
B. observer D. onlooker

Clues: **bystander (N)** one who looks on or observes a person present but not taking part.

☞ **opposite:** active participant

Answer: A

271. **CASUAL**

A. haphazard
B. offhand C. accidental
D. formal

Clues: **casual (adj)** happening by chance or on an irregular basis; showing little concern, informal.

☞ **opposite: formal**, serious, premeditated, intentional.

Answer: D

272. **MELANCHOLY**

A. depressed
B. dejected C. cheerful
D. gloomy

Clues: **melancholy (adj)** sad, gloomy, unhappy

☞ **antonyms;** merry, happy, **cheerful**, joy, elation

Answer: C

274. **HILARIOUS**
A. highly amusing
B. boring
C. side-splitting
D. extremely funny

Clues: **hilarious (adj)** extremely funny, causing loud amusement.

☞ **antonyms; boring**, dull, humorless, heartending.

Answer: B

275. **DOWNTRODDEN**
A. mistreated
B. ground underfoot
C. oppressed
D. unlifted

Clues: **downtrodden (adj)** treated unfairly and cruelly, oppressed.

☞ **opposite: unlifted**, liberated

Answer: D

276. **ERODE**
A. corrode B. abrade C. nurture D. collide

Clues: **erode (v)** to wear away, eat away (corrode, abrade)

☞ **opposite: nurture, promote, encourage**

Answer: C

277. **GRAPHIC**
A. unrealistic
B. lively
C. colorful
D. descriptive

Clues: **graphic (adj)** lifelike, vivid; relating to the pictorial arts. **(See B C D)**

☞ **opposite:** dull, boring, unrealistic, colorless.

Answer: A

278. **ORDEAL**
A. test B. hardship C. pleasure D. painful

Clues: **ordeal (n)** a difficult or painful experience, atrial.

☞ **opposite: pleasure**, cinch

Answer: C

279. **PARCH**
A. dry up B. **saturate** C. **desiccate** D. **dehydrate**

Clues: **parch (v)** to make dry and thirsty; to shrivel with heat

☞ **opposite:** soak, drench, **saturate**, waterlog

Answer: B

279. **REGAL**

A. majestic B. stately C. lowly D. princely

Clues: **regal (adj)** royal, kinglike, fit for a king (majestic, august).

☞ **opposite: lowly**, humble, abject, servile

Answer: C

280. **VITAL**

A. nonessential C. fundamental

B. indispensable D. living

Clues: **vital (adj)** having life, living, necessary to life, essential; key, crucial.

☞ **opposite: nonessential**, unnecessary

Answer: A

281. **BELLOW**

A. yell B. whisper C. bawl D. holler

Clues: **below (v)** to make a sound similar to that of a bull, roar **(See A C D)**

bellow (N) aloud, angry roar

☞ **opposite: whisper**, murmur

Answer: B

282. **CLUTTER**

A. order B. litter C. confusion D. disorder

Clues: **clutter (v)** to fill or cover in a disorderly way:

cutter (N) confusion

☞ **antonyms: order**, tidiness, neatness

Answer: A

283. **DILAPIDATED**

A. indisrepair C. shipshape

B. deteriorated D. gone to seed

Clues: **dilapidated (adj)** falling apart or ruined, run-down. **(See A B D)**

☞ **opposite: shipshape**, trim

Answer: C

284. **DISMANTLE**

A. take away C. far apart

B. put together D. disassemble

Clues: **dismantle (v)** to take apart; to strip of something.

☞ **opposite: put together**, assemble, construct.

Answer: B

386. GRUELING
A. exhausting
B. easy
C. punishing
D. taxing

Clues: **grueling (adj)** very tiring, calling for an extreme effort. **(See A C D).**

☞ **antonyms: easy**, effortless **Answer: B**

387. HOSPITABLE
A. gracious B. chilly C. cordial D. courteous

Clues: **hospitable (adj)** offering friendly or generous treatment to guests; open to anything new or strange **(See A C D)**

☞ **antonyms:** unfriendly, cold, icy, **chilly**. **Answer: B**

388. LAVISH
A. excessive
B. profuse
C. stingy
D. extravagant

Clues: **lavish (adj)** overly generous, extravagant; abundant.
lavish (v) to spend or give freely or without limit.

☞ **antonyms: stingy**, meager **(v)** begrudge, stint, deny **Answer: C**

389. SURPLUS
A. shortage
B. glut, surfeit
C. overage
D. excess

Clues: **surplus (N)** an amount beyond what is required; excess.

☞ **opposite: shortage**, lack, dearth, paucity **Answer: A**

390. TIMIDITY
A. fearfulness
B. faintheartedness
C. frightened
D. intrepidity

Clues: **timidity (N)** the state of being easily frightened.
(See A B C)

☞ **antonyms:** fearlessness, boldness, **intrepidity**. **Answer: D**

391. GIGANTIC
A. tiny B. enormous C. colossal D. mammoth

Clues: **gigantic (adj)** huge, giant, immense **(See B C D)**
☞ **Opposite: tiny**, infinitesimal, diminutive. **Answer: A**

291. **HAVOC**

A. peace and quiet
B. devastation
C. harm
D. disarray

Hints: **havoc (N)** very great destruction, ruin; great confusion and disorder **(See B C D)**
☞ **antonyms:** calm, order **Answer: A**

292. **DIALOGUE**

A. exchange of ideas
B. mologue
C. soliloquy
D. B and C

Clues: **dialogue (N)** a conversation between two or more people; in interchange of opinions and ideas, free discussion.
☞ **antonyms: monologue, soliloquy** **Answer: D**

293. **INNUMERABLE**

A. countless
B. beyond rack
C. without num
D. countable

Clues: **innumerable (adj)** too many to count; without number.
☞ **opposite: countable**, few in number **Answer: D**

294. **LAX**

A. remiss
B. vigilant
C. slack
D. negligent

Clues: **lax (adj)** not strict, careless, lacking discipline, not tense, relaxed.
☞ **opposite:** strict, **vigilant**, conscientious, scrupulous **Answer: B**

295. **IMPLORE**

A. demand forcefully
B. entreat
C. beseech
D. pray

Clues: **implore (v)** to beg earnestly for **(see B C D)**
☞ **opposite: demand forcefully**, clamor for **Answer: A**

LEISURELY

A. slow B. hasty C. relaxed D. unhurried

Clues: **leisurely (adj)** unhurried, taking plenty of time.
(adv) in an easy going or unhurried way.
antonyms: hasty, hurried, rushed, hectic **Answer: B**

TURMOIL

A. peace and quiet C. tumult
B. unheaval D. chaos

Clues: **turmoil (N)** a state of great confusion or disorder, mental strain or agitation.

antonyms: **peace and quiet**, order **Answer: A**

PIECEMEAL

A. bit by bit C. all at once
B. gradually D. all

Clues: **piecemeal (adj)** one piece at a time, gradually, bit by bit.

antonyms: all at once **Answer: A**

MALADY

A. health C. ailment
B. well-being D. A and B

Clues: **malady (N)** a sickness, illness, disease, disorder, indisposition. **Answer: D**

SECLUSION

A. aloneness C. solitude
B. solitariness D. unite

Clues: choice A B and C are its **synonyms.** **Answer: D**

RANDOM

A. deliberate C. arbitrary
B. haphazard D. by chance

Clues: **random (adj)** by chance, not planned, or prearranged; irregular.

antonyms: planned, **deliberate**, systematic. **Answer: A**

302. **FLUCTUATE**

A. oscillate
B. remain unchanged
C. wave
D. see saw

Clues: **fluctuate (v)** to change continually, to move up and down. **(see A C D).**

☞ **opposite:** stay put, remain unchanged

Answer: B

303. **ENCHANT**

A. delight B. thrill C. bewitch D. bore

Clues: **enchant (v)** to please greatly; to charm, put under a magic spell.

☞ **antonyms: bore**, nauseate, disgust

Answer: D

304. **COUNTENANCE**

A. support
B. disapprove of
C. condone
D. a face

Clues: **countenance (N)** a face, facial expression; **countenance (v)** to tolerate or approve

☞ **antonyms:** disapprove of

Answer: B

305. **MAGNITUDE**

A. smallness
B. extent
C. immensity
D. enormity

Clues: **magnitude (N)** the great size or importance of something.

(See B C D)

☞ **antonyms: smallness**, unimportance, insignificance

Answer: A

306. **MASSIVE**

A. bulky B. huge C. immense D. flimsy

Clues: **massive (adj)** large and heavy; great in size or scope.

(See A B C)

☞ **antonyms: flimsy**, frail, thin

Answer: D

307. **REVERE**

A. admire B. esteem C. cherish D. mock

Clues: **revere (v)** to love and respect deeply, honor greatly **(See A B D)**

☞ **antonyms:** scorn, disdain, mock, deride

Answer: D

308. **STODGY**

A. forward-looking
B. stuffy
C. tiresome
D. blah

Clues: **stodgy (adj)** dull, boring; old-fashioned, hidebound; lumpy, thick **(See B C D)**

☞ **antonyms:** forward-looking, avant-grade, progressive

Answer: A

309. **MATERNAL**

A. protective
B. motherly
C. paternal
D. sympathetic

Clues: **maternal (adj)** of or like a mother. **(See A B D)**

☞ **antonyms:** fatherly, paternal

Answer: C

310. **COSMOPOLITAN**

A. global
B. provincial
C. international
D. polished

Clues: **cosmopolitan (ad)** found in most part of the world; having many fields of interest; of worldwide scope; sophisticated **(see A C D)**

☞ **antonyms:** narrow, unsophisticated, **provincial**

Answer: B

311. **AFFLICTION**

A. blessing B. illness C. woe D. torment

Clues: **affication (N)** a physical ailment; a cause of pain or trouble, misfortune, anguish **(See B C D)**

Answer: A

☞ **antonyms: blessing**, boon, joy

312. **HEED**

A. ignore B. disregard C. listen to D. A and B

Clues: **heed (v)** to pay careful attention to, notice to be guided by, or attend.

Answer: D

☞ **antonyms: ignore, disregard**

313. **CYNICAL**

A. skeptical
B. sarcastic
C. contemptuous
D. optimistic

Clues: **cynical (adj)** inclined to believe the worst of people; bitterly mocking or sneering.

☞ **antonyms:** hopeful, **optimistic**

Answer: D

314. **SAGACIOUS**

A. silly B. smart C. clever D. astute

Clues: **sagacious (adj)** shrewd; wise in a keen, practical way. **(See B C D)**

☞ **antonyms: silly**, foolish, ill-advised, dopey

Answer: A

315. **MIRTH**

A. merriment B. glee C. gaiety D. gloom

Clues: **mirth (N)** merry fun, gaiety; laughter, glee.

☞ **opposite: gloom**, sadness, sorrow

Answer: D

316. ***LEGITIMATE***

A. lawful B. unauthorized C. legal D. right

Clues: **legitimate (adj)** lawful, rightful, reasonable, justifiable, proper, genuine.

☞ **opposite:** unlawful, illegal, improper, **unauthorized**

Answer: B

317. **MEANDER**

A. proceed directly
B. begin abruptly
C. treat harshly
D. distribute unevenly

Clues: to **meander** is to wander about. Its opposite is to **flow/ proceed directly**

Answer: A

318. **ALLEVIATE**

A. transpose
B. fortify
C. exacerbate
D. rectify

Answer: C

Review Exercise

Choose the word or phrase that is most nearly <u>opposite/ antonym in meaning to the word in capital letter.</u>

MOURNFUL
A. joyous
B. informal
C. appropriate
D. sympathetic

MERCENARY
A. ruthless
B. altruistic
C. unworthy
D. capricious

INACCURACY
A. calculation
B. moderation
C. novelty
D. exactness

ABUSIVE
A. prompt
B. inconsiderate
C. cordial
D. indecisive

5. **SUMMARIZE**
A. recount in detail
B. express poetically
C. submit as evidence
D. question closely

6. **GRANDLOSE**
A. docile
B. unlike to occur
C. uncommunicative
D. simple and unimposing

7. **STAGNATION**
A. humility
B. growth
C. distraction
D. remoteness

8. **INTRENCH**
A. defy B. oust C. extinguish D. squander

9. **LACKLUSTER**
A. vibrant
B. superficial
C. complex
D. abundant

10. **PERPETUATE**
A. release
B. elaborate
C. minimize
D. extinguish

11. **GINGERLY**
A. honestly
B. fragrantly
C. recklessly
D. blandly

12. **CENSURE**
A. eradicate
B. commend
C. reform
D. argument

13. **UNREMITTING**
A. occasional
B. functional
C. impending
D. extensive

14. **DESICCATE**
A. hallow
B. lengthen
C. saturate
D. exonerate

15. **TRANSIENCE**
A. lack of caution
B. permanence
C. original nature
D. slowness

16. **DISTRIBUTE**
A. lend
B. interference
C. gather
D. represent

17. **HIDEOUS**
A. attractive
B. authentic
C. open to view
D. broadminded

18. **EXTINGUISH**
A. redicule
B. restore to life
C. invite interest
D. decline

19. **PROTRUSION**
A. strong dislike
B. illusion
C. deep recess
D. growing scarcity

20. **TEMPER WITH**
A. testify against
B. postpone indefinitely
C. study purposely
D. leave strictly alone

21. **CONSIDERATE**
A. tactless
B. implausible
C. artificial
D. inconspicuous

22. **ENTICE**
A. repel
B. authorize
C. baffle
D. misplace

23. **ORTHODOXY**
A. renown
B. unconventionality
C. remoteness
D. inquisitiveness

SUMPTUOUS
A. jovial
B. frugal
C. open
D. retrained

RAZE
A. erect
B. temper
C. accumulate
D. manipulate

CARNAL
A. chronic
B. agile
C. spiritual
D. knowledgeable

REMONSTRATE
A. advocate
B. mistake
C. flaunt
D. deliberate

DISSOLUTION
A. futility
B. persuasion
C. compliance
D. establishment

IRK
A. soothe
B. pry
C. convince
D. tinge

LACHRYMOSE
A. profuse
B. tearless
C. unpoetic
D. verbose

LIMBER
A. stiff
B. orderly
C. gloomy
D. durable

DISINFECT
A. operate
B. contaminate
C. diagnose
D. arrange

SECEDE
A. dilute
B. regress
C. join
D. repeal

DETACHEMENT
A. involvement
B. prosperity
C. termination
D. resemblance

INFALLIBLE
A. overly ambitious
B. capable of error
C. widely know
D. exposed to danger

16. **SLUR**
A. sensitivity
B. sacrifice
C. commendation
D. challenge

17. **HAGDOG**
A. immortal
B. hungry
C. painful
D. confident

18. **UNSWERVING**
A. aggressive
B. remorseful
C. tedious
D. faltering

19. **ENERVATE**
A. invigorate
B. narrate
C. accomplish
D. enrage

20. **NIGGARDILY**
A. appropriate
B. generous
C. ongoing
D. radiant

21. **EGRESS**
A. entry
B. simplicity
C. accuracy
D. revolution

22. **HEDGE**
A. act on impulse
B. make a foolish comment
C. state without qualification
D. establish a connection

23. **EXACERBATE**
A. all eviate
B. signify
C. ratify
D. transgress

24. **VOLUBILITY**
A. expansion
B. terseness
C. fluctuation
D. incapability

25. **INATTENTIVE**
A. futile
B. energetic
C. reasonable
D. watchful

26. **CURRENT**
A. outmoded
B. vital
C. improper
D. private

27. **INDUSTRY**
A. practicality
B. cleanliness
C. laziness
D. humility

48. AB
A.
B.

49. BA
A.
B.

50. SP
A.
B

51. S
A
B

52. N
A
E

53. S

54.

55.

56.

57.

58.

59.

48. **ABRUPT**
A. candid
B. gradual
C. typical
D. conclusive

49. **BAN**
A. punish
B. surrender
C. sanction
D. divest

50. **SPUNK**
A. success
B. timidity
C. grow awareness
D. loss of prestige

51. **SHY AWAY**
A. except
B. advance
C. bedazzle
D. embolden

52. **MONOTONY**
A. variety
B. strength
C. loyalty
D. gloominess

53. **SAGE**
A. braggart
B. tyrant
C. miser
D. fool

54. **ADMONITION**
A. premonition
B. hallucination
C. commendation
D. trepidation

55. **DIVULGE**
A. conceal
B. unite
C. indulge
D. reconceal

56. **STUPEFY**
A. bend
B. enliven
C. talk nonsense
D. lie

57. **APPOSITE**
A. severe
B. healthy
C. long-lived
D. inappropriate

58. **COGENT**
A. contemplative
B. qualification
C. unpersuasive
D. expository

59. **ODIUM**
A. conviction
B. qualification
C. veneration
D. hesitance

60. **DETERIORATE**
A. improve
B. prevent
C. relate
D. permit

61. **FICKLE**
A. urgent
B. industrious
C. spotless
D. loyal

62. **COMPLY**
A. unit
B. rebel
C. appreciate
D. simplify

63. **MISGIVING**
A. lack of honesty
B. gratitude
C. absent of doubt
D. wish to reform

64. **REJUVENATE**
A. judge
B. tolerate
C. deny
D. age

65. **CREDIT**
A. believe false
B. forget
C. treat as equal
D. underemphasize

66. **UNFEGNED**
A. remarkable
B. pretended
C. gentle
D. brief

67. **STILTED**
A. informal
B. verbose
C. tentative
D. senseless

68. **UNGAINLY**
A. inaccurate
B. unnoticed
C. restrained
D. graceful

69. **AFFABILITY**
A. grumpiness
B. frugality
C. vigilance
D. obstinacy

70. **QUIXOTIC**
A. benevolent
B. grave
C. pragmatic
D. slow

71. **PARIAH**
A. general favorite
B. innocent
C. unexpected guest
D. persistent heckler

BANEFUL
A. reckless
B. obscure
C. conceited
D. begin

DISCURSIVE
A. handwritten
B. concise
C. endemic
D. malevolent

GAMELY
A. fearfully
B. clumsily
C. gloomy
D. diligently

STOIC
A. whiner
B. victor
C. sevant
D. bystander

SOPHOMORIC
A. deeply insightful
B. easily excited
C. gregarious
D. anti-intellectual

HIGH-HANDED
A. hard-headed
B. dynamic
C. reasonable
D. dejected

SWALLOW
A. reduce in bulk
B. distinguish
C. make peace
D. disbelieve

EMBROIL
A. disengage
B. remonstrate
C. consider genuinely
D. reconsider

INFLUX
A. movement outward
B. consistency
C. dynamic
D. dejected

FLAMBOYANT
A. disheveled
B. credible
C. unadorned
D. suken

GAMBOL
A. monotony
B. purification
C. digress
D. plod

RAUCOUS
A. mellifluous
B. buoyant
C. absorbent
D. orderly

84. **OPACITY**
A. firmness
B. transparence
C. poverty
D. slum

85. **METHODICAL**
A. crucial
B. unsystematic
C. affectionate
D. synthetic

86. **SEEDY**
A. irregular
B. intricate
C. elegant
D. slow

87. **FETTER**
A. diminish
B. enervate
C. liberate
D. return

88. **DIVOTION**
A. intention
B. lack of piety
C. animation
D. rude awakening

89. **ANTAGONISM**
A. prosperity
B. symmetry
C. indifference
D. friendliness

90. **SLAKE**
A. rough
B. active
C. liberal
D. dependent

91. **DORMANT**
A. immaterial
B. accommodating
C. alert
D. severe

92. **FAWN**
A. move sedately
B. embellish
C. disdain
D. meddle

93. **MEEKNESS**
A. evasion
B. freedom
C. talkativeness
D. ferocity

94. **CRASS**
A. boastful
B. cheerful
C. extensive
D. refined

95. **SUBDUE**
A. infiltrate
B. increase in value
C. fail to conquer
D. sabotage

Antonyms

96. **NOVEL**
A. conclusive
B. unoriginal
C. irrelevant
D. inclusive

97. **ZENITH**
A. sum total
B. reduction in value
C. horizon
D. lowest point

98. **MORIBUND**
A. vital
B. profound
C. abhorrent
D. scare

99. **RUTHLESS**
A. slow
B. progressive
C. meager
D. merciful

100. **NEFARIOUS**
A. hypocritical
B. lackadaisical
C. corrigible
D. exemplary

Answer Key

1. A	11. C	21. A	31. C	41. A
2. B	12. B	22. A	32. B	42. C
3. D	13. A	23. B	33. C	43. A
4. C	14. C	24. D	34. A	44. B
5. A	15. B	25. A	35. B	45. D
6. D	16. C	26. C	36. C	46. A
7. B	17. A	27. A	37. D	47. C
8. B	18. B	28. D	38. D	48. B
9. A	19. C	29. A	39. A	49. C
10. D	20. D	30. B	40. B	50. B
51. B	61. D	71. A	81. C	91. C
52. A	62. B	72. D	82. D	92. C
53. D	63. C	73. B	83. A	93. D
54. C	64. D	74. A	84. B	94. D
55. A	65. A	75. A	85. B	95. C
56. B	66. B	76. A	86. C	96. B
57. D	67. A	77. C	87. C	97. D
58. C	68. D	78. D	88. B	98. A
59. C	69. A	79. A	89. D	99. D
60. A	70. C	80. A	90. B	100. D

UNIT TWO

SYNONYMS

SYNONYMS:

When you write or speak, you want to express your meaning exactly. English is so rich in words that you can often choose among words with similar **meanings** to find just the right one.

- ✓ A word that has **the same** or **nearly the same meaning**, is therefore, called **synonym**.
- ✓ Although **synonyms** mean about the same thing, they often convey slightly different shades of meaning **(see antonyms – unit one).**

Here are some examples

felt **drowsy:** A. ill B. content C. sleepy D. livery

Clues: Although all the words make sense when used with **felt**, only one word is close in meaning to **drowsy**. Notice that you might feel **drowsry** because you are ill, but 'ill' doesn't mean the same thing. Notice that content and lively are not **synonyms** of **drowsy**. The answer is **C, sleepy.**

Words	Synonyms	Words	Synonyms
eat	consume	panic	crowds
hurt	injure	source	beginning
big	large	accurate	correct
clash	conflict	occasion	event
fir	discharge	crude	rough
slim	slender	apparently	seemingly
enormous	huge	dynamic	energetic
fortune	lucky	frustrate	foil

responsibility	duty	utmost	greatest
artificial	fake	prominent	well known
benefit	gain	reluctant	hesitant
pattern	design	tremble	shake
notion	idea	tedious	boring
imitate	copy	refrain	resist
ritual	ceremony		

UNIVERSITY ENTRANCE EXAMINATION AND OTHER QUESTION WITH DITAILED EXAMPLES AND EXPLANATIONS (UEE/EHEECE 2001 – 2009)

1. **PARTISAN**

A. neutral B. biased C. objective D. impartial

Clues: **partisan (adj)** someone who is partisan strongly supports a particular person or cause, often without thinking carefully about the matter.

Biased (adj) if someone is biased, they refer one group of people to another, and behave **unfairly** as a result.

Answer: B

2. **NEUTRAL**

A. unkind B. precious C. mean D. indifferent

Clues: **Neutral (adj)** if a person or a country adopts **neutral** position or remains **neutral**, they don't support anyone in disagreement, war, or contest.

Answer: D

3. **CONCEAL**

A. reveal B. hide C. ignorant D. respected

Clue: **conceal (v)** - to cover or hide something carefully.
- to keep from sight.

Answer: B

4. **EXCEED**

A. outstrip B. magnify C. delimit D. offset

Clue: **exceed (v)** if something **exceeds** a particular amount or number, it is greater or larger than that amount or number.
- to pass beyond the measure of something, to surpass

Answer: C

GAURANTEE

A. safety B. treaty C. security D. warranty

Clue: **guarantee and warranty**

Warranty (N) is a written or printed guarantee, especially one given to the buyer of an article, promising to repair or replace it if necessary within a specific period of time.

Answer: D

SPORADIC

A. frequent
B. separated
C. unmanageable
D. irregular

Clue: **sporadic (adj):** happening or seen only occasionally or in a few places.

- occurring at intervals that are not regular.

Answer: D

7. **ELUCIDATE**

A. unit B. escape C. sharpen D. clarify

Clue: **elucidate (v)** – to make something clear.
– to explain something clearly. **Answer: D**

8. **WARY**

A. hesitant B. veritable C. certain D. confident

Clue: **wary (adj)** – cautious because of aware of fearing possible danger or problems.

hesitant (adj) – feeling uncertain or unwilling.

Answer: A

9. **SALTY**

A. brackish B. chosen C. tough D. wet

Clue: **salty (adj)** is to mean brackish (= somewhat salty).

Answer: A

10. **EXITINGUISH**

A. distinguish
B. ignite
C. switch off
D. turn on

Clue: **extinguish (v)** - to cause to stop burning
- to bring to an end (as by destroying in fire).

Answer: C

11. **PRESEVERANCE**

A. continuity of effort
B. frequency suffering
C. lack of energy
D. severity of disease

Clue: **perseverance (N)** - is the act of persisting, or going on, or undertaking something inspite of difficulties.

Answer: A

12. **OUTSPOKEN**

A. anxious B. candid C. experienced D. sallow

Clue: **outspoken (adj)** - is a direct and open in speech or expression (= frank and straight forward).

Answer: B

13. **DEVASTATE**

A. encourage .
B. destroy totally
C. decline
D. dangerous

Clue: **devastate (v):** completely demolish /disappear

Answer: B

14. **FOLIAGE**

A. bacteria
B. a big tree
C. vegetation
D. mammalian

Clue: **foliage (N):** which means plant leaves, or vegetation.
apathetic (adj): uncaring

Answer: C

15. **HINDER**

A. block B. encourage C. handle D. control

Clue: **hinder (v):** to block, obstruct, be in the way of

Answer: A

16. **PREVIEW**

A. try B. foretaste C. revise D. preserve

Clue: **preview (N)** – something seen in advance (foretaste)
preview (v) – to view beforehand

Answer: B

17. **INIMITABLE**

A. matchless
B. incomparable
C. unique
D. all are right

Clue: **inimitable (adj)** – means not capable of being copied or imitated.

Answer: D

18. **ANONYMOUS**

A. synonymous
B. indifferent
C. nameless
D. familiar

Clue: **anonymous (adj)** – unnamed, without the name of the person involved (writer, composer, etc); unknown; lacking individuality or character.

Answer: C

19. **MAKESHIFT**

A. stop gap
B. substitute
C. permanent
D. A and B

Clue: **makeshift (N)** – is a temporary substitute for something else; **(adj)** – crude, flimsy, or temporary.

☞ **opposite: permanent, durable, solid, sturdy**

Answer: D

20. **PRESUME**

A. accuse B. surmise C. perceive D. guess

Clues: **presume (v)** – to take for granted, assume or suppose; to dare, take upon oneself, take liberties **(trespass, infringe)**

Answer: B

21. **PERJURY**

A. false witness
B. telling truth
C. speech tone
D. strength

Clues: **perjury (N)** – the act of swearing to a lie. **Answer: A**

22. **REMNANT**

A. remainder
B. residue
C. left over
D. fragment
E. All

Clues: **remnant (N)** – a small part remaining behind. **Answer: E**

23. **SWERVE**

A. a slow act
B. sheer off
C. shake up
D. lose hope

Clues: **swerve (v)** – to turn a side sharply **(veer, digress)**

swerve (N) – a sharp or sudden turn

Answer: B

24. **PRUDENT**

A. wary
B. sensible
C. judicious
D. all are

Clues: **prudent (adj)**- cautious, careful, showing good sense.

Answer: D

25. **INSUBORDINATE**

A. obedient
B. submissive
C. disobedient
D. docile

Clues: **insubordinate (adj)** – disobedient, rebellious, defiant, unruly, mutinous.

Answer: C

26. **OUTSTRIP**

A. outdo
B. outperform
C. outdistance
D. surpass
E. All are

Clues: **outstrip (v)** – to get a head of, do better than, exceed.

Answer: E

27. **AGITATION**

A. uneasiness
B. peace of mind
C. composure
D. calm

Clues: **agitation (N)** – is a violent stirring or movement; noisy confusion, excitement. **(disquiet, uneasiness, upset)**

Answer: A

28. **PATRONIZE**

A. do business with
B. refuse to deal with
C. agree fairly
D. speak angrily

Clues: **patronize (v)** – is to give one's business to regularly as a customer; to support, provide financial help, to treat someone as an inferior while making a show of being kind or gracious.

(do business with, deal with, trade with)

Answer: A

29. **OFFICIATE**

A. chair B. preside C. moderate D. all are

Clues: **officiate (v)** – to perform the duties of an office; to conduct a religious ceremony; to referee

Answer: D

31. **MIRTH**

A. merriment B. gloom C. sadness D. sorrow

Clues: **mirth (N)** – merry fun, gaiety; laughter, glee **Answer: A**

32. **PARTIAL**

A. just B. complete C. biased D. fair

Clues: **partial (adj)** – not complete; favoring one side over another; showing a strong liking for someone or something. **Answer: C**

33. **CLAMOR**

A. get happy B. uproar C. peaceful D. nonsense

Clues: **clamor (n)** – a public outcry; any loud and continued.
clamor (v) – to call for by loud, continued outcries (uproar, din, racket, cry out for). **Answer: B**

34. **BANKRUPT**

A. insolvent C. solvent

B. financially sound D. calminess

Clues: **bankrupt (adj)** – is to mean in the state of financial ruin.
bankrupt (v) – to ruin financially.
bankrupt (N) – one who has been ruined financially.

(insolvent, flat broken)

Answer: A

35. **COINCIDE**

A. agree B. concur C. match D. all are

Clues: **coincide (v)** – to be in full agreement; to be the same in nature, character, or function; to happen at the same time.

Answer: D

36. **AUTHORITATIVE**

A. unofficial B. reliable C. unreliable D. illegal

Clues: **authoritative (adj)** – official, coming from a source that calls for obedience or beliefs; dictatorial. **Answer: B**

36. **LEGACY**

A. law B. rule C. heritage D. principle

Clues: **legacy (N)** – an inheritance; something handed down fr an ancestor or from the past **(bequest, heritage)**.

Answer:

37. **DESPOT**

A. dictator C. strongman

B. autocratic D. all are

Clues: **despot (N)** – a ruler who oppresses his or her subjects tyrant. **(dictator, autocrat, strong man)**

Answer:

38. **HARMONIOUS**

A. harsh C. compatible

B. discordant D. grating

Clues: **harmonious (adj)** – able to get along together we combining different elements that blend pleasingl melodious **(agreeable, tuneful)**

Answer: C

39. **INDISPOSED**

A. ailing B. unwell C. reluctant D. all are

Clues: **indisposed (adj part)** – slightly ill; disinclined to do something.

Answer:

40. **WAN**

A. bloodless C. ruddy

B. rosy D. blooming

Clues: **wan (adj)** – unnaturally pale or sickly looking; lackin vitality; dim, faint; weak, ineffectual **(ashen, pasty, pallid, gaunt)**

Answer:

41. **NUTRITIOUS**

A. healthful C. unwell

B. wholesome D. A and B

Clues: **nutritious (adj)** – nourishing, valuable and satisfying as food.

Answer:

42. **OPPRESS**

A. free B. mistreat C. liberate D. pamper

Clues: **oppress (v)** – to govern or rule cruelly or unjustly; to weight heavily upon **(coddle)**. **Answer: B**

VANQUISH

A. succumb to
B. fill with
C. conquer
D. yield to

Clues: **vanquish (v)** – to defeat in a battle or cotest, overthrow; to overcome a feeling or condition.)

Answer: C

REPUTABLE

A. reliable
B. respectable
C. trustworthy
D. all are

Clues: **reputable (adj)** – well thought of, having a good reputation.

Answer: D

PALL

A. bore B. weary C. fascinate D. A and B

Clues: **pall (v)** – to lose in interest, attraction, or effectiveness; to become tiresome.

pall (N) – a dark covering, something that conceals.

Answer: D

FLUCTUATE

A. oscillate
B. stay put
C. remain unchanged
D. delay

Clues: **fluctuate (v)** – to change continually; to move up and down **(waver, seesaw, oscillate).**

Answer: A

INDIGNANT

A. quench
B. inflame
C. extinguish
D. put out

Clues: **ignite (v)** – to set on fire, cause to burn; to heat up, exist **(light, kindle)**

Answer: B

DIMINISH

A. increase B. enlarge C. decrease D. augment

Clues: **diminish (v)** – to make or become smaller, reduce in size. **(lessen, dwindle).**

Answer: C

49. **FOSTER**

A. nurture B. support C. cultivate D. all are

Clues: **foster (v)** – to bring up, give care to; to promote, encourage;

foster (adj) – in the same family but not related by birth.

Answer: D

50. **ENCHANT**

A. delight B. nauseate C. bore D. disgust

Clues: **enchant (v)** – to please greatly, to charm, put under a magic spell **(thrill, bewitch)**

Answer: A

51. **IMPARTIAL**

A. one-sided
B. neutral
C. prejudiced
D. biased

Clues: **impartial (adj)** – just, unbiased, fair, note talking sides (disinterested, objective, neutral).

Answer: B

52. **INFLATE**

A. flatten B. diminish C. deflate D. exaggerate

Clues: **inflate (v)** – to fill with air or gas; to swell or puff out; to make something appear larger than it is. **(blow up, pump up, enlarge)**

Answer: D

53. **PELT**

A. spend B. sweep C. bombard D. relieved

Clues: **pelt (v)** – to throw a stream of things; to strike successively; to hurry. **(shower pepper).**

Answer: C

54. **BUFFOON**

A. fool B. wise C. educated D. active

Clues: **button (N)** – a clown; a coarse, stupid person, jester

Answer: A

57. **CONTROVERSIAL**

A. arguable C. agreeable
B. debatable D. A and B

Clues: **controversial (adj)** – arousing argument, dispute, or disagreement.

Answer: D

58. **CONSOLIDATE**

A. scatter B. strengthen C. disperse D. separate

Clues: **consolidate (v)** – to combine, unite, to make solid, or firm **(firm up, merge)**

Answer: B

59. **DOCILE**

A. disobedient C. teachable
B. intractable D. wayward

Clues: **docile (adj)** – easily taught, led, or managed; obedient. **(manageable, teachable, pliant)**

Answer: C

60. **CANVASS**

A. solicit B. throw C. control D. analyise

Clues: **canvass (v)** – to go through an area in order to procure votes, sales, or opinions; to go over in detail; to discuss **(poll, survey)**

Answer: A

61. **FLOUNDER**

A. wallow B. struggle C. settle D. A and B

Clues: **flounder (v)** – to thrash about in a clumsy or ineffective way.

Answer: D

62. **MELANCHOLY**

A. happy B. depressed C. cheerful D. merry

Clues: **melancholy (adj)** – depressed, dejected
melancholy (N) – depression, dejection

Answer: B

63. **FUTILE**

A. vain B. successful C. effective D. achieved

Clues: **futile (adj)** – not successful, failing to have any result; useless, unimportant, frivolous, fruitless, ineffective.

Answer: A

62. **MORBID**

A. depressed
B. unsound
C. sick
D. all are

Clues: **morbid (adj)** – unhealthy mental state, extremely gloomy; caused by or related to disease, unwholesome.

Answer: B

63. **NOTORIOUS**

A. unknown
B. obscure
C. disgraceful
D. respectable

Clues: **notorious (adj)** – widely known because of bad conduct **(disgraceful, infamous, disreputable)**

Answer: C

64. **MISDEMEANOR**

A. petty offense
B. transgression
C. mislead
D. all are

Clues: **misdemeanor (N)** – a crime or offense that is less serious than a felony; any minor misbehavior or misconduct.

Answer: D

65. **STALEMATE**

A. victory over
B. defeat
C. deadlock
D. come

Clues: **stalemate (N)** – standoff, draw
stalemate (v) – deadlock

Answer: C

66. **NARRATIVE**

A. tale
B. poem
C. rhythm
D. mood

Clues: **narrative (n)** – a story, detailed report.
narrative (adj) – having the quality or the nature of a story. **(tale, chronicle)**.

Answer: A

67. **CHRONOLOGICAL**

A. in time sequence
B. consecutive
C. unorganized time
D. A and B

Clues: **chronological (adj)** arranged in order of time of occurrence.

68. **TEDIOUS**

A. tiresome B. easy C. interesting D. specific

Clues: **tedious (adj)**- wearisome, or boring to do. **Answer: A**

69. **OPTIMIST**

A. unpromised C. hopeful

B. useless D. unpredictable

Clues: **optimist (adj)** – someone who hopes that a good thing /time will come in the future.

☞ **opposite: pessimist (=hopeless)** **Answer: C**

70. **WANDER**

A. roam without purpose C. read silently

B. think seriously D. do intentionally

Clues: **wander (v)** – to roam without purpose and direction, proceed by an indirect road. **Answer: A**

71. **WAYFARER**

A. a travel on bus C. a travel by car

B. a travel on foot D. a travel by boat

Clues: **wayfarer (n)** – is someone who travels on foot.

Answer: B

72. **VAIN**

A. useless B. hopeless C. conceited D. all are

Answer: D

73. **PRODIGY**

A. exceptional talents C. common course

B. special foods D. less in number

Clues: **prodigy (adj)** – if someone is prodigy, they have special talents. **Answer: A**

74. **RESTRAIN**

A. check B. hold back C rebook D. A and B

Clues: **restrain (v)** – check and holdback; resist performance of a behavior.

Answer: D

75. **SACRIFICE**

A. offering B. aggressive C. occurring D. gaining

Clues: **sacrifice (N)** – offering.

sacrifice (v) – offer as a sacrifice; sell or give away at loss.

Answer: A

76. **HALLOW**

A. impure C. underestimate

B. honor D. discourage

Clues: **hallow (v)** – sanctify, honor, or make holy

Answer: B

77. **IMMENSE**

A. narrow C. boundless

B. limited D. tiny

Clues: **immense (adj)** – extremely large, vast, enormous.

Answer: C

78. **HOSTILITY**

A. partnership C. antagonism

B. gloomy D. friendship

Clues: **hostility (n)** – resentfulness, anger, antagonism.

Answer: C

79. **INCREDULOUS**

A. unbelieving C. hopeful

B. skeptical D. A and B

Clues: **incredulous (adj)** – is to mean in a state of disbelieve (incredible).

Answer: D

80. **INANIMATE**

A. lifeless B. living C. grief D. all are

Hints: **inanimate (adj)** – means without life or spirit

Answer: A

81. **LAUDABLE**
A. laughable
B. excellent
C. melancholy
D. useless
Clues: **laudable (adj)** – worthy of praise; excellent; commendable.
Answer: B

82. **GARGANTUAN**
A. very small
B. gigantic in size
C. tiny
D. little
Hints: **gargantuan (adj)** – incredibly large, gigantic in size.
Answer: B

83. **MISERY**
A. amazement
B. carefulness
C. comfortable
D. wretchedness
Clues: **misery (n)** – state of extremely suffering. **Answer: D**

84. **LUXURY**
A. badness
B. very easy
C. expensive living
D. bagary
Clues: **luxury (n)** – high or expensive living; an unnecessary comfort or enjoyment. **Answer: C**

85. **INSTIGATE**
A. urge on
B. encourage
C. imitate
D. A and B **Answer: D**

86. **GRIEVOUS**
A. harmonious to hear
B. sweet to eat
C. difficult to hear
D. ugly to face
Clues: **grievous (adj)** – filled with grief, offensive **Answer: C**

87. **CHARMING**
A. attractive
B. pleasing
C. pretty
D. all are **Answer: D**

88. **CONTEMPORARY**
A. modern
B. ancient
C. outdate
D. permanent

Clues: **contemporary (n)** on living in the same era.
contemporary (adj) – modern, current, of the same era.

Answer: A

89. **DECREE**

A. edict B. order C. decision D. all are

Clues: **decree (n/v)** – issue an order

Answer: D

90. **CATASTROPHE**

A. caltivate B. disaster C. discern D. disorder

Clues: **catastrophe (n)** – calamity such as flood, fire, or famine.

Answer: B

91. **DISTINGUISHED**

A. ignored B. famous C. estimated D. all are

Clues: **distinguished (adj)** – eminent, noted, well-known, famous

Answer: B

92. **ENDEAVOR**

A. try hard
B. work toward a goal
C. an attempt
D. all are

Answer: D

93. **DISGUST**

A. dislike
B. make guess
C. disagree
D. dismiss

Clues: **disgust (v)** – distaste, cause loathing or contempt.

Answer: A

94. **CENSURE**

A. think seriously
B. take care of
C. critize sharply
D. support fairly

Clues: **censure (v)** – expression of blame or disapproval.

Answer: C

95. **CONSULT**

A. give or seek advice
B. exchange views
C. speak rudely
D. A and B

Answer: D

Synonyms

ARDUOS

A. very easy
B. strenuous
C. previous
D. ambitions

Clues: **arduos (adj)** difficult, filled with hardships.

Answer: B

ANXIOUS

A. worried B. excited C. surprised D. disastrous

Clues: **anxious (adj)** – nervously expectant; desirous

Answer: A

ANGUISH

A. an astonishing state
B. serve physical or mental pain
C. hardship time
D. good attention

Hints: **anguish (n /v)** – feel or cause to feel anguish.
– serious mental depression **Answer: B**

BEREAVEMENT

A. deprivation
B. loss
C. gain
D. A and B **Answer: D**

AESTHETIC

A. involving beauty
B. the artistic sense
C. the study of athletics
D. A and B

Clues: **aesthetic (adj)** – having a feeling for what is a beautiful.

Answer: D

ASSUAGE

A. multiply
B. falsely
C. satisfy
D. identify

Clues: **assuage (v)** – ease; soothe, satisfy **Answer: C**

CRUMPLE

A. jump B. collage C. decrease D. plant

Clues: **crumple (v)** – to wad up as paper; to fall down or collapse.

Answer: B

103. **DEPICT**

A. to make vague
B. describe clearly
C. discharge
D. elicit

Hints: **depict (v)** – precise description; portray

Answer: B

104. **VISIONARY**

A. hopeless
B. specialist
C. idealist
D. visible

Clues: **visionary (n)** – dreamer, idealist
visionary (adj) – dreamy, imaginary, impractical

Answer: C

105. **SURREPTITIOUS**

A. stealthy B. sneaky C. surpass D. A and B

Answer: D

106. **STRENOUS**

A. walk slowly
B. needing great effort
C. lack energy
D. all are

Clues: **strenuous (v)** – requiring great exertion.

Answer: B

107. **VERILY**

A. in truth B. assuredly C. credibly D. all are

Answer: D

108. **THREATEN**

A. take care of
B. strive
C. try on
D. endanger

Clues: **threaten (v)** – endanger, state intent to do harm.

Answer: D

109. **WINDFALL**

A. unexpected good fortune
B. well-being
C. good will of one self
D. disagreement

Clues: **windfall (n)** – unexpected pain

Answer: A

110. **UNIVERSAL**

A. limited to
B. worldwide
C. common
D. specific

Hints: **universal (adj)** having general application; common to several purpose; worldwide, international.

Answer: B

STOUT

A. heavy B. gloom C. light D. easy

Clues: **stout (adj)** – heavy or robust; resolute or bold in character.

Answer: A

PERILOUS

A. not safe
B. precious
C. peace and quiet
D. obligatory

Clues: **perilous (adj)** – dangerous, not safe, put at risk.

Answer: A

NOSTALGIC

A. wistful
B. sentimental towards
C. think affectionately about the past.
D. all are

Answer: D

OVERWHELM

A. overpower
B. over weight
C. overtake
D. over up

Clues: **overwhelm (v)** if a group of people. **overwhelm** a place or another group, they gain complete control or victory over them.

overwhelming (adj) if something is **overwhelming,** it affects you very strongly, and you don't know how to deal with it **(smother, overpower, overthrow, bury)**

Answer: A

PORTABLE

A. which is easily eaten
B. which is easily read
C. which is easily understood
D. capable of being carried or moved

Clues: **portable (adj)** – a portable machine or device is designed to be easily **carried or moved.**

Answer: D

116. **PORTER**

A. someone whose job is to carry things- luggage
B. someone whose job is to be in charge of the entrance of a building – a hotel
C. someone who plays all types of games
D. A and B

Answer: B

117. **PREVATION**

A. a state of living in poverty
B. going without life's basic needs
C. living a luxurious life
D. A and B

Answer: B

118. **VAGUE**

A. clear
B. obscure
C. obvious
D. descriptive

<u>Clues</u>: **vague (adj)** – dimly outlined, lacking in clarity, obscure.

Answer: B

119. **NUMB**

A. null
B. motive
C. active
D. dull or unfeeling state

<u>Clues</u>: **numb(n)** – insensible, as from cold.

Answer: D

120. **RELIEVE**

A. ease B. unhealthy C. devastate D. react

<u>Clues</u>: **relieve (v)** – ease or lessen, especially of pain.

Answer: A

121. **OCTOGENARIAN**

A. too young person
B. too old person
C. too fat person
D. too thin person

<u>Clues</u>: **octogenarian (n)** – a person between eighty and ninety years old. (too old person)

Answer: B

122. **PROLIFERATION**

A. slow in number
B. decrease in amount
C. rapid reproduction
D. grow steadily

Clues: **proliferation (N)** – rapid reproduction and increase in number, grow fast.

Answer: C

123. **RAPACIOUS**

A. greedy B. voracious C. rapid D. A and B

Clues: **rapacious (adj)** – describing someone or you disapprove of their greedy or selfish behavior.

Answer: D

124. **BEESECH**

A. request
B. agreement
C. refusal
D. invite

Clues: **beseech (v)** – request, plea, implore

Answer: A

125. **ADVERSITY**

A. trouble
B. difficult circumstance
C. misfortune
D. all are

Hints: **adversity (n)** – a disastrous event.

Answer: D

126. **COLLEAGUE**

A. an associate
B. a college pupil
C. common property
D. all are

Clues: **colleague (n)** – an associate, often a person in a work place.

Answer: A

127. **FASTIDIOUS**

A. fussy
B. particular
C. painstaking
D. all are

Clues: **fastidious (adj)** – if you describe that someone is **fastidious**, you mean that they pay great attention to detail because they like everything to be very clear, neat, accurate and in good order.

Answer: D

128. **BEEF**

A. meat from a cow
B. meat from a sheep
C. meet from a goat
D. meat from chicken

lamb: **a flesh of a young sheep.**

Answer: A

129. **PERMANENT**

A. unlimited length of time
B. temporary event
C. occurrence
D. sequence of time

Answer: A

130. **MODEST**

A. unexpected
B. understated
C. modern
D. boldness

Clues: **modest (adj)** – understated, unpretentious, shy; not large.

Answer: B

131. **MONOTONOUS**

A. repetitive
B. dull
C. unvarying in sound or tone
D. all are

Clues: **monotonous (adj)** – something that is **monotonous** is very boring because it has a regular repeated pattern which never changes.

Answer: D

132. **MISCONCEPTION**

A. mistaken belief or idea
B. erroneous perception
C. wrongly realized
D. all are

Answer: D

133. **INCULCATE**

A. incomplete
B. calculate
C. instruct
D. incorrect

Clues: **inculcate (v)** – to instruct, teach; instill ideas.

Answer: C

134. **INDIFFERENT**

A. unconcerned
B. inanimate
C. not different
D. differ

Clues: **indifferent (adj)** – unconcerned; without preference; mediocre

Answer: A

135. **INTUITION**

A. famous B. impression C. insight D. caution

Clues: **intuition (N)** – knowledge by instinct; hunch

Answer: C

136. **MELLOW**

A. fellow B. mature C. insight D. caution

Clues: **mellow (adj)** – ripe, mature, golden; not easily ruffled by life's ups and down.

Answer: B

137. **IMPOSTOR**

A. cheat B. empower C. impose D. decline

Clues: **impostor (n)** – a swindler, deceiver; one who uses a false name or character in order to cheat.

Answer: A

138. **PEDESTRIAN**

A. commonplace C. driver /rider

B. prosaic D. A and B

Clues: **pedestrian (n)** – one who goes on foot.

pedestrian (adj) – relating to walking; on foot; ordinary, dull, unimaginative.

Answer: D

139. **RITE**

A. participation C. involvement

B. observance D. kite

Clues: **rite (n)** – a ceremony; the customary form of a ceremony; any formal custom or practice

(observance, ritual, liturgy)

Answer: B

140. **DOMINATE**

A. control B. govern C. overlook D. all are

Clues: **dominate (v)** – to rule over by strength or power, control; to tower over, command due to height.

Answer: D

141. **NURTURE**

A. raise B. resist C. eradicate D. demolish

Clues: **nurture (v)** – to boring up, care for, train nourish.

nurture (v) – rearing, training, upbringing.

Answer: A

142. **ONSLAUGHT**

A. assault C. foray E. all are

B. charge D. onset

Clues: **onslaught (n)** – a violent attack; a sudden rush of something.

Answer: E

143. **TARNISH**

A. remove shine C. taint

B. become dull D. all are

Answer: D

144. **VEHEMENCE**

A. violence
B. establishment
C. carefulness
D. hardy

Clues: **vehemence (n)** – forceful and emotional expression.

Answer: A

145. **VICINITY**

A. clarity
B. neighborhood
C. similarity
D. far away

Clues: **vicinity (n)** – nearby or surrounding; locate

Answer: B

146. **SEETHE**

A. bubble B. churn C. agitate D. all are

Answer: D

147. **SAVORY**

A. open appetite
B. make favour
C. make hungry
D. remove pain

Clues: **savory (adj)** – appetizing to the sense of taste or smell.

Answer: A

148. **SPUNK**

A. enthusiasm
B. courage
C. desire to achieve
D. all

Clues: **spunk (n)** – is to mean courage, spur, stimulate, urge on

Answer: D

149. **TERMINATE**

A. depart B. begin C. end D. hasten

Clues: **Terminate** (v) is the opposite of **begin /start.**

Answer: C

150. **HEARTFELT**

A. sincere
B. interesting
C. upsetting
D. shocking

Clues: **heartfelt (adj)** felt or believed deeply and sincerely (genuine, sincere, true).

Answer: A

Review Exercise

Circle or choose the word that most nearly express the meaning of the word in CAPITAL.

ABNORMAL
A. typical B. terrible C. funny D. unusual

LUBRICATE
A. start B. oil C. stop D. repair

ULTIMATE
A. temporary B. hasty C. sincere D. final

VENGEANCE
A. mercy B. revenge C. daring D. fate

PROFICIENT
A. in experienced C. slow
B. awkward D. skillful

FUGITIVE
A. criminal B. runaway C. jailer D. victim

PRIOR
A. earlier B. intelligent C. necessary D. careful

HOMICIDE
A. killing B. lying C. stealing D. cheating

PROMINENT
A. shady B. wealthy C. leading D. foreign

WRATH
A. troubles B. rage
C. self-steem D. temperature

RELUCTANT
A. experienced C. unwilling
B. awkward D. paid

ININVATE
A. claim B. deny C. admit D. imply

13. **LEGITIMATE**
A. lawful B. foreign C. new D. tyrannical

14. **SAGACIOUS**
A. nasty B. humorous C. kind D. wise

15. **LEGIBLE**
A. detailed B. recentC. readable D. thorough

16. **HUMDRUM**
A. excising B. poor C. unusual D. dull

17. **DISPUTATIOUS**
A. likable
B. annoying
C. logical
D. argumentative

18. **VIGILANT**
A. careless B. carefree C. merciless D. alert

19. **PROCURE**
A. seek B. offer C. reject D. obtain

20. **INDULGE**
A. coddle B. discipline C. mistreat D. feed

21. **MISCELLANEOUS**
A. worthless B. expensive C. various D. similar

22. **INFLICT**
A. deal out B. cure C. study D. fear

23. **VICIOUS**
A. gentile B. trained C. savage D. useful

24. **FLOURISH**
A. hold B. seize C. destroy D. wave

25. **SODDEN**
A. little B. wet C. huge D. messy

26. **MELANCHOLY**
A. typical B. angry C. strange D. sad

27. **INTERROGATE**
A. question B. believe C. punish D. listen to

28. **HOAX**
A. fraud B. nobody C. thief D. problem

Synonyms

1. **LOOM**
   A. see B. notice C. discover D. appear
2. **CUSTOMARY**
   A. wrong B. profitable C. clever D. usual
3. **LITERATE**
   A. ignorant B. unlettered C. educated D. messy
4. **ANIMATED**
   A. lively B. dull C. zoological D. lifeless
5. **ORATION**
   A. eating B. listening C. celebration D. speech
6. **BARREN**
   A. fertile B. arid C. lush D. productive
7. **INFLAMMABLE**
   A. fire proof C. calm
   B. fire-resistant D. combustible
8. **DYNAMIC**
   A. explosive C. lethargic
   B. energetic D. sturdy
9. **UTMOST**
   A. strangest C. greatest
   B. high-powered D. least
10. **RELUCTANT**
    A. hesitant C. undecided
    B. willing D. delighted
11. **FRUISTRATE**
    A. assist B. foil C. settle D. mislead
12. **CONTROVERSIAL**
    A. patriotic B. debatable C. agreeable D. star
13. **SUBSTANTIAL**
    A. important B. prosperous C. tangible D. affordable

42. **CATASTOROPHE**
A. triumph B. success C. tragedy D. battle

43. **INCENTIVE**
A. pressure C. hindrance
B. restraint D. inducement

44. **PARCHED**
A. saturated B. rugged C. enclosed D. dry

45. **REGAL**
A. tall B. hardy C. majestic D. lowly

46. **CUSUAL**
A. occasional C. unconcerned
B. personal D. accidental

47. **PUNY**
A. sickly C. undersized
B. unimportant D. off the wall

48. **FLAGRANT**
A. youthful C. serious
B. petty D. unexplained

49. **DILAPIDATED**
A. messy C. confused
B. shipshape D. run-down

50. **BENEFICIARY**
A. official B. recipient C. target D. enemy

51. **MULL**
A. consider C. make empty
B. reflection D. A and B

52. **LEISURELY**
A. hasty B. unhurried C. hectic D. playful

53. **SULLEN**
A. silent B. sluggish C. morose D. surly

54. **VOID**
A. annulment C. vacancy
B. cancellation D. promotion

Answer Key

D	11. C	21. C	31. C	41. A	51. D
B	12. D	22. D	32. A	42. C	52. D
D	13. A	23. C	33. D	43. D	53. A
B	14. D	24. D	34. B	44. D	54. C
D	15. C	25. B	35. D	45. C	
D	16. D	26. D	36. B	46. C	
A	17. D	27. A	37. C	47. C	
A	18. B	28. A	38. A	48. C	
C	19. D	29. A	39. B	49. D	
10. B	20. A	30. D	40. B	50. B	

UNIT THREE

ANALOGIES

What are analogies?

- A **verbal analogy** expresses a relationship or comparison between sets of words.
- A complete analogy compares the two pairs of words and makes a statement about them. It asserts that the relationship between the first pair of words is the same as the relationship between the second.
- Generally, **analogies** require you to define how two sets of words are **alike** or **different**. Sometimes those words express a **similar** relationship, and sometimes the words express an **opposite** relationship. (i.e to choose the pair of words that best matches or parallels the relationship of key, or given, pair of words.)

Here are two examples

1. **maple** is to **tree** is as
   a. acorn is to oak.
   b. hen is to rooster
   c. rose is to flower
   d. shrub is to lilac
2. **joyful** is to **gloomy** as
   a. cheerful is to happy
   b. strong is to weak
   c. quick is to famous
   d. hungry is to starving

Clues: In order to find the correct answer to exercise **1**, you must first determine the relationship between the two key words, **maple** and **tree**. In this case, that relationship might be expressed as "a **maple** is a kind (or type) of tree." The next is to select from choices **a, b, c** and **d** the pair of words that best reflects the same relationship.

Clearly, the correct answer is **(C)**; it is the only choice that parallels the relationship of the key words:
A **rose** is a kind (or type) of flower, just as a **maple** is a kind (or type) of tree. The other choices do not express the same relationship.

In exercise **2,** the relationship between the key words can be expressed as **"joyful** means the opposite of **gloomy."** Which of the choices best represents the same relationship? The answer, of course, is **(b)**; **"strong"** means the opposite of **"weak."**

☞ **Follow also the following techniques with its patterns.**

✧ We have said that **analogies** are very helpful in understanding word **relationships**. It is also a partial similarity between things that are somewhat different.

☞ **see these words:**

dog : cat

✧ These two words can be used to start an **analogy**. the sign [:] means "**is to**". The analogy is continued by adding another sign **[::]** meaning "**as**" Let's add these two signs to our two words:

dog : cat ::

(dog is **to** cat **as** cat is **to**)

✧ What you are looking at so far is expressed like this. As indicated above, **"dog is to cat as..."** but now we need a work which expresses a relationship with cat. So continuing, the pattern will appear like this:

dog : cat :: cat :

✧ Dog and cat are natural enemies. The analogy can be completed by inserting the name of an animal that is the cat's natural enemy. The completed analogy will look like this.

dog : cat :: cat : rat

✧ You can probably see how **analogies, synonyms,** and **antonyms** go hand in hand. It is easy to see that just knowing whether key words are the same or opposite can be very helpful. In this book is

a typical analogy problems (questions), the kind of problem you will be dealing with as you work your way through this book. What answer would you choose to complete the following analogy?

pleasure : pain :: win : _____?

a. victory b. joy c. lose

These pair of words have the opposite relationship **(pleasure = happiness; pain = ailness).**

Answer: (

Analogy Questions
(General Tactics)

✧ As earlier mentioned, in the **analogy question,** you are given a **pair of words** which have some kind of relationship. You are asked to select from four and sometimes five pairs of words the pair which has the same relationship as the first two words. Follow the following **strategies**:

1. Before you look at the choices, try to state the relationship between the **CAPITALIZED WORDS** in a good sentences. Then use the word pairs from the answer choices in the same sentence. Frequently, only one will make sense, and you'll have the correct answer.
2. Don't be misled if the choices are from different fields or areas, or seem to deal with different items, from the given pair. Study the capitalized words until you see the connection between them; then search for the same relationship among the choices.

Examples

BOTANIST: MICROSCOPE :: CARPENTER : HAMMER

even though the two workers may have little else in common besides their uses of tools.

3. If more than one answer fits the relationship in your sentence, look for a narrower approach.

Examples

MITTEN : HAND ::

A. bracelet : wrist
B. belt : waist
C. muffler : neck
D. ring : finger
E. sandal : foot

you make up the sentence, "you wear a **mitten** on your **hand**." Unfortunately, all the answer choices will fit that sentence. So you say to yourself "why do you wear a **mitten**? You wear a **mitten** to keep your hand warm." Now when you try to substitute, only choice (C) works, so you've your answer.

Watch out for errors steaming from grammatical reveals. Ask yourself *who* is doing what to whom.

Examples

FUGITIVE : FLEE is not the same as **LAUGHINGSTOCK: MOCK**.

- A **fugitive** is a person who **flees**. A **laughingstock** is a person who **is mocked**.

Be familiar with the whole range of common analogy types. Know the usual ways in which pairs of words on the **SAT** are linked.

Examples

DAUNTLESS: COURAGEOUS

- **dauntless** (fearless) and **courageous** are synonyms.

DAUNTLESS: COWARDLY

- Someone **dauntless** doesn't exhibit cowardice.

POET: SONNET

- A **poet** creates a **sonnet**.

PAINTER: BRUSH

- A **painter** uses a **brush**.

SAW: WOOD

- A **saw** cuts **wood**.

CROWBAR: PRY

- A **crowbar** is a tool used to **pry**.

NOD: ASSENT

- A **nod** is a sign of **assent** (agreement).

STAMMER: TALK

- To **stammer** is to **talk** in a halting manner.

LUKEWARM: BOILING

- **Lukewarm** is less intense than **boiling.**

WHALE: MAMMAL

- A **whale** is a member of the class know as **mammal.**

TIGER: CARNIVOROUS

- A **tiger** is by definition a **carnivorous** (meat-eating) animal.

ARCHIPELAGO: ISLAND

- An **archipelago** (chain of islands) is made up of many islands.

DOE: STAGE

- A **doe** is a female deer; a **stag**, a male deer.

DOVE: PEACE

- A **dove** is the symbol of peace.

Common Relationship found in analogy Questions

☞ There are many different kinds of relationships represented in the **analogy questions** you will find in this book.

The common relationships include:

Type of Analogy		Examples
actions to object	→	play: clarinet
causes to effect	→	sun: sunburn
item to category	→	iguana: reptile
object to its material	→	curtains : cloth
object to its function	→	pencil; writing
part to whole	→	page : book
time sequence	→	recent: current
type to characteristic	→	dancer : agile
word to antonym	→	help: hinder
word to synonym	→	provisions: supplies
worker and creation	→	artist: sketch
worker and tool	→	lumberjack: saw

More common Analogy Types

definition
REFUGE : **SHELTER**

A refuge (place of asylum) by definition **shelters**

NOMAD : **WANDER**

A nomad by definition **wanders.**

HAGGLER : **BARGAIN**

A haggler, a person who argues over prices, by definition bargain.

Defining Characteristic

TIGER : **CARNIVOROUS**

A tiger is defined as a carnivorous or meat eating animal.

ENTOMOLOGIST : **INSECTS**

An **intomologist** is defined as a person who studies insects.

HIVE : **BEE**

A hive is defined as a home for bees.

C. Class **and member**

RODENT : **SQUUIRREL**

A **squirrel** is a kind rodent.

SOFA : **FURNITURE**

A **sofa** belongs to the category known as furniture.

SONNET : **POEM**

A **sonnet** is a kind of poem.

D. Group **and Member**

DANCER : **ENSEMBLE**

A **dancer** is a member of an ensemble or troupe.

LION : **PRIDE**

A **lion** is a member of a pride or company.

GAGGLE : **GEESE**

- A **gaggle** is group or flock of geese.

E. Antonyms

✧ **Antonyms** are words that are opposite in meaning. Both words belong to the same part of speech.

CONCERNED : INDIFFERENT

- indifferent means **unconcerned.**

WAX : WANE

- wax, to grow larger, and wane, to dwindle, are opposites.

ANARCHY : ORDER

- **Anarchy** is the opposite of **order.**

F. Antonym Variants

In an **antonym variant**, the words are not strictly antonyms; however, their meanings are opposed. Take the adjective "nervous." A strict antonym for the adjective **nervous** would be the adjective **poised.** However, where an Antonym would put the adjective poised. An Antonym variant puts the noun poise. It looks like this:

NERVIOUS : POISE

- Nervous means lacking in **poise.**

WICKED : VIRTUE

- Something **wicked** lacks **virtue.** It is the opposite of **virtuous.**

WILLFUL : OBDIENT

- Willful means lacking in obedience. It is the opposite of **obedient.**

G. Synonyms

Synonyms are words that have the same meaning. Both words belong to the same part of speech.

MAGNIFICENT : GRANDIOSE

- Grandiose means magnificent.

NARRATE : TELL

- To **narrate** is to **tell.**

EDIFICE : BUILDING

- An **edifice** is a building.

H. Synonym Variants

synonym variant. the words are not strictly synonyms; however. ...t meanings are similar. For example, take the adjective "**willful**" a ...t synonym for the adjective willful would be the adjective **unruly.** ...ever, where a synonym would put the noun **unruliness**. It looks ...e this:

WILLFUL: UNRULINESS

- Willful means exhibiting unruliness.

VERBOSE : WORDINESS

- Someone verbose is **wordy;** he or she exhibits wordness.

FRIENDLY : AMICABILITY

- Someone friendly is amicable; he or she shows **amicability**.

I. **Degree of Intensity**

LUKEWARM : BOILING

- **Lukewarm** is less extreme than boiling.

FLURRY : BLIZZARD

- A flurry or shower of snow is less extreme than a blizzard.

ANNOYED : FURIOUS

- To be **annoyed** is less intense an emotion than to be **frious.**

J. **Part to whole**

ISLAND : ARCHIPELAGO

- Many islands make up an archipelago.

LETTER : ALPHABET

- The English alphabet is made up of 26 letters.

FINGER : HAND

- The finger is part of the hand.

K. **Part to whole**

ASYLUM : REFUGE

- An **asylum** provides **refuge** or protection.

FEET : MARCH

- A function of **feet** is to <u>march</u>.

LULL : STORM

- A <u>lull</u> temporarily interrupts a <u>storm</u>.

L. Manner

MUMBLE : SPEAK

- To **mumble** is to <u>speak indistinctly</u>, that is, to <u>speak</u> in an <u>indistinct</u> manner.

STRUT : WALK

- To **strut** is to **walk** proudly, that is, to <u>walk</u> in a <u>proud manner</u>.

STRAINED : WIT

- **Wit** that is <u>strained</u> is forced in <u>manner</u>.

M. Worker and Article created

POET : SONNET

- A **poet** <u>creates</u> a **sonnet**.

ARCHITECT : BLUEPRINT

- An **architect** designs a **blueprint**.

MASON : WALL

- A **mason** <u>builds</u> a **wall**.

N. Worker and Tool

PAINTER : BRUSH.

- A <u>painter</u> uses a <u>brush</u>.

GOLFER : CLUB

- A **golfer** uses a **club** to strike the ball.

CARPENTER : VISE

- A <u>carpenter</u> uses a <u>vise</u> to hold the object being worked on.

N. Worker and Tool

PAINTER : BRUSH

- A <u>painter</u> uses a <u>brush</u>.

GOLFER : CLUB

- A **golfer** uses a **club** to strike the ball.

CARPENTER : VISE

A carpenter uses a vise to hold the object being worked on.

Worker and Action

ACROBAT : CARTIWHEEL

An **acrobat** performs a cartwheel.

FINANCIER : INVEST

A financier invests.

TENOR : ARIA

A tenor sings an aria.

Worker and workplace

TEACHER : CLASSROOM

A teacher works in a classroom.

SCULPTOR : STUDIO

A sculptor works in a studio.

DRUGGIST : PHARMACY

A **druggist** works in a pharmacy.

Tool and object it acts upon

KNIFE : BREAD

A knife cuts bread.

PEN : PAPER

A pen writes on paper.

RAKE : LEAVES

A rake gathers leaves.

Tool and its Action

SAW : CUT

Saw is a tool used to cut wood.

CROWBAR : PRY

A **crowbar** is a tool used to pry things apart.

SIEVE : SIFT

- A **sieve** is a tool used to strain or sift.

S. Action and its Significance

HUG : AFFECTION

- A hug is a sign of affection.

NOD : ASSENT

- A nod signifies assent or agreement.

WINCE : PAN

- A wince is a sign that one feels pain.

T. Cause and effect

VIRUS : INFLUENZA

- A virus causes influenza.

FIRE : ASHES

- Fire causes ashes.

U. Time Sequence

FIRST : LAST

- **First** and **last** mark the beginning and end of a sequence.

V. Spatial sequence

ATTIC : BASEMENT

- The attic is the highest point in the house; the basement, the lowest point.

W. Gender

DOE : STAG

- A doe is a female deer, a stag, a male deer.

X. Age

COLT : STALLION

- A **colt** is a young stallion.

Y. Symbol and Abstraction it represents

DOVE : PEACE

- A **dove** is a symbol of peace.

Practice

For each of the following Paris of words, name the common analogy type to which it belongs.

1. SURGEON : SCALGEL _____
2. BARK : TREE _____
3. FLOWER : PEONY _____
4. DRILL : BORE _____
5. MINUTE : HOUR _____
6. COW : HERBIVOROUS _____
7. TENT : SHETER _____
8. EWE : RAM _____
9. SHOAT : PIG _____
10. LAUREL : VICTORY _____
11. FAWN : DEER _____
12. MEAL : LUNCH _____
13. SCULPTOR : STATUE _____
14. TELLER : BANK _____
15. DISPERSE : ASSEMBLE _____
16. DRENCHED : MOIST _____
17. ADORE : LOATHE _____
18. SCULPTOR : MALLET _____
19. VERACIOUS : TRUTHFUL _____
20. ANGLER : FISH _____
21. BUTCHER : CLEAVER _____
22. RIDDLE : CRYPTIC _____
23. KAYAK : BOAT _____
24. BROOM : SWEEP _____
25. SPORIFIC : SLEEP _____

Answer for the above analogy type

1. Worker and tool
2. Part of whole
3. Class and member
4. Tool and its action. A drill is a tool used to bore holes.
5. Part to whole. A minute is part of an hour.
6. Defining characteristic. A cow is defined as herbivorous.
7. Class and member. A tent is a kind of shelter.
8. Defining characteristics. An ewe is a female sheep: a ram, a male sheep.
9. Defining characteristic. A shoat is a young pig.
10. Symbol and what it represents. The laurel is the symbol of victory.
11. Defining characteristic. A fawn is a young deer.
12. Class and member. One example of a meal is lunch.
13. Worker and article created. A sculptor creates a statue.
14. Worker and workplace. A teller works in a bank.
15. Antonyms disperse (scatter) and assemble are opposites.
16. Degree of intensity. Drenched means extremely wet; moist, only moderately so.
17. Antonyms. Adore and loathe (hate) are opposites.
18. Worker and tool. A sculptor uses a mallet.
19. Synonyms. Veracious and truthful have the same meaning.
20. Function. An angler tries to catch fish.
21. Worker and tool. A butcher uses clever.
22. Defining characteristic. A riddle is by definition cryptic (mysterious).
23. Class and member. A kayak is a kind of boat.
24. Tool and object. A broom is a tool used to sweep.
25. Cause and effect. Something soporific induces sleep.

UNIVERSITY ENTRANCE EXAMINATION AND OTHER DIFFIRENT QUESTIONS WITH DETAILED EXPLANATIONS AND EXAMPLES (UEE/EHEECE 2001 – 2009 E.C.)

DIRECTION: Each question below consists of a related pair of words or phrases, followed by four lettered pairs of words or phrases. Study and try to understand, particularity the way how to perform the analogy questions.

MASON : WALL

A. artist : easel
B. author : book
C. fisherman : trout
D. congressman : senator

Clues: A **mason** is stone worker.
He creates a wall.
An **author** creates a book.

Answer: B

STUDENT : KNOWLEDGE

A. hypocrite : truth
B. prospector : gold
C. disciple : discipline
D. actor : rehearsal

Clues: A **student** seeks knowledge.
A **prospector** seeks gold.

Answer: B

3. **FIRE : ASHES**

A. accident : delay
B. wood : splinters
C. regret : melancholy
D. event : memories

Clues: A **fire** leaves behind (results in) **ashes.**
An **event** leaves behind **memories.**

Answer: D

4. **GOOSE : GANDER**

A. dusk : drake
B. hen : chicken
C. sheep flock
D. dog : kennel

Clues: A **gander** is a male **goose.**
A **drake** is a male **duck.**
☞ The relationship between the capitalized pair of words is **gender** (maleness or femaleness).

Answer: A

5. **QUOTATION : QUOTATION MARKS**

A. remark : colon
B. aside : parentheses
C. sentence : period
D. clause : semicolon

Clues: The beginning and end of a **quotation** (group of words repeated from a book or speech) are indicated by **quotation marks**. The beginning and end of an **aside** (parenthetical mark; temporary digression from the main subject) are indicated by **parenthesis.**

Answer: B

6. **PREDICT : FORETELL**

A. procrastinate : expedite
B. lie : prevaricate
C. prophesy : vindicate
D. anticipate : participate

Clues: Someone who can **predict** the future can be said to **foretell** it.

- Someone who is known to **lie** can be said to **prevaricate** or **speak falsely**.

Answer: B

7. **CARPENTER : SAW**

A. seamstress : scissors
B. runner : sneakers
C. lawyer : brief
D. painter : brush

Clues: A **carpenter** uses a **saw** to cut things.

A **seamstress** uses a **scissors** to cut things.

Note: **seamstress** ; a woman who sews, especially, as a job.

Answer: A

8. **CAPTAIN : SHOALS**

A. corporal : sergeant
B. soldier : bush
C. doctor : hospital
D. pilot : radar

Clues: A **captain** must beware of **shoals.**

shoals: a shallow areas or sandbanks in the water.

A **solider** must beware of an **ambush.**

Note: **ambush :** a surprise attack by people lying in wait.

Answer: B

9. **WARM : FEVERISH**

A. tepid : frozen
B. angry : irate
C. industrious
D. moist : soaked

Clues: **warm** is less extreme in heat than **feverish.**

moist is less extreme in wetness than **soacked.**

Answer: D

2. **VISIBLE** is to **SEE** as

A. strong : weak
B. health : sick
C. audible : hear
D. clever : lazy

Clues: If something is **visible**, you can be definition **see** it, just as if something is **audible**, you can by definition **hear** it.

Answer: C

3. **BRAVE** is to **FAVORABLE** as

A. cowardly unfavorable
B. trained: skilled
C. student : teacher
D. sleep : dream

Clues: The tone of **brave** is **favorable** just as the tone of **cowardly** is **unfavorable**.

Answer: A

4. **BUSYBODY** is to **NOSY** as

A. lion : roar
B. klutz : clumsy
C. might : gloom
D. sound : cry

Clues: A **busybody** is by definition someone who is **nosy**, just a **klutz** is by definition someone who is **clumsy**.

Note: **cumsy** awkward in movement or performance.

Answer: B

5. **LIAR** is to **TRUTHFUL** as

A. deceit : deny
B. real : actual
C. reluctant : hesitant
D. bigot : fair-minded

Clues: A **liar** is by definition not likely to be **truthful**, just as a **bigot** is by definition not likely to be **far-minded**.

Answer: D

6. **COWARDALY** is to **COURAGE** as

A. clean : spotless
B. inanimate : lively
C. happy : sad
D. awkward : grace

Clues: Someone who is **cowardly** lacks **courage**, just as someone who is **awkward** lacks **grace**.

Answer: D

7. **BIG** is to **LARGE** as

A. narrow : wide
B. worried : nervous
C. big : small
D. tall : short

Clues: **Big** means the same thing as **large**, just as **little** means the same thing as **small** (similar or synonym relationship).

Answer: C

16. **FLEET : SHIPS**

A. pack : wolves
B. shelf : books
C. team : coaches
D. committee : meetings

Clue: A **fleet** is a group of **ships**.
A **pack** is a group of **wolves**.

Answer: A

17. **BANK : MONEY**

A. ring : diamond
B. body : germ
C. transfusion : blood
D. cask : wine

Clues: One stores /keeps **money** in a **bank**.
One stores /keeps **wine** in **cask**.

Answer: D

18. **FLIMSY : STRENGTH**

A. fancy : beauty
B. hazardous:
C. wary : caution
D. clumsy : grace

Clues: **flimsy** (adj) means **weak**, lacking **strength**.
clumsy (adj) means **awkward**, lacking **grace**.

Answer: D

19. **REFUGEE : ASYLUM**

A. tutor : education
B. hermit : solitude
C. convict : prison
D. judge : courtroom

Clues: A **refuge** seeks **asylum** (means shelter)
A **hermit** seeks **solitude** (means isolation).
Note: **asylum** (n) -- is a protection from danger.

Answer: B

20. **LIBRETTO : POET**

A. blueprint : draftsman
B. handwriting : author
C. scalpel : surgeon
D. exercise : gymnast

Clues: A **poet** creates a **libretto**.
A **draftsman** creates a **blueprint**.
Note: **libretto** – is the word of an opera.

Answer: A

21. **APPROVAL : APPLAUSE**

A. delight
B. provocation : tears
C. censure : boos
D. anger : wince

Clues: **Applause** indicates **approval**.
Boos indicates **censure** (means strong disapproval)

Answer: C

INANE : MEANING

A. aesthetic : beauty
B. random : plan
C. vacant : space
D. affluent : wealth

Clues: Something **inane (senseless)** by definition lacks **meaning**.
Something **random** (haphazard) by definition lacks **plan**.

Answer: B

ADVERTISING : SALES

A. incense : aroma
B. vaccination : disease
C. salary : bonus
D. fertilizer : crops

Clues: **Advertizing** is used to increase the growth of **sales**.
Fertilizer is used to increase the growth of **crops**.

Answer: D

MEANDER : JOURNEY

A. ramble : speech
B. soar : flight
C. clarify : point
D. indigent : wealth

Clues: To meander is to wander **aimlessly** during a **journey**.
To **ramble** is to wander aimlessly during a **speech**.

Answer: A

LACKADAISICAL : VIGOR

A. hostile : animosity
B. mendacious : improvement
C. indigent : wealth
D. devoted : dedication

Clues: **lackadaisical** means listless or lacking in **vigor, indigent** means penniless or lacking in **wealth**.

Answer: C

RAZOR : BEARD

A. barber : hair
B. sickle : grass
C. student : class
D. carburetor : gasoline

Clues: A **razor** is used to cut a **beard**.
A **sickle** is used to cut **grass**.

Answer: B

HORNS : BULL

A. mane : lion
B. wattlers : turkey
C. antlers : stag
D. hoofs : horse

Clues: A **bull** has sharp **horns** on its head.
A **stag** (a male deer) has sharp **antlers** on its head.

Answer: C

28. **TIGER : ZOOLOGY**

A. insect : entomology
B. tiger lily : botany
C. granite : biology
D. butterfly : anthropology

Clues: **zoology**, the study of animals, would include a particular animal, the **tiger**.
Botany, the study of plants, would include a particular plant, **tiger lily**.

Answer: B

29. **CHAIRPERSON : BOARD**

A. mascot : gang
B. captain : team
C. foe : battle
D. politician : plant form

Clues: The **chairperson** is the head of the **board**.
The **captain** is the head of the **team**.

Answer: A

30. **FROG : AMPHIBIAN**

A. whale : mammalian
B. otter : curst ocean
C. dog : loyal
D. cow : herbivorous

Clues: **Frog** is a member of the **amphibian** class, adapted to live both on the land and in water.
Whale is a member of the **mammalian** class, adapted to feed its young with milk from the mother.

Answer: A

31. **JUDGE: COURTHOUSE**

A. lawyer : brief
B. surgeon : hospital
C. carpenter : bench
D. architect : blue print

Clues: A **judge** works in a **courthouse**.
A **surgeon** works in a **hospital**.

Answer: B

32. **PAUPER : MONEY**

A. banker : debtors
B. teacher : school
C. moron : intelligence
D. author : publisher

Clues: A **pauper** has very little **money**.
A **moron** has very little **intelligence.**

Answer: C

33. **ALARM : TRIGGER**

A. tunnel ; dig
B. criminal ; corner
C. prison : escape
D. trap : spring

Clues: To **trigger** an **alarm** is to release it or set it off.
To **spring** a **trap** is release it or set it off.

Answer: D

CARELESSNESS : JEOPARADIZE

A. carefulness : security
B. crowding : discomfort
C. failure : discouragement
D. neglect : endanger

Clues: **Jeoparadize (v)** – to cause something to be harmed, to put something in **danger** of this happening.
Neglect (v) – to give not enough care or attention to something. (neglect ends in danger).

Answer: D

SHELTER : PROTECTION

A. incentive : motivation
B. shield : injury
C. antiseptic : cure
D. inundation : food

Clues: **shelter** is used for **protection**.
antiseptic is used for **cure**.

Answer: C

INFER is to DEDUCE as IMPLY is to _____.

A. conclude
B. generalize
C. insinuate
D. wrap up

Clues: **infer (v)** – to reach an opinion based on information (= to arrive at conclusion).
Deduce (v) – to arrive at fact, theory by reasoning
IMPLY (v) – to suggest something, indirectly rather than stating it directly.
Insinuate (v) – to suggest something in indirect way

Answer: C

INDUCTION : DISCHAGE

A. qualification : empty
B. therapy : confuse
C. calamity : repair
D. detention : release

Clues: **induction** – from specific to general
discharge – sent out from its source (general to specific)
(these two words are **antonyms**).
therapy – treatment designed to rehabilitate a person
(state of being embarrass).

Answer: B

38. **AMBIGULTY : LUCID**

A. vague : lucid
B. doubt : uncertainty
C. candid : sincere
D. open : change

Clues: **ambiguity** – unclear (not clear)
lucid – clear, easily understood
☞ The two words are **antonym (opposite)**.
vague – not clearly understood.

Answer: A

39. **BLOOD : CLOT**

A. wine : decant
B. milk : curdle
C. water : flow
D. oil : change

Clues: **clot :** is a thick mass of a semi-liquid substance, especially **blood**, when it is exposed to air.
curdle: cause to form curds or lumps, especially milk.

Answer: B

40. **SMELL : SENSE**

A. success : ambition
B. indifferent : passion
C. thirst : drive
D. inebriety : excess

Clues: **smell** – is to use one of the five **senses**.
success – is reaching at using one's **ambition.** **Answer: A**

41. **HEPTAGON : OCTAGON**

A. rectangle : hexagon
B. parallelogram : square
C. triangle : nonagon
D. quadrilateral : pentagon

Clues: **Heptagon:** with seven straight sides and seven angles.
Octagon: with eight straight sides and eight angles.
Quadrilateral: with four sides and angles.
Pentagon: with five sides and angles. **Answer: D**

42. **OBSOLETE : UPTODATE**

A. primitive : old-fashioned
B. archaic : advanced
C. complete : modern
D. outdated : useless

Clues: **obsolete** – no longer in use, or outdated
uptodate – fashioned, new, modern
The two words are completely **opposite**.
archaic – antiquated, ancient, very old
advanced – new, latest

Answer: B

Which of the following pair of words is the most **opposite** in meaning?

A. rugged : practical
B. practical : indifferent
C. breast plate : chest
D. pendant : neck

Clues: **indifferent** – careless, unresponsible, unconcerned
diligent – careful, responsible, concerned

Answer: D

HELMET : HEAD

A. pedal : foot
B. gun : hand
C. breast plate : chest
D. pendant : neck

Clues: A **helmet** protects one's **head**.
A **breastplate**, piece of armor, protects one's chest.

Answer: C

GULLIBLE : DEPED

A. malleable : molded
B. credible : cheated
C. careful : cautioned
D. articulate : silenced

Clues: A person who is **gullible** is easily fooled or **duped**.
A person who is **malleable** (impressionable) is easily **molded**, or **influenced**.

Answer: A

FOLLY : SENSE

A. plagiarism : books
B. insolvency : funds
C. beauty : beholder
D. anxiety : care

Clues: A person engaged in **folly** or **foolishness** lacks **sense**.
A person is a state of **insolvency** or **bankruptcy** lacks **funds**.

Answer: B

MOCK : CONTEMPT

A. falsety : mimicry
B. anticipate : fear
C. scold : disapproval
D. imitate : respect

Clues: A person **mocks** shows **contempt**.
A person who **scolds** shows **disapproval**.

Answer: C

48. **CONDIMENT : FOOD**

A. brochure : book
B. additive : milk
C. tenement : building
D. wit : conversation

Clues: A **condiment** enhances **food**.
- wit inhances **conversation.**

Answer: D

49. **DUNGEON : CONFINEMENT**

A. church : chapel
B. school : truant
C. asylum : refugee
D. hospital : mercy

Clues: A **dungeon** by definition is a place of **confinement**.
An **asylum** by definition is a place of **refugee**.

Answer: C

50. **GRIDIRON : FOOTBALL**

A. diamond : baseball
B. net : tennis
C. round : boxing
D. puck : hockey

Clues: People play **football** on a field called a **gridiron**.
People play **baseball** on a field called a **diamond**.

Answer: A

51. **HERMIT : GREGARIOUS**

A. coward : pusillanimous
B. miser : penurious
C. scholar : literate
D. ascetic : hedonistic

Clues: A **hermit**, who chooses to live all alone, is by definition not **gregarious** (companionable and outgoing).
An **ascetic,** who chooses a life of self-denial, is by definition not **hedonistic** (devoted to pleasure).

Answer: D

52. **CLANDESTINE : OVERT**

A. thorough : complete
B. limited : unrestrained
C. warm : feverish
D. circular : oval

Clues: An activity which is **clandestine** (secret) is not **overt** or unconcealed.
An activity which is **limited** (within boundaries) is not **unrestrained** or free of restriction.

Answer: B

PINE : YEARN

53. A. whisper : shout
B. amaze : astonish
C. collect : scatter
D. meander : march

Clues: To **pine** (to long for or to languish) means to **yearn**.
To **amaze** (to astound) means to **astonish**. **Answer: B**

WINE : VINTER

54. A. beer : brewer
B. tobacco : smoker
C. oil : masseur
D. water : plumber

Clues: A **vinter** is the person who makes **wine**.
A **brewer** is the person who makes **beer**. **Answer: A**

MENDACITY : HONESTY

55. A. courage : cravenness
B. truth : beau
C. strength : fortitude
D. unsophistication : ingenousness

Clues: **Mendacity** or **untruthfulness** is the opposite of **honsesty**.
Courage is the opposite of **cravenness** or **cowardice**.
(opposite relationship): **Answer: A**

PREFACE : BOOK

56. A. presage : folly
B. prologue : play
C. preamble : poem
D. appendix : text

Clues: The **preface** is an introduction at the beginning of a **book**.
The **prologue** is an introduction at the beginning of a **play**. **Answer: B**

LAUREL : VICTORY

57. A. fig leaf : license
B. black cat : defeat
C. flag : triumph
D. olive branch : peace

Clues: A wreath made from the **laurel** tree symbolizes **victory**.
- An **olive branch** symbolizes **peace**. **Answer: D**

EXPERIENCE : KNOWLEDGE

58. A. beauty : cosmotics
B. growth : investment
C. purgative : disease
D. poison : death

Clues: **Experience** leads to **knowledge**.
poison leads to **death**. **Answer: D**

59. **LOAF : BREAD**

A. wheel : cheese
B. yolk : egg
C. pod : pea
D. cream : milk

Clues: **Bread** is produced in the shape of a **loaf**.
Cheese is produced in the shape of a **wheel**.

Answer: A

60. **BOOK : CHAPTER**

A. painting : frame
B. building : story
C. tree : root
D. movie : scenario

Clues: A **book** consists of several **chapters**.
A **building** consists of several **stories.**

Answer: B

61. **TANK : OXYGEN**

A. carton : milk
B. automobile : gasoline
C. salt : sodium
D. molecule : atom

Clues: **oxygen** is stored in a **tank**.
milk is stored in a **carton**.

Answer: A

62. **TRANSLATE : INTERPRETER**

A. prosecute : culprit
B. edit : reviewer
C. officiate : philosopher
D. expurgate : censor

Clues: An **interpreter translates** (turns something from one language into another).
- A **censor expurgates** (removes morally offensive or objectionable material).

Answer: D

63. **EXERCISE : STRONG**

A. decide : astute
B. drink : quench
C. read : knowledge
D. watch : alert

Clues: An **exercise** makes one fit, healthy and strong.
Reading makes one knowledgeable, confident.

Answer: C

64. **AESTHETICS : BEATY**

A. epistemology : knowledge
B. ethics : society
C. logistic : truth
D. rhetoric : music

Clues: **Aesthetics** is the branch of philosophy that studies the principles of **beauty** especially in art.
- **Epistemology** is the part of philosophy that deals with **knowledge**.

Answer: A

179

MAIL : MAILBOX

A. coat : factory
B. gasoline : pump
C. medicine : treatment
D. toothpaste ; cleanliness

Clues: **mail** – is the post that the official system used for sending and delivering letters, packages.

mailbox – is a letter box or small box near the main door of building or by the road, which **mail** is delivered to.

Gasoline - is liquid substance used for machine engine (vehicles).

pump – is a machine used to force liquid, gas or air into out of something (a petrol, or a gas **pump**).

☞ Thus, **mail** is through **mailbox** where as **gasoline** is through **pump**.

Answer: B

67. **FOUNDATION : HOUSE**

A. ground : soil
B. mountain : tunnel
C. pedestal : picture
D. pole : banner

Clues: **foundation** – is the base /bottom, the starting point when house is built; house is supported by its foundation.

pole – is a long thin straight piece of wood or metal, especially one with the end placed in the ground used as support.

banner – is a long piece of cloth with a message on it that is carried between two poles.

Answer: D

68. **DIVIDE : MULTIPLY**

A. drive : ride
B. enumerate : count
C. leave : enter
D. speak : communicate

Clues: The capitalized pair of words are **antonyms** choice (C) has the same relationship with it.

Answer: C

69. **COMMUTE : SENTENCE**

A. aggregate : wealth
B. execute : mandate
C. mitigate : punishment
D. set : penalty

Clues: **commute** – means to replace punishment with another that is less server.

sentence – is sever punishment. (eg. death sentence, life imprisonment sentence).

Thus, **commute** is to lessen this hard punishment.

mitigate – to make something less harmful, serious.

Thus, to **mitigate** means to make punishment less harmful.

70. **ENDORSEMENT : POLICY** **Answer: C**

A. consignment : invoice
B. purchase : receipt
C. rider : bill
D. violation : ordinance

Clues: **endorsement : -** a public statement or action showing that you support something. E.g. the election victory is clear **endorsement** of their **polices** (we **endorse policy**).

71. **BLAST : WIND** **Answer: A**

A. bursi : sound
B. ebb : tide
C. glimmer : light
D. finder : fire

Clues: **blast** – a violent gust or wind; sound made by a wind instrument.

burst – a sudden outbreak of **sound**.

72. **KINDLE : FLAME** **Answer: A**

A. shoot : battle
B. incite : riot
C. kidnap : ransom
D. emend : book

Clues: To start a **flame**, you **kindle** it.

To start a **riot**, you **incite** it.

73. **WASTEFUL : SQUANDER** **Answer: B**

A. intrepid : fear
B. circumspect : fascinate
C. caution : procrastinate
D. querulous : complain

Clues: Someone **wasteful (extravagant)** by definition **squanders (wastes)**.
- Someone **querulous** or **peevish** (habitually **complaining**) by definition **complains**.

SKIRMISH : BATTLE

A. quiz : examination
B. injury : scar
C. detour : journey
D. ambush : retreat

Answer: D

Clues: A **skirmish** (minor military engagement) is less important than a **battle**.
A **quiz** (minor test) is less important than **examination**.

Answer: A

PARAGON : STANDARD

A. author : publisher
B. painting : landscape
C. imitation : copy
D. colleague : rival

Clues: **paragon** (model of excellence) and **standard** or **ideal** are **synonyms**.
Similarly, **imitation** and **copy** are **synonyms**.

Answer: C

HOSTILE : FRIENDSHIP

A. inimical : opposition
B. intolerant : bias
C. bombastic : grandiloquence
D. traitorous : loyalty

Clues: A **hostile (antagonist)** act shows a lack of **friendship**.
A **traitorous ((disloyal)** act shows a lack of **loyalty**.

Answer: D

PENITENET : CONTRITION

A. hedonist : indulgence
B. skeptic : gullibility
C. pragmatist ; resignation
D. stoic : arrogance

Clues: A **penitent** (person who repents) exhibits **contrition** (remorse).
A **hedonist** (person devoted to the pursuit of pleasure as a way of life) exhibits **indulgence** (gratification of one's desires).

Answer:

78. **MARATHON : STAMINA**

A. relay: independence
B. sprint: celerity
C. jog : weariness
D. hurdle ; perseverance

Answer: B

Clues: A **marathon** is a very long race.
It requires **stamina** or **endurance**.
A **sprint** is a very short race.
It requires **celerity** or **speed**.

79. **NAIVE : INGENUE**

A. urbane : sophisticate
B. crafty : artisan
C. modest : braggart
D. ordinary : genius

Answer: A

Clues: An **ingénue** or **inexperienced** young person is **naïve** (innocent, frank).
A **sophisticated** or **wordily** person is **urbane** (suave, elegantly polite).

80. **HULKING : MASSIVE**

A. furtive : ironic
B. pert : polite
C. raucous : harsh
D. brutal : compulsive

Answer: C

Clues: **Hulking** means bulky or **massive.**
Raucous means grating **harsh.**

81. **DISASTER : LUGUBRIOUS**

A. success : saturnine
B. tragedy : complacent
C. failure : gay
D. fruition : satisfied

Answer: C

Clues: A person is **lugubrious** or **mournful** after a **disaster.**
A person is **satisfied** after the **fruition**.

82. **ASSEMBLE : GATHER**

A. select : collect
B. feign : dissemble
C. proclaim : confide
D. dismiss : hasten

Answer: D

Hints: To **assemble** things is to **gather** them.
- To **feign** or **deceive** is to **dissemble.**

Answer: B

83. **RETOUCH : PHOTOGRAPH**

A. compose : melody
B. hang : painting
C. finger : fabric
D. refine : style

Clues: To **retouch** a **photograph** is to alter its appearance, to free it from imperfection.

- To **refine** a **style** is to free it from imperfections as well.

Answer: D

84. **HONESTY : HYPOCRISY**

A. pacifism : belligerence
B. treachery : duplicity
C. integrity : righteousness
D. magnanimity : greatness

Clues: **Honesty** is the opposite of **hypocrisy**.

hypocricy – pretending to be what one isn't.

pacifism – a belief in peace. It is the opposite of **belligerence** (=being warlike, eager to fight).

Answer: A

85. **INDIGENT : WEALTH**

A. emaciated : nourishment
B. contented : happiness
C. smug : complacency
D. aristocratic : stature

Clues: Someone who is **indigent** or **poor** lacks **wealth.**

Someone who is **emaciated** or **abnormally thin** lacks **nourishment.**

Answer: A

86. **SHALE : GEOLOGIST**

A. obelisk : fireman
B. reef : astrologer
C. aster : botanist
D. love : philologist

Clues: A **geologist** is someone who studies rocks, deals with **shale** (= a particular kind of rock).

- A **botanist** is someone who studies plants, deals with **aster** (= a particular kind of flower).

Answer: C

87. **FLEETING : DURATION**

A. glancing : distance
B. rapid : leisure
C. concise : length
D. fragrant : smell

Clues: Something **fleeting** is brief in **duration**.
- Something **concise** is brief in **length**.

Answer: C

88. **DIDACTIC : TEACH**

A. satric : mock
B. apologetic : deny
C. sophomoric : learn
D. chaotic : rule

Clues: **Didactic** by definition means inclined to **teach**.
- **Satric** by definition means inclined to **mock**.

Answer: A

89. **SOPORIFIC : SLEEP**

A. calorific : hunger
B. worrisome : anxiety
C. obtuse : insight
D. honorific : imbarrassment

Clues: **Soporific** means inducing or causing **sleep**.
- **worrisome** means inducing or causing **anxiety**.

Answer: B

90. **HACKNEYED : ORIGINAL**

A. withdrawn : reserved
B. evasive : elusive
C. derivative : traditional
D. mature : juvenile

Clues: **Hackneyed** (worn, trite) is the opposite of **original**
Mature is the opposite of **juvenile**.

Answer: D

91. **AUGER : CARPENTER**

A. awl : cobbler
B. apron : chef
C. seam : seamstress
D. studio : sculptor

Clues: An **auger** or bit is a tool used by a **carpenter**.
An **awl** (tool used for piercing a **cobbler** or shoemaker.

Answer: A

92. **MUSTER : CREW**

A. cheer : team
B. demobilize : troops
C. convene : committee
D. dominate : opposition

Clues: To **muster** a ship's **crew** is to assemble or gather them together.
To **convene** a **committee** is to assemble or gather its members together.

Answer: C

93. **DWELL : DENIZEN**

A. inherit : heir
B. shun : outcast
C. obey : autocrat
D. patronize : protege

Clues: A **denizen** (resident) by definition **dwells** or resides in a region.
- An **heir** by definition **inherits** or receives a legacy from someone who has died.

Answer: A

94. **HAZARDOUS : PERIL**

A. innovative : novelty
B. competitive : pride
C. supercilious : modesty
D. venerable : immaturity

Clues: Something **hazardous** (dangerous) is characterized by **peril** (danger).
- Something **innovative** (new in form or design) is characterized by **novelty** (newness, originality)

Answer: A

95. **LAUDABLE : PRAISE**

A. fastidious : taste
B. enamored : love
C. fragile : stability
D. contemptible : scorn

Clues: **laudable** means **praiseworthy** or deserving praise
contemptible means despicable or deserving **scorn**.

Answer: D

96. **OPTHALMOLOGIST : EYES**

A. pediatrician : bones
B. podiatrist : feet
C. cardiologist : brain
D. bacteriologist : atoms

Clues: An **ophthalmologist** specializes in the care of the **eyes**.
A **podiatrist** specializes in the care of the **feet**.

Answer: B

97. **MEANDERING : DIRECT**

A. affable : permissive
B. circuitous : round about
C. aboveboard : open
D. digressive : concise

Clues: **meandering** (proceeding by an indirect course) and **direct** are antonyms.
digressive (departing from the main subject) and **concise** (keeping brief and to the point) are antonyms.

Answer: D

98. **IRON : RUST**

A. bronze : patina
B. yeast : mold
C. cake : icing
D. stone : gravel

Clues: The covering of **iron** caused by oxidation is called **rust** and **alertness (N)** are antonyms.
- The covering of **bronze** is called a **patina**.

Answer: A

99. **EPHEMERAL : LAST**

A. competitive : contend
B. insignificant : matter
C. perishable : die
D. transient : travel

Clues: Something **ephemeral** by definition doesn't **last**.
- Something **insignificant** by definition doesn't **matter**.

Answer: B

100. **DONOR : GIFT**

A. braggart : attention
B. mourner : sympathy
C. advice : counsel
D. prophet : prediction

Clues: A **donor** or **benefactor** makes a **gift**.
A **prophet** or **seer** makes a **prediction**.

Answer: D

101. **ACT : DRAMA**

A. grade : course
B. chapter : essay
C. platoon : company
D. aria : opera

Clues: A **drama** or play is divided into **acts**.
A military **company** is divided into **platoons**.

Answer: C

CEMENT : CROWEL

A. paint : brush
B. floor : polish
C. lawn : rake
D. wallpaper : ladder

Clues: **cement** is applied with a **trowel**.
paint is applied with a **brush**.

Answer: A

PIGHEADED : YIELD

A. dogged : pursue
B. lionhearted : retreat
C. lily-livered : flee
D. eye-eyed : discern

Clues: Someone **pigheaded** or **stubborn** is disinclined to **yield (give in)**.
Someone **lionhearted** or **brave** is disinclined to **retreat**.

Answer: B

UNCTUOUS : SINCERITY

A. frivolous : gravity
B. fatuous : ambivalence
C. unclean : impunity
D. hypocritical : virtue

Clues: An **unctuous** or excessively plous manner is affected, having a false appearance of **sincerity**.
A **hypocritical** or dissembling manner is feigned, having a false appearance of **virtue**.

Answer: D

PIRATE : BUCCANEER

A. captain : admiral
B. sailor : beachcomber
C. cynic : flatterer
D. wanderer : nomad

Hints: **pirate** and **buccaneer** are synonyms, as are **wanderer** and **nomad**.

Answer: D

LAW is to **ANARCHY** as **MEDICINE** is to _____.

A. Illness
B. recovery
C. health
D. death

Clues: **Law** is very essential to control **anarchy**
(anarchy = unlawful, illegal, against law).

Medicine is very essential to control **illness** /**disease** /**ailment**.

***Note*: health** is a general word; it includes illness and wellbeing.

107. **UNCLE** is to **NEPHEW** as **AUNT** is to _____ **Answer:**

A. cousin
B. daughter
C. sister
D. niece

***Clues*: nephew** is your sister's or brother's **son**.
(= You're an **uncle** for him).
niece: is your sister's or brother's **daughter**.
(= You're an **aunt** for her).

108. **ETYMOLOGY : DERIVATION** **Answer: D**

A. ecology : urbanization
B. taxonomy : classification
C. autonomy : authorization
D. economy : rationalization

***Clues*: etymology** : is the study of the origins and historical development of words.

derivation : words origins /source

taxonomy : is the process of naming and **classification** of things such as animals and plants into groups with in a large system according to their similarities and differences.

109. **TONE : SCALE** **Answer: B**

A. noise : wave
B. colour : spectrum
C. waves : sound
D. verse : rhyme

***Clues*: Tone:** is a difference in pitch between two musical notes equal to two semitones.

Scale: in music, **scale** is a fixed sequence of musical.

In a **verse** (a group of lines forming a unit in a poem or song). Words have or end with the same sound as another which call **rhyme**.

Answer: D

10. **WORD : SENTENE**

A. picture: pencil
B. writing: paper
C. note : bar
D. word : letter

Clues: Words make a **sentence.**
(A **sentence** is made of **words**).
letters make a **word.**
(a group of **letters** make a **word**)

Answer: D

11. **YAWN : BOREDOME ::**

A. dream : sleep
B. anger : madness
C. smile : amusement
D. face : expression

Clues: When you get **bored,** you will **yawn.**
When get **amused,** you will be **smiled.**

Answer: C

12. **LOAF : BREAD**

A. yolk : egg
B. wheel : cheese
C. icing : cake
D. cream : milk

Clues: **Bread** is produced in the form of **loaf.**
Cheese is produced in the form of **wheel.**

Answer: B

13. **TREE : FOREST ::**

A. lion : jungle
B. camel : desert
C. blossom : bouquets
D. lawn : garden

Clues: A **tree** is a part of **forest.**
A **blossom** is part a of **bouquest.**

Answer: C

14. **INAUGURATE : PRESIDENT ::**

A. install : officer
B. campaign : candidate
C. promote : student
D. succeed : governor

Clues: To **inaugurate** a **president** is to introduce him or her into office.
- To **install** an **officer** is to do the same.

Answer: A

115. **SONG : VOCALIST**

A. soliloquy : director
B. monologue : speaker
C. melody : composer
D. forecast: prestidigitator

Clues: A **song** is delivered by a **vocalist (singer).**
A **monologue** is delivered by a **vocalist.**

Answer: B

116. **BUOY : SHIP**

A. law : society
B. bell : channel
C. radar : obstruction
D. cockpit : plane

Clues: A **buoy** indicates the correct path for **ship** to follow.
Law indicates the correct path for **society** to follow.

Answer: A

117. **GRATEFUL : INGRATITUDE**

A. despotic : tyranny
B. hopeful : despair
C. melancholy : gloom
D. pessimistic : rejection

Clues: A person filled with **ingratitude** (ungratefulness) is not **grateful.**
A person filled with **despair** is not **hopeful.**

Answer: B

118. **TEPID : BOILING**

A. cold : scorching
B. pleasant : useful
C. indifferent : apathetic
D. frugal : parsimonious

Clues: The relationship involves a matter of degree.
Tepid (lukewarm) **indicates the presence of some heat; boiling** indicates the presence of a far greater degree of heat.
Similarly, **frugal** (economical) indicates the practice of some **thrift; parsimonious (stingy)** indicates the practice of a far greater degree of thrift.

Answer: D

119. **MAGNANIMITY : KINDLY**
A. anger: ironic
B. Ire: choleric
C. charity : fortuitous
D. friendship : frivolous
Clues: A person filled with **magnanimity** is forgiving and **kindly**.
A person filled with **ire** (anger) is **choleric** or wrathful.
Answer: B

120. **HYPOCRISY : HONESTY**
A. Duplicity: deceit
B. literature: philosophy
C. pugnacity : pacifism
D. authenticity : quality
Clues: **Hypocrisy** (pretense of something untrue) involves a lack of **honesty**.
- **Pugnacity** (belligerence; hostility) involves a lack of **pacifism** (opposition to war).
Answer: C

121. **HOARY : YOUTHFULNESS**
A. innovative : modernity
B. evanescent : transience
C. somnolent : alertness
D. energetic : vigor
Clues: Someone **hoary** (white with age) lacks **youthfulness**.
- Someone **somnolent** (drowsy, sleepy) lacks **alertness**.
Answer: C

122. **BEES : APIARY**
A. geese : gaggle
B. rabbits : hutch
C. snakes : herpetology
D. silkworms : silk
Clues: **Bees** are kept in an **apiary**.
Rabbits are kept in a **hutch**.
Answer: B

123. **MINNOW : FISH**
A. fang : snake
B. flock : sheep
C. marrow : bone
D. poodle : dog
Clues: A **minnow** is a type of **fish**.
A **poodle** is a type of **dog**.
Answer: D

124. **JAR : RECEPTACLE**
A. trunk: lid
B. rifle: ammunition
C. cottage : building
D. book : story
Clues: A **jar** is a type of **receptable**.
A **cottage** is a type of **building**.
Answer: C

125. **RUNNERS : SLED**

A. wheels : skateboard
B. dancer : stage
C. chalk : blackboard
D. bricks : wall

Clues: A **sled** moves across the ground on its **runners**.
A **skateboard** moves across the ground on its **wheels**.

Answer: A

126. **APPRAISER : BUILDING**

A. critic : book
B. miser : gold
C. gourmet : caviar
D. judge : courtroom

Clues: An **appraiser's** job is to evaluate a **building**.
A **critic's** job is to evaluate a **book**.

Answer: A

127. **ECSTATIC : JOYFUL**

A. melancholy : exuberant
B. despondent : discouraged
C. marked : significant
D. emotional : spiritual

Clues: **Ecstatic** means very **joyful**.
Despondent means very **discouraged**.

Answer: B

128. **STAMPS : PHILATELY**

A. countries : alliance
B. rockets : pyrotechnics
C. coins : numismatics
D. words : lexicon

Clues: **philately**, by definition, is the study and collection of **stamps**.
- **Numismatics**, by definition, is the study and collection of **coins**.

Answer: C

129. **CALCULATED : SPONTANEITY**

A. vivid : color
B. verbose : glibness
C. familiar : novelty
D. rehearsed : performance

Clues: **calculated** or planned means lacking in **spontaneity** (naturalness, being without forethought).
- **Familiar** or usual means lacking in **novelty** (newness, originality).

Answer: C

130. **LEXICOGRAPHER : WORDS**

A. archaeologist : fossils
B. surveyor : seeds
C. photographer : charts
D. gardner : weeds

Clues: A **lexicographer** (dictionary maker), by definition, studies **words**.

- An **archaeologist**, by definition, studies prehistoric remains, such as **fossils**.

Answer: A

132. **EVANESCENT : PERMANENT**

A. wealthy : affluent
B. docile : stubborn
C. fleeting : timorous
D. glittering : icy

Clues: **Evanescent** means transient or momentary, the opposite of **permanent**.

- **Docile** means easy to control or lead, the opposite of **stubborn**.

Answer: B

133. **HEINOUS : INIQUALITY**

A. egregious: excellence
B. notable: benevolence
C. notorious : levity
D. ingenious : innocence

Clues: **Heinous** means very evil, having the quality of **iniquity** (wickedness).

- **ingenuous** means frank and trusting having the quality of **innocence**.

Answer: D

134. **MUTTON : SHEEP**

A. veal : calf
B. colt : horse
C. kannel : dog
D. gander : goose

Clues: **mutton** is the meat of a **sheep**.
veal is the meat of a **calf**.

Answer: A

135. **CLATTER : POTS**

A. shatter : windowpane
B. collect : coins
C. rustle : leaves
D. wrinkle : clothes

Clues: **pots** in contrast with each other **clatter**.
leaves in contrast with each other **rustle**.

Answer: C

136. **HOOF : CAMEL**

A. paw : dog
B. wing : pigeon
C. horn : rhinoceros
D. tail : squirrel

Clues: The foot of a **camel** is a **hoof**.
The foot of a **dog** is a **paw**.

Answer: A

136. **BRIDGE : GAP**

A. cleanse : wound
B. reconcile : estrangement
C. enter : doorway
D. return : favor

Clues: **Bridging** a **gap** spans a physical distance between two sides, making a connection between them.

Reconciling (setting right) an **estrangement** (loss of affection, separation) spans an emotional distance between them.

Answer: B

137. **POACHER : GAME**

A. pickpocket : wallet
B. commuter : train
C. traitor : country
D. composer : melody

Clues: A **poacher** steals **game** (animals).

A **pickpocket** steals a **wallet.**

Answer: A

138. **COMET : TAIL**

A. star : galaxy
B. river : bank
C. ear : headlight
D. vessel : wake

Clues: Behind a **comet** stretches its glowing **fail.**

Behind a **vessel** (ship) stretches its **wake**, a trail in water.

Answer: D

139. **EPICURE : FOOD**

A. chef : pastry
B. fanatic : religion
C. connoisseur : art
D. professional : salary

Clues: An **epicure**: knows and cares about **food.**

A **connoisseur** knows and cares about **art.** **Answer: C**

140. **CIRCUMSPECT : WARY**

A. respectful : pert
B. reckless : foolhardy
C. meaningful : inane
D. wicked : abstemious

Clues: **Circumspect** (cautious) and **wary** are synonyms, as are **reckless** and **foolhardy.**

Answer: B

141. **COPIOUS : SCARCE**

A. expeditious : slow
B. ambitious : earnest
C. durable : massive
D. extinct : rare

Clues: **Copious** or plentiful is the opposite of **scarce.**

Expeditious or quick is the opposite of **slow.**

Answer: A

142. **INDEPENDENT : AUTONOMY**

A. coy : silence
B. prodigal : economy
C. nervous : equanimity
D. inventive : resourcefulness

Clues: Someone who is **independent** has **autonomy** or **freedom**. Someone who is **inventive**, or **creative** has **resourcefulness** (cleverness at solving problems).

Answer: D

143. **ATLAS : MAP**

A. dictionary : synonyms
B. catalogue : dates
C. lexicon : numbers
D. glossary : words

Clues: **Atlas** is a book of maps or a list of **maps**.
Glossary is a list of **words**.

Answer: D

144. **DETAIN : SUSPECT**

A. ambush : patrol
B. confine : prisoner
C. testify : observer
D. trace : escape

Clues: **Suspect** is a person who is **detained,** or kept in an official place for a short time.
Prisoner is a person who **confined**, or kept in an official and legal place for sometimes.

Answer: B

145. **MUTE : SOUND**

A. reverse : direction
B. test : commitment
C. dampen : enthusiasm
D. distract : attraction

Clues: **Mute** is making no sound, **dampen** is to make something damp or less strong.
So, **dampen** something sprits **enthusiasm** expectation.

Answer: C

146. **GENERATOR : CRURRET**

A. brain : cortex
B. organ : kidney
C. gland : enzyme
D. muscle : spasm

Clues: **Generator** produces **current.**
Gland produces **enzyme.**

Answer: C

147. **PROPONENT : THEORY**

A. attorney : law
B. adherent : belief
C. nonbeliever : sin
D. traitor : country

Clues: **Proponent** is a person who supports a cause **theory**, etc.
Adherent is a supporter of a party or set of ideas.

Answer: B

148. **ORATOR : ARTICULATE**

A. judge: unbiased
B. celebrity : talented
C. novice : unfamiliar
D. soldier : merciless

Clues: **Orator** is a person who **articulates** his speech in good manner. **Orator** is almost the same as **articulate**.
Novice is a person who is new or **unfamiliar** for something; it is almost synonyms to **unfamiliar**.

Answer: C

149. **TITLED : NOBLE**

A. commissioned : officer
B. deposed : ruler
C. elected : candidate
D. acclaimed : artist

Clues: **Noble** is having high rank or **titled**.
Officer is a person who appointed to **command** others.

Answer: A

150. **COOP : CHICKENS**

A. sty : pigs
B. school : fish
C. flock : geese
D. forest : deer

Clues: A **coop** is an enclosure for **chickens**.
A **sty** is an enclosure for **pigs**.

Answer: A

151. **SPRINGBOARD : DRIVER**

A. club : golfer
B. trampoline : gymnast
C. target : archer
D. helmet : cyclist

Clues: A **driver** uses **springboard** or gain height for a **drive**.
A **gymnast** uses a **trampoline** to gain height for tumbling.

Answer: B

MASON : STONE

A. dentist : drill
B. tailor : fabric
C. actor : stage
D. miser : gold

Clues: A **mason** (stone worker) constructs things from **stone**.
A **tailor** constructs things from **fabric.** **Answer: B**

153. **CLAUSTROPHOBIA : CONFINEMENT**

A. acrophobia : height
B. kleptomania : staling
C. hydrophobia : dogs
D. egomania : self

Clues: **Claustrophobia** is a fear of enclosed places, or **confinement**.
Acrophobia is a fear of high place or **height**. **Answer: A**

154. **SAGE : HERB**

A. leaf : tree
B. poodle : dog
C. salt : meal
D. bud : flower

Clues: **sage** is a kind of **herb**.
A **poodle** is a kind of **dog**. **Answer: B**

155. **JITTERY : FIDGET**

A. drowsy : lull
B. mulish : roam
C. foolish : weep
D. touchy : snap

Clues: Someone **jittry** or nervious **tends to fidget** or move uneasily.
Someone **touchy** or irritable **tends to snap** or retort angrily. **Answer: D**

156. **SHARD : POTTERY**

A. rift : rock
B. handle : pitcher
C. splinter : bone
D. veneer : wood

Clues: A **shard** is a broken fragment of **pottery**.
A **splinter** is a broken fragment of **bone**. **Answer: C**

157. **ANARCHIC : ORDER**

A. warped : distortion
B. authoritative : command
C. meticulous : detail
D. aimless : objective

Clues: Something **anarchic** or **chaotic** by definition lacks **order**.
- Something **aimless** or purposeless by definition lacks an **objective** or goal.

Answer: D

158. **DEVOUT : PIETY**

A. compliant : submissiveness
B. fallacious : accuracy
C. restless : energy
D. superstitious : zeal

Clues: To be **devout** to religious demonstrates **piety** (dutifulness in religion).
- To be **compliant** to yielding demonstrates **submissiveness**.

Answer: A

159. **IMPECUNIOUS : HOVEL**

A. prosperous : mansion
B. affluent : tent
C. indigent : castle
D. defenseless : fortress

Clues: A **hovel** or hut is a poor dwelling for someone **impecunious** or penniless.
- A **mansion** is an imposing dwelling for someone sperous or well-to-do.

Answer: A

160. **WHISPER : SHOUT**

A. foretell : predict
B. wish : desire
C. lessen : increase
D. rich : poor

Clues: To w**hisper** is to mean to **be silent /quiet** (It is the opposite of **shout**).
- To **lessen** is to mean to **decrease** (make low in volume) (It is the opposite of **increase (rise up)**

Answer: C

161. **DRAB : DULL**

A. wealth : poverty
B. joy : sorrow
C. wisdom : intelligence
D. cry : laugh

☞ **Except (A)** the rest options have similar meaning.

Answer: A

CONSUMER : SELLER ::

A. boy : male
B. animal : pet
C. man : woman
D. student : teacher

Clues: **celler** provides goods and services for a **costomer** /**buyer**/ **consumer**. A **teacher** provides knowledge for students.

Answer: D

163. **FLURRY : BLIZZARD ::**

A. drizzle : downpour
B. current : wave
C. hurricane : typhoon
D. deluge : drought

Clues: A **flurry** is a shower of snow that is less intense than a **blizzard**.
A **drizzle** is a shower of rain that is less intense than a **downpour**.

Answer: A

164. **WRIGGLE : EEL ::**

A. roost : hen
B. leap : frog
C. muzzle : dog
D. gobble : turkey

Clues: In moving, an **eel** characteristically **wriggles** or twists.
A **frog** characteristically **leaps** or jumps.

Answer: B

165. **BLINDFORD : VISION ::**

A. bandage : wound
B. gag : speech
C. fog : highway
D. megaphone : hearing

Clues: A **blindfold** hampers or blocks one's **vision**.
A **gag** hampers or blocks one's **speech**.

Answer: B

166. **PHOTOGRAPHER : DARKROOM ::**

A. artist : exhibit
B. novelist : plot
C. chemist : laboratory
D. sailor : lighthouse

Clues: A **photographer** works in **darkroom**.
A **chemist** works in a **laboratory**.

Answer: C

167. **SHORTCUT : METHOD ::**

A. conclusion : procedure
B. outcome : event
C. detour : route
D. abridgement : text

Clues: A **shortcut** is an abbreviated or shortened **method**.
An **abridgement** is an abbreviated or shortened **text**.

Answer: D

168. **EGGSHELL : FRAGILITY ::**

A. barbell : weight
B. rainbow : mobility
C. peapod : variety
D. dewdrop : grief

Clues: An **eggshell** is characterized by delicacy or **fragility**
A **barbell** is characterized by **weight**.

Answer: A

169. **WILLFUL : OBEDIENT ::**

A. outspoken : candor
B. fretful : patience
C. careful : caution
D. tuneful : harmony

Clues: Someone **willful** (=stubborn or contrary) lacks **obedient**
Someone **fretful** (irritable or impatient) lacks **patience**.

Answer: B

170. **FLAUNT : DISPLAY ::**

A. rehearse : perform
B. cancel : initiate
C. transform : modify
D. brag : declare

Hints: To **flaunt** something is to **display** it ostentatiously or boastfully.
To **brag** about something is to **delare** it ostentatiously or boastfully.

Answer: D

171. **APPLAUD : APPROBATION ::**

A. jeer : derision
B. sneer : apathy
C. hiss : apprehension
D. cancel : initiate

Clues: To **applaud** is to indicate **approbation** or a approval.
To **jeer** is to indicate **derision** or mockery.

Answer: A

172. **ERRATA : BOOKS ::**

A. filters : water
B. clips : films
C. excerpts : novels
D. flaws : jewels

Hints: Detects in **books** are called **errata**.
Detects in **jewels** are called **flaws**.

Answer: D

173. **PIRATE : BUCCANEER**

A. wanderer : nomad
B. puritan : virtuoso
C. cynic : flatterer
D. sailor : beachcomber

Clues: **pirate** and **buccaneer** are synonyms, as are **wanderer** and **nomad**.

174. **MEND : TORN ::**

A. raze: worn
B. repair: broken
C. replace: old
D. empty: full

Clues: Something **torn** needs to be **mended**.
Something **broken** needs to be **repaired**.

Answer: B

175. **PRIDE : LIONS ::**

A. honor : thieves
B. arrogance : kings
C. lair : bear
D. gaggle : geese

Clues: **pride** is a group of **lions**.
gaggle is a group of **geese**.

Answer: D

176. **HORSE : CORRAL ::**

A. dog : muzzle
B. deer : stag
C. pig : sty
D. sheep : flock

Clues: **Horses** are kept in **corral** (house for horses).
Pigs are kept in **sty** (house for pigs).

Answer: C

177. **DOG : CANINE ::**

A. cat : leonine
B. deer : herbivorous
C. tiger : feline
D. sailor : marine

Clues: **Canine** means relating to **dogs** (dog family).
Feline means belonging or reliant to the **cat** family. (e.g. **tiger**)

Answer: C

178. **BUOYANT : FLOAT ::**

A. feeble : collapse
B. flexible : bend
C. rugged : shred
D. hard : shatter

Clues: Something **buoyant** can easily **flaut** in the air or on water.
Something **flexible** can easily **bend**.

Answer: B

179. **AIRY : STUFFY**

A. slight : bulky
B. foolish : stupid
C. keen : sharp
D. meager : sparse

Clues: **Airy** (a lot of fresh air inside) and **stuffy** (unpleasant and warm air) are **antonyms**.
Bulky (large and heavy) is also the **opposite** of **slight** (small and light).

Answer: A

180. **EDIT : TEXT**

A. print: negative
B. cut : film
C. coin : phrase
D. weight : decision

Clues: The author **edits** a **text** such as an article, or a book by correct and adapt it.

The director of a **film** says **"cut"** when they want the actors and the camera crew to stop filming.

Answer: B

181. **TADPOLE : FROG ::**

A. nest : robin
B. poultry : hen
C. donkey : mule
D. caterpillar : butterfly

Clues: **Tadpole** develops into **frog**.

Caterpillar develops into **butterfly or moth**.

Answer: D

182. **ARRIVAL : DEPARTURE ::**

A. welcome : thanks
B. presentation : gift
C. introduction : conclusion
D. flight : landing

Clues: **Arrival** getting to somewhere) is the opposite of **departure** (leaving out somewhere /place).

As **introduction** (the beginning of something) and **conclusion** (the end of something).

Answer: C

183. **ARID : MOISTURE ::**

A. insipid : taste
B. frail : weakness
C. severe : harshness
D. erect : posture

Clues: A place is desired as **arid** (no **moisture**).

Food or drink is desired as **insipid** (no **taste**). **Answer: A**

184. **EXCESSIVE : MODERATION**

A. expensive : cost
B. impulsive : reflection
C. extensive : duration
D. arbitrary : courage

Answer: B

185. **CONSTELLATION : STARTS**

A. choir : angles
B. tree : leaves
C. mine : diamonds
D. archipelago : islands

Clues: A group of **starts** is called **constellation**.

A group of small **island** is called **archipelago**.

Answer: D

CLAMOR : NOISE ::

A. glamor : gloom
B. renown : fame
C. author : text
D. skirmish : battle

Clues: **Clamor** is a loud confusion **noise**.
Renown is an extremely **fame** (for person). **Answer: B**

REPERTORY : PLAY ::

A. title : novel
B. landscape : painting
C. menu : dish
D. recital : concert

Clues: **Repertory** is the performance by a company various **plays**, operas, etc.
Menu is a list of various **dishes**. **Answer: C**

SPECTATOR : GAME

A. observer : experiment
B. spy : deception
C. surgeon : operation
D. soldier : battle

Clues: **spectator** is someone who watches a **game**.
An observer is a person who watches an **experiment**. **Answer: A**

RESPIRATOR : LUNGS ::

A. thermometer : temperature
B. hypodermic : skin
C. pacemaker : heart
D. nostril : nose

Clues: **Respirator** is a piece of equipment that makes it possible for somebody to breath over along period when there are unable to do so naturally.
- connected with breathing **(lungs) pacemaker** is an electronic device that is put regularly relating to **heart**. **Answer: C**

CLUMSY : AWKWARD ::

A. gauche : bungling
B. hopeless : optimistic
C. reluctant : unruly
D. ardroit : tactless

Clues: **Clumsy (awkward**, lacking coordination, not graceful, not dexterous.
gauche (awkward or lacking in social graces; **bumbling, bungling**. **Answer: A**

191. **PUNY** is to **GIGANTIC** as

A. foremost is to prominent
B. lethargic is to prominent
C. grueling is to graphic
D. vicious is to malignant

Clues: **Puny** is to mean **inferior (small)** in size.
Gigantic is to mean **superior (huge)** in size.
Lethargic is to mean **torpid, lazy, slow**
Prominent is to mean **standing out**, conspicuous, **eminent**

Answer: B

192. **DESPONDENT** is to **MELANCHOLY as**

A. indifferent is to enthusiasm
B. peevish is to courtesy
C. sullen is to joy
D. indignant is to wrath

Clues: Someone **despondent** is affected with great sadness or depression **(melancholy).**
indignant is showing **anger** or **indignation.**
wrath is showing great **anger (fury, ire).**

Answer: D

193. **INNUMERABLE** is to **COUNT** as

A. indispensable is to employ
B. interminable is to begin
C. illegible is to read
D. inflammable is to kindle

Clues: As we say **innumerable** is to **count**. we say **illegible** is to **read (illegible** = something which can be read easily).

Answer: C

194. **BLINDNESS** is to **affliction** as

A. homicide is to ordeal
B. embezzlement is to catastrophe
C. pneumonia is to malady
D. perjury is to misdemeanor

Clues: **Affliction** causes pain such as **blindness, deafness.**
pneumonia (ailment, disease, disorder, illness...) can cause **malady**.

Answer: C

FRUITLESS is to **BARREN** as

A. substantial is to trivial
B. grimy is to immaculate
C. adjacent is to distant
D. transparent is to clear

Clues: **Barren (arid) land** is described as **fruitless** or **unproductive.**
Something **clear** is described as **transparent** (**transparent** allow heat and light through it).

Answer: D

BEWILDER is to **UNCERTAIN** as

A. irritate is to peevish
B. control is to wayward
C. procure is to reluctant
D. dominant is to hostile

Clues: **To be wilder** means to make someone **uncertain** (to confuse, puzzle or befuddle someone).
To irritate is to make someone **peevish (complaint, fretful, whining).**

Answer: A

MISSILE is to **PELT** as

A. feather is to maul
B. net is to snare
C. firebrand is to douse
D. hose is to singe

Clues: **pelt** is to **bombard**. as with **missiles.**
(e.g They **pelted** the attacking army with bullets).
(to pelt = to hurl missile)
To snare means to catch in **a net.**

Answer: B

ERADICTE is to **NURTURE** as

A. germinate is to flourish
B. browse is to maul
C. stifle is to foster
D. dupe is to hoodwink

Clues: **Eradicate (remove, destroy)** is the antonym word with **nurture (grow, develop).**
Stifle (suppress) is the opposite of **foster** (encourage, develop).

Answer: C

199. **INCENTIVE** is to **GOAD** as
A. potential is to mortify
B. quibble is to dishearten
C. flaw is to entice
D. hazard is to dissuade

Clues: **Incentive** is something that encourages action or effort.
Goad is a stimulus to action (v) — to provoke to action.
To mortify is to reduce **potency** of **(potential)**

Answer: A

200. **SCALE : LADDER ::**
A. mail : letter
B. build : wall
C. climb : slope
D. weight : parcel

Clues: **Scale** to climb to the top of using a **ladder**.
Climb to tend steadily upward **(slope)**.

Answer: C

201. **CONFUSION : APLOMB ::**
A. suavity : urbanity
B. uncertainly : reason
C. mistake : error
D. anarchy : order

(the explanation is left for you)

Note: [: means **"is to"** :: **means "as")**

Answer: D

202. **LOGIC : REASONING**
A. ethics : behaviour
B. sensitivity : morality
C. creativity : enthusiasm
D. bravery : charisma

Clues: Use **logic** to **reasoning** (logical reasoning)
Use **ethics** to study **behaviour** (goodness /badness) of somebody **(ethical behaviour)**

Answer: A

203. **WEALTH : INCOME**
A. health : medicine
B. appetite : hunger
C. feeling : affection
D. salary : performance

Clues: If a person owns /earns a lot of **income**, he is said to be a **wealthy** person (a rich person).

- **income** results in **wealthy** as **performance** results in **salary.**

Answer: D

204. **HEAT : CALORIES**

A. liquid: gallons
B. lamp: watts
C. thermometer : degree
D. steam : pressure

Clues: **Heat** (energy) is measured by **calories** as **lamp** is measured by **watts**.

Answer: B

205. **COHABIT : RESIDE**

A. secret : truth
B. infiltrate : influence
C. conspire : plot
D. coincide : contradict

Clues: The two original words are **synonyms**. Both mean "**live together**".

Conspire means to secretly **plot** or make plans together.

Answer: C

206. **SAND : GLASS**

A. cotton : wool
B. seed : flower
C. stove : oven
D. tree : paper

Clues: **Glass** is made of **sand** whereas **paper** is made of **tree**.

Answer: D

207. **SEARCH : FIND**

A. sleep : wake
B. draw : paint
C. explore : discover
D. think : relate

Clues: **Search** and **find** are synonyms as **explore** and **discover** are.

Answer: C

208. **DOCTOR : DISEASE**

A. psychiatrist : maladjustment
B. broker : stocks
C. charlatan : truth
D. pediatrician : senility

Clues: A **doctor** attempts to treat a **disease**. A **psychiatrist** attempts to treat a **maladjustment.**

Answer: A (function)

209. **FISH : SCALES**

A. bird : feathers
B. snake : fangs
C. cat : claws
D. plane: wings

Clues: The body of a **fish** is covered with **scales**. The body of a **bird** is covered with **feathers.**

Answer: A (defining characteristics)

210. **SCISSORS : SEVER**

A. barrel : roll
B. scales : average
C. eraser : smudge
D. millstone : grind

Clues: **Scissors** by definition cut or **sever**. A **millstone** by definition **grinds**.

Answer: D (Tool and action)

211. **ANARCHY : GOVERNMENT**

A. penury : wealth
B. chaos : disorder
C. verbosity : words
D. ethics : philosophy

Clues: **Anarchy** is the absence of **government. Penury (poverty)** is the absence of **wealth**.

Answer: A

212. **ROBIN : NEST**

A. animal : cave
B. alligator : swamp
C. rabbit : burrow
D. horse : stall

Clues: A **robin** constructs a **nest** to live in. A **rabbit** digs out a **burrow** to live in.

Answer: C (Defining characteristic)

213. **TEAM : ATHLETES**

A. game : series
B. alliance : nations
preachers
C. term : holidays
D. congregations :

Clues: A **team** is made up of **athletes.** An **allians** is made up of **nations**.

Answer: B (Group and member)

214. **DIMMED : LIGHT**

A. bleached : texture
B. muffled : sound
C. measured : weight
D. fragrant : smell

Clues: **Light** that is **dimmed** is lessened in brightness.
Sound that is **muffled** is lessened in volume.

Answer: B (manner)

CLOCK: TIME

A. odometer: speed
B. watch: wrist
C. radio: sound
D. yardstick: distance

Clues: A **clock** measures **time.** A **yardistick** measures **distance.**

Answer: D (Function)

STETHOSCOPE : PHYSICIAN

A. transit : surveyor
B. studio : sculptor
C. plot : author
D. ruthless : compassion

Clues: A **stethoscope** is a tool of a **physician**. A **transit** (measuring instrument) is the tool of a **surveyor.**

Answer: A (worker and tool)

GUSTATORY : TASTE

A. hortatory : hearing
B. palpable : touch
C. kinesthetic : slight
D. olfactory : smell

Clues: **Gustatory** by definition means related to the sense of **taste.**

Olfactory by definition means related to the sense of **smell.**

Answer: D (Defining characteristic)

FISH : TROUT

A. ocean: wave
B. mammal: whole
C. bird: aviary
D. antenna: insect

Clues: A **trout** is a kind of **fish**. A **whole** is a kind **mammal.**

Answer: B (class and member)

HELMET : HEAD

A. amulet: shoulder
B. gauntlet: hand
C. planet: sun
D. scepter: crown

Clues: A **helmet** protects a **head**. A **gauntlet** (armored glove) protects a hand.

Answer: B (Function)

SOLDIER : REGIMENT

A. dancer: balletomane
B. gauntlet: hand
C. singer: chorus
D. colonel: martinet

Clues: A **solider** is part of a **regiment** (military unit)
A **singer** is part of a **chorus**

Answer: C (part of whole)

221. **INDUSTRY : BEAVER**

A. forgetfulness: elephant
B. indolence: wolf
C. frocity: lion
D. pride: peacock

Clues: The **beaver** is a symbol of **industry**. ("busy as a beaver").
The **peacock** is a symbol of **pride**. ("proud as a peacock").
(symbol and Abstraction it represents).

Answer: D

222. **STATIC : MOVEMENT**

A. humdrum : excitement
B. chronic : timeliness
C. diligent : industry
D. prestigious : wealth

Clues: Something **static (unmoving)** lacks **movement**.
Something **humdrum (dull)** lacks **excitement**.

Answer: A (Antonym variant)

223. **SPOKE : WHEEL**

A. book : shelf
B. rung : ladder
C. balance : lever
D. square : circle

Clues: A **spoke** is a part of a **wheel**. A **rung** is part of a **ladder**.

Answer: B (Part to whole)

224. **PAN : CAMERA**

A. tune : radio
B. rotate : periscope
C. charge : battery
D. ban : book

Clues: To **pan** a **camera** is to **rotate** it to get a comprehensive view. To **rotate** a **periscope** is to turn it to get a comprehensive view.

Answer: B

225. **VERTEX : CONE**

A. peak: mountain
B. pod: seed
C. whirlpool: pond
D. perimeter: rectangle

Clues: The **vertex** is defined as the top part of a **cone**.
A **peak** is defined as the top part.

Answer: A (Part of whole)

DISCONSOLATE : GRIEF

A. fatuous: weight
B. incurable: disease
C. solitary: confinement
D. explicit: statement

Clues: A **grief** that is **disconsolate** is not able to be eased or consoled. A **disease** that is **incurable** is not be able to be cured.

Answer: B (manner)

227. **DEGREE : TEMPERATURE**

A. ounce : weight
B. light : heat
C. time : length
D. mass : energy

Clues: A **degree** is a measure of **temperature**. An **ounce** is a measure of **weight**.

228. **INNING : BASEBALL**

A. round: boxing
B. serve: tennis
C. touchdown: football
D. puck: hockey

Clues: An **inning** is a division of a **baseball** game. A **round** is a division of a **boxing** match.

Answer: A (part to whole)

229. **HOBBLE : WALK**

A. gallop : run
B. stammer : speak
C. sniff : smell
D. stumble : fall

Clues: To **hobble** is to **walk** laboriously and with difficulty. To **stammer** is to **speak** laboriously and with difficulty.

Answer: B (manner)

230. **GEOLOGIST : FELDSPAR**

A. metrologist : orbit
B. botanist : zinnia
C. architeck : monolith
D. cosmetologist : space

Clues: A **geologist** studies rocks; **feldspar** is a kind of rock.
A **botanist** studies plants; a **zinnia** is a kind of plant.

Answer: B (Defining characteristic)

231. **IMPROMPTU : REHERSAL**

A. practiced: technique
B.
C. offhand: premeditation
D. glib: fluency

Clues: Something **impromptu** or improvised is performed without **rehearsal**. Something **offhand** is said or done without **premeditation** or advanced thought.

Answer: C (Antonym pattern)

232. **TOLERANCE : BIGOTRY**

A. prodigality: ribaldry
B. magnanimity: parsimony
C. emulation: rivalry
D. patience: conformity

Clues: **Tolerance** is the opposite of prejudice or **bigotry**. **Magnanimity** (greatness of spirit: generosity) is the opposite **parsimony** or stinginess.

Answer: C (Antonym pattern)

233. **WORDS : WRITER**

A. honor: thief
B. mortar: bricklayer
C. child: teacher
D. batter: baker

Clues: A **writer** creates a book out of **words**.
A **baker** creates a cake out of **batter**.

Answer: D (worker and material)

234. **GRAIN : SILO**

A. tree : acorn
B. seed : plant
C. water : bucket
D. druggist : doctor

Clues: **Grain** is kept in a **silo**.
Water is kept in a **bucket**.

Answer: C

235. **STANZA : POEM**

A. flag : anthem
B. story : building
C. mural : painting
D. rhyme : prose

Clues: A **stanza** is a subdivision of a **poem**. A **story** is a subdivision of a **building**.

Answer: B (Part to whole)

236. **HEART : PUMP**

A. lungs : collapse
B. appendix : burst
C. stomach : digest
D. intestine : twist

Clues: The function of **heart** is to **pump**: the function of **stomach** is to **digest**.

Answer: C (function)

237. **MASTER: SERVANT**

A. judge: jury
B. monarch: subject
C. serf : peasant
D. capital : labor

Clues: By definition, a **servant** is bound to do the bidding of his or her **master;** a **subject** is bound to do the bidding of his /her **monarch**.

Answer: B (Function)

238. **AMUSING : UPROARIOUS**

A. puzzling : dumbfounding
B. quiet : noisy
C. petty : narrow minded
D. exhausted : weary

Clues: Something **uproarious** is extremely **amusing**; something **dumbfounding** is extremely **puzzling**.

Answer: A (Degree of intensity)

239. **AIRPLANE : HANGAR**

A. sheep : channel
B. jet : runway
C. helicopter : pad
D. automobile : garage

Clues: A **hangar** is a place for servicing and storing **airplane**.
A **garage** is a place for servicing and storing **automobile**.

240. **THEROMEMETER : HEAT**

A. filament : light
B. chronometer : color
C. odometer : waves
D. geiger counter : radiation

Clues: A **thermometer** measure temperature or **heat**; a **geiger counter** measures **radiation.** **Answer: D (Function)**

241. **VIOLA : INSTRUMENT**

A. color : sound
B. spectrum : shade
C. fiddle : bass
D. chisel : tool

Clues: A **viola** is a kind of **instrument;** a **chisel** is a kind of **tool.**

Answer: D (member of class)

242. **PRIDE : LION**

A. lair : bear
B. bevy : quail
C. horn : unicorn
D. ensign : banner

Clues: A **pride** is a company or group of **lions**. Similarly, a **bevy** is a flock or group of birds, especially **quails**.

Answer: B (Group and member)

243. **ROOSTER : HEN**

A. duck: drake
B. dog: cat
C. gander : gosling
D. gander : goose

Clues: The **rooster** is the male of the species; the **hen**, the female. Likewise, the **gander** is the male of its specie; the **goose**, the female.

Answer: D (Gender)

244. **CHAIRMAN : GAVEL**

A. conductor: baton
B. violinist: bow
C. orator : dais
D. teacher : blackboard

Clues: A **chair person /man** uses a **gavel** to conduct meetings. A **conductor** uses a **baton** to conduct musical performance.

Answer: A (worker and tool)

245. **COBBLER : SHOES**

A. mechanic: automobile
B. carpenter: saw
C. painter: easel
D. interrogator: questions

Clues: A **cobbler** repairs **shoes** and a **mechanic** repairs **automobiles.**

Answer: A (Function)

246. **ASSURANCE : FEAR**

A. opiate : pain
B. insurance : premium
C. cigarette : cough
D. narcotic : drug

Clues: **Assurance** will allay (calm or quiet) **fear,** an **opiate** will allay **pain**.

Answer: A (class and member)

247. **BARBER : SEHARS**

A. banker : batter
B. dentist : drill
C. patient : prescription
D. architect : blueprint

Clues: The **barber** uses his **shears** (scissors) to trim hair; the **dentist** uses his **drill** to bore holes in teeth.

Answer: B (worker and tool)

248. **MENTOR : COUNSEL**

A. poet : criticism
B. plea : mercy
C. bodyguard : protection
D. judge : lawyer

Clues: A **mentor** (wise and trusted counselor) provides one with **counsel** (advice). A **bodyguard** provides one with **protection**.

Answer: C (Definition)

315

249. **AUTOMOBILE : GASOLINE**

A. train : caboose
B. cow : milk
C. disease : germ
D. man : food

Clues: **Gasoline** provide the energy for an **automobile,** and **food** is the energy for **man**.

Answer: D (Function)

250. **COMPOSER : SYMPHONY**

A. poet : sonnet
B. director : movie
C. actor : comedy
D. conductor : orchestra

Clues: **A symphony** is written by a **composer,** a **sonnet** is written by a **poet**.

Answer: A (worker and creation)

251. **DAMPEN : DRENCH**

A. glide : drift
B. gambol : play
C. simmer : boil
D. stagnate : flow

Clues: To **dampen** something is less intense than to **drench** it;
To **simmer** something is less intense than to **boil** it.

Answer: C (Degree of intensity)

252. **EXERTION : FATIGUE**

A. school : graduation
B. exercise : atrophy
C. sedation : tranquility
D. effort : result

Clues: **Exertion** (vigorous effort) causes **fatigue**.
Sedation (administration of a tranquilizing drug) causes **tranquility** or calm.

Answer: C (Cause and effect)

253. **CARDIOLOGY : HEARTH**

A. pathology : maps
B. apology : sorrow
C. pharmacology : drug
D. orthography : religion

Clues: **Cardiology** is the study of the **heart**.
Pharmacology is the study /science of **drugs**.

Answer: C (Definition)

254. **BACTERIUM : COLONY**

A. microbe : disease
B. fish : shoal
C. virus : immunization
D. sovereign : kingdom

Clues: A **bacterium** is a member of a **colony** (group of bacteria growing together).
A **fish** is a member of a **shoal** or school.

Answer: B (member and group)

255. **VISIONARY : PRACTICAL**

A. dilettante : amateurish
B. retainer : loyal
C. braggart : modest
D. rebel : revolutionary

Clues: A **visionary** (dreamer) is not **practical**.
A **braggart** (boaster) is not **modest**.

Answer: C (Antonym variant)

256. **JOURNALIST : TYPEWRITER**

A. surgeon : bones
B. carpenter : lumber
C. electrician : pliers
D. poet : beauty

Clues: A **typewriter** is a tool used by a **journalist**.
A pair of **pliers** is a tool used by **electrician**.

Answer: C (worker and tool)

257. **MUTTON : SHEEP**

A. bleat : lumb
B. sow : pig
C. hide : buffalo
D. beef : steer

Clues: The meat of a **sheep** is called **mutton**. The meat of a **steer** is called **beef**.

Answer: D (Definition)

258. **ENTRY : DIARY**

A. sonnet : ballad
B. paragraph : prose
C. missive : epistle
D. episode : serial

Clues: An **entry** is one day's cord that is part of a **diary**.
An **episode** is one separate performance that is part of a **serial**.

Answer: D (part of whole)

259. **BLIND : SIGHT**

A. diabetic : sugar
B. indigent : tact
C. amnesiac : memory
D. benevolent : charity

Clues: A **blind** person suffers from a loss of **slight**. Similarly, an **amnesiac** suffers from a loss of **memory**.

Answer: C (Antonym variant)

260. **STAGE : ACTOR**

A. rink : skater
B. safe : banker
C. library : lecture
D. baton : conductor

Clues: An **actor** performs on **stage**.
A **skater** performs at a **rink**.

Answer: A (worker and work place)

INVENTORY : MERCHANDISE

A. repertory : theatre
B. roster : members
C. gadget : profits
D. dormitory : college

Clues: An **inventory** lists **merchandise** or assets.
A **roster** lists **members**.

Answer: B (Defining characteristic)

INTERSET : FASCINATION

A. dislike : abhorrence
B. delusion : gullibility
C. exertion : fatigue
D. science : witch craft

Clues: **Interest** is a less intense feeling than **fascination.**
Dislike is a less intense feeling than **abhorrence** (loathing detestation).

Answer: A (Defining characteristic)

263. **SCOTCH : RUMOR**

A. divert : traffic
B. broach : topic
C. suppress : riot
D. spread : gossip

Clues: To **scotch** or block a **rumor** is to crush it.
To **suppress** or quell a **riot** is to crush it.

Answer: C (Purpose)

264. **AREA : SQUARE**

A. diagonal : rectangle
B. volume : cube
C. angle : triangle
D. base : cylinder

Clues: **Area** measures the size of a **square.**
Volume measures the size of **cube.**

Answer: B (defining characteristic)

265. **EULOGY : PRAISE**

A. elegy : death
B. slander : disparagement
C. paean : anger
D. reproof : confirmation

Clues: A **eulogy** is an expression of **praise**. A **slander** (statement damaging someone's reputation) is an expression of **disparagement** (depreciation, scorn).

Answer: B (Definition)

Review Exercise

Select the lettered pair that best express a relationship similar to that expressed in the original pair.

1. **TECTONICS : GEOLOGY ::**
   A. calculus : mathematics
   B. traffic : pathology
   C. mass : velocity
   D. astronomy : meteorology

2. **APORCRAPHAL : AUTHENTICITY ::**
   A. deceptive : artifice
   B. hypocritical : integrity
   C. nefarious : wickedness
   D. dogmatic : assertiveness

3. **REVERIE : DREAMER ::**
   A. soliloquy : ingenue
   B. pedantry : novice
   C. contrition : penitent
   D. modesty:braggart

4. **UNDERSTATEMENT : HYPERBOLE ::**
   A. digression : irony
   B. mainutrition : food
   C. metaphor : simile
   D. frugality : extravance

5. **BACKFIRE : MISCARY ::**
   A. entangle : mire
   B. decide vacillate
   C. launch : abort
   D. moisten : drench

6. **FLIGHT : FUGITIVE ::**
   A. trial : prisoner
   B. masquerade : impostor
   C. home : burglar
   D. espionage : detective

7. **LOOKOUT : WARNING ::**
   A. fire fighter: prevention
   B. guard : alertness
   C. sentinel : prediction
   D. meteorologist : forecast

8. **MEDLEY : MUSIC ::**
   A. collage : art
   B. harmony : sound
   C. choreography : dance
   D. repertory : theater

9. **FLEDGLING : BIRD ::**
   A. bevy : quail
   B. doe : deer
   C. foal : horse
   D. shoal : fish

10. **CARAPACE : TORTOISE ::**
    A. shell : crab
    B. roe : salmon
    C. fin :shark
    D. chameleon : lizard

SARONG : GARMENT ::

A. avocado : salad
B. marble : statue
C. hammock : furniture
D. hassock : footstool

DUB : KNIGHTHOOD ::

A. doubt : victory
B. christen : name
C. swear : oath
D. engage : battle

13. **TELLER : CAGE ::**

A. judge : gavel
B. clerk : register
C. witness : stand
D. doctor : scalped

14. **SOFA : FURNITURE ::**

A. squash : sport
B. file : drawer
C. bedroom : horse
D. ammunition : weapon

15. **BILLFOLD : DOLLAR ::**

A. blindfold : eye
B. briefcase : lunch
C. receptacle : jar
D. beak : eagle

16. **SIGNATURE : ILLUSTRATION ::**

A. note : scale
B. credit : purchase
C. byline : column
D. alias : charge

17. **COMPAS : DIRECTION ::**

A. radar : accuracy
B. altimeter : elevation
C. accelerator : speed
D. radio : volume

18. **WATER : PICHER ::**

A. gasoline : hose
B. oil : cruet
C. wine : vinegar
D. radio : volume

19. **CUB : LION ::**

A. tiger : tigress
B. pack : wolf
C. kit : fox
D. dam : beaver

20. **NAIL : PUNCTURE ::**

A. scalpel : incision
B. sword : scabbard
C. easel : painter
D. needle : magnet

21. **DOCTOR : DISEASE ::**

A. teacher : pupil
B. scholar : knowledge
C. psychiatrist : maladjustment
D. lawyer : law

22. **TIGER : CARVIVOROUS ::**
A. man : omnivorous
B. lion : quadruped
C. dog : herbivorous
D. fish : aquatic

23. **FERTILIZER : GROWTH ::**
A. nitrogen : grass
B. advertisement : sales
C. stimulant : stimulus
D. analysis : development

24. **SCALES : JUSTICE**
A. torch : liberty
B. balance : right
C. laws : courts
D. weights : measures

25. **SURPRISE : EXCLAMATIONS ::**
A. happiness : grimace
B. difference : nod
C. dismay : groan
D. insolence : bow

26. **DISCORDANT : HARMONIOUS ::**
A. cacophonous : euphonious
B. chronic : pacific
C. irate : choleric
D. critical : peaceful

27. **REST : COUCH ::**
A. relaxation : psychiatrist
B. exercise : gymnasium
C. fatigue : effort
D. instruction : correction

28. **APOSTATE : RELIGION**
A. traitor : country
B. jailer : law
C. teacher : education
D. potentate : kingdom

29. **NUMISMATIST : COINS ::**
A. magician : tricks
B. mathematician : money
C. prophet : wisdom
D. philatelist : stamps

30. **CHIP : PLAT ::**
A. scour : pot
B. snag : sweater
C. scoop : spoon
D. peel : carrot

31. **WHISPER : SPEAK**
A. brush : touch
B. skip : walk
C. listen : talk
D. request : ask

32. **BALLERINA : PIROUETTE ::**
A. actor : stage
B. catcher : mitt
C. jockery : mount
D. physician : disease

BRITTLE : BREAK ::

A. elastic : stretch
B. steady : lean
C. valuable : steal
D. buoyant : sink

EVACUATE : BUILDING ::

A. evaluate : project
B. abandon : ship
C. launder : garment
D. analyze : argument

DEADHEAD : CONTRIBUTE ::

A. diehard : resist
B. killjoy : dishearten
C. holdout : participate
D. windbag : talk

EDUCABLE : LEARN ::

A. vulnerable : wound
B. vindicable : judge
C. predictable : foretell
D. flexible : adapt

PERT : SAUCY ::

A. inventive : bored
B. generous : wasteful
C. wistful : listless
D. plain : homely

ELUSIVE : CAPTURE ::

A. headstrong : control
B. sensible : decide
C. gullible : trick
D. persuasive : convince

LINEAGE : PERSON ::

A. foliage : tree
B. derivation : word
C. title : book
D. landscape : portrait

IMPANEL : DUST ::

A. testify : witness
B. envolve : bystander
C. enroll : student
D. accuse : defendant

PULVERIZE : JUROR ::

A. vaporize : mist
B. analize : argument
C. solidify : fluid
D. permeate : odor

VISTA : PROSPECT

A. facsimile : edition
B. misfortune : luck
C. magnitude : extent
D. opportunity : signal

43. **PECCADILLO : TRIFLING ::**
A. squabble : petty
B. persuit : trivial
C. notion : farfetched
D. diagonisis : accurate

44. **SCRIMSHAW : IVORY ::**
A. kiln : ceramic
B. whittling : wood
C. collage : glue
D. wire : copper

45. **RIND : LEMON ::**
A. steam : tomato
B. shell : almond
C. pit : peach
D. aroma : vinegar

46. **BYSTANDER : UNINVOLVED ::**
A. newcomer : familiar
B. informer : nervous
C. wrongdoer : guilty
D. commuter : impatient

47. **PIPELINE : OIL ::**
A. vein : blood
B. goblet : wine
C. keg : beer
D. candle : wax

48. **GOOSE :GADER ::**
A. pigeon : flock
B. sheep : ram
C. hen : chicken
D. lion : mane

49. **LIZARD : REPTILE ::**
A. alligator : swamp
B. chameleon : color
C. bee : insectivore
D. man : primate

50. **PAARCHED : MOISTURE ::**
A. listless : energy
B. frail : delicacy
C. erect : posture
D. feverish : warmth

51. **TITLE : CARDCATALOG ::**
A. phone number : directory
B. register : roster
C. subject : essay
D. chapter : novel

52. **YEAR : TIME**
A. orbit : space
B. light-year : distance
C. ruler : length
D. calorie : weight

53. **RAVINE : EROSION**
A. reef : navigation
B. sandbar : precipitation
C. blight : conflagration
D. crater : explosion

54. **EXPEDITIOUS : SPEED ::**
A. decorous : impropriety
B. astute : wisdom
C. portable : frailty
D. haggard : sturdiness

55. **IRREFUTABLE : DISPROVED ::**
A. irreparable : mended
B. incomparable : applauded
C. intolerable : biased
D. insoluble : iradicated

56. **BROKER : STOCKS ::**
A. obtometrist : eye
B. bibliophile : books
C. realtor : land
D. historian : events

57. **MOROSE : EXULTANT ::**
A. subtle : sensitive
B. timorous : brave
C. lively : animated
D. tractable : docile

58. **ANNOTATE : TEXT ::**
A. caption : photograph
B. censor : film
C. enact : law
D. abridge : novel

59. **DRUDGERY : IRKSOME ::**
A. ambivalence : suspicious
B. dissonance : harmonious
C. journey : wearisome
D. encumbrance : burdensome

60. **RUBBER : ELASTIC ::**
A. paper : brittle
B. metal : heavy
C. diamond : hard
D. satin : sheer

61. **REAM : PAPER ::**
A. cord : wood
B. ink : pen
C. tablet : clay
D. carton : milk

62. **FLOOR PLAN : BUILDING ::**
A. flow chart : region
B. bargraph : prison
C. weather map : weather
D. eye chart : vision

63. **COLG : DRAIN**
A. clear : forest
B. time : race
C. water : lawn
D. obstract : highway

64. **SUSPENDERS : TROUSERS ::**
A. hem : skirt
B. cord : telephone
C. cables : bridge
D. wrappings : gift

65. **VALISE : CLOTHING ::**
A. hamper : food
B. crate : eggs
C. sofa : furniture
D. tank : helium

66. **HANDCUFFS : WRISTS ::**
A. gloves : hands
B. goggles : eyes
C. earrings : ears
D. leg-irons : ankles

67. **STARE : GLANCE ::**
A. scorn : admire
B. gulp : sip
C. hunt : stalk
D. confide : tell

68. **PERFORATE : HOLES ::**
A. filter : water
B. speckle : spots
C. evaporate : perfume
D. decorate : rooms

69. **LUMBER : MOVE ::**
A. tumble : handle
B. soar : fly
C. chatter : speak
D. rumba : dance

70. **POISON : ANTIDOTE ::**
A. acid : alkali
B. iodine : tincture
C. cause : effect
D. snakebite : serum

71. **PUGNACIOUS : BATTLE ::**
A. loquacious : drink
B. tenacious : persist
C. timorous : beg
D. voracious : lie

72. **CLEARSIGHTED : PERSPICACITY ::**
A. severe : clemency
B. transparent : opacity
C. daring : temerity
D. reserved : impulsiveness

73. **PLEAD : SUPPLIANT ::**
A. disperse : rioter
B. revere : elder
C. translate : temerity
D. shun : outcast

74. **EPIGRAM : PITHY**
A. allegatory : lengthy
B. proverb : modern
C. elegy : satiric
D. saga : heroic

75. **BARK: REDWOOD::**
A. peel : grapefruit
B. grass : lawn
C. bulb : tulip
D. date : palm

76. **TORNADO: WINDSTORM::**
A. whirlpool : eddy
B. ice : snowstorm
C. hailstone : sleet
D. lighting : thunder

77. **HARD HAT : HEAD ::**
A. cough drop : throat
B. wristwatch : wrist
C. hearing aid : ear
D. goggles : eyes

78. **CURIOUS : WONDER ::**
A. ignorant : heed
B. observant : notice
C. active : lead
D. just : justify

79. **REFEREE : JUDGE ::**
A. grumbler : complain
B. culprit : condemn
C. tutee : teach
D. player : object

80. **TELESCOPE : MAGNIFICATION ::**
A. computer : innovation
B. elevator : capacity
C. filter : purification
D. horoscope : truth

81. **KEY : LOCK ::**
A. switch : current
B. diskette : drive
C. tab : zipper
D. link : chain

82. **POCKED : PITS ::**
A. folded : layers
B. locked : cages
C. riddled : holes
D. split : peas

83. **UNKEMPT : COMBING**
A. filthy : cleansing
B. sick : ailing
C. undisturbed : resting
D. disheveled : housing

84. **FLIPPER : SEAL**
A. talon : hawk
B. blubber : whale
C. gill : salmon
D. wing : seagull

85. **UNDERGIRD : SEAL**
A. overwork : endeavor
B. undermine : weaken
C. expand : contract
D. support : adhere

86. **BUFFOON : DIGNITY ::**
A. gambler : risk
B. imposeter: identification
C. blackguard : strength
D. braggart : modesty

87. **ROUT : DEPEAT ::**
A. memory : oblivion
B. triumph : failure
C. ovation : applause
D. grief : loss

88. **METAPHOR : FIGURATIVE ::**
A. precept : instructive
B. irony : dramatic
C. epic : literal
D. fable : contemporary

89. **CALUMNY : ASPERSIONS**
A. expectation : threats
B. approbation : praise
C. slander : mockery
D. approval : criticism

90. **CAKE : CUPCAKE ::**
A. pie : tart
B. bean : pea
C. cracker : cookie
D. dough : bread

91. **DESERT : SAND ::**
A. marsh : mud
B. beach : ocean
C. tree : branch
D. valley : rock

92. **LIGHTBULB : SOCKET ::**
A. handle : knob
B. bolt : nut
C. hook : eye
D. wheel : car

93. **FLURRY : BLIZZARD ::**
A. hail : snow
B. sprinkle : drizzle
C. ripple : tidal wave
D. wind : current

94. **GIRDER : SUPPORT ::**
A. shield : protect
B. desk : write
C. vehicle : enclose
D. banner : droop

95. **DROWSY : DOZE ::**
A. hasty : think
B. lazy : work
C. dreamy : smile
D. anxious : fret

97. **STILTS : HEIGHT ::**
A. wheels : slope
B. padding : bulk
C. pounds : weight
D. blocks : distance

98. **ASH : VOLCANO ::**
A. mud : ditch
B. waves : cyclone
C. water : geyser
D. ice : glacier

99. **LOOKOUT : WARNING ::**
A. backup : retreat
B. sentinel : attack
C. seer : prophecy
D. scout : ambush

100. **CHAMPION : CAUSE ::**
A. defeat : reason
B. vanquish : misgiving
C. admit : injury
D. espouse : principle

101. **VACILLATE : IRRESOLUTE ::**
A. belittle : short
B. decide : tentative
C. modify : changeless
D. appease : conciliatory

102. **DEMAGOGUE : OFFICIAL ::**
A. tyrant : despot
B. tax collector : civil servant
C. leader : aide
D. pirate : sailor

103. **CLOISTER : SECLUSION ::**
A. theater : excitement
B. fortress : solitude
C. museum : monument
D. heaven : refuge

104. **DIMINUENDO : SOUND ::**
A. encouragement : progress
B. deceleration : speed
C. repayment : debt
D. clarity : vision

105. **LAST : SHOE ::**
A. pattern : glove
B. calf : trousers
C. strap : slip
D. buckle : belt

106. **MUTTON : SHEEP ::**
A. calf : cow
B. colt : filly
C. leash : dog
D. venison : deer

106. **DIET : WEIGHT ::**
A. bleach : colour
B. calculate : odds
C. coppy : pattern
D. measure : length

107. **SHEEP : WOOL ::**
A. plant : cotton
B. otter : fur
C. elephant : task
D. horse : fodder

108. **REPOSE : WEARY ::**
A. nourishment : hungry
B. shelter : thirsty
C. clothing : meek
D. protection : poor

109. **FOX : CANNING ::**
A. dog : playful
B. hyena : amusing
C. vixen : cute
D. beaver : industrious

110. **ENROLL : STUDENT ::**
A. evaluate : counselor
B. enlist : soldier
C. enact : lawyer
D. dismiss : employee

111. **SOUVENIR : MEMENTO ::**
A. prologue : play
B. apparel : attire
C. carousel : fair
D. journey : departure

112. **MEAT : DECAYED ::**
A. odor : aromatic
B. food : meager
C. fish : live
D. bread : stale

113. **EPAULET : SHOULDER ::**
A. medal : chest
B. sash : window
C. sword : scabbard
D. decoration : uniform

114. **LUNGS : BLOOD ::**
A. heart : circulation
B. balloon : helium
C. carburetor : gasoline
D. glands : secretions

115. **ADAMANT : UNYIELDING ::**
A. incoherent : sensible
B. timorous : intrepid
C. permanent : transient
D. warning : admonitory

116. **PERJURY : OATH ::**
A. testimony : court
B. embezzlement
C. jury : row
D. plagiarism : authority

117. **MARSUPIAL : KANGAROO**
A. reptile : shark
B. crustacean : lobster
C. mammal : platypus
D. amphibian : whale

118. **EULOGY : BLAME ::**
A. benediction : curse
B. simile : likeness
C. elegy : loss
D. satire : mocking

119. **DANCER : TERPSICHOREAN ::**
A. tailor : piscatorial
B. laborer : lercluean
C. singer : histrionic
D. actor : thespian

120. **CHECKER : SUPERMARKET ::**
A. teller : bank
B. pilot : airplane
C. surgeon : hospital
D. coach : gymnasium

121. **KAYAK : BOAT ::**
A. school : fish
B. tent : shelter
C. bumper : automobile
D. hangar : airplane

122. **BUTCHER : CLEVER ::**
A. baker : pastry
B. lumberjack : chainsaw
C. dentist : chair
D. architect : blueprint

123. **MOISTEN : DRENCH ::**
A. heat : chill
B. prick : stab
C. enclose : confine
D. disregard : ignore

124. **WITCH : COVEN ::**
A. fairly : spell
B. doctor : medicine
C. actor : trouge
D. seer : prophecy

125. **CONTINENT : ISLAND ::**
A. ocean : lake
B. cape : cove
C. river : canal
D. plateau : plain

126. **WINE : GRAPES ::**
A. bread : chest
B. rind : cheese
C. vodka : potatoes
D. milk : curds

127. **SKINFLIENT : STINGY ::**
A. spendthrift : weak
B. diehard : stubborn
C. daredevil : alert
D. blackguard : protective

128. **WHET : KNIFE ::**
A. grind : ax
B. home : strop
C. cut : scissors
D. polish : lens

129. **COWARDICE : BRAVE ::**
A. timidity : young
B. stature : tall
C. penury : wealthy
D. valor : intrepid

130. **STORY : BULDING ::**
A. foundation : skyscraper
B. cable : elevator
C. plot : outline
D. rung : ladder

131. **DEED : PURCHASE ::**
A. charity : donation
B. bequest : gift
C. license : marriage
D. citation : guilt

132. **PULMONARY : LUNG**
A. cardiac : heart
B. arterial : vein
C. neurotic : brain
D. livid : kidney

133. **ISOLATIONIST : NEUTRAL**
A. bigot : tolerant
B. braggart : modest
C. pessimist : lugubrious
D. altruist : selfish

134. **CANONIZE : SAINT ::**
A. deify : sinner
B. humanize : scholar
C. guard : dignitary
D. lionize : celebrity

135. **PORTHOLE : VESSEL ::**
A. jar : container
B. gate : fence
C. window : building
D. radar : car

136. **BIRD : FEATHER ::**
A. cat : whiskers
B. snake : rattles
C. lizard : claws
D. fish : scales

137. **CHAINS : CLANK ::**
A. bells : chime
B. birds : flutter
C. boards : warp
D. glasses : shatter

DICTIONARY : DEFINITION ::

A. cookbook : recipe
B. bank : interest
C. album : record
D. prediction : future

KNOT : WOOD ::

A. pane : glass
B. wart : skin
C. bow : ribbon
D. rind : cheese

ACTING : STAGE ::

A. singing : choreography
B. dancing : pirouette
C. skating : rink
D. painting : gallery

SNICKER : DISRESPECT ::

A. whimper : impatience
B. chortle : glee
C. sneer : detachment
D. glower : cheerfulness

MAP : CARTOGRAPHER ::

A. chart : captain
B. photograph : topographer
C. scheme : surveyor
D. blueprint : draftsman

FISHERMAN : NET ::

A. hunter : prey
B. astronomer : galaxy
C. physician : stethoscope
D. physicist : atom

REPROBATE : VILLAIN ::

A. disciple : follower
B. zealot : witness
C. hero : rogue
D. traitor : coward

MYTH : LEGENDARY ::

A. anecdote : witty
B. sermon : lengthy
C. fable : didactic
D. epic : comic

FATUOURS : INANE ::

A. clever : humane
B. complacent : dissatisfied
C. portentous : significant
D. ambiguous : clearcut

DERIVATION : LEXICOGRAPHER ::

A. evolution : biologist
B. adaptation : heretic
C. revolution : anarchist
D. renovation : contractor

148. **ANTHOROPOLOGY : MAN ::**
A. anthology : book
B. ichthyology : fish
C. geology : atmosphere
D. etymology : insects

149. **TIRADE : ABUSIVE ::**
A. aphorism : boring
B. critique : insolent
C. prologue : conclusive
D. encomium : laudatory

150. **DOG : LEASH ::**
A. horse : rein
B. ox : plow
C. bird : nest
D. monkey : trapeze

151. **KNIGHT : ARMOR ::**
A. king : throne
B. dancer : leotard
C. astronaut : spacesuit
D. soldier : platoon

152. **NOVEL : CHAPTERS**
A. poem : images
B. essay : arguments
C. opera : arias
D. play : acts

153. **AIR PRESSURE : BAROMETER ::**
A. distance : kilometer
B. blood pressure : artery
C. heat : thermometer
D. weight : altimeter

154. **LETTERHEAD : STATIONERY ::**
A. monogram : handkerchief
B. mailbox : envelop
C. ream : paper
D. circulation : newspaper

155. **CONSTELLATION : STAR ::**
A. league : federation
B. satellite : planet
C. candle : lamp
D. union : member

156. **TRAILER : MOTION PICTURE ::**
A. commercial : product
B. edition : novel
C. theater : play
D. truck : cargo

157. **CHISEL : SCULPTOR ::**
A. paint : painter
B. acid : etcher
C. paper : author
D. plane : carpenter

158. **SCHOOL : TUITION ::**
A. library : fine
B. church : tithe
C. lawyer : bail
D. hospital : insurance

159. **CHAUVINIST : COUNTRY ::**
A. patriot : loyalty
B. tourist : travel
C. miser : money
D. runaway : family

160. **MORPHINE : ALIEVIATE ::**
A. tobacco : fumigate
B. oil : smear
C. liquor : intoxicate
D. consolation : soothe

161. **INTREPID : COURAGE ::**
A. fearful : temerity
B. sagacious : wisdom
C. different : acuity
D. false : honor

162. **BOOTLESS : SERVICEABLE ::**
A. bottomless : profound
B. reckless : caution
C. grave : earnest
D. boundless : unlimited

163. **POLYGLOT : LANGUAGE ::**
A. controversial : argument
B. polychrome : time
C. repetitive : reinforcement
D. polygamous : spouse

164. **DEBATE : FORENSIC ::**
A. drama : histrionic
B. opera : choreographic
C. concerto : harmonic
D. argument : domestic

165. **EVICT : LANDLORD ::**
A. rule : monarch
B. edit : author
C. elect : mayor
D. expel : principal

166. **CUE : BILLIARDS ::**
A. club : gulf
B. ring : boxing
C. net : tennis
D. puck : hockey

167. **YACHT : HARBOR ::**
A. cruiser : ocean
B. ring : boxing
C. truck : trailer
D. limousine : garage

168. **SPRINKLE : RAIN ::**
A. simmer : boil
B. vanish : appear
C. soak : drench
D. wander : roam

169. **GIRDER : SUPPORT ::**

A. card : shuffle
B. axe : sharpen
C. ladder : lean
D. winch : hoist

170. **LAUNCH : ROCKET ::**

A. predict : weather
B. win : battle
C. kickoff : campaign
D. guide : missile

171. **PALOMINO : HORSE ::**

A. fawn : deer
B. collie : dog
C. sow : pig
D. pride : lion

172. **ROBE : JUDGE ::**

A. brief : lawyer
B. vestments : priest
C. helmet : quarterback
D. medal : hero

173. **DEPAULT : DEBT ::**

A. break : promise
B. borrow : loan
C. mistake : identify
D. dismiss : charge

174. **POACHER : GAME ::**

A. organizer : labor
B. jobholder : employment
C. golfer : sport
D. plagiarist : text

175. **SCHOONER : BEER ::**

A. cup : tea
B. boat : water
C. cask : wine
D. tanker : oil

176. **GROVEL : SERVILITY ::**

A. fume : anger
B. titter : arrogance
C. yawn : civility
D. preen : modesty

177. **PRODIGAL : FRUGALITY ::**

A. skeptic : incredulity
B. pessimist : hopefulness
C. zealot : enthusiasm
D. ascetic : self-denial

178. **DELICATE : FASTIDIOUS ::**

A. hardworking : diligent
B. deceptive : fallacious
C. altruistic : mercenary
D. demonstrative : effusive

179. **RICOCHET : BULLET ::**

A. carom : ball
B. deceptive : fallacious
C. soar : falcon
D. pierce : dart

180. **BARN : CATTLE ::**
A. flight : geese
B. track : train
C. flock : sheep
D. hanger : aircraft

181. **MUSEUM : DISPLAY ::**
A. factory : produce
B. office : supervise
C. court : dismiss
D. hospital : prevent

182. **PEN NAME : WRITER ::**
A. consume : cheerleader
B. disguise : journalist
C. address : pen pal
D. alias : agent

183. **PREACHER : SERMON ::**
A. lawyer : libel
B. doctor : treatment
C. coach : pep talk
D. teacher : quiz

184. **MATCHELESS : EQUAL ::**
A. heedless : action
B. aimless : treatment
C. fruitless : dessert
D. speechless : surprise

185. **ORDER : CHOAS ::**
A. wealth : poverty
B. conception : birth
C. share : distribution
D. organization : structure

186. **BACKUP : REINFORCEMENT ::**
A. counselor : advice
B. coach : excellence
C. upstart : intelligence
D. student : probation

187. **SCENERY : CONTEMPLATE ::**
A. fragrance : savor
B. summary : compose
C. vista : reveal
D. landscape : despoil

188. **INFIDEL : HEATEN ::**
A. wretch : pagan
B. ruler : monarch
C. miser : skinflint
D. coward : warrior

189. **DIGRESS : POINT ::**
A. detour : path
B. sharpen : edge
C. indicate : direction
D. support : argument

190. **ABDICATE : THRONE ::**
A. celebrate : victory
B. compete : trophy
C. resign : position
D. remove : threat

191. **INDOLENT : SLOTH ::**
A. covetous : enigma
B. gluttonous : loyalty
C. arrogant : acuity
D. wrathful : ire

192. **GROGGY : SLEEP ::**
A. greedy : food
B. heartless : encouragement
C. dreamy : revrie
D. penniless : cash

193. **STUDIO : SCULPTOR ::**
A. rehearsal : conductor
B. greenhouse : painter
C. museum : architect
D. courtroom : judge

194. **ANCHOR : BOAT ::**
A. steer : bicycle
B. gallop : horse
C. park : car
D. board : train

195. **SINGER : OPERA ::**
A. director : film
B. dancer : ballet
C. artist : studio
D. actor : rehearsal

196. **LAWYER : CLIENT ::**
A. judge : clerk
B. doctor : nurse
C. psychiatrist : patient
D. guard : prisoner

197. **ACCOUNTANT : LEDGERS ::**
A. publisher : magazine
B. principal : school
C. proofreader : text
D. lawyer : will

198. **CONTRACT : DISEASE ::**
A. incur : obligation
B. refer : patient
C. expand : organization
D. diminish : problem

199. **BLUNT : SHARPEN ::**
A. study : support
B. dull : polish
C. hard : indure
D. chilly : cool

200. **INCUBATOR : INFANT ::**
A. henhouse : chicken
B. hive : bee
C. cooler : wine
D. greenhouse : plant

201. **PRESEVERANCE : SUCCESS ::**
A. reasoning : logic
B. trepidation : failure
C. assiduity : fruition
D. monstrosity : fear

202. **DIAGRAM : SENTENCE ::**
A. eye chart : diagnosis
C. thesis : essay

197. **WOLF : DOG::**
A. tiger : cat
B. bear : cup
C. whale : salmon
D. lion : leopard

198. **LABORIOUS : EASE ::**
A. diligent : industry
B. evasive : frankless
C. cynical : suspicion
D. meticulous : accuracy

199. **COFFEE : BEAN ::**
A. coal : caffeine
B. beer : keg
C. tea : leaf
D. milk : cream

200. **MOLE : EARTH ::**
A. termite : wood
B. dog : manger
C. horse : hay
D. snake : rock

201. **RANGE : MOUNTAINS ::**
A. sea : rivers
B. novel : short stories
C. atlas : maps
D. string : beads

208. **INFRACTION : LAW**
A. interruption : continuity
B. renovation : structure
C. establishment : order
D. punishment : crime

209. **LIGHT : PHYSICS ::**
A. chemistry : biology
B. optics : mechanics
C. weight : gravity
D. zoology : biology

210. **PROOF : ALCOHOL ::**
A. octane : gasoline
B. tanker : oil
C. pulp : juice
D. canteen : water

Answer Key

1. A
2. B
3. C
4. D
5. A
6. B
7. D
8. A
9. C
10. A
11. C
12. B
13. C
14. A
15. A
16. C
17. B
18. B
19. C
20. A
21. C
22. A
23. B
24. D
25. C
26. A
27. B
28. A
29. D
30. B
31. A
32. A
33. A
34. B
35. C
36. D
37. D
38. A
39. B
40. C
41. A
42. C
43. A
44. B
45. B
46. C
47. A
48. B
49. D
50. A
51. A
52. B
53. D
54. B
55. A
56. C
57. B
58. A
59. C
60. C
61. A
62. C
63. D
64. C
65. A
66. D
67. B
68. B
69. A
70. D
71. B
72. C
73. C
74. D
75. A
76. A
77. D
78. B
79. A
80. C
81. B
82. C
83. A
84. D
85. C
86. D
87. C
88. A
89. B
90. A
91. A
92. B
93. C
94. A
95. A
96. B
97. C
98. C
99. D
100.D
101.D
102.D
103.B
104.A
105.D
106.A
107.B
108.A
109.D
110.B
111.B
112.D
113.A
114.C
115.D
116.B
117.B
118.A
119.D
120.A
121. B
122. B
123. B
124. A
125. A
126. C
127. B
128. A
129. C
130. D
131. C
132. A
133. C
134. D
135. C
136. D
137. A
138. A
139. B
140. C
141. B
142. D
143. C
144. A
145. C
146. C
147. A
148. B
149. D
150. A
151.C
152.D
153.C
154.A
155.D
156.A
157.B
158.B
159.C
160.D
161.B
162.B
163.D
164.A
165.D
166.A
167.D
168.A
169.D
170.C
171.B
172.B
173.A
174.D
175.A
176.A
177.B
178.D
179.A
180.D
181. A
182. D
183. C
184. B
185. A
186. A
187. A
188. C
189. A
190. C
191. D
192. D
193. D
194. C
195. B
196. C
197. C
198. A
199. B
200. D
201. C
202. B
203. A
204. B
205. C
206. A
207. D
208. A
209. D
210. A

UNIT FOUR

SENTNCE COMPLETION

- In this type of question, you are given a sentence with **one** or **two words** omitted. You are also given **four** possible choices. You have to select the best of the **four** possible answers provided.

Basic Strategies

1. Before you look at the choices, read the sentence and think of a word that makes sense. The word you think of may not be the exact word that appears in the answer choices, but it will probably be similar in meaning to the right answer.
2. Look at all the possible answers before you make your final choice. You are looking for the word that **best** fits the meaning of the sentence as a whole. In order to be sure you have not bean hasty in making your decision, substitute all the answer choices for the missing word. That way you can satisfy yourself that you have come up with the answer that best fits.
3. In double-blank sentence, go through the answers, testing the **first** word in each choice (and eliminating those that don't fit). Read through the entire sentence. Then, insert the first word of each answer pair in the sentence's first blank. Ask yourself whether this particular word makes sense in this blank. If the initial word of an answer pair makes no sense in the sentence, you can eliminate that answer pair.
4. Use your knowledge of word parts and context clues to get at the meanings of unfamiliar words. If a word used unknown to you, look at its context in the sentence to see whether the context provides a clue to the meaning of the word. Often authors will use an unfamiliar word and then immediately define it within the same sentence. Similarly, look for familiar word parts **-prefixes, suffixes,** and **roots-** in unfamiliar words.

5. Watch out for negative words and words signaling frequency or duration. Only a small change makes these two sentences very different in meaning.

   They were not lovers.

   They were not often lovers.

6. Look for words which indicate that the omitted portion of the sentence continues a thought developed elsewhere in the sentence. Examples are **and, moreover, in addition,** and **furthermore** In such cases, a <u>synonym</u> or <u>near-synonym</u> for another words in the sentences should provide the correct answer.

7. Look for words or phrases which indicate a contrast between one idea and another.

   Examples are **but, nevertheless, although, despite, however, eventhough, even though,** and **on the other hand**. In such cases, an <u>antonym</u> or <u>near-antonym</u> for another word in the sentence should provide the correct answer.

8. Look for words or phrases that indicate that one thing causes another –words like **because, since, therefore, thus, consequently, accordingly, hence, for**, etc.

UNIVERSITY ENTRANCE EXAMINATION AND OTHER QUESTIONS WITH DETAILED EXPLANATIONS. (2001 – 2009)

1. My _____ were _____ when I heard his explanation I was convinced that he was telling the truth.

   A. doubts… dispelled
   B. suspicious… confirmed
   C. fears…distracted
   D. misgivings…aroused

   <u>Clues</u>: If the listener believed the speaker, any **doubts** that he might have had, must have been **dispelled** (made to disappear).

   Answer: A

2. The owners of the spa advertised that their _____ were especially _____ for the arthritic.

   A. waters … healthful
   B. mountains…beneficial
   C. water…deleterious
   D. spring … toxic

   <u>Clues</u>: A spa is a mineral spring. Its **waters** are thought to be **healthful** (conductive to health).

   Note: The difference in meaning between **healthful** and **healthy** (having good health).

   Answer: A

3. Warner's emotion power and her magic as an author stem from her choice of details that _____ emotions rather than simply describing them.

A. evoke B. deny C. disdain D. diminish

Clues: The writer gains emotional force by choosing details that **evoke** or **call up** emotions. **Answer: A**

4. Excited and unafraid, the _____ child examined the stranger with bright-eyed curiosity.

A. drowsy B. inquisitive C. timorous D. hesitant

Clues: A curious child is by definition **inquisitive**.

Answer: B

5. Left to endure a penniless old age, the _____ man lives to regret his _____ youth.

A. wasteful….thrifty
B. miserly….friendless
C. improvident…prodigal
D. careful…prudent

Clues: A person who has spent a **prodigal** (wastefully extravagant) youth may live to regret his **improvident** (lacking foresight) ways. **Answer: C**

6. The word **tephra**, from the Greek word meaning ash, has come into use among geologists to describe the assortment of fragments, ranging from blocks of material to dust, that is _ into the air during a volcanic eruption.

A. repressed B. ejected C. amassed D. articulated

Clues: In a volcanic eruption, ash and other matter is **ejected** or forced out of the volcano. **Answer: B**

7. The strike was settled when the outside _____ brought the reluctant representatives of labor and management together to hear their _____ views.

A. auditors…benal
B. opportunists…tentative
C. media…irreconcilable
D. arbitrators…conflicting

Clues: **Arbitrators** (=people chosen to decide disputes) labor to reconcile people with **conflicting** views. **Answer: D**

8. Although the poet Stevie Smith always envied children their happy, simple existence, believing they alone has the ideal life, her own childhood was far from _____.

A. neutral B. solitary C. idyllic D. indifferent

Clues: Despite Stevie Smith's belief in an ideal childhood, her child hood wasn't **idyllic** or charmingly simple.

Answer: C

9. Although its publicity has been _____, the film itself is intelligent, well-acted, handsomely produced, and altogether

A. extensive…moderate C. perfect…spectacular

B. tasteless…respectable D. sophisticated…amateur

Clues: The sentences has the sense:

"The publicity has been **senseless**, but the film itself is …**respectable**.

Answer: B

10. Those people who are interested in _____ today are concerned about the effect of toxic wastes on our environment.

A. ecology C. geology

B. sociology D. entomology

Clues: **Ecology** is the study of the environment and its interrelationships with organisms; therefore, the ecologists would be fearful about the effect that toxic material would have on the environment.

Answer: A

11. In America, you have a _____ speech that is neither American, Oxford English, nor English but a _____ of all three.

A. nasal … mutilation

B. mangled … conglomeration

C. motley… conflagration

D. hybrid … combination

Clues: **Hybrid** means derived from mixed sources. **Answer: D**

12. Voting local, state, and national elections is the _____ of every American citizen.

A. duty B. limit C. emotion D. respect

Clues: Every citizen must fulfill the **obligation** or **duty**, of choosing the leaders of the country.

Answer: A

13. The doctor have ordered additional tests because the patient has not _____ as rapidly as had been expected.

A. Behaved
B. Qualified
C. recuperated
D. deteriorated

Clues: **Recuperate** means recover health. **Answer: C**

14. Even though his work is often immature, Fanta's latest effort is not _____; it is an intelligent report.

A. sophomoric
B. redundant
C. theoretical
D. atypical

Clues: **Sophomoric** means **immature**. **Answer: A**

15. The jury was convinced that some of the witnesses were guilty of _____ since _____ of conflicting testimony had been heard.

A. subornation… a refutation
B. lying … a minimum
C. falsification…an absence
D. perjury… a plethora

Clues: **Plethora** means an overabundant amount. If conflicting testimony exists in large amounts, there is always a possibility that some of the witness were guilty of **perjury** (lying under oath). **Answer: D**

16. The Science of _____ was largely influenced by Audubon and his magnificent paintings of American birds.

A. ornithology
B. virology
C. entomology
D. endocrinology

Clue: The study of birds is called **ornithology**. **Answer: A**

17. Although I do not wish to _____ your activities, I must insist that you _____ this assignment before you start anything else.

A. enlarge…cancel
B. body : germ
C. circumscribe …complete
D. compromise…accept

Clue: **Circumscribe** means limit the scope of. **Answer: C**

18. John Ferguson was widely _____ for his contributions to scientific research and was _____ for the noble prize.

A. castigated...proposed
B. acclaimed ... nominated
C. renowned...lauded
D. known...condemned

Clues: Only a person **acclaimed** (praised) for his work would be **nominated** (named) for a prize. **Answer: B**

19. Promotion in that company was based all too frequently on favoritism rather on _____.

A. ability B. loyalty C. nepotism D. seniority

Clues: A contrast to promotion by favoritism is promotion on the basis of **ability**. **Answer: A**

20. It is no uncommon occurrence, in the _____ world of theatre, to encounter extreme prosperity one week and financial ____ the next.

A. idealistic...greed
B. uncertain...betterment
C. dramatic...solvency
D. chancy...ruin

Clues: Someone in a financially uncertain or **chancy** profession would be likely to face extremes of prosperity and **ruin**.

Answer: D

21. He displayed his _____ nature when he contributed only 50 cents at a time while everyone else was donating at least five dollars.

A. sordid
B. penurious
C. specious
D. lethargic

Clues: **Penurious** means **stingy**. **Answer: B**

22. The orator delivered a _____ speech that was _____ by the more sophisticated in the audience who were not swayed by the pompous language.

A. fiery...ignored
B. eulogistic...derived
C. bombastic ...ridiculed
D. patriotic...cheered

Clues: A pompous speech may be described as **bombastic** such a speech would be **ridiculed** (laughed at) by those not influenced by pomposity. **Answer: C**

23. Such _____ act cannot be condoned; we must find the culprits and punish them.

A. a heinous
B. a sporadic
C. a fortuitous
D. an impromptu

Clues: **Heinous** means reprehensible, wicked.

Answer: A

24. I've no _____ in bringing this matter before this group; I am only interested in seeing that justice prevails.

A. obligation
B. preference
C. animus
D. predilection

Clues: **Animus** (=hostile feeling) is the only choice that fits this sentence.

Answer: C

25. Because your solution to this problem is so _____ I must oppose its adoption by this committee.

A. diabolical
B. practical
C. enigmatic
D. gratuitous

Clues: A **diabolic** (fiendish) solution would be opposed.

Answer: A

26. The trite and _____ situations in many television programs alienate many viewers.

A. strange
B. gruesome
C. novel
D. predictable

Clues: The key to this question is **'and.'** We need a synonym for **trite**. This is provided by **predictable**.

Answer: D

27. She should be _____ for her ruthless attack on the reputation of her hapless opponent.

A. admire B. affronted C. censured D. extolled

Clues: A **ruthless** (merciless) attack on a hapless (unfortunate) victim should be blamed or **censured**.

Answer: C

28. The somber news from the flood stricken area does not justify the attitude which you are displaying.

A. lugubrious
B. optimistic
C. belligerent
D. gloomy

Clues: Bad news never justifies an **optimistic** attitude.

Answer: B

29. We were annoyed by her _____ reply for we had been led to expect definite assurances of her approval.

A. vehement
B. articulate
C. noncommittal
D. caustic

Clues: Anyone expecting definite assurances would be irritated by a **noncommittal** (=non committing oneself to a definition course of action) reply.

Answer: C

30. Propaganda may influence the sophisticated and urbane as well as the more _____ members of the community.

A. gullible
B. blasé
C. worldly
D. knowledgeable

Clues: In this sentence, the use of **as well** indicates that we are looking for a contrast to the words **sophisticated** and **urbane**. This contrast is provided only by **gullible**. Propaganda will fool the **gullible** (=easily deceived).

Answer: A

31. Pacifists argued that our _____ attitude toward our former allies could only result in a more strained relationship.

A. placid
B. uncomfortable
C. truculent
D. harmonious

Clues: A **truculent** (=aggressively hostile) attitude will result in a **strained** (=tense) relationship.

Answer: C

32. Ato Habtamu has a bad habit of making _____ remarks which cause us to forget the gist of what he is saying.

A. awkward
B. tentative
C. telling
D. digressive

Clues: **Digressive** means wandering away. Such remarks can make a listener forget the topic.

Answer: D

33. The sales tax has been called _____by many; it presents a greater hardship to the _____ members of the state.

A. regressive…poorer
B. fair …. wealthier
C. liberal … marginal
D. retrospective …affluent

Clues: A tax is called **regressive** when it taxes unfairly those least able to pay. The **poorer** members of society pay a larger proportion of their income than the wealthier under a sales tax.

Answer: A

34. The child was not _____ in his response; he showed his _ openly.

A. candid …. anger
B. quick…eagerness
C. inhibited…emotions
D. slow…hesitation

Clues: **inhibited** means holding back. **Answer: C**

35. If you come to the conference with such _____ attitude, we can't expect to reach a _____ agreement.

A. an obdurate…harmonious
B. a complicated…conclusive
C. an indolent…satisfactory
D. a subservient….passive

Clues: **Obdurate** means stubborn and unyielding. **Answer: A**

36. I have no _____ in this matter. I am forced to follow the guidelines set forth in this manual.

A. grudge
B. prudence
C. latitude
D. wisdom

Clues: **Latitude** means freedom of action.

Answer: C

37. The _____ qualities of this salve will provide you with temporary relief from the pain which you now suffer.

A. obscure
B. soporific
C. obvious
D. analgesic

Clues: **Analgesic** means causing relief from pain.

Answer: D

38. I am seeking an _____ solution to this dispute, one which will be fair and acceptable to both sides.

A. infamous
B. equitable
C. obvious
D. indifferent

Clues: **Equitable** means just and fair. **Answer: B**

39. I object to this contract because it is _____; it stipulates my responsibilities but not yours.

A. universal
B. questionable
C. unilateral
D. indifferent

Clues: **unilateral** means one-sided. **Answer: C**

40. Before embarking on his first trip to Europe, Amana discussed his _____ with his friends and his travel agent; he left little to chance.

A. escapade
B. desires
C. itinerary
D. challenge

Clues: **Itinerary** means a detailed plan of journey. **Answer: C**

41. Because she was a _____ writer, she was able to complete four novels by the end of year.

A. controversial
B. prolific
C. veritable
D. voracious

Clues: **prolific** (=producing many works) best describes this writer. **Answer: B**

42. Your term paper is too wordy: try to make it more _____

A. succinct
B. succulent
C. verbose
D. superficial

Clues: **Succinct** means compact, free of unnecessary verbiage. **Answer: A**

43. Intolerance among _____ groups is deplorable and usually is based on lack of information about the cultural groups involved.

A. alien
B. aesthetic
C. cohesive
D. ethnic

Clues: **Ethnic** means pertaining to a race or nationality possessing common cultural qualities. **Answer: D**

44. We were amazed at the _____ of the culprits; they were not at all embarrassed by their arrest.

A. humility
B. brazenness
C. iniquity
D. impotence

Clues: If the culprits display no sense of shame or embarrassment they are guilty of **brazenness**.

Answer: B

45. In such _____ weather, you must dress properly in order to avoid catching a cold or the flu.

A. inclement
B. incompatible
C. salubrious
D. balmy

Clues: **inclement** means severe or stormy.

Answer: A

46. Many in the audience were yawning because the speeches were, to say the least _____

A. maudlin
B. lachrymose
C. soporific
D. salacious

Clues: **Soporific** means causing sleep.**Answer: C**

47. Though financially successful, the theatre season, once again, is more noted for its _____ than for its original productions.

A. revivals
B. musicals
C. failures
D. rehearsals

Clues: **Revivals** (=new productions of old players) contrast with original productions.

Answer: A

48. Paradoxically, while it is relatively easy to prove a fraudulent work of art is a fraud, it often virtually impossible to prove that an authentic one is _____.

A. genuine
B. dubious
C. objective
D. impartial

Clues: It is ironic that it is easier to prove something a take than it is to prove it **genuine** or real.

Answer: A

49. In their new collections of lighthearted, provocative dresses, French Fashion designers are gambling that even _____ professional women are ready for a bit of _____ in style.

A. serious ... frivolity
B. elegant ...tradition
C. strict ... reticence
D. modern ... harmony

Clues: There is a chance that **serious** women may not be attracted by an inappropriate **frivolity** or lightheartedness in style.

Answer: A

50. Using novel concepts and techniques previously unknown in commercial advertising, the _____ advertising campaign broke new ground in the filed of marketing.

A. imitative
B. innovative
C. interminable
D. inadequate

Clues: A campaign that breaks new ground is by definition **innovative**.

Answer: B

51. In observing the ceremonies and rituals of worship, we must not make a show of our faith: the challenge of religion is to be _ without becoming _____.

A. reverent...relevant
B. indulgent ... obvious
C. pious...sanctimonious
D. heretical...caught

Clues: Someone **sanctimonious** makes a show of religious faith; someone truly **pious** or devout does not.

Answer: C

52. In judging his guilt, the question remains whether he acted out of purely _____ motives or whether he acted with thoughts of his own _____ in mind.

A. disinterested...advantage
B. selfish...profit
C. malicious...cleverness
D. benevolent...impartiality

Clues: A **disinterested** (=unselfishly motivated) act would not be motivated by selfish thoughts of one's own **advantage**.

Answer: A

53. In the course of learning to deal with the world of ides, the adolescent gradually becomes able to express himself or herself in _____ such as courage and philosophy.

A. abstractions
B. proverbs
C. epigrams
D. hypotheses

Clues: Words like "courage" and "philosophy" are **abstractions** (=words representing ideas rather than concrete objects).

Answer: A

54. According to wildlife specialists, the few remaining pandas are so romantically _____ that their very survival.

A. regarded C. displayed

B. protected D. disinclined

Clues: Animals that were "romantically disinclined" might therefore not be inclined to reproduce. **Answer: D**

55. In order that they may be able to discriminated wisely among the many conflicting arguments put before them, legislators must be trained to _____ the truth.

A. discern B. disturb C. condone D. confuse

Clues: To make the correct decisions, the lawmakers must be able to **discern** or recognize the truth. **Answer: A**

56. It was only the first day of summer vacation, but his nerves were already _____ by the constant clamor of the children.

A. frayed B. vacillated C. eliminated D. alleviated

Clues: Nerves would be **frayed** or strained by constant clamor (noise).

Answer: A

57. Immediately after the Revolution, the bold men who had faced and beaten the British Empire were _____ by their ____ of the power of a strong central government and their fear of its becoming a tyranny.

A. bewildered…loss C. comforted…acquisition

B. paralyzed…distrust D. threatened…diversion

Clues: The combination of **distrusting** a powerful central government and fearing that is would be beyond their power to control **paralyzed** the founding fathers and left them powerless to act.

Answer: B

58. It is not _____ for a politician to _____ a mother on the beauty of her plain child; it is merely common sense.

A. wisdom…reprimand
B. hypocrisy…compliment
C. sensitivity…evaluate
D. kindness…ridicule

Clues: The statement is paradoxical. In **complimenting** the child of a potential voter, the politician isn't so much being **hypocritical** as being practical (showing political common sense)

Answer: B

59. Strindberg mistrusted the modern woman, believing she needed to _____ man and thereby invalidate his masculinity.

A. dominate B. abstract C. civilize D. comfort

Clues: Strindberg believed that by **dominating** or mastering man, woman would invalidate his masculinity.

Answer: A

60. One's contemporaries know one best; a man who cannot win honor in his own _____ will have a very small chance of winning it from posterity.

A. field B. right C. age D. way

Clues: If one's contemporaries don't give one honor in one's own **age**, one probably will not receive it from **posterity** (= future ages).

Answer: C

61. The health inspector reported back to the department that the _____ of the apartment was unbelievable is was difficult to realize men and women could live in such filth.

A. squalor
B. barrenness
C. propriety
D. availability

Clues: **Squalor** or filth would be criticized by a health inspector.

Answer: A

62. Because the periodic nature of her complaints had begun to consult a doctor about these _____ attacks.

A. trivial
B. tentative
C. superficial
D. recurrent

Clues: **Recurrent** or periodic attacks could well alarm a family.

Answer: D

63. She dropped her libel suit after the newspaper finally published a _____ of its original defamatory statement.

A. repetition
B. retraction
C. glossary
D. summary

Clues: A **retraction** or withdrawal of a libelous statement might satisfy the person with complaint. **Answer: B**

64. To take a _____ attitude often is to _____ any chance of favorable relation.

A. kindly…deny
B. promising…negate
C. patronizing…eliminate
D. sincere…reduce

Clues: A **patronizing** or condescending attitude may often others and **eliminate** good relations. **Answer: C**

65. He had long ago abandoned the moral principles of his youth and now led a debauched, _____life.

A. rudimentary
B. incontinent
C. circumspect
D. celibate

Clues: An **incontinent** or dissolute life would be described as debauched. **Answer: B**

66. Our mood swings about the economy grow more erratic; when things go well, we become _____; when things go poorly, _____ descends.

A. euphoric…gloomy
B. restive…anxiety
C. ascetic…misery
D. prudent…benevolence

Clues: The contrast is between **euphoric** or **ecstatic** moods and **gloomy** ones. **Answer: A**

67. Rather than allowing these dramatic exchanges between her characters to develop fully, Ms.Tesema unfortunately tends to _____ the discussions involving the two women.

A. augment
B. elaborate
C. exacerbate
D. truncate

Clues: Instead of allowing the exchanges to develop fully, the playwright cuts short or **truncates.** **Answer: D**